Short-Reach Optics for AI Compute

An Optical Systems Engineering Handbook: Design, Validation, Qualification, Manufacturing, and Fleet Debug

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Short-Reach Optics for AI Compute

From First Principles to the State of the Art

A concise technical overview of the optics that wire together large-scale AI systems. Compiled from public sources (IEEE, OIF, MSA specifications, vendor disclosures, and industry literature). Figures and third-party announcements are cited to their sources; any errors are the author's own.

Preface

Artificial intelligence has become an infrastructure problem. Training and serving frontier models at scale is no longer limited only by the accelerator at the center of the rack, but by how many accelerators can be wired together efficiently, reliably, and within a fixed power envelope. That wiring is increasingly optical, and the optics (especially the lasers inside them) have become a first-order lever on the cost, power, and reliability of the whole system.

This book is a concise technical overview of that layer: the short-reach optical interconnects that stitch together AI datacenters, from in-package optical I/O out to intra-rack links, deliberately setting aside the 2–10 km campus links that belong to coherent optics (Section 3.3). It concentrates on the subjects that decide whether these links work at scale: IM/DD physics and vocabulary; lasers and external light sources; WDM and wavelength locking; quantitative noise and sensitivity models; validation from bench to fleet; reliability qualification; and manufacturing validation at volume. Two chapters bracket those fundamentals: the orientation from optical science to a shippable interconnect (Chapter 1), and how AI datacenter networks are built where optics dominate cost and power (Chapter 10).

How to use this handbook. The book supports three modes. Pick one and stay in it until you switch deliberately.

Deep learning Read the body chapters from requirements downward (Chapter 1 through Chapter 10), then the failure-analysis handbook (Chapter 11). Every major chapter asks: How does it work? How is it measured, and what uncertainty does the measurement remove? How does it fail? How is it debugged?

Interview preparation Follow the one-week plan (Section A.13), practice cases (Chapter B), drill thirty-second answers and tradeoff questions (Chapter C), review the Top 25 index (Chapter E), and read how Staff judgment works (Chapter F).

Incident / problem solving Open the wall-chart trees (Chapter D), failure handbook (Chapter 11), case (Chapter B), Staff judgment (Chapter F), validation ladder

(Chapter 7 and table 7.2), qualification (Chapter 8), or manufacturing validation (Chapter 9).

Start from the requirement, not the component. Move downward from system requirements to architecture, subsystem, component, and only then to the physics needed to make the decision. A choice at one layer constrains the next. The aim is operational judgment: reduce uncertainty enough to make the next decision, then name the control that prevents recurrence.

On sources. The industry moves quickly. Where the text cites specific products or figures (co-packaged-optics programs, per-lane roadmaps, energy-per-bit trends) it draws on public disclosures current as of early 2026, cited in the references. Where a claim is an inference rather than an established fact, the text says so. History and trend notes are included where they help explain why today's defaults exist, not as a full chronology of the field.

Key idea. The goal of an optical systems engineer is not to know every component. It is to make good engineering decisions under uncertainty using measurements, physics, and evidence. At gigawatt, multi-generation scale, the optical interconnect and its lasers are a first-order lever on power, cost, and reliability. Everything here serves that argument.

Content freeze: this handbook is frozen for expansion except structural relocation of existing material. Prefer cuts, cross-refs, and errata over new frameworks.

Contents

Chapter 1

From Optical Physics to a Shippable Interconnect

Why system architecture, validation, reliability, manufacturing, and fleet operation belong in one book.

An optical interconnect is not merely a laser, modulator, fiber, and photodiode connected in sequence. It is a system that must satisfy bandwidth, reach, power, thermal, cost, reliability, manufacturing, interoperability, and operational requirements simultaneously.

A researcher may demonstrate an excellent modulator on a probe station. A laser engineer may produce high optical power and low relative intensity noise. A communications engineer may close a simulated link with sophisticated equalization. None of those achievements alone demonstrates that the resulting product can be cooled in a dense rack, assembled at acceptable yield, tested economically, operated for years, exchanged between suppliers, managed by firmware, or debugged after deployment.

That gap is the subject of this book.

The goal is not to replace textbooks on electromagnetic theory, semiconductor lasers, fiber communications, or digital signal processing. The goal is to show how those disciplines interact when an engineering organization must choose an architecture, close its budgets, validate real hardware, qualify credible failure mechanisms, manufacture at scale, and learn from the fleet.

This book teaches the layer between optical science and an operating product. It is written for technically strong readers who may know Maxwell's equations, semiconductor lasers, modulators, waveguides, or coherent communications deeply, but who are not necessarily exposed to complete product development: architecture selection under system constraints, margin allocation, validation ladders, lifetime qualification, manufacturing measurement systems, supplier control, and fleet learning.

The book therefore follows the life of an optical interconnect. It begins with the system need and the energy required to move a bit. It then develops the physical and quantitative models needed to understand IM/DD links. From there it examines architectural choices: copper or optics, multimode or single-mode, direct modulation or external modulation, one wavelength or several, lightweight equalization or substantial DSP. The later chapters address the questions that determine whether the design becomes a product: validation, reliability qualification, manufacturing validation, networking integration, failure analysis, and fleet learning.

The central lesson is that no technology choice is free. Every decision moves burden somewhere else. A lower-cost laser may demand more thermal control. More optical margin may consume power and lifetime. More DSP may improve tolerance while increasing power, latency, firmware complexity, and validation scope. A design that performs beautifully in the laboratory may be difficult to manufacture. A design that passes qualification may still fail through an uncontrolled supplier process or an unanticipated system interaction.

Optical systems engineering is the practice of seeing those connections early enough to make responsible decisions.

1.1 The gap between a device demonstration and a product

Evidence grows in levels. Success at one level does not guarantee success at the next.

Device demonstration Does the component produce the desired physical effect? Does a laser lase? Does a modulator achieve sufficient extinction? Does a detector have adequate responsivity? Does a waveguide have acceptable loss?

Link demonstration Can bits be transmitted and recovered? What BER is achieved, at what optical power and operating condition, and how sensitive is the result to temperature, noise, loss, and reflections?

Product validation Does the complete module meet intended system requirements with the host, peer, fiber plant, firmware, and thermal environment? Where are the operating cliffs, and how much margin remains (Chapter 7)?

Reliability qualification Which mechanisms may degrade the product? Does time or environmental exposure permanently change performance, and what lifetime claim does the evidence support (Chapter 8)?

Manufacturing validation Can the factory reproduce the qualified result? Can production measurements distinguish acceptable and unacceptable units? Are yield, variation, traceability, and supplier controls understood (Chapter 9)?

Fleet operation Does deployed behavior match the release model? Are failures clustering by lot, host, site, firmware, temperature, or age, and what changes after an escape (Chapter 11)?

Work from the system downward, then close the loop in the fleet:

```
Requirements
   |
Architecture
   |
Validation
   |
Deployment
   |
Fleet (telemetry / RMA / FIT)
   |
Feedback into requirements
```

Inside architecture, descend only as far as the requirement forces: requirements $\rightarrow$ architecture $\rightarrow$ subsystem $\rightarrow$ component $\rightarrow$ needed physics. Freeze an architecture class only when reach, lane rate, power, lifetime, cost, and manufacturing volume have named owners. Do not pick a laser die before those constraints rule out the other paths. Chapter 7 expands the validation-to-fleet segment into the full numbered lifecycle (Table 7.2 and section D.2).

1.2 Begin with the system need, not a preferred technology

Architecture should not begin with "we should use silicon photonics," "we should use coherent," "we should use VCSELs," "we should add DSP," or "we need more optical power." It should begin with requirements: bandwidth, lane rate, reach, topology, fiber availability, connector count, latency, power, cooling, cost, reliability, service model, supplier strategy, and expected production volume.

The correct technology is the one whose complete burden best fits the system, not necessarily the component with the strongest isolated performance. Architecture choices redistribute burden among signal margin, power, thermal management, complexity, reliability, manufacturing, test, control software, and field service. That redistribution is the recurring perspective of this book.

1.3 Why compare copper and optics?

The book discusses electrical interconnects even though its subject is optics because copper and optics compete at the system boundary. The decision depends on reach, data rate, insertion loss, equalization burden, connector and package loss, power, latency, density, cooling, routing, serviceability, and cost.

Copper may be preferable when reach is short, channels are well controlled, electrical loss is manageable, and low cost and simplicity dominate. Optics becomes attractive when electrical loss or equalization power becomes excessive, reach or routing becomes difficult, density grows, isolation matters, or the architecture benefits from moving the electrical-to-optical boundary closer to compute.

That boundary is not fixed. It moves as baud rates increase, packages change, SerDes improve, optical engines become denser, and cooling or service constraints change. The energy and channel arguments in Chapter 2 explain why the boundary moves and why optics becomes necessary before it becomes easy.

1.4 Why IM/DD rather than coherent, and when that changes

IM/DD (intensity modulation with direct detection) encodes information primarily in optical intensity. The receiver detects optical power rather than recovering the full optical field. Benefits for short reach include comparatively simple optics, lower DSP burden, lower latency, and often lower power and cost. Costs include sensitivity to intensity noise, limited access to phase information, dispersion and bandwidth constraints, and tighter dependence on eye quality and receiver performance.

Coherent recovery of amplitude and phase (often with polarization) buys spectral efficiency, stronger dispersion compensation, longer reach, and richer formats. It also buys a local oscillator, a coherent receiver, ADC/DAC burden, DSP, power, latency, calibration, and more qualification and manufacturing complexity.

The question is not whether coherent is more sophisticated. The question is whether its added capability is worth its system cost for the intended reach and bandwidth. Chapter 3 develops the short-reach IM/DD baseline; Chapter 10 notes where coherent begins moving inward on the roadmap.

1.5 Why multimode versus single-mode matters

Fiber choice selects more than fiber. Multimode systems may enable VCSEL-based sources, relaxed coupling in some implementations, established short-reach ecosystems, and potentially lower component cost, but they introduce modal bandwidth, modal dispersion, launch dependence, reach constraints, and fiber-plant assumptions.

Single-mode systems enable longer reach, reduced modal dispersion, WDM, silicon-photonic integration, and broader architectural scaling, but they introduce tighter optical alignment, coupling loss, wavelength control, reflection sensitivity, and more demanding packaging.

Choosing the fiber selects an ecosystem of sources, coupling methods, packaging tolerances, measurements, and failure mechanisms. Source and modulation decisions in Chapter 5 and wavelength control in Chapter 6 follow from that ecosystem choice.

1.6 Why VCSEL, DML, EML, or silicon photonics?

These are not merely competing devices. They allocate light generation, modulation, wavelength control, coupling, packaging, thermal control, manufacturing, and test differently.

VCSEL Short-reach multimode ecosystem, array integration, direct modulation, and wafer-level test opportunities, with modal behavior, bandwidth and reach limits, temperature dependence, and multimode-fiber constraints.

DML Compact direct modulation and fewer optical elements, with chirp, bandwidth, extinction, temperature behavior, and feedback or RIN sensitivity as typical burdens.

EML High-speed modulation and improved chirp or eye behavior relative to some DML paths, with bias control, thermal sensitivity, integration cost, and aging of both laser and modulator functions.

Silicon photonics with external or integrated laser Integration of modulators, routing, WDM, and monitoring across lanes and wavelengths, with coupling, laser attach or external-laser interfaces, ring or MZM control, heater power, calibration, process variation, and packaging complexity.

The device choice determines which problems disappear and which problems move into control, packaging, DSP, qualification, or manufacturing. Detail lives in Chapter 5.

1.7 Why DSP, or why not DSP?

DSP is neither inherently good nor evidence of a bad optical design. It can provide equalization, clock-recovery assistance, compensation for bandwidth limits, adaptation across units and temperature, improved tolerance to channel variation, FEC, and telemetry. It also costs power, latency, silicon area, firmware complexity, startup and adaptation behavior, interoperability risk, additional validation corners, and potentially harder failure isolation.

Minimal-DSP or analog-heavy designs place more burden on optical quality, electrical bandwidth, packaging, and channel control. Moderate equalization balances hardware quality and adaptive correction. DSP-heavy designs may tolerate greater physical impairment but create a more complex system whose adaptation, telemetry, and failure behavior must be validated.

DSP does not remove impairment; it changes how the impairment is managed and where the engineering risk resides (Chapters 3 and 4).

1.8 Why WDM and wavelength control?

Adding wavelengths raises aggregate bandwidth per fiber and can reduce fiber count, with packaging and routing advantages. It also creates a wavelength grid, laser drift, filter passbands, thermal crosstalk, locking, capture versus hold, mux/demux loss, adjacent-channel crosstalk, and service or replacement behavior.

WDM trades fiber simplicity for wavelength-control complexity. That burden is developed in Chapter 6.

1.9 Why quantitative models matter

The book develops optical power, OMA, extinction ratio, receiver sensitivity, noise, RIN, BER, dispersion, bandwidth, link budgets, and thermal response because a quantitative model lets you predict behavior, identify dominant terms, allocate margin, choose a measurement, detect double counting, and know when a result is surprising.

Models are bounded by assumptions. A model is valuable when its assumptions are visible and its prediction can be checked against hardware (Chapter 4 and section 7.7).

1.10 Why validation, reliability, and manufacturing are separate

A product working once does not establish distribution, corner behavior, margin, interoperability, lifetime, manufacturability, or field readiness. Validation turns requirements into evidence. The canonical lifecycle in Chapter 7 is:

1. Define requirements.
2. Review architecture.
3. Bring up hardware.

4. Characterize behavior.
5. Validate margin and interoperability.
6. Qualify reliability.
7. Validate manufacturing.
8. Run a controlled pilot.
9. Ramp production.
10. Monitor the fleet.
11. Feed learning back.

Do not treat those Steps as interchangeable.

Validation asks whether the product works for its intended use now. Reliability qualification asks whether named mechanisms threaten its ability to continue working over time and environmental exposure: laser aging, thermal fatigue, humidity, corrosion, ESD, vibration, connector durability, acceleration models, sample confidence, and acceptance criteria. Qualification is not a collection of harsh tests. It is a confidence argument connecting a requirement, mechanism, stress, observable, and decision (Chapter 8).

A qualified engineering unit does not prove that a factory can reproduce the result. Manufacturing validation covers production-reference freeze, representative builds, genealogy and traceability, measurement-system analysis, first-pass yield, ATP, gauge R&R, process capability, SPC, supplier controls, change control, and ramp decisions. Reliability qualification asks whether the design survives. Manufacturing validation asks whether the production system can reproduce, measure, control, and protect it (Chapter 9).

1.11 Why networking and fleet learning belong in the loop

An optical module exists inside switch or accelerator architecture, topology, radix and bandwidth requirements, rack power, cooling, cable plant, redundancy, and deployment or service procedures. A component metric may not map directly to system value: lower module power may reduce cooling burden; additional reach may simplify topology; higher radix may reduce stages; a serviceable pluggable may beat a denser but hard-to-repair architecture; an optical technology may shift failures from replaceable modules into board-level assemblies (Chapter 10).

Shipment is not the end of validation. The fleet may reveal rare interactions, process escapes, supplier correlations, installation problems, aging, environmental corners, firmware effects, and cohort differences. Failure analysis teaches how to preserve evidence, scope the population, separate symptom from mechanism, choose discriminating measurements, confirm the mechanism, contain risk, and change a recurrence control. A failure is not fully resolved when the unit works again. It is resolved when the affected population is understood and an effective recurrence control changes (Chapter 11 and section 7.12).

1.12 How to read this book

Chapter 1 Orientation: what complete problem are we solving?

Chapter 2 Energy and physical limits: why moving bits becomes difficult, and when optics becomes attractive.

Chapter 3 IM/DD: how the short-reach optical link carries information.

Chapter 4 Noise, RIN, and BER: what limits signal quality quantitatively.

Chapter 5 Sources and modulation: which transmitter architecture best fits the system.

Chapter 6 WDM and wavelength control: how multiple wavelengths scale capacity, and what control burden they create.

Chapter 7 Validation: how to build evidence from requirements through fleet deployment.

Chapter 8 Reliability qualification: what evidence supports life and environmental claims.

Chapter 9 Manufacturing validation: whether the factory can reproduce, measure, and control the qualified design.

Chapter 10 Networking: how the optical product affects datacenter architecture.

Chapter 11 Failure analysis: how to investigate escapes and feed learning back.

The Preface describes reading modes (deep learning, interview preparation, and incident response). This chapter maps the engineering story those modes walk through. For a design drill, pick one link style (retimed 800G DR, LPO, or CPO WDM) and trace it through Sections 3.2, 10.3 and 10.10.

1.13 A recurring example: an 800G short-reach link

Illustrative scenario only. Suppose the organization needs an 800G short-reach link between AI compute and a switch. Reach and density make a purely electrical solution difficult. The team compares multimode and single-mode: single-mode enables a longer plant and WDM but increases coupling and wavelength-control burden. A DML may reduce component count but may not close the bandwidth and chirp budget. An EML or silicon-photonic modulator may improve signal quality but adds bias, thermal, packaging, or control complexity. DSP may recover bandwidth margin but increases power and validation scope. The architecture closes on paper. Hardware is characterized and challenged across realistic system corners. Reliability qualification targets laser, package, humidity, and interface mechanisms. Manufacturing validation establishes test correlation, yield, traceability, and process control. A controlled pilot checks assumptions in deployment. Fleet evidence feeds the next revision. The later chapters are the detail behind each of those sentences.

1.14 What the reader should learn

This book aims to teach you to:

- reason from system requirements rather than preferred devices;
- compare architectures through their complete burden;
- build quantitative link and margin arguments;
- distinguish power from signal quality;
- select measurements based on uncertainty reduction;
- separate characterization, validation, qualification, and production testing;
- connect reliability stress to failure mechanisms;
- validate manufacturing measurements and process controls;
- interpret fleet failures as population and mechanism questions;
- communicate evidence, confidence, decisions, and remaining risk.

You are not expected to memorize every table. The desired outcome is: given an unfamiliar optical-system problem, identify the important constraints, choose the next useful evidence, and make a defensible engineering decision.

1.15 Why the interconnect matters at AI scale

The orientation above is general. The pressure that makes it urgent in this book comes from AI compute: vertically integrated, purpose-built silicon at gigawatt scale, with networking named as a first-order design axis beside compute and memory.

A representative public example is *Jalapeño*, a purpose-built LLM inference accelerator Broadcom and a hyperscaler partner announced in 2026 as a blank-slate design, the first chip in a multi-generation compute platform.[1] Public features of the class include Broadcom silicon and networking (including *Tomahawk*), Celestica board and rack integration, a roughly nine-month design-to-tape-out cadence, and gigawatt-scale multi-generation deployment planned from late 2026. The design thesis is to reduce data movement and balance compute, memory, and networking so realized utilization approaches theoretical peak. Once networking sits on that line, the optical interconnect is how the system scales past a single package, and laser quality plus IM/DD validation become infrastructure problems rather than module afterthoughts.

[1] Source: Broadcom and partner, "LLM-optimized inference chip," 24 June 2026.

Inference is not training. Prefill is highly parallel and compute-bound; decode is autoregressive and memory-bandwidth-bound (Chapter 10 and section 10.6). Frontier models are sharded, so every generated token triggers collective communication: all-reduce for tensor parallelism, all-to-all for mixture-of-experts routing, point-to-point for pipeline stages. The interconnect therefore sits on the latency critical path of inference, not merely the plumbing between training runs.

1.15.1 The shifting bottleneck

Each generation of AI infrastructure has been limited by a different resource. The progression explains why optics moved from a commodity NIC accessory to a first-order design axis:

Compute-limited Early deep learning (pre-2016). GPUs were scarce; models fit on one card; the network barely mattered.

Memory-limited Larger models and batches (2016–2020). HBM bandwidth set training throughput; the network carried gradients but was rarely the gate.

Network-limited Sharded frontier models (2020–present). Collectives put fabric bandwidth and tail latency on the critical path (Sections 10.6 and 10.7).

Power-limited Gigawatt-class deployments (emerging). Site megawatts cap total capacity; every pJ/bit the interconnect saves is a watt returned to compute (Section 10.13).

The bottleneck did not replace the previous one; it stacked on top. A modern cluster is simultaneously memory-bandwidth-bound in decode, network-bound in collectives, and power-bound at the site. The interconnect sits at the intersection of the last two.

1.15.2 AI clusters are communication machines

An accelerator does useful work only while its operands arrive on time. Three traffic patterns set the pressure:

All-reduce combines partial results across a group; the slowest path can hold the whole group (Section 10.7).

All-to-all moves different payloads between every pair, as in mixture-of-experts routing; it tests bisection bandwidth and path balance.

Point-to-point carries pipeline stages and storage traffic; tail latency and retries can matter more than average throughput.

More compute increases the traffic the system can inject. It does not increase fabric capacity. Public large-model training data show why this is also an availability problem: network faults are only one failure bucket, but a synchronous job pays for every interruption across the whole allocation [23]. Compute scales faster than communication unless the network, optics, and software are designed as one system.

1.16 The debugging pyramid

When a link fails, work from the top down. The pyramid is a scope order inside the Staff loop (Section A.1), not a second philosophy. Apply the power-versus-quality fork early (Sections D.4 and 4.8) and organize lost margin with the five ledgers (Section 5.19).

```
System symptom
   |
Signal quality (BER / FEC / eye)
   |
Link budget (power / noise / timing / spectrum)
   |
Subsystem bisect (Tx / channel / Rx)
   |
Confirmed mechanism (evidence)
   |
Decision + recurrence control
```

Layer 1: System Workload symptom: BER, throughput, latency, collective stall. The symptom often rules out entire subsystems.

Layer 2: Signal quality Eye, jitter, noise, equalization margin, FEC error distribution (Sections 7.8 and 7.12).

Layer 3: Link budget Optical power, sensitivity, insertion loss, extinction ratio, ORL (Section 7.7).

Layer 4: Subsystem Laser, modulator, driver, photodiode, TIA, DSP, connector, fiber, or host SerDes; bisect with loopbacks and golden swaps (Sections 7.9 and 7.10).

Layer 5: Confirmed mechanism Leading hypothesis until evidence confirms; then FA or 8D (Section 9.2).

Do not skip layers. A mechanism story without a confirmed system symptom and a localized subsystem is a story, not a close.

Engineering heuristic. Scope before mechanism. A fleet-wide sudden event is almost never solved by staring at one connector.

1.17 Interview takeaway

Key idea. An optical interconnect is a chain of coupled engineering decisions. Copper versus optics, multimode versus single-mode, IM/DD versus coherent, source and modulation architecture, DSP, WDM, packaging, qualification, manufacturing, and fleet operation cannot be optimized independently. This book develops the physics required to understand those choices, but its larger purpose is to show how the choices become evidence, products, and operating systems.

Three opening questions.

1. Why can the best-performing optical component produce the wrong system architecture?
2. What is the difference between demonstrating a link, validating a product, qualifying reliability, and validating manufacturing?
3. When an architecture removes one impairment, where might it move the engineering burden?

Chapter 2

First principles: the energy of moving a bit

Before any device, modulation format, or standard, one quantity governs short-reach interconnect design: the energy required to move a bit from one place to another. David Miller's work on optical interconnects to silicon lays out the clearest first-principles framework for this,[1] and although the specific numbers are years old, the scaling arguments are what matter, and they still decide where the optics go.

[1] D. A. B. Miller, "Device requirements for optical interconnects to silicon chips," *Proc. IEEE*, 2009; and "Attojoule optoelectronics for low-energy information processing and communications," *JLT*, 2017.

2.1 The electrical baseline: charging a wire

To send a bit down an electrical line you charge and discharge its capacitance through a voltage swing. To order of magnitude,

$$E_{\text{elec}} \approx \tfrac{1}{2}\, C\, V^2,$$

and the line capacitance grows with length, $C \approx c' L$ (with c' on the order of a hundred-odd femtofarads per millimeter, depending on the medium).[2] So the energy to move a bit electrically *rises with distance*, and the resistive-capacitive delay and the equalization needed to fight it rise with both distance and data rate.

[2] The exact prefactor depends on the signaling scheme (voltage-mode, current-mode, terminated transmission line) but the *length dependence* is the first principle that matters here.

2.2 The optical alternative: energy at the ends

An optical link spends its energy differently. The dominant costs are the *conversions at the two ends* (driving the laser or modulator, and the receiver that turns light back into charge) plus the wall-plug inefficiency of the laser. Over short reaches, waveguide and fiber loss are small, so the energy per bit is, to first order, *independent of* *distance*.

Key idea. Electrical energy per bit grows with length; optical energy per bit is set at the endpoints and is roughly flat with length. That single contrast is the seed of everything else.

2.3 The break-even distance, and why optics moves toward the chip

Put the two together and there is a cross-over length: below it, electrical wins; above it, optical wins. The decisive observation is what happens as data rates climb: the electrical cost at a given length rises (more charging events per second, worse loss at higher frequency, more equalization), so the *break-even distance shrinks*.

That is the whole history of the field in one sentence. As rates went from gigabits to hundreds of gigabits per lane, the cross-over marched inward: campus, then rack, then board, then package, and now die-to-die. It is the first-principles reason co-packaged optics and in-package optical I/O exist (Sections 6.6 and 10.10), and the reason the scope of this book is the *shortest* links (Section 3.3).

2.4 The receiver, capacitance, and attojoule targets

Miller's sharpest point concerns the receiver. To register a bit, the photocurrent must develop a detectable voltage on the receiver's input node, so the detection energy again looks like a CV^2 on that node's capacitance. Minimize the photodetector and input capacitance (by integrating the detector tightly with the first transistor, eliminating parasitic pads and wires) and you can detect a bit with *fewer photons and less energy*. This is the argument for close electronic–photonic integration, and it is what makes sub-100 fJ/bit (and, in principle, attojoule-class) devices conceivable.

[3]

[3] This is why integration is not a packaging convenience but an energy strategy: shortening the electrical path between detector and amplifier lowers capacitance, which lowers both the energy and the optical power a link needs.

2.5 The floors: photons per bit and noise

Two physical floors bound how far this can go. A real receiver needs enough photons per bit for adequate signal-to-noise: today hundreds, with ideal detection in the handful-of-photons range. That sets a floor on received optical power, and hence on laser energy. Separately, the kT/C noise on the receiver node (not the Landauer $kT \ln 2$ limit of logic, which is far smaller) is the practical noise floor for communication, and it too rewards small capacitance.

Miller is careful to separate the energy of *logic* from the energy of *communication*; short-reach interconnect is overwhelmingly a communication-energy problem, dominated by the endpoints described above.

2.6 Recent progress toward the limits

Miller's numbers are years old, but the framework has aged well: the last few years have been a steady march of experiments toward the floors it predicts, and they validate rather than overturn it. Three threads are worth knowing.

First, **low-capacitance 3D integration is delivering the receiver energy Miller argued for**. A 2025 demonstration integrated an 80-channel transceiver in three dimensions, stacking the photonics directly on the CMOS to minimize the receiver-node capacitance, and reported roughly 120 fJ$/$bit.[4] Tellingly, it reaches that number not by pushing per-lane rate but by using *many slow channels* (about 10 Gb$/$s each) so each receiver stays in its most sensitive, lowest-energy regime, with WDM providing the aggregate bandwidth. That is Miller's prescription almost verbatim.

[4] S. Daudlin *et al.*, "Three-dimensional photonic integration for ultra-low-energy, high-bandwidth interchip data links," *Nature Photonics* 19:502–509, 2025.

Second, **the capacitance argument is directly measurable**. A co-designed 12 nm-FinFET-on-silicon-photonics transceiver using direct-bond interconnect cut input parasitic capacitance by about 75 %, which bought ~ 6 dB of receiver sensitivity and pushed link energy to a few hundred fJ/bit.[5] The CV^2 story is not a metaphor; you can watch the sensitivity move as the capacitance drops.

[5] P.-H. Chang *et al.*, "A 3D integrated energy-efficient transceiver … co-designed 12 nm FinFET and silicon photonic ICs," *JLT* 41(21):6741–6755, 2023.

Third, **WDM receivers with near-zero-power wavelength control are approaching sub-pJ/bit at terabit scale**: a single-chip 32-channel WDM PAM4 receiver reached 1.024 Tb$/$s on one fiber at under 0.38 pJ$/$bit with no DSP or equalization.[6] And on the device side, alternative emitters are being pushed into the same regime, a 2025 microLED link demonstrated 200 fJ$/$bit transmitter energy at BER < 10^{-12} with no FEC,[7] exactly the LED branch of Miller's device-by-device comparison.

[6] A. Pirmoradi *et al.*, "A single-chip 1.024 Tb$/$s silicon photonics PAM4 receiver," arXiv:2507.12452, 2025.

[7] Avicena, ECOC 2025 demonstration of the LightBundle microLED interconnect.

> **Key idea.** The 2009/2017 framework has not been superseded; it has been confirmed in hardware. The winning recipe in every recent result (low-capacitance co-/3D-integration plus many WDM channels at modest per-lane rate) is precisely what Miller's energy accounting recommends. What has changed is only the vertical axis: from theoretical attojoule targets toward demonstrated hundreds-of-fJ/bit links.

2.7 The energy ladder

A useful system model orders operations by how far information must move:

$$\begin{array}{c}\text{arithmetic operation}\\ \downarrow\\ \text{local SRAM access}\\ \downarrow\\ \text{HBM access}\\ \downarrow\\ \text{package crossing}\\ \downarrow\\ \text{board crossing}\\ \downarrow\\ \text{rack crossing}\\ \downarrow\\ \text{datacenter crossing.}\end{array}$$

The exact energy at each rung depends on process, signaling, utilization, and reach. The ordering is the point: capacitance, equalization, conversion, and cooling costs grow as the bit leaves the compute core. Miller's endpoint model explains why the optical crossover moves inward as electrical rate rises [2, 3]. A design review should therefore count bytes moved and distance crossed, not only arithmetic throughput.

2.8 Debug discussion: why more compute can make training slower

When a larger accelerator count produces worse step time, treat it as a scaling failure, not proof that the compute is slow.

1. **Observe.** Compare useful compute time, collective time, queueing, retransmits, and idle time at each scale point.

2. **Hypothesize.** Check for a saturated fabric tier, uneven routing, stragglers, memory pressure, or a synchronization barrier that grows with group size.

3. **Measure.** Plot collective latency and delivered bandwidth against accelerator count. Correlate link error and retry counters with the tail of the step-time distribution.

4. **Bisect.** Hold the model fixed and vary topology, group size, message size, or route. A roofline-style split between compute and data movement keeps the diagnosis tied to measured limits [71].

5. **Act.** Fix the limiting rung: reduce communication, rebalance the collective, add fabric capacity, shorten the electrical reach, or move the optical conversion closer to the package.

Debug story

Observed. Training throughput fell when the accelerator count doubled.
Investigation. Compute kernels held the same efficiency, while all-reduce tail latency and link retries rose on one fabric tier.
Finding. The added workers spent more time waiting at the collective barrier than computing.
Root cause. The network tier was oversubscribed for the new group size, and one marginal link amplified the tail.
Resolution. The team rebalanced routes, removed the weak link, and set the next scale gate on measured collective latency rather than peak FLOPs.

2.9 Engineering lens

2.9.1 How it works

Electrical energy per bit grows with distance; optical energy is spent at the endpoints and is flat with distance. So as lane rates rise and the electrical cost climbs, the crossover where optics wins marches inward toward the chip. Tight electronic-photonic integration lowers receiver capacitance, cutting both energy and the optical power a link needs.

2.9.2 How it is measured

Energy per bit (pJ/bit) is the headline metric, measured end to end from host SerDes through the optical link. Break it into electrical (SerDes, driver, TIA), electro-optic (modulator, laser wall-plug), and thermal (cooling overhead, PUE). Compare against the roofline: bytes moved times energy per byte against the power envelope.

2.9.3 How it fails

A link that meets its energy target on a quiet bench can exceed it in the field when equalization depth increases, retransmits rise, or thermal control draws extra current. A poor co-packaging yield or high coupling loss forces higher laser power and erases the integration advantage. A firmware update that increases DSP activity raises module watts without changing the optics.

2.9.4 How it is debugged

Compare measured link power against the design budget, split by block. If the total exceeds the target, identify which block moved: SerDes, driver, laser bias, TEC, or DSP. Correlate power anomalies with BER, temperature, and equalizer state. A link that draws more power than expected is often also running at reduced margin.

Engineering heuristic. A power rise after a firmware update is often deeper equalization, retries, or thermal control, not a new laser physics. Check what the software asked the hardware to do.

Junior mistake: quote faceplate watts without a measured energy-per-bit split, or treat peak FLOPs as proof the fabric is not the limit (Chapter 10 and section 10.7).

2.10 Interview takeaway

Key idea. Short-reach optics is, at bottom, an energy-per-bit optimization. Optical energy is spent at the endpoints and is flat with distance; electrical energy grows with distance and rate; so rising data rates push optics ever closer to the chip. Miller's framework is the lens the rest of this book looks through.

Three questions to test yourself.

1. Why does electrical energy per bit grow with distance while optical does not?
2. At what reach does the optical crossover occur for your target lane rate, and what assumptions drive that number?
3. More accelerators were added but training throughput fell. Where on the energy ladder is the bottleneck?

2.11 Why this framework anchors the book

Everything that follows is an effort to approach these floors at the required data rate and reliability. Laser wall-plug efficiency (Chapter 5) sets how much optical power you can afford. Modulation and FEC (Chapter 3 and sections 3.6 and 3.12) trade SNR for reach and rate. WDM (Chapter 6) amortizes one laser source across many wavelengths. Receiver noise and sensitivity (Chapter 4, section 4.4.3, and section 7.7) decide whether that power is enough. Co-packaging and energy-per-bit trends (Section 10.13) show the same first principle playing out as copper reach collapses and optics move onto the interposer.

Chapter 3

Intensity modulation, direct detection

3.1 What IM/DD means

Almost every short-reach datacenter link still uses the same basic deal with physics: put data on optical *power*, and recover it with a photodiode. That is *IM/DD*: *intensity modulation, direct detection*. The transmitter encodes bits as changes in optical intensity; the receiver is a photodiode plus a transimpedance amplifier (TIA) that turns photocurrent back into voltage. There is no local oscillator and no coherent mixing.[1] Detection is square-law (photocurrent proportional to optical power), so phase is discarded. That limitation is acceptable inside the rack and across a few hundred meters of fiber, where cost, power, and latency matter more than spectral efficiency. It is why IM/DD, not coherent, wires AI compute fabrics today.

[1] Contrast with *coherent* detection, which mixes the signal against a local-oscillator laser to recover amplitude *and* phase. Coherent wins for long-haul spectral efficiency; IM/DD wins on cost and power for short reach.

3.2 The IM/DD transceiver chain

Every pluggable module and every co-packaged engine is a rearrangement of the same chain. Once you can name the blocks, you can place equalization, FEC, and validation measurements without getting lost in form-factor jargon.

```
Modulation / platform choice
   |
Driver / EQ placement
   |
Test points (faceplate, host, optical)
   |
FEC / retimer burden
   |
Validation and ATP surface
```

Transmit laser or CW source → modulator (EML, MZM, ring) → driver → fiber coupling (Chapter 5 and section 3.14.3).

Channel fiber, connectors, MUX (Section 7.2.2 and section 6.3).

Receive photodiode → TIA (optional CTLE) → SerDes/CDR or linear ADC to host (Sections 3.6 and 4.5).

Digital KP4 FEC in host or retimer (Section 3.12); module DSP optional (Section 10.3, section 10.5.1, and table 10.4).

Exit when you can name the blocks, the test planes, and where FEC and equalization sit before diving into a modulator family. **Decision unlocked:** place measurements and ATP surface for this link class, or reopen the platform choice in Chapter 5.

The rest of this chapter fills in modulation physics, equalization, FEC, and the modulator platforms. Noise math and measurement practice live in Chapters 4 and 7.

3.2.1 A pluggable link, end to end

The block list above is one module. A working rack-to-rack link chains two of them back to back across a fiber, and it helps to trace a single bit through the whole path once. Figure 3.1 follows that path as a folded loop: the transmit side runs left to right across the top, the fiber turns the corner on the right, and the receive side runs back along the bottom into the far switch.

Start at the switch ASIC in rack A. It builds Ethernet frames, runs the reconciliation and coding sublayers, and encodes KP4 FEC (Sections 3.8 and 3.12), then hands parallel bit streams to the host *SerDes*. The SerDes serializes each stream to a 112 GBd PAM4 lane, applies transmit FFE, and drives the host PCB and the module cage connector. That electrical hop, host to module, is the AUI (an OIF VSR-class channel; Sections 3.3 and 3.14). Inside the module, an optional DSP or retimer reshapes the lane, or for a linear pluggable (LPO) nothing does and the host SerDes owns the whole electrical budget (Section 3.6 and section 3.14.2). The driver then swings a modulator, an EML, Mach–Zehnder, or ring (Section 3.14.3), and electrons become photons. That is the first domain crossing, marked E→O in the figure.

The fiber plant is the quiet middle: duplex or parallel single-mode fiber, connectors, and patch panels carrying the light a few meters to a few hundred meters (Section 7.2.2 and section 3.3). At the far module the light hits a photodiode and TIA and becomes current again (O→E, the second crossing; Section 4.5). An optional module DSP cleans it up, and the rack B host SerDes recovers timing, equalizes the lane, and decodes KP4. The switch ASIC reassembles frames and forwards them into the fabric, out a NIC, and over the in-node link (PCIe or an NVLink-class fabric) into the destination GPU or CPU. Every pluggable link is this shape; form factors and CPO only rearrange where the boundaries fall.

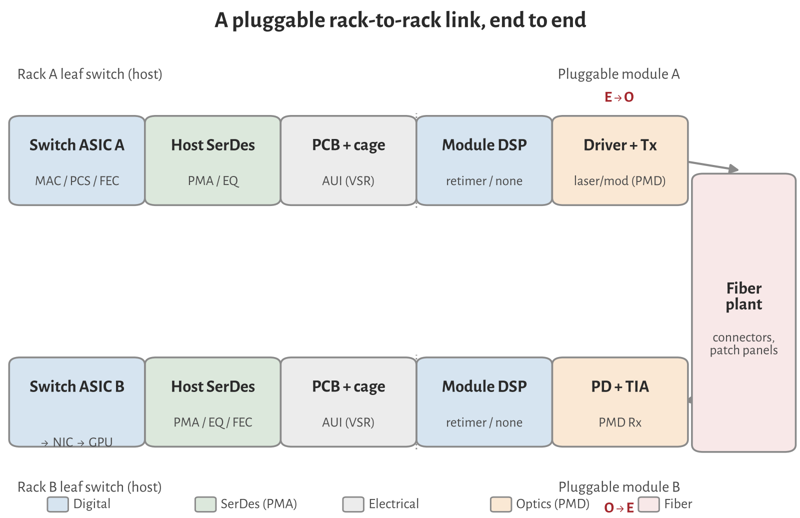

Figure 3.1: A pluggable rack-to-rack link, drawn as a folded loop. Transmit runs left to right (switch ASIC ⟶ host SerDes ⟶ cage/AUI ⟶ module DSP ⟶ driver and optics), the fiber turns the corner, and receive runs back into the far switch, then on to the NIC and GPU. Colors mark the domain of each block; the two E↔O crossings sit at the optics.

This is the *scale-out* path, the rack-to-rack Ethernet fabric where optics already dominate. The *scale-up* fabric that ties GPUs together (NVLink- or UALink-class; Section 10.1) is a different link. Its reach is much shorter, so most of it is still copper: direct-attach cable or backplane under roughly a meter, with no optics in the path at all. Optics only appears once rack densification pushes those links past the copper wall, and even then it borrows the same IM/DD building blocks over a different link and protocol layer. So the chain in Figure 3.1 is the one that carries the optical volume today; scale-up is where optics is arriving next.

3.3 Reach regimes, and the scope of this book

"Optical interconnect" is a wide phrase. A die-to-die link of a few millimeters and a 10 km campus span both move bits on light, but they solve different problems with different physics. AI compute fabrics live at the short end of that spectrum, where lasers, IM/DD, and validation dominate cost, power, and reliability. This book stays there and largely sets aside the 2–10 km campus and data-center-interconnect links that have moved toward coherent optics.

The dividing line is roughly where a link leaves the compute fabric. Inside that boundary, from millimeters in-package out to a few hundred meters, IM/DD is the workhorse and the laser is the component that gates scale. That is the territory this book treats.

Regime	Distance	Notes
In-package / die-to-die	mm–cm	optical I/O chiplets (e.g. Section 6.6)
Chip-to-chip / co-packaged	cm	CPO engines on the switch/XPU substrate
Scale-up (rack scale)	$\sim$1–10 m	XPU-to-XPU; copper below $\sim$1–2 m, optics beyond
Intra-rack / intra-row	up to $\sim$100–500 m	SR over MMF; DR single-mode
Campus / DCI *(out of scope)*	2 km (FR), 10 km (LR), and beyond	increasingly coherent, not IM/DD

Table 3.1. Reach regimes. This book focuses on the top four.

3.4 Where coherent takes over

3.4.1 The two detection schemes, in one paragraph

IM/DD encodes bits as changes in optical intensity and recovers them with a square-law photodiode. Phase and polarization are discarded at detection. Coherent detection mixes the received field against a narrow-linewidth local-oscillator (LO) laser on a balanced photodiode pair, recovering amplitude and phase on both polarizations. The receiver digitizes in-phase and quadrature samples and runs heavy DSP: carrier recovery, polarization demux, chromatic dispersion compensation, and soft-decision FEC. The coherent transmitter typically needs an I/Q modulator (or equivalent) under the same linewidth discipline. That stack buys spectral efficiency and reach tolerance. It costs extra photonic parts, ADC/DSP die area, power, and latency.

3.4.2 Why short reach stays IM/DD

Inside the compute fabric, from millimeters in-package out to a few hundred meters of single-mode fiber, loss and dispersion are modest: standard G.652.D fiber runs about 0.2 dB/km at 1550 nm with a zero-dispersion wavelength near 1310 nm, so a few hundred meters costs well under a decibel [91]. A PAM4 IM/DD link with equalization, FEC, and a reasonable laser closes without coherent mixing. The energy argument from Chapter 2 applies directly: IM/DD keeps the expensive digital work at the electrical ends where process scaling helps, and the optical path is passive fiber. Rack and row runs also lean on bend-insensitive G.657 fiber in patch panels and shelves, and the SR/DR reach classes below map onto the OM4/OM5 multimode and OS1/OS2 single-mode cabling classes that ISO/IEC 11801 and TIA-

492 define for the datacenter plant [92]. Retimed 800G pluggables run roughly 14–18 W; LPO trims to roughly 7–9 W by deleting module DSP [72]. A 400ZR-class coherent pluggable is built around a coherent DSP that alone draws roughly 5–8 W inside a 15 W module budget [85], landing near 45 pJ/bit at 400 Gb/s. Over a 100 m rack link that extra DSP power does not buy enough margin to justify the BOM and service complexity (Section 10.13).

3.4.3 What coherent buys, and its price

Coherent wins where the channel is harder: kilometers of fiber, amplified DWDM, and dispersion that would crush a directly modulated PAM4 eye. Polarization multiplexing and higher-order QAM raise bits per symbol. Electronic dispersion compensation removes the need for tight wavelength and chirp control on the transmit laser. LO mixing gives a sensitivity boost that IM/DD only matches with more launched power. The price is upfront: linewidth requirements on both the transmit laser and the LO (often well below 100 kHz for long reach), I/Q modulators or integrated coherent PICs, high-speed ADCs, and a DSP that dominates module power. OIF standardized 400ZR coherent pluggables with a 15 W module budget for DCI spans to roughly 120 km [85]. ZR+ variants push reach further at similar or higher power. That trade is right for campus and metro DCI. It is not the default inside an AI rack.

3.4.4 The crossover, and where it is moving

Today the practical line sits near 2 km: intra-datacenter DR and shorter hops stay IM/DD (Section 3.3 and table 3.13); campus FR/LR and DCI have moved to coherent 400ZR/ZR+ (Table 3.1). Pressure pushes coherent inward on per-lane rates: at 224G and especially 448G the IM/DD SNR and TDECQ margin tighten, and some proposals explore coherent-lite or coherent receivers for the longest scale-out hops [10, 14]. Pressure keeps IM/DD dominant for AI compute: LPO and CPO attack the power wall (Section 10.13 and section 3.14.2) by stripping module DSP and shortening electrical reach, not by adding an LO and ADC chain. For inference fabrics at in-package to a few hundred meters, IM/DD remains the workhorse through the 224G generation. The open question is 448G and beyond: whether the last scale-out meters stay PAM4/PAM6 IM/DD or whether a stripped coherent option appears for links that already run retimed modules today (Section 3.14.3).

3.5 PAM4: more bits per symbol

Through the 10G and early 25G eras, most short-reach optics used two-level NRZ: one eye, one decision threshold, simple receivers. As per-lane rates climbed past about 50 Gb/s, keeping NRZ would have demanded more electrical bandwidth than connectors and SerDes could afford. The industry answer was *PAM4*: four amplitude levels carrying two bits per symbol. Line rates below are in gigabaud (*GBd*); host I/O is built in *SerDes* blocks.

GBd Gigabaud: 10^9 symbols per second. Line rate = baud $\times$ bits per symbol.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

Table 3.2: Per-lane rates and symbol rates (PAM4). Ethernet lane rates and OIF CEI electrical classes are related but not interchangeable; see Sections 3.12 and 3.14.

Per-lane rate	Symbol rate	Context
100G/lane Ethernet	≈53.125 GBd	IEEE 802.3df-class PMD/AUI
200G/lane Ethernet	≈106.25 GBd	IEEE 802.3dj-class PMD/AUI
CEI-224G electrical	≈112 GBd	OIF CEI host/module SerDes class

PAM4 halves the required bandwidth versus NRZ for a given bit rate. At equal outer swing and equal additive noise, adjacent PAM4 levels are separated by one-third of the binary separation, an ideal spacing penalty of $20\log_{10}(3) \approx 9.54$ dB. That is not every implementation's total sensitivity penalty; equalization, DSP, and FEC are part of the architecture (Section 3.6). Looking ahead, the 448G debate is partly about whether to stay on PAM4 at still higher baud or move to PAM6/PAM8 to ease the electrical channel (Section 3.14.3).

3.6 Equalization and clock recovery

Once PAM4 became the default, the electrical channel stopped looking like a wire and started looking like a filter. PCB traces, connectors, and cables low-pass the signal; the receiver must undo that inter-symbol interference (ISI) before slicing. Three equalizer classes appear in every short-reach link, plus clock recovery when the bit clock is rebuilt.

ISI Inter-symbol interference: energy from neighboring symbols overlapping the current unit interval.

CTLE Continuous-time linear equalizer: analog high-frequency boost in SerDes or TIA.

CTLE (continuous-time linear equalizer) provides analog high-frequency boost, often a zero-pole pair in the SerDes front-end, the TIA, or a redriver chip. CTLE is cheap and low latency but fixed: it cannot adapt tap-by-tap to an arbitrary channel.

FFE Feed-forward equalizer: FIR filter. TDECQ applies a bounded FFE as the reference equalizer.

FFE (feed-forward equalizer) is a finite-impulse-response filter with pre- and post-cursor taps. Host SerDes and module DSPs use FFE (often with CTLE ahead of it) to open the eye. Transmitter and dispersion eye closure quaternary (TDECQ) applies a *bounded* reference FFE when scoring transmitters (Section 7.4).

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

DFE (decision-feedback equalizer) cancels post-cursor ISI using past symbol decisions. CEI electrical eye budgets reference an 8-tap DFE at the slicer (Section 10.5.2). DFE creates feedback-loop timing complexity and error propagation; system latency depends on implementation and any additional block processing or retiming. It recovers more ISI than FFE alone at high loss.

DFE Decision-feedback equalizer: cancels post-cursor ISI using past symbol decisions.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

CDR (clock-data recovery) extracts bit timing from the data stream and re-samples the eye. A *retimer* includes CDR plus equalization and regenerates a clean output; a *redriver* has CTLE/VGA but no CDR (Section 10.5.1).

Where these blocks sit depends on the module style (Section 10.3 and chapter 10):

- **Fully retimed pluggable:** host SerDes → connector → module DSP (FFE/DFE/CDR) → optical engine. The module cleans a bad electrical channel.
- **Redriver / ACC:** CTLE (+ VGA) in the cable or mid-channel; host SerDes still owns CDR and heavy EQ.
- **LPO:** no module DSP/CDR. Host SerDes EQ and FEC must survive the full path; module may add only CTLE in the TIA and a linear driver (Section 10.5.1). Optical TDECQ and electrical margin both tighten.

Figure 3.2 shows the correct SerDes equalizer order, then where those blocks live in a retimed module versus LPO.

> **Key idea.** CTLE boosts analog bandwidth; FFE/DFE cancel ISI digitally; CDR rebuilds timing in retimers. LPO deletes the module-side safety net, so host equalization and KP4 margin (Section 3.12) become the whole electrical story.

3.7 SerDes and DSP: who does the work

The equalizers above have to run somewhere, and "SerDes" and "DSP" get used loosely for that somewhere. They are not the same thing, and at 112 GBd the line between them has mostly dissolved. Pinning down the vocabulary makes the retimed versus LPO versus CPO argument (Section 10.5.1 and chapter 10) concrete instead of a wall of acronyms.

3.7.1 What the SerDes is

SerDes is the high-speed electrical I/O macro on an ASIC: switch silicon, a NIC, an accelerator, or a module chip. On transmit it serializes wide, slow parallel data from the MAC/PCS into one PAM4 lane on the wire; on receive it deserializes the sampled lane back to parallel words. That is the literal serializer/deserializer job. Everything else the SerDes does is in service of getting a clean symbol stream across a lossy channel.

Figure 3.2: Equalization chains. (A) Correct order on one electrical lane: Tx FFE, channel, CTLE, Rx FFE, DFE, then slicer/CDR. (B) Fully retimed pluggable: module DSP owns FFE/DFE/CDR on both sides of the optics. (C) LPO: no module DSP; the host SerDes must close the full EQ and FEC path.

3.7.2 Analog SerDes versus DSP-based SerDes

How the SerDes conditions the signal splits into two design styles. An *analog* (mixed-signal) SerDes equalizes in the analog domain (CTLE, a few analog FFE/DFE taps) and slices directly. It is low power and low latency but limited in how much loss it can undo and how flexibly it adapts. A *DSP-based* SerDes puts a high-speed ADC on the receive lane, digitizes the eye, and does FFE, DFE, and timing recovery numerically, with a DAC and digital pre-distortion on transmit. It burns more power and adds latency, but it closes far more channel loss and adapts tap-by-tap. Above roughly 50 GBd the DSP-based architecture won: 112 GBd (224G) host SerDes and module retimers are ADC/DSP designs. So the "DSP" that cancels ISI at 224G usually lives *inside* the SerDes, not in a separate chip.

DSP SerDes DSP-based SerDes: ADC on the Rx lane, digital FFE/DFE and timing recovery, DAC on Tx. Standard above $\sim$50 GBd; the "DSP" that cancels ISI at 224G lives inside the SerDes.

3.7.3 Two things people call "the DSP"

That is the first meaning of DSP: the digital equalization and clock recovery inside a modern SerDes. The second meaning is *the module DSP*, a distinct retimer chip in a pluggable that has its own ADC/DSP SerDes on the host side, an FFE/DFE/CDR core, a gearbox (Section 3.14.3), and often the FEC engine, then drives the optics. When the book says a retimed module "has a DSP" and LPO "deletes the DSP" (Section 10.5.1), it means this second chip. Deleting it does not delete DSP from the link; it moves the equalization burden back onto the *host* SerDes DSP and onto FEC.

3.7.4 Where FEC sits

FEC is a third block, and it is usually not part of the SerDes proper. KP4 encode and decode live in the PCS/FEC layer on the host (or in the module DSP when one is present), operating on the recovered symbol stream (Section 3.12). This is why a link can have a healthy SerDes eye and still fail on FEC, or ride an ugly eye that KP4 cleans: the SerDes recovers symbols, FEC fixes what is left. Validation reads pre-FEC BER exactly at that SerDes-to-FEC boundary.

3.7.5 Mapping it back to module styles

Table 3.3 sorts the blocks by where they physically run for the three module styles. Figure 3.2 shows the same split as signal chains. Read the LPO column as the design point that matters most for AI power budgets: the host SerDes DSP and host FEC carry the entire electrical channel, because the module has no retimer to hide behind (Section 3.14.2).

Block	Retimed	LPO	CPO (XSR)
Serialize / PAM4	Host SerDes	Host SerDes	Host SerDes
Rx EQ (FFE/DFE)	Module DSP	Host SerDes DSP	Host SerDes DSP
CDR / retiming	Module DSP	Host SerDes	Host SerDes
KP4 FEC	Host (or module)	Host	Host
Optical drive	Module DSP out	Linear driver	On-package engine

Table 3.3: Where each block runs, by module style.

Key idea. The SerDes is the ASIC's electrical PHY; at 224G its equalization and clock recovery are done in DSP inside the SerDes. "The module DSP" is a separate retimer chip. LPO removes that retimer, so the host SerDes DSP plus KP4 FEC (Section 3.12) own the whole electrical channel. Optics begin only after that handoff.

3.8 The Ethernet PHY stack: where optics attaches

The SerDes, the FEC, and the optics are not loose parts. IEEE 802.3 stacks them into named sublayers, and knowing the stack tells you exactly where the optical transceiver plugs in and which block owns each impairment. Ethernet lives at the bottom two OSI layers: the *MAC* (data link) builds frames and carries the addressing, and the *PHY* (physical) turns those frames into signals on the wire. Everything in this book sits inside the PHY.

MAC Media access control: Ethernet data-link sublayer; builds frames, addresses, CRC. MAC rate is payload throughput.

PHY Physical layer: PCS plus PMD; maps MAC frames to the line code.

3.8.1 The sublayers, top to bottom

The PHY is itself a stack. Table 3.4 lists the sublayers a 400G/800G port walks through on transmit (receive runs in reverse).

Table 3.4: IEEE 802.3 PHY sublayers, transmit order.

Sublayer	Function	Domain
MAC	Frame, address, CRC	Digital, host
RS sublayer	Reconciliation Sublayer: adapt MAC to the PHY across the xMII	Digital, host
PCS	64B/66B, 256B/257B transcode, scramble, RS-FEC encoding	Digital
PMA	Serialize, lane mux, clock recovery (SerDes)	Mixed-signal
PMD	Modulate/detect light: laser, driver, PD, TIA	Optical

Do not confuse the RS *sublayer* (Reconciliation) with *RS-FEC* (Reed–Solomon forward error correction) inside the PCS. Same letters, different jobs.

Two interface names sit between these blocks and cause most of the confusion. The *xMII* (for example 400GMII) is a wide parallel bus inside the chip between MAC and PCS. The *AUI* (for example 400GAUI-4) is the electrical serial lane set that leaves the host and crosses the faceplate connector to the module. The AUI is the CEI electrical channel your SerDes has to survive (Section 3.7 and chapter 10). The optical modulation and detection function belongs to the *PMD*. A physical module may also contain PMA, PCS, FEC, gearbox, retiming, and management functions on either side of the AUI.

xMII Media-independent interface: a wide parallel bus between MAC and PCS inside the chip (e.g. 400GMII).

AUI Attachment unit interface: the electrical serial lanes between host and module (e.g. 400GAUI-4, at the faceplate cage).

PMD Physical medium dependent: the optical transceiver (Tx/Rx) in IEEE 802.3.

3.8.2 Where the optics attaches

Optics is the PMD, and only the PMD. Everything above it is electrical or logical and is medium-independent: the same MAC, RS, PCS, and PMA feed a copper DAC, a multimode SR module, or a single-mode DR module. That is the whole point of the layering. When IEEE names `400GBASE-DR4` or `800GBASE-DR8`, it is naming a PMD (Section 3.13); the digital stack above is shared. It is also why the retimed-versus-LPO-versus-CPO argument is really about *where the PMA and PCS/FEC physically sit* relative to the AUI, not about changing the optics (Section 3.7).

3.8.3 Line-rate accounting

Rate names stack in five domains. Keep them separate when reading a datasheet or interview question:

```
MAC payload rate
        |
PCS coded (line) rate
        |
AUI electrical lane rate
        |
PMD optical lane rate
        |
Symbol rate (baud)
```

The MAC-to-line gap (Table 3.6) is the arithmetic you will reuse. Take one finalized 400G Ethernet port. The MAC delivers 400 Gb/s of payload. The PCS transcodes it (256B/257B adds about 0.4%) and then RS(544,514) FEC adds 30 parity symbols per 514 data symbols, a $544/514$ expansion of roughly 5.8% relative to payload (equivalently 5.5% of the coded line, Section 3.12). The two multiply out to $257/256 \times 544/514 = 1.0625$, so the PCS coded line carries $400 \times 1.0625 = 425$ Gb/s, split as four AUI lanes at 106.25 Gb/s (53.125 GBd PAM4). The optical PMD carries that same 425 Gb/s aggregate. So "400G Ethernet" is a 400 Gb/s MAC rate riding a 425 Gb/s coded line; the extra 25 Gb/s is transcode plus FEC, not wasted bandwidth.

OIF *CEI-224G* is an electrical-interface class near 112 GBd PAM4. IEEE 200 Gb/s-per-lane Ethernet (802.3dj) uses related but not identical rate accounting (typically 106.25 GBd PAM4 on the AUI/PMD for a 200G Ethernet lane). Do not treat the CEI class name and the Ethernet lane name as interchangeable.

3.8.4 What the PCS actually does

Inside the PCS, the MAC's 64-bit words are first wrapped in 64B/66B blocks: a 2-bit sync header (`01` for data, `10` for control) lets the receiver find block boundaries in the serial stream. Four 66-bit blocks are then transcoded into one 257-bit block, which compresses the four 2-bit sync headers to a single bit and, crucially, produces a 256-bit payload that aligns cleanly with 10-bit RS-FEC symbols

PCS Physical coding sublayer: 64B/66B line coding, 256B/257B transcoding, scrambling, and RS-FEC. Maps MAC data to the coded line stream.

(lcm$(256, 10) = 1280$, so five blocks make exactly 128 symbols with no remainder). The RS(544,514) encoder then adds parity (Section 3.12), and the PMA distributes the coded symbols across logical lanes, bit-multiplexes them down to the physical AUI lanes, and serializes each to PAM4. The payload bits themselves are never altered by transcoding; only the framing overhead is squeezed.

Where these blocks physically live is an implementation choice, not a layer rule. The PCS/FEC can sit in the host switch ASIC or in a module retimer DSP; LPO deletes the module DSP and forces the host to own PCS/FEC and all of the PMA equalization (Section 3.7 and section 10.5.1). The layer names stay the same regardless of which chip runs them.

Key idea. Ethernet is MAC (frames) plus PHY. The PHY is RS sublayer, PCS (including RS-FEC), PMA, and PMD. Optical modulation and detection belong to the PMD; a physical module may host more than the PMD. The MAC-to-line-rate gap is transcode $(257/256)$ times FEC $(544/514)$, giving 425 Gb/s on a 400G port. The module-style debate is about which side of the AUI runs the PMA and PCS/FEC.

3.9 Test points: where the link is measured

Every metric in this book is taken *somewhere*. A datasheet line like "TDECQ $\leq$ 3.4 dB" or "stressed sensitivity -3.1 dBm" means nothing until you name the reference plane it was measured at. The standards name those planes TP0 through TP5, plus the host-referenced pair TP1a and TP4a, and they run in order from the transmit silicon to the receive silicon. Learning them turns a scattered pile of numbers into a map: each spec belongs to exactly one plane, and each plane is owned by one document. Figure 3.3 walks the chain; Table 3.5 is the lookup card.

The mistake to avoid is treating TP2 and TP3 as electrical points just past a connector. They are *optical* planes at the fiber. TP2 is the light launched into the fiber, measured after the module's electrical-to-optical conversion (driver plus laser or modulator). TP3 is the light arriving at the far module, before its optical-to-electrical conversion (photodiode plus TIA). The electrical planes are the ones on either side of those two crossings.

READ THE CHAIN IN ORDER. At the transmit end, **TP0** is the transmit SerDes output at the die pad, the silicon designer's reference. **TP1a** is the same host signal referenced at the module cage, the *AUI* the module must accept; its electrical eye quality is *EECQ*, the electrical analog of TDECQ. **TP1** is the module's electrical input on the far side of the mated connector. The module then converts electrical to optical, and **TP2** is the optical launch at the transmitter *MDI*, where TDECQ, TECQ, $\mathrm{OMA}_{\mathrm{outer}}$, ER, and RIN are specified (Chapter 7 and section 7.4). After the fiber, **TP3** is the optical input at the receiver MDI, where stressed receiver sensitivity (SECQ)

EECQ Electrical eye closure quaternary: the electrical analog of TDECQ, quoted at the host-referenced points TP1a and TP4a.

MDI Medium dependent interface: the optical connector where a module meets the fiber. TP2 (Tx launch) and TP3 (Rx input) sit here.

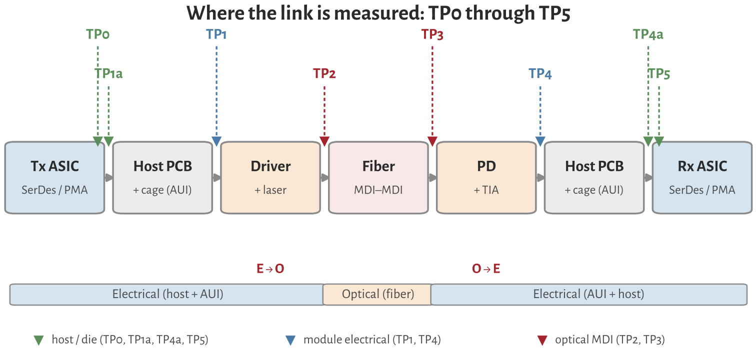

Figure 3.3: Test points on a one-way IM/DD link. Green marks the host and die-pad electrical planes (TP0, TP1a, TP4a, TP5); blue the module's electrical connector (TP1, TP4); red the optical planes at the fiber *MDI* (TP2, TP3). The two domain crossings, E⟶O in the module transmitter and O⟶E in the module receiver, sit between the electrical and optical planes.

is specified. The module converts back to electrical at **TP4** (its electrical output), **TP4a** is the host-referenced input near the receive SerDes under worst-case module output, and **TP5** is the receive die pad.

Who owns which plane. The split is not arbitrary. IEEE 802.3 optical PMD clauses define the transceiver planes: the optical TP2 and TP3, and the module's electrical PMD input and output at TP1 and TP4 [8]. The OIF Common Electrical I/O chip-to-module work owns the electrical planes deeper into the host: the die pads TP0 and TP5, and the host-referenced compliance points TP1a and TP4a, which are measured through host and module compliance boards to de-embed the fixture [66]. The LPO MSA then binds both sides into a single product contract for a DSP-less module, with the six-point ladder (TP1a, TP1, TP2, TP3, TP4, TP4a) tabulated in Table 10.3 [79].

The numbering is a family of conventions, not one universal set. IEEE optical PMD clauses commonly define only TP1 through TP4, with the optical measurements at TP2 and TP3; the die pads TP0/TP5 and the host-referenced TP1a/TP4a come from the chip-to-module electrical world. Fibre Channel historically used its own reference points. Match the labels to the document you are reading rather than assuming a shared TP0-to-TP5 spine.

Where compliance testing concentrates. Optical transmitter compliance lives at TP2 (TDECQ, OMA, ER); optical receiver compliance lives at TP3 (stressed sensitivity). Electrical host and module compliance concentrate at TP1a and TP4a, which is why a module that passes optical TDECQ at TP2 can still fail interop if it misses EECQ at TP1a. TP0 and TP5 sit inside the package and are informative for silicon design but hard to probe in a finished system. A retimed module recovers the signal before TP4; a linear (LPO) module does not, so at 224G its host EQ and FEC carry the whole electrical channel (Section 10.5.1 and section 3.13).

Point	Domain	Location	Principal measurement	Owning spec
TP0	Electrical	Tx ASIC/SerDes die pad (host)	Silicon Tx quality; design reference, hard to probe once packaged	OIF CEI C2M
TP1a	Electrical	Host output at the cage, host-referenced (SerDes output via a host compliance board)	Host Tx eye, EECQ; the AUI the module must accept	OIF CEI C2M; LPO MSA
TP1	Electrical	Module electrical input (module side of the mated connector)	Module input stressor calibration	IEEE 802.3 AUI; OIF CEI
TP2	Optical	Transmitter MDI, fiber launch (after E⟶O)	TDECQ, TECQ, $\mathrm{OMA}_{\mathrm{outer}}$, ER, $\mathrm{RIN}_x\mathrm{OMA}$	IEEE 802.3 PMD; LPO MSA
TP3	Optical	Receiver MDI, fiber input (before O⟶E)	Stressed receiver sensitivity, SECQ	IEEE 802.3 PMD; LPO MSA
TP4	Electrical	Module electrical output (module side of the connector)	Module Rx electrical output, EECQ	IEEE 802.3 AUI; OIF CEI
TP4a	Electrical	Stressed host input, host-referenced (near the Rx SerDes)	Host Rx under worst-case module output	OIF CEI C2M; LPO MSA
TP5	Electrical	Rx ASIC/SerDes die pad (host)	Silicon Rx recovery; design reference, hard to probe	OIF CEI C2M

Table 3.5. Test-point reference planes on a short-reach IM/DD link, transmit to receive. The optical planes TP2 and TP3 carry the transmitter- and receiver-quality specs; the electrical planes carry the host and module eye specs. See Table 10.3 for the concrete LPO MSA assignments.

Key idea. The plane makes the number. TP2 and TP3 are optical, at the fiber MDI, and carry transmitter and receiver quality (TDECQ, stressed sensitivity). TP1 and TP4 are the module's electrical connector; TP1a and TP4a are host-referenced near the SerDes; TP0 and TP5 are the die pads. IEEE 802.3 owns the optical planes, OIF CEI owns the die and host-referenced electrical planes, and the LPO MSA binds all of them into one DSP-less product contract (Table 10.3).

3.10 The link-budget vocabulary

These terms are the language of both datasheets and debug. Learn them as a ledger, not as a glossary quiz: when a link fails, you are almost always arguing about one of them.

OMA (optical modulation amplitude) $P_1 - P_0$: the real signal swing. For PAM4 the "outer OMA" spans the outermost levels.

Extinction ratio (ER) P_1/P_0, in dB. Trades against insertion loss and against *TDECQ* (Chapter 7).

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

RIN (relative intensity noise) the laser's own amplitude noise; sets a noise floor that matters more as OMA shrinks.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

Chirp the unwanted frequency modulation that accompanies intensity modulation. Large in directly modulated lasers; small and controllable in externally modulated lasers. See Section 3.11.

Chromatic dispersion penalty dispersion $\times$ chirp $\times$ distance produces inter-symbol interference. This drives the wavelength plan and the choice between directly and externally modulated lasers (Section 3.11).

Receiver sensitivity the minimum OMA needed to hit a target bit-error ratio; "stressed" sensitivity adds a calibrated impairment for margin.

[2]

[2] Start with transmitter OMA, subtract connector and fiber loss, then subtract only penalties not already in the compliance numbers you use next. Do not debit TDECQ again when the Tx OMA limit or SECQ-stressed sensitivity already includes that quality allocation (Sections 4.4, 7.4 and 7.7 and chapter 4).

Engineering heuristic. Argue about one ledger term at a time. A fight that mixes OMA, ER, RIN, and TDECQ without naming which debit moved is not a debug.

3.11 Chirp, dispersion, and the IM/DD penalty

IM/DD detects power, not phase, but phase still matters on the fiber. Intensity modulation couples to frequency through the laser or modulator *chirp* parameter α: a change in output power shifts the instantaneous optical frequency. Single-mode fiber then converts that frequency wander into intensity distortion via chromatic dispersion (different λ travel at slightly different speeds).

The penalty grows with chirp $\times$ dispersion $\times$ distance. At O-band (~1310 nm) dispersion is near zero ($|D| \lesssim 3$ ps/(nm·km)), which is why silicon photonics and many datacenter SMF links use 1310 nm class wavelengths. At C-band (~1550 nm) dispersion is larger; chirpy sources pay more per kilometer.

SMF Single-mode fiber: 9 μm core; O-band (1310 nm) or C-band (1550 nm). G.652.D and G.657 for datacenter plant.

Source choice sets chirp (Chapter 5, table 3.12, and section 3.14.3):

- **DML:** large α; fine for tens of meters of MMF or uncooled SR, poor for FR-class SMF unless rate and reach stay short.
- **EML / external MZM / ring:** low chirp; default for DR/FR and CPO.
- **Reflections:** light reflected back into the laser raises effective chirp and RIN; optical return loss (ORL) specs exist for this reason (Section 7.2.2).

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

For the distances this book treats (in-package through a few hundred meters), dispersion is often secondary to TDECQ, connector loss, and receiver noise. It becomes decisive when a low-chirp assumption fails (aging EAM bias, DML on SMF, or FR edge cases). TDECQ embeds dispersion in its name because the reference receiver includes a dispersion penalty model for the standardized test channel.

3.12 Forward error correction

Modern PAM4 links do not meet raw bit-error-ratio targets on their own; they lean on *FEC*. The datacenter workhorse is *KP4*, a Reed–Solomon code RS(544,514) operating on 10-bit symbols. The name comes from `100GBASE–KP4` in IEEE 802.3bj: K for backplane, P for PAM4, 4 for four lanes [9]. Today "KP4" means that FEC code on many PAM4 Ethernet PHYs (optical DR, related SerDes generations, and backplane), not only the original backplane PHY. For a named KP4-class optical PMD, the PHY is specified to a *pre-FEC* BER objective on the order of 2.4×10^{-4} under the clause's random-error model. *Post-FEC* quality is stated as residual BER, uncorrectable codeword rate, or frame-loss ratio with a confidence interval, not as "effectively error-free."

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

KP4 IEEE 802.3 Clause 91 FEC for PAM4: Reed–Solomon RS(544,514) on 10-bit symbols, 5.5% overhead. Pre-FEC BER target 2.4×10^{-4}.

RS(544,514) Reed–Solomon block code: 544 symbols out, 514 payload, 30 parity. Corrects up to 15 symbol errors per codeword.

Two consequences matter in practice:

1. Validation targets are stated in pre-FEC BER, and FEC symbol-error histograms are a rich debug signal (they reveal *how* a link is failing, not just that it is).
2. Emerging *linear-drive* optics (*LPO*/*LRO*, Chapter 10) remove the DSP/retimer to save power, which tightens the link budget and leans even harder on FEC and on transmitter quality (Section 3.6).

LPO Linear pluggable optics: no retimer/DSP in the module; host SerDes drives the modulator directly.

LRO Linear retimed optics: retimer in the module but no heavy DSP; lighter than fully retimed pluggables.

KP4 in practice. *KP4* maps to Reed–Solomon RS(544,514) on 10-bit symbols: 514 payload symbols, 30 parity, up to 15 correctable symbol errors per codeword. Coding overhead is $30/544 \approx 5.5\%$, so the *coded line rate* exceeds the MAC/info rate. Table 3.6 uses a finalized 400G Ethernet port as the arithmetic example. For 200G/lane Ethernet and CEI-224G electrical classes, keep the two documents separate: IEEE states MAC and PMD/PCS rates; OIF CEI states electrical interface classes near 112 GBd PAM4.

A named KP4-class optical PHY is qualified to a *pre-FEC* BER objective, typically 2.4×10^{-4} under that clause's assumptions. That corresponds to Q ≈ 3.5 on the binary Gaussian model of Section 4.1 (an equivalent-quality approximation, not a full PAM4 treatment). Post-FEC targets are residual BER or uncorrectable-codeword / frame-loss metrics in the 10^{-12} to 10^{-15} class, with stated confidence. Validation reports pre-FEC BER during bring-up; post-FEC confirms the decoder meets the named metric.

FEC symbol-error *histograms* are a debug tool: clustered errors point to burst impairments (reflections, power droop); sparse errors point to Gaussian noise margin. At 448G/lane, CEI notes that KP4 at 10^{-4} pre-FEC may not close on 40 dB-class chan-

Term	Meaning	Example
MAC rate	User Ethernet throughput at MAC	400 Gb/s aggregate
PCS coded rate	Bits after transcode and RS-FEC	425 Gb/s aggregate
AUI / PMD lane	Per-lane electrical or optical rate	106.25 Gb/s
Symbol rate	Baud on each PAM4 lane	53.125 GBd

Table 3.6: Rate names on one finalized 400G Ethernet port (four lanes).

nels without stronger codes (Table 3.10): MLC plus higher overhead RS, or hard-decision FEC in demos (Section 3.14.3).

3.13 Ethernet optical PMDs and reach classes

IEEE 802.3 names the optical transceiver (*PMD*) separately from the MAC rate. Short-reach IM/DD classes that matter for AI fabrics:

SR (short reach) multimode fiber, VCSEL at 850 nm or evolving MMF solutions; rack and row distances; modal noise and bandwidth limit legacy SR.

DR (datacenter reach) single-mode, typically 500 m class at 1310 nm; PAM4, KP4, TDECQ limits; the mainstream 400G/800G/1.6T module reach.

FR (far reach) single-mode, 2 km class; same optics family as DR with tighter transmitter specs; chirp and dispersion matter more at the margin.

SR Short reach: multimode fiber class (typically 100 m over OM4); VCSEL at 850 nm. Legacy rack optics.

MMF Multimode fiber: 850–940 nm VCSEL links (OM3/OM4/OM5); reach and modal BW limit to SR distances.

DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.

FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.

Naming examples: 400G-DR4 (four lanes), 800G-DR8, 800G-FR8 (eight λ, 2 km). 802.3dj adds 200G/lane Ethernet; the 400G/lane study group targets the next generation (Table 3.13 and section 3.14.3). Campus LR and coherent DCI sit beyond this book's scope (Table 3.1).

3.14 The per-lane roadmap: 224G and beyond

Per-lane rate is the axis the whole industry advances along, because doubling it roughly doubles switch and module capacity for the same lane count. The electrical I/O roadmap is set by the OIF's Common Electrical I/O (CEI) projects, and optics tracks it closely.

CEI class	Electrical lane	Modulation	Often paired Ethernet
CEI-112G	112 Gb/s	PAM4 (≈56 GBd)	100G/lane class
CEI-224G	224 Gb/s	PAM4 (≈112 GBd)	200G/lane class
CEI-448G	448 Gb/s	TBD (PAM4/6/8)	400G/lane class (study)

Table 3.7: OIF CEI electrical per-lane roadmap (not an Ethernet PMD table). Related IEEE Ethernet lane rates are listed only as the generation usually paired in products.

Figure 3.4 puts that ladder on a longer clock: OIF CEI has roughly doubled the rate per differential pair every four to five years since the early 2000s. Open markers for 224G and 448G follow the demo deck's "202X" placement; 224G is shipping now, 448G is still pathfinding [66].

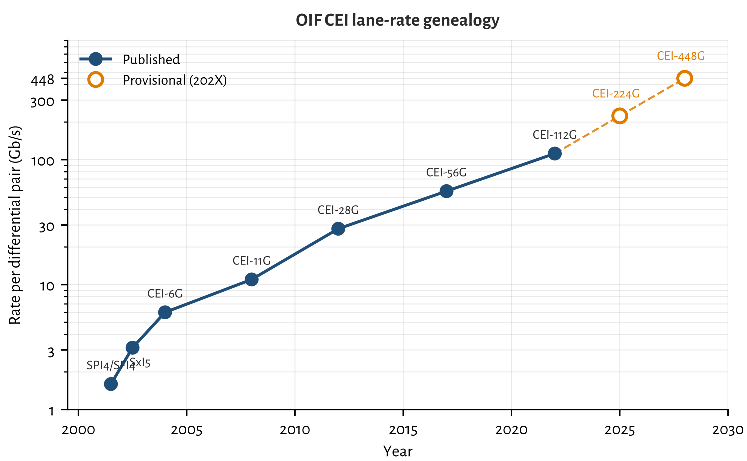

Figure 3.4: OIF CEI rate per differential pair versus year (log scale). Data from the OFC 2025 CEI interoperability demo genealogy; 224G/448G years are provisional placements for the deck's 202X entries, and 448G's rate is the naming target (PAM order still open).

3.14.1 224G is settled; the frontier is deployment

CEI-224G kicked off in 2022 as the electrical follow-on to CEI-112G. It kept PAM4 and roughly doubled the baud, to about 112 GBd per lane, with reach projects from XSR through LR now maturing. That choice is no longer the open debate; it is the shipping baseline. Eight 224G lanes make a 1.6 Tb/s port, and the same SerDes generation feeds 102.4 Tb/s-class switch silicon (Section 10.10). Copper reach has collapsed to about a meter, DSP-based SerDes are assumed, and a *CEI-224G-Linear* variant defines linear operation without a DSP/retimer in the optical module: the electrical foundation for LPO (Chapter 10). The remaining work is deployment: closing LPO margins, qualifying ELSFP banks for CPO, and holding yield at volume, not inventing a new modulation alphabet.

Figure 3.5 is the reach map those projects sit on: *XSR*/XSR+ inside the package (die-to-die and die-to-optical-engine), *VSR* from ASIC to a faceplate pluggable, *MR* chip-to-chip on the board (PCB or twinax), and *LR* for backplane and longer copper cables [66]. The media labels (host PCB traces, twinax, optical fiber, optical module) are where the loss budget actually lives; the CEI class name is the electrical recipe for that hop. Table 3.8 is the matching lookup card from the CEI-224G project map (OFC 2025 demo framing) [67][66].

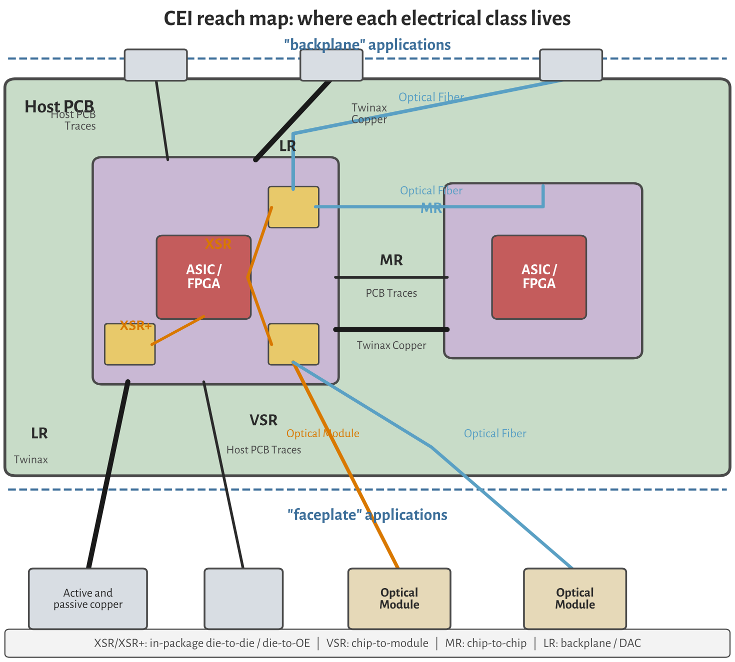

Figure 3.5: CEI reach map on a host PCB (redrawn). Backplane applications sit above the board; faceplate cages and modules sit below. XSR/XSR+ are in-package; VSR is chip-to-module; MR is chip-to-chip; LR covers backplane and active/passive copper.

One SerDes core may not cover XSR through LR efficiently: short reaches want simple, low-power equalization, while LR burns DSP to close tens of dB of loss. That is why Linear is its own project rather than "VSR with the DSP deleted," and why CPO (XSR) and LPO (Linear) show up as different power/serviceability bets (Section 3.14.2 and sections 10.3 and 10.10).

3.14.2 Deploying 224G: LPO, COM, and TDECQ corners

At 224G the alphabet is settled (PAM4 + KP4). What fails in the field is the *margin stack*: electrical channel operating margin (COM) on the host side, transmitter and dispersion eye closure quaternary (TDECQ) on the optical side, and the production corners that couple them (Section 7.9). Retimed modules still dominate 1.6T faceplate ports because their DSP absorbs host-channel sin. LPO and LRO exist to delete that DSP for power and latency; they only ship when both ledgers close without it [3][72][66][4][37].

[3] **??**IEEE EPS, Linear Pluggable Optics overview (2026).

[4] Semtech 224G/lane MZM drivers + TIAs (GN1834 family) for LPO/LRO/CPO (OFC 2026).

The two ledgers that must close together. A retimed module can hide a bad host PCB behind module EQ. An LPO module cannot. You therefore run two go/no-go tests

Class	Hop	Nominal reach / channel	Notes
XSR	Die-to-die or die-to-OE	$\lesssim$50 mm package substrate	CPO / chiplet optics; light EQ
VSR	Chip-to-pluggable module	$\sim$200 mm host + $\sim$20 mm module, 1 connector	Classic faceplate retimed path
MR	Chip-to-chip / midplane	$\sim$500 mm, 1 connector	Board-scale copper
LR	Backplane or copper cable	$\sim$1000 mm host+daughter, 2 connectors	DAC/ACC/AEC territory
Linear	Chip-to-linear module	Same cages as VSR-class ports; no module DSP	LPO foundation; host EQ + FEC

Table 3.8. CEI-224G electrical classes (project map). Post-FEC residual is typically 10^{-15} class when the IA allows FEC; quote the named pre-FEC objective from the active draft. Draft LR/MR IAs were member-review as of OFC 2025; Linear is the separate full-linear module track.

as one program:

Electrical COM After the CEI reference TX, channel, and Rx (CTLE + 8-tap DFE), COM ≥ 3 dB is the usual pass line at the slicer (Section 10.5.2). At 112 GBd the residual eye after that equalizer is only a few millivolts tall; KP4 then cleans pre-FEC BER near 2.4×10^{-4} down to 10^{-15} class (Section 3.12).

COM Channel operating margin: statistical SNR of the fully equalized link. CEI go/no-go is typically COM $\geq$3 dB.

Optical TDECQ / TECQ Outer OMA, RLM, and TDECQ (or *TECQ* without the test fiber) score the transmitter after a bounded reference FFE (Section 7.4). LPO MSA language for 100G/lane DR already couples OMA to max(TECQ, TDECQ) and caps TDECQ near 3.4 dB; 224G-class linear modules inherit the same philosophy even while CEI-224G-Linear and IEEE 802.3dj freeze the exact limits [5][79][6][8][66].

TECQ Transmitter eye closure quaternary: same measurement as TDECQ without the test fiber. Used with TDECQ in LPO MSA OMA tables.

[5] LPO MSA 100G-DR-LPO v1.0: 53.125 GBd PAM4 SMF to 500 m; TDECQ/TECQ $\lesssim$3.4 dB; builds on 802.3 + CEI-112G-LINEAR.

[6] IEEE P802.3dj: 200/400/800/1600G Ethernet (200G/lane; SA ballot 2026).

CEI-224G-Linear defines the host/module electrical test points (TP1/TP1a, TP4/TP4a) so a linear module can sit between a DSP host SerDes and the fiber without its own CDR [66]. Commercial 224G driver/TIA families advertise that interface explicitly (tunable swing, on-chip CTLE, CEI-224G-Linear host EQ) Semtech 224G. If either ledger is soft, prefer retimed or LRO until the host PCB and module linearity improve; do not "fix" LPO with more FEC alone.

Where 224G LPO programs actually break. The failure modes are familiar once you stop treating LPO as a cheaper OSFP:

- **Host FIR / CTLE mistuned.** Taps pegged or CTLE boost too aggressive raises COM loss and looks like a bad module. Golden-swap the module first (Section 7.9).

- **Connector and package return loss.** VSR/MR channels already sit near the edge at 112 GBd; a resonant stub or long bondwire eats the few mV of slicer margin (Section 10.5 and section 10.5.1).
- **Module nonlinearity.** Driver or TIA compression wrecks RLM; TDECQ climbs even when average power looks fine. Linear-optics parts exist because retimed DSP no longer hides this (Section 3.14.3 and section 4.5).
- **ORL / RIN feedback.** Dirty fiber or a bad isolator raises effective RIN and floors pre-FEC BER while LIV still looks healthy (Section 4.3 and section 7.2.2).
- **Chassis thermal + neighbor load.** Faceplate case temperature and adjacent lanes move bias, TEC, and (for rings) lock; TDECQ and unlock show up together (Table 7.5 and section 6.5).

Half-retimed LRO as the pragmatic middle. When full LPO will not close on the target host, *LRO/TRO* (retimed TX, linear RX) keeps roughly half the DSP power and still relaxes the Tx eye into the fiber [72]. Many AI clusters take that path first: spend watts on the harder electrical→optical direction, keep the receive path linear into the host. The validation split is the same: COM and stressed host input on the linear side, TDECQ on the retimed Tx side (Sections 7.5 and 10.3).

CPO at the same SerDes generation. Co-packaged engines shipping in 2025–26 typically run *200 Gb/s per optical channel* on 100G/200G SerDes into microring banks with external lasers (Section 10.10): same CEI-224G-class shoreline as faceplate 224G, but the lossy pluggable connector is gone and the laser service model moves to ELSFP (Section 5.14). Deployment corners shift from cage thermals to FAU mate, lock hold under neighbor heaters, and ELS hot-swap (Table 7.5 and chapter 6). The electrical alphabet is still 224G PAM4; the hard part is packaging and wavelength control.

> **Key idea.** 224G deployment is a margin problem, not a modulation problem. Close COM and TDECQ together under production-representative corners; use LRO when full LPO will not; treat CPO as the same SerDes generation with a shorter electrical path and a harder laser/lock service model.

3.14.3 448G is where the modulation debate lives

The next step, *CEI-448G* (framework published in late 2025, targeting 3.2 Tb/s systems from 2026 onward), is where the long PAM4 consensus finally comes under pressure [7] [14]. A full CEI-448G glossary of terms is in Chapter G. The debate is not whether 448 Gb/s *per lane* is useful (eight lanes make a 3.2 Tb/s OSFP-class port), but what *symbol rate* and *modulation order* each part of the link must run at to get there.

[7] OIF, *Next Generation CEI-448G Framework* (OIF-FD-CEI-448G-01.0, 2025).

OSFP Octal Small Form-factor Pluggable: high-power faceplate cage for 800G/1.6T/3.2T modules; larger thermal envelope than QSFP-DD.

Line rate, symbol rate, and where each applies. *448G/lane* in CEI means roughly 448 Gb/s of payload on one differential electrical pair between host and module (or between dies in a package). Ethernet maps the same lane count to aggregate rates (400G, 800G, 1.6T, 3.2T). FEC and coding overhead sit on top of the 448 Gb/s target, as they did at 112G and 224G.

For PAM-M, the relationship is

$$R_{\text{line}} \approx \log_2(M)\, R_{\text{sym}},$$

with R_{sym} the baud rate and the Nyquist frequency $f_N \approx R_{\text{sym}}/2$. At 448 Gb/s the OIF framework tabulates the options in Table 3.9 [14]. PAM4 needs 224 GBd and $f_N \approx 112$ GHz. PAM6 drops to 173 GBd and $f_N \approx 87$ GHz. PAM8 drops further to 149 GBd and $f_N \approx 75$ GHz. Each step down in baud buys channel bandwidth at the cost of finer amplitude levels, lower SNR margin, and heavier FEC/DSP.

The critical split is *where* on the board those numbers apply:

- **In-package (*XSR*/XSR+, die-to-die, die-to-OE):** trace lengths of millimeters to a few centimeters. Package bandwidth can reach $\gtrsim$115 GHz with $\leq$0.5 mm BGA pitch and skip-layer routing. 448G-PAM4 at 224 GBd is the working assumption here, including linear drive from host SerDes straight into a co-packaged optical engine [14].

XSR CEI extra-short reach: die-to-die or die-to-OE, mm-scale traces.

- **Host PCB plus pluggable connector (*VSR*/*MR*/*LR*):** meters of PCB, a module connector, and often a cable. Measured end-to-end channel bandwidth is limited to about 90 GHz today, mostly by connector pin stubs and resonance notches above that frequency [14]. At 90 GHz, PAM4 at 112 GHz Nyquist does not close; PAM6 or PAM8 is the fallback unless connector technology moves the roll-off past 112 GHz.

VSR CEI very short reach: host PCB plus one module connector ($\sim$200 mm).

MR CEI medium reach: longer host plus module channel than VSR.

LR CEI long reach: longest CEI copper class before optics take over.

- **Optical lane (fiber):** IM/DD at 448 Gb/s still targets PAM4 at ≈224 GBd per wavelength. The fiber channel is not connector-limited in the same way; TDECQ, chirp, dispersion, and receiver bandwidth dominate instead of copper insertion loss [14][8] [30].

[8] St-Arnault *et al.*, 3.2 Tb/s IM/DD with TFLN modulators (arXiv:2503.24147, preprint).

Figure 3.6 is the packaging map behind that split: co-packaged copper (*CPC*), co-packaged optics (CPO), and faceplate pluggables (retimed, LPO/LRO, AEC/DAC) all leave the host IC at different places, which is why connector bandwidth and host EQ budgets change together [14].

CPC Co-packaged copper: copper I/O at the package edge (OIF map); feeds faceplate cages or short copper without an on-package optical engine.

Industry snapshot — mid-2026 (224G electrical / 200G Ethernet). Before the 448G fork, faceplate and copper products already pair CEI-224G-class electrical lanes with IEEE 200G/lane Ethernet PMDs. Faceplate ports commonly run eight ≈112 GBd PAM4 electrical lanes into 1.6 Tb/s OSFP-class modules (retimed DSP modules remain common; LPO and LRO appear where host COM and TDECQ close,

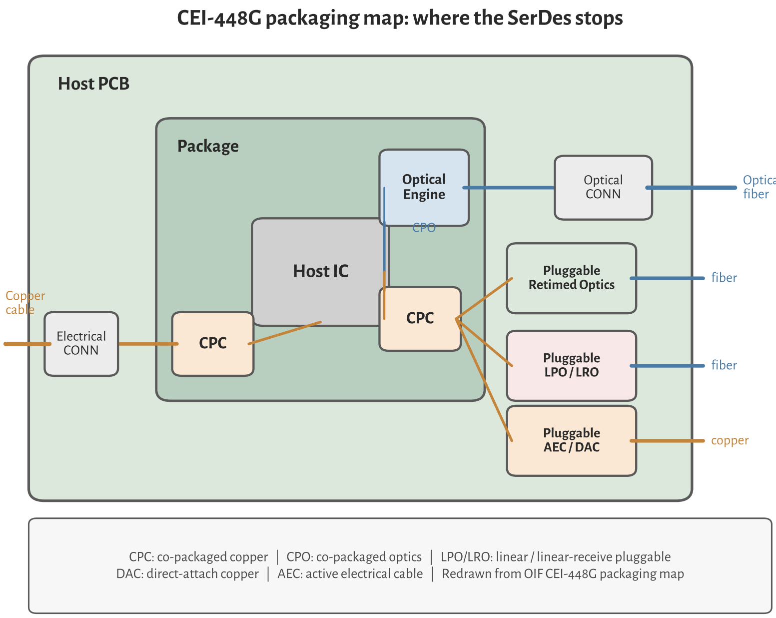

Figure 3.6: CEI-448G packaging map (redrawn). CPC brings copper to the package edge; CPO puts an optical engine in the package; faceplate cages take retimed optics, LPO/LRO, or AEC/DAC. Same host IC, different SerDes reach and EQ burden.

Section 3.14.2). Copper inside the rack uses DAC, ACC, and AEC at the same SerDes generation. Public CPO engines with Tomahawk 6 and Quantum-X class switches are often *200 Gb/s per optical channel* on 100G/200G SerDes into microring banks with external lasers (Section 10.10): same packaging map, one generation behind the 448G electrical debate. KP4 FEC, CEI-224G-Linear for LPO, and KP4-class pre-FEC BER near 2.4×10^{-4} are the usual contract (Sections 3.12 and 3.14 and section 10.5.2). The rest of this section is about what breaks when you try to double that electrical lane rate on the same shoreline.

So the "448G debate" is really two linked questions: can the *electrical* shoreline run PAM4 at 224 GBd, and can the *optical PMD* modulate and detect at the same symbol rate? Figure 3.7 maps the three dominant architectures and the baud rate at each hop.

Scheme	Symbol rate	f_N	UI	SNR penalty vs. PAM4
PAM4	224 GBd	112 GHz	4.5 ps	0 dB
PAM6	173 GBd	87 GHz	5.8 ps	≈2 dB
PAM8	149 GBd	75 GHz	6.7 ps	≈4–5 dB

Table 3.9: PAM-*M* options at 448 Gb/s/lane (OIF CEI-448G framework, pre-FEC line-rate target).

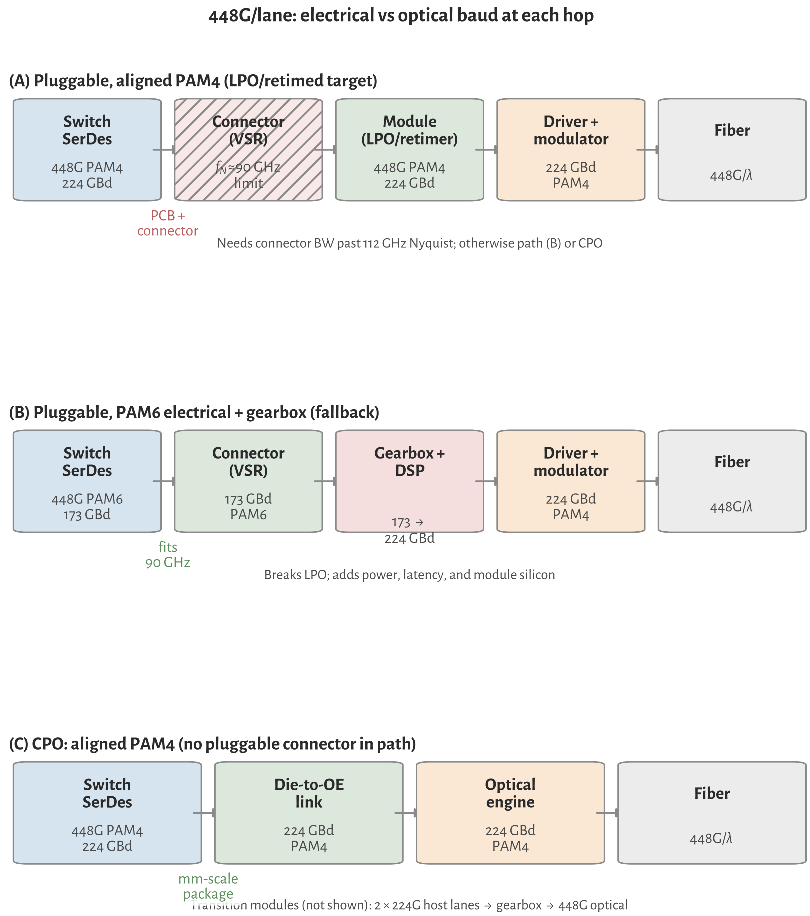

Figure 3.7: 448G/lane signal paths: aligned pluggable PAM4 (connector-limited), PAM6 electrical with module gearbox (LPO incompatible), and CPO or gearboxed 224G transition.

FEC overhead and the SNR budget. At the 200G/lane Ethernet generation, IEEE optical PHYs lean on *KP4* FEC: Reed–Solomon RS(544,514) from 802.3 Clause 91, with coding overhead $30/544 \approx 5.5\%$ and a pre-FEC BER objective near 2.4×10^{-4} for named optical PHYs [8][14]. Post-FEC residual BER targets are 10^{-15} class under the clause assumptions. A 200 Gb/s Ethernet lane typically runs PAM4 near 106.25 GBd on the AUI/PMD; the CEI-224G electrical class sits near 112 GBd. Do not treat those baud numbers as the same document. The coded line rate exceeds the MAC rate; after KP4 the payload is rounded to 200 Gb/s at the MAC.

448G pushes FEC harder. CEI-448G lists pre-FEC BER 10^{-4} as the working target (same as prior CEI generations) but notes that 10^{-4} at $\geq$40 dB channel loss may not close without a *stronger* code than KP4 [14]. PAM6 adds $\sim$2 dB SNR penalty on top, so workshop assumptions often pair PAM6 with *MLC* and higher-overhead FEC ($\sim$12%) for a fair comparison to PAM4 with KP4 [9] [28]. Table 3.10 summarizes the landscape (OH = coding overhead); 448G codes are not finalized (see caption for sources).

MLC Multi-level coding: coded modulation paired with PAM6/PAM8. OIF 448G workshop models often add a strong outer RS code; $\sim$12% overhead vs. KP4 at 5.5%.

[9] OIF 448G AI Workshop: native modulation vs. gearboxed 224G (2025).

OH Coding overhead: fraction of line rate spent on parity symbols (KP4 $\approx$5.5%).

Context	FEC	OH	Role at 448G/lane
802.3dj / 224G	KP4 RS(544,514)	5.5%	Baseline; pre-FEC BER near 2.4e-4.
CEI-448G, PAM4 elec.	KP4 or stronger	5.5%+	Target pre-FEC 1e-4; KP4 may fail on 40 dB-class channels.
CEI-448G, PAM6 elec.	MLC + strong RS	12%	Offsets PAM6 SNR penalty; adds latency (workshop assumption).
448G optical demos	HD-FEC / product	10–15%	3.2T TFLN demo threshold; not necessarily host FEC.
FEC architecture	end-to-end vs. terminated	n/a	448G may need dedicated electrical inner code.

Table 3.10. FEC at 448G/lane (448G codes provisional).

Sources: 802.3dj/KP4 baseline [8]; CEI-448G electrical targets [14]; PAM6 FEC assumption [28]; optical demo [30].

HDFEC Hard-decision FEC: high-overhead codes used in optical demos and long-reach products (10–15% OH), not the host KP4 budget.

Higher FEC overhead feeds back into the modulation debate: every extra parity bit raises the line rate and Nyquist frequency, which hurts a bandwidth-limited copper channel. That is one reason PAM6 (lower baud) and stronger FEC get paired, and why CPO (shorter electrical path) keeps PAM4 attractive.

Electrical SerDes: feasible, but the channel is the gate. 448G SerDes themselves are widely treated as feasible on advanced CMOS ($\leq$N3/N2): ADC/DAC DSP transmitters and receivers demonstrated at 224 Gb/s are the template, and coherent-optics SerDes with $\sim$100 GHz class AFE at 200 GBd is cited as encouraging precedent for doubling to 448G-PAM4 [14]. Synopsys and others have published channel simulations showing margin for both PAM4 and PAM6 on 224G-class channels when equalization and FEC scale with the modulation order[10] [27]. Power targets cluster around 0.5–2.5 pJ/bit at 448G, similar in spirit to 224G scaling.

[10] Synopsys, *Designing for 448G* white paper (2025).

The blocker is not the PLL or the DAC; it is the *passive channel*. Connector bandwidth near 90 GHz forces a fork:

1. **Fix the channel for PAM4:** new high-density connectors, shorter reach, skip-layer PCBs, cabled-host internal twinax. Simulations show passive CPC reach up to $\sim$1.2 m at 448G-PAM4 under optimistic connector assumptions [14]. This path preserves electrical/optical format alignment and keeps LPO/LRO architectures viable (Chapter 10).
2. **Keep the channel, change modulation:** PAM6 at 173 GBd fits a 90 GHz channel

but adds ∼2 dB SNR penalty and pushes the host toward stronger FEC than KP4. PAM8 relaxes bandwidth further at ∼4–5 dB penalty [14].

3. **Bypass the bad channel:** co-packaged optics with millimeter-scale die-to-OE links. Host 448G-PAM4 SerDes drives the optical engine directly; the pluggable connector never sees 224 GBd [14].

If electrical PAM6 wins on the PCB but optics stay at PAM4, the module needs a *gearbox* (rate/format conversion) in the retimed path. That adds power, latency, and silicon area, and it breaks linear pluggable optics: LPO requires the host electrical waveform to match what the optical modulator expects [14][72]. First 448G optical modules may therefore ship with gearboxed 224G SerDes (two 224G lanes per 448G optical lane) before native 448G electrical I/O is ready [28].

Optical modulation at 224 GBd: hard, but no longer hypothetical. On the optics side, the industry assumption is still IM/DD PAM4 at ≈224 GBd for 448 Gb/s per λ [14][30]. That requires roughly 112 GHz class electro-optic bandwidth in the modulator (and comparable photodiode/TIA bandwidth in the receiver), plus a driver with enough linear swing and RF BW. By 2025–26 that stack exists in demos and, increasingly, in announced commercial parts: silicon rings and MZMs with peaking, TFLN MZMs past 100 GHz EO bandwidth, EMLs still owning volume at 200G/lane, and SiGe drivers / Ge receivers catching up. The snapshot below is the state of play; device sections later unpack each family.

- **Silicon microring modulators:** resonant, compact modulators for dense WDM and CPO (Section 3.14.3). OFC 2026 demos reach 224–416 Gb/s PAM4 with inductive peaking [11] [42][12] [43].
- **Silicon Mach–Zehnder modulators:** the broadband, non-resonant counterpart to microrings (Section 3.14.3). OFC 2026 demos reach 400G/lane PAM4 with SiGe drivers [13] [45].
- **EML:** integrated DFB + EAM remains the default through 200G/lane for DR/FR pluggables: one chip, low chirp, mature supply chain.
- **TFLN MZM:** thin-film lithium niobate Mach–Zehnder modulators with CW lasers exceed 100 GHz EO bandwidth and are the leading path to native 224 GBd IM/DD (Section 3.14.3). A 3.2 Tb/s system (eight ×225 GBd PAM4 lanes) with 3 nm CMOS SerDes and TFLN modulators over 2 km *FR8*/*DR8* was reported in 2025 [30].
- **Drivers:** commercial modulator drivers with >120 GHz RF BW for 400G PAM4 EML/MZM/TFLN platforms appeared in 2026 [14] [36].
- **Receivers:** Ge/Si PIN and APD photodiodes above 100 GHz and 224G TIAs exist (Section 4.5); the receive side is not the long pole relative to modulator/driver bandwidth at 448G.

[11] Lin *et al.*, 90-GHz Si microring, 224-Gb/s PAM4, inductive tuning (OFC 2026 Th2A.14).

[12] Peng *et al.*, Si microring >400 Gbps PAM4, T-coil peaking (OFC 2026 M2A.2).

[13] Dong *et al.*, 400G/lane Si MZM PAM4 with SiGe driver (OFC 2026 postdeadline Th4A.4).

FR8/DR8 802.3 800G single-mode reaches: FR8 = 2 km, DR8 = 500 m (eight lanes per direction).

[14] MACOM 448G/lane PAM4 drivers (MAOM-025408/022404): >120 GHz RF BW (OFC 2026).

Silicon microring and microdisk modulators. Skim the platform notes below for device vocabulary. Requirements-led choice, ATP, and ownership live in Chapter 5; return here when you need ring, MZM, TFLN, or driver physics after the path is frozen.

A microring modulator (MRM) wraps a phase-shifter waveguide into a closed loop coupled to a straight *bus* waveguide. A *microdisk* is the same idea in disk form: a pillar cavity evanescently coupled to the bus, often with a wider free spectral range (FSR) in a smaller footprint. Both are resonant filters as well as modulators: when the input wavelength sits on resonance, drop-port power is high; off resonance it is rejected. Data modulation shifts the resonance (carrier depletion or injection in an embedded pn junction) or detunes the laser relative to a fixed ring, mapping voltage to intensity at the through or drop port.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

FSR Free spectral range: wavelength span between adjacent resonances of a ring or etalon. Sets WDM channel packing density.

The central design trade is *photon lifetime versus bandwidth*. A high-Q ring stores photons longer, which improves modulator efficiency (V_π) but narrows the optical passband and creates an electrical bandwidth ceiling through the RC-limited junction. Coupling strength, ring radius, and whether the device operates in undercoupled or over-coupled regime set Q, extinction, and the electro-optic (EO) rolloff. That trade does not appear in broadband Mach–Zehnder modulators (Section 3.14.3).

Three constraints dominate ring modulator design at 100–400G per λ. First, EO bandwidth: the junction RC roll-off is often below the target Nyquist frequency, so inductive peaking (T-coils on the drive path), optimized detuning from resonance, and compact RLC co-design extend EO BW without widening the ring so much that Q collapses [42][43]. Production CPO rings target 50–90 GHz; conference demos report 90–110+ GHz with aggressive peaking [42][43][15][44]. Second, wavelength alignment: ring resonance drifts roughly 10 GHz/°C in silicon, so each λ in a WDM bank needs the laser or the ring tuned onto the modulator resonance (Chapter 6 and section 6.4). Thermal crosstalk from neighbors shifts resonances in dense arrays, which is why fleet validation must include corner temperature and adjacent-channel heating (Section 6.5 and chapter 7). Third, optical loss and FSR: ring radius sets FSR ($\Delta\lambda$ between adjacent resonances). Microdisk and *Euler* ring layouts widen FSR so more channels fit under one free spectral range without collisions [44], while residual coupling loss and on-resonance insertion loss (often 1–3 dB class per modulator) eat link budget.

[15] OFC 2026 M2A.1: Euler Si microring, 256-Gb/s PAM4, >67-GHz EO BW, 3-THz FSR.

Integration is where rings win. A single SOI die can pack dozens of ring modulators and filters for CW-WDM (Chapter 6 and section 6.6), each fed by one wavelength from an external comb or multi-wavelength ELS (Chapter 5). The photonic engine co-packaged with a switch ASIC (Broadcom Bailly/Davisson, NVIDIA Quantum-X/Spectrum-X, Section 10.10) uses microring modulators at 200 Gb/s per channel today. Fiber count stays low because many λ share one waveguide; area and modulator count scale with WDM order rather than with faceplate port count.

State of the art in 2025–26 shows how far that peaking path has been pushed. Inductive and wavelength tuning carried silicon MRMs to 224 Gb/s PAM4 at 90 GHz EO BW [42]. T-coil-peaked designs report 416 Gb/s PAM4 ($\approx$208 GBd) with >110 GHz 1-dB EO BW and TDECQ 2.88 dB at 1 Vpp [43]. Euler microdisk rings show 256 Gb/s PAM4 with >67 GHz EO BW and 3 THz FSR in O-band [44]. These are lab and conference results, but they match the 200G/lane CPO shipping point and probe 448G-class lane rates when paired with sufficient electrical drive.

The platform choice is a packaging and control decision as much as a bandwidth one. Prefer rings when many λ must fit on one PIC and modulator area dominates: CPO WDM engines, scale-up optical I/O, and any architecture that already budgets for wavelength locking (Chapters 6 and 10). Prefer a silicon MZM for single-λ DR/FR where a flat passband avoids lock loops (Section 3.14.3). Prefer TFLN when you need native 224 GBd in a pluggable and ring thermal control at fleet scale looks harder than hybrid assembly (Section 3.14.3).

Silicon Mach–Zehnder modulators. Silicon photonics builds the Mach–Zehnder modulator (MZM) as a push-pull interferometer on a silicon-on-insulator (SOI) rib or strip waveguide. Phase shifters in each arm use *carrier depletion* in an embedded pn junction: reverse bias pulls carriers out of the waveguide core, lowering refractive index and shifting optical phase. Intensity modulation comes from recombining the arms at a 3-dB coupler, so chirp stays low compared with directly modulated lasers (Chapter 5).

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

SOI Silicon on insulator: substrate for silicon photonics; 220 nm Si layer on buried oxide confines the optical mode.

Three constraints mirror every high-speed MZM, but silicon's weak electro-optic coefficient (Δn per volt is far smaller than lithium niobate) sets the numbers. The optical S_{21} response must span the Nyquist frequency ($\approx$56 GHz at 112 GBd, 112 GHz at 224 GBd); junction capacitance, series resistance, and traveling-wave electrode (TWE) microwave loss set the roll-off. Production Si MZMs for 100–200G/lane modules typically quote 70–100+ GHz 3-dB BW; differential-drive layouts and compact 300-mm platforms report 80–95 GHz class results [16] [47] [17] [46]. $V_\pi L$ for carrier-depletion Si is often $\approx$1.5–2.5 V·cm, so millimeter-scale devices need 2–4 V peak-to-peak drive at the target baud inside the linear range of a SiGe or BiCMOS modulator driver (448G-class drivers exceed 120 GHz RF BW) [36]. Unlike resonant rings, an MZM is broadband, so WDM channels do not fight thermal lock (Chapter 6); the cost is length: mm-scale shifters and splitters add 2–4 dB on chip, and the device is far larger than a ring, which matters in dense CPO tiles.

[16] OFC 2026 W3E.5: differential-drive Si MZM, 81.8-GHz 3-dB BW, 100-GBd PAM8 eyes.

[17] OFC 2026 M2A.6: 500-μm Si MZM, 94.7-GHz median EO BW, 2.4-dB IL (300-mm platform).

Integration is the main reason Si MZM survives competition from EML and TFLN. The modulator shares a die with Ge photodiodes, fiber grating couplers or edge couplers, monitors, and (in WDM products) MUX/de-MUX filters, all in a CMOS foundry-compatible flow. An external CW DFB or ELS (Chapter 5) couples in through a spot-size converter; the RF driver is usually a separate die wire-bonded or 2.5D packaged next to the PIC, the same assembly style as TFLN modules but without bonding a second optical material stack.

State of the art in 2025–26: Si MZMs are the default modulator in 100G/lane and 200G/lane DR/FR silicon-photonics transceivers. Pushing to 400G/lane IM/DD was open until OFC 2026, when Coherent reported 400 Gb/s PAM4 per lane with a Si MZM and commercial SiGe driver (2.5 V swing) [45]. The same conference cycle showed a 500 μm compact MZM on a 300-mm platform with 94.7 GHz median EO BW and 2.4 dB on-chip insertion loss [46], and a differential-drive MZM with 81.8 GHz 3-dB BW with eyes to 100 GBd PAM8 [47]. These are lab and conference demos, not shipping modules, but they close the headline lane-rate gap with ring modulators while keeping a flat passband that rings only match with tight wavelength control (Section 3.14.3).

SiPh Silicon photonics: waveguides and modulators on SOI; CMOS fab-compatible. Mainstream for DR/FR and CPO.

Thin-film lithium niobate Mach–Zehnder modulators. *TFLN* refers to a submicrometer slice of lithium niobate bonded to a silicon or silica handle wafer. Compared with bulk LN, the tight optical mode confinement shrinks electrode gap and interaction length, which cuts the half-wave voltage–length product $V_\pi L$ while keeping a wide *electro-optic* (EO) bandwidth. The modulator is a Mach–Zehnder interferometer (MZM): a *Pockels* phase shifter in each arm, driven in push-pull, converts RF voltage on a *traveling-wave electrode* (TWE) into intensity at the output coupler.

TFLN Thin-film lithium niobate: sub-μm LN on Si handle. >110 GHz EO BW; leading path to native 224 GBd PAM4.

EO Electro-optic: conversion from voltage to optical phase or intensity. EO bandwidth sets modulator speed.

TWE Traveling-wave electrode: RF transmission line on a modulator that velocity-matches the optical mode for high BW.

Three design constraints set whether a TFLN MZM can run at 224 GBd PAM4 ($f_N \approx 112$ GHz). EO bandwidth, measured as the S_{21} roll-off of optical response versus RF drive, must clear that Nyquist: production-class devices quote $\gtrsim$110 GHz 3-dB BW, and research devices with low-*k* underfill or advanced TWE layouts extrapolate toward 200+ GHz [18] [32] [19] [OFC 2026 W1A.3]. $V_\pi L \approx 1$–2 V·cm is typical, so a 5–7 mm device gives $V_\pi \approx 1.5$–2 V that must fit inside the linear swing of a SiGe or InP driver at 112 GHz class [36] [20] [31]. Along the TWE, microwave and optical group indices must match or bandwidth collapses; low-loss underfill, narrow-gap CPW or co-planar layouts, and transparent conductive oxide (TCO) electrodes trade insertion loss against efficiency [31][32].

[18] He *et al.*, dual-band TFLN MZM 390-Gb/s PAM8, 220-GHz extrapolated BW (Laser Photonics Rev. 2025; arXiv:2411.15037).

[19] OFC 2026 W1A.3: 200-mm TFLN platform, 110-GHz EO, $V_\pi L = 1.9$ V·cm.

[20] Liu *et al.*, TFLN MZM 224-Gb/s PAM4, 108-GHz EO BW (OFC 2025 M3K.2; arXiv:2311.05119).

Demonstrated IM/DD results on TFLN MZMs include 224 Gb/s PAM4 at 108 GHz EO BW (O-band, $V_\pi L = 1.02$ V·cm) [31] and 390 Gb/s PAM8 on the same dual-band chip (extrapolated 220 GHz BW, sub-fJ/bit in lab) [32]. System-level proof came with eight $\times$225 GBd PAM4 lanes over 2 km using 3 nm SerDes and packaged TFLN modulators [30]. Commercial suppliers (HyperLight, Lumiphase, and foundry lines on 200-mm silicon) now ship 110 GHz-class packaged MZMs aimed at 200–240 GBd signaling [21] [35][OFC 2026 W1A.3].

[21] HyperLight, 110-GHz packaged TFLN IQ modulators for 240-GBaud class (Sept. 2025 announcement).

Integration looks unlike monolithic silicon photonics. A CW DFB or ELS feeds the TFLN chip through a Si/SiN coupler; the RF driver sits on a separate die, wirebonded or flip-chip mounted with matched 50 Ω lines. Hybrid Si–LN platforms (silicon waveguides, LN overlay) were demonstrated early for 100G/lane and remain a template for co-packaged assemblies [22] [34]. The laser is not on the TFLN chip, so alignment, fiber attach, and thermal bias of the MZM quadrature point become

[22] He *et al.*, hybrid Si–LN integration (Nature Photonics 2019).

validation items (Chapter 7).

EML and the electro-absorption modulator. An *EML* integrates a DFB laser with an *EAM* (electro-absorption modulator) on one InP chip. The EAM is a reverse-biased absorption region: voltage shifts the band edge (Franz-Keldysh or quantum-confined Stark effect), attenuating light with far less chirp than direct current modulation (Section 3.11).

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

EMLs dominate 100G/lane and 200G/lane DR/FR pluggables because they are single-chip, mature in supply chain, and match ∼70–100 GHz class EO bandwidth (Table 3.12). The design limits that show up in validation are EAM bias and aging (bias sets extinction and chirp; aging drifts the curve and shows up as TDECQ/RLM creep, Chapters 7 and 8), driver swing (a few volts inside the linear absorption region; 448G-class EML drivers appeared alongside MZM drivers in 2026 [36]), and thermal headroom (uncooled datacom is standard; slope efficiency and bias must stay in range across case temperature).

EML wins on cost and integration through 200G/lane. Above that, external modulators (Si MZM, TFLN, rings with CW laser) chase bandwidth and chirp headroom (Section 3.14.3).

Modulator drivers: requirements, records, outlook. Every external modulator (Si MZM, TFLN, ring, EAM) needs an RF path that delivers enough *linear* swing at the target baud. That path is either a dedicated SiGe/BiCMOS *modulator driver* die, or (for LPO) the host SerDes itself. Laser *bias* drivers are a different circuit and a different noise budget (Section 5.8). Do not share the modulator driver's switching returns with the CW bias rail.

SiGe Silicon-germanium BiCMOS: high-f_T process for TIAs and modulator drivers at 100–200+ GHz bandwidth.
BiCMOS Bipolar CMOS: process combining bipolar (high f_T) and CMOS on one die; used for high-speed TIAs and drivers.

What the driver must deliver. At 224 GBd PAM4 the symbol Nyquist frequency is 112 GHz, so a usable driver is not just "fast enough on paper." It needs RF bandwidth roughly $\gtrsim$100–120 GHz (3 dB) with flat gain and low group-delay ripple, output swing matched to V_π or EAM drive (often ∼1.5–3 V peak-to-peak, differential or single-ended, part- and modulator-dependent), linearity good enough that RLM and TDECQ stay inside the PMD budget when the optical path adds its own compression (Section 7.4), and return loss / matching into 50 Ω (or the designed modulator impedance) so bondwire and package reflections do not eat the bandwidth you paid for on the die. Distributed (traveling-wave) SiGe drivers dominate the high-BW niche: many cascoded gain cells along an artificial transmission line. Lumped drivers win on area and power at lower baud. Flip-chip bumped die and short wirebonds matter as much as f_T: a 120 GHz die behind a long bond loop is a 70 GHz module.

Record and commercial snapshot (2025–26). Table 3.11 separates research demos (often with offline DSP) from shipping or announced commercial parts aimed at 1.6T/3.2T modules. Commercial anchors: MACOM's MAOM-025408 MZM and MAOM-022404 EML drivers (>120 GHz RF BW, OFC 2026) [36]; Semtech's GN1877/GN1887 224G quad/octal family for LPO/LRO/CPO across SiPh, InP, and TFLN [37]. Research anchors: a 130 nm SiGe distributed driver at 105.7 GHz BW and 2.25 V swing running 232 GBd PAM4 into TFLN (offline DSP) [23] [38]; OFC 2026 co-designed engines at 210 GBd / 420 Gb/s PAM4 (TFLN TOSA) [24] [39], 180 GBd PAM4 (InP MZM + EML-class driver) [25] [40], and differential SiGe+TFLN at 140 GBd PAM8 / 1.4 pJ/bit [26] [41]; plus the Si MZM postdeadline with a commercial SiGe driver at 400 Gb/s/lane and 2.5 V swing [45].

[23] RFIC 2026: 105.7 GHz SiGe distributed driver; 232 GBd PAM4 into TFLN (research).

[24] OFC 2026 W4J.4: 210 GBd / 420 Gb/s PAM4 TFLN TOSA (co-designed driver).

[25] OFC 2026 W3E.6: 180 GBd PAM4 InP MZM + 224G-class EML driver.

[26] OFC 2026 M4D.4: differential SiGe+TFLN; 140 GBd PAM8 at 1.4 pJ/bit.

Part / paper	Process	RF BW	Rate / swing	Note
MACOM MAOM-025408 (MZM)	SiGe	>120 GHz	448G-class PAM4	Commercial; SiPh MZM
MACOM MAOM-022404 (EML)	SiGe	>120 GHz	448G-class PAM4	Commercial; EML / TFLN
Semtech GN1877 / GN1887	—	224G-class	224 Gb/s/lane	Quad/octal; LPO path
RFIC 2026 distributed drv	130 nm SiGe	105.7 GHz	232 GBd; 2.25 V	Research; TFLN; offline DSP
OFC 2026 Si MZM + SiGe	commercial SiGe	(drv-limited)	400 Gb/s/lane; 2.5 V	Postdeadline Th4A.4
OFC 2026 TFLN TOSA	co-designed	—	210 GBd / 420 Gb/s	Driver+TFLN engine
OFC 2026 InP MZM + EML drv	InP + SiGe-class	76 GHz MZM	180 GBd PAM4	Co-packaged engine
Nokia/Bell Labs hybrid	SiGe + TFLN	—	140 GBd PAM8; 1.4 pJ/bit	Diff. drive; low $V_\pi L$

Table 3.11. Modulator-driver snapshot (c. 2026). Commercial rows are vendor announcements (bandwidth/swing as published); research rows often use offline DSP and short RF interconnects. "448G-class" means aimed at ∼400–448 Gb/s/lane PAM4 modules, not a CEI compliance claim. Sources cited in the paragraph above.

LPO and the host-as-driver path. For *LPO*, the host SerDes (or a linear driver in the module with no retimer) is the modulator driver. Waveform fidelity is end to end: host FFE, connector ISI, driver/modulator nonlinearity, and TIA all land in TDECQ and pre-FEC BER (Sections 3.6 and 7.4 and section 10.5.1). That is why linear-optics driver families (e.g. Semtech's 224G set) advertise CEI-224G-Linear host EQ and tunable swing: the module cannot clean up what the host launches Semtech 224G. Retimed modules hide some of this behind DSP; they also burn the watts LPO was meant to save.

Outlook. Driver roadmaps are no longer waiting on papers alone. Commercial 448G-class parts are shipping as dies: MACOM's >120 GHz MZM and EML drivers (OFC 2026) are the clearest public 400G/lane announcement MACOM, while research benches already run past 200 GBd on short RF paths. Expect a short period where optics and drivers lead host SerDes, so gearboxed 224G electrical into 448G optical remains common (Section 3.14.3). The hard problems are packaging (bondwire, FAU, faceplate connectors), co-design of peaking with modulator S_{21}, and LPO cases where driver linearity and host COM sit beside TDECQ as first-order validation items (Section 10.5.2 and section 7.9). After the driver, the next ceilings are modulator EO bandwidth and the PD/TIA noise stack (Section 3.14.3 and section 4.5).

Key idea. A 448G-class modulator driver is a >100 GHz linear SiGe amp with swing matched to V_π, not a SerDes pin. Commercial dies now claim >120 GHz for MZM/EML/TFLN; research shows $\sim$230 GBd on co-designed benches. Package and modulator match set the real ceiling; LPO makes the host the driver.

Gearboxes: when electrical and optical rates diverge. A *gearbox* converts lane rate or modulation format between electrical and optical domains. The 448G transition often uses *two* 224G electrical lanes into one 448G optical lane (Figure 3.7 and section 3.14.3) when host SerDes or connectors lag optics.

Gearbox Rate/format converter between electrical and optical lanes; common in 448G transition; incompatible with strict LPO.

Gearboxes add power, latency, and silicon area. They also break strict LPO: the host waveform no longer matches what the optical lane carries, so a retimed or half-retimed module is required [14][72]. Expect gearboxed 3.2T modules before native 448G electrical I/O and matched VSR connectors land in volume.

Putting the platforms side by side, Table 3.12 summarizes the trade space at 100–400G per λ; Section 3.14.3 expand each path. Through 200G/lane DR, EML still leads on cost and single-chip integration. Silicon rings and MZMs lead in CPO where area and CMOS fab matter, with Si MZM preferred for flat single-λ DR/FR and rings preferred for dense WDM. TFLN leads when you need $\gtrsim$100 GHz EO BW with low chirp for native 224 GBd PAM4 in a pluggable or NPO module and silicon cannot close the 112 GHz Nyquist without heavy peaking.

Platform	EO BW	Per-λ demo	Chirp	Integration	Typical use
Si microring	90–110 GHz	224–416 Gb/s PAM4	low	monolithic SiPh	CPO, WDM
Si MZM	70–100+ GHz	400 Gb/s PAM4	low	monolithic SiPh	DR/FR, CPO
TFLN MZM	108–110+ GHz	224–390 Gb/s PAM4/8	very low	hybrid + driver	400G/lane pluggable, FR
EML	70–100 GHz	200 Gb/s PAM4	low	single InP chip	DR/FR to 200G/lane

Table 3.12. IM/DD transmitter platforms at 100–400G per λ (2026 snapshot).

The honest summary: *modulating* at 224 GBd PAM4 is demonstrated in multiple material systems when the electrical drive path is short and the driver is dedicated (not a lossy meter of PCB plus OSFP connector). *Modulating* at that rate from a switch ASIC through a pluggable module is the harder system problem. Silicon photonics rings without peaking still sit below the 112 GHz Nyquist needed for 448G-PAM4; TFLN, EML, and peaked Si rings are the near-term paths. WDM (Chapter 6) and CPO remain the architectural escape hatches: more aggregate bits without forcing every electrical lane to 224 GBd on a lossy shoreline.

Where the standards conversation stands. Electrical and Ethernet standards are climbing the same ladder on slightly different clocks. OIF CEI owns the electrical I/O recipe; IEEE 802.3 owns how Ethernet names the lane and the optical PMD. Neither has locked 448G modulation yet.

OIF CEI (electrical I/O). No modulation order is locked yet in CEI-448G. PAM4 remains attractive if connectors reach $\gtrsim$112 GHz and if optics stay on PAM4 (alignment for LPO, lower FEC overhead, backward compatibility with 224G gear). PAM6 is the leading compromise for bandwidth-starved host channels if connector progress stalls [14][27][29]. Electrical IAs target a 2026–28 window [14].

[27] Cignal AI on OIF 448G workshop (April 2025).

IEEE 802.3 (Ethernet PHY/MAC). Ethernet standards trail CEI slightly but follow the same lane-doubling cadence. Table 3.13 is the map as of mid-2026.

Table 3.13: IEEE 802.3 lane-rate generations.

Project	Status	Eth. lane	PHY line
802.3df	Feb 2024	100 Gb/s	≈106.25 Gb/s PAM4 (53.125 GBd)
802.3dj	SA ballot 2026	200 Gb/s	≈212.5 Gb/s PAM4 (106.25 GBd)
802.3 400G/lane SG	Mar 2026	400 Gb/s	≈448G class (study)

802.3df (approved February 2024) defined 200/400/800 Gb/s Ethernet using **100 Gb/s lanes** (≈53.125 GBd PAM4 on the AUI/PMD). **802.3dj** is the active 200 Gb/s-lane amendment: 200/400/800/1600 Gb/s Ethernet with PAM4 near 106.25 GBd, KP4 FEC, and a large PHY portfolio (copper and IM/DD optics). The related OIF electrical class is CEI-224G near 112 GBd; keep the IEEE and CEI documents separate. The draft cleared 802.3 working-group recirculation ballot in February 2026 and entered IEEE Standards Association ballot; approval is expected around late 2026 [8][28][10].

[28] IEEE 802.3 Ethernet for AI ad hoc (400G/lane CFI, 2025–26).

400 Gb/s per lane is the next Ethernet milestone. The *Ethernet for AI* (E4AI) ad hoc incubated demand for 3.2 Tb/s ports and 400G/lane signaling [10]. A call for interest in March 2026 led to an IEEE 802.3 **400 Gb/s/lane Signaling Study Group** (chartered 13 March 2026) to draft a PAR for electrical interconnects and single-mode fiber reaches up to 500 m [29][11]. The proposed scope has two tracks: a fast track for 400/800/1600 Gb/s on 400G/lane copper and IM/DD optics (scale-up, intra/inter-rack), and a follow-on phase for **3.2 Tb/s** and additional PHYs [10][11].

[29] IEEE 802.3 400 Gb/s/lane Signaling Study Group (chartered March 2026).

That aligns with OIF CEI-448G and industry 3.2T module work, with one naming difference: IEEE quotes **400 Gb/s per lane** (MAC/info rate); CEI quotes **448 Gb/s** (line rate including FEC overhead). They refer to the same generation.

Until native 448G host SerDes and matched connectors land, expect a transition generation that gearboxes 224G electrical lanes into 448G optical lanes (Figure 3.7), then a consolidation generation with 400G/lane Ethernet PHYs where electrical and optical both run PAM4 at 224 GBd end to end.

3.15 Engineering lens

3.15.1 How it works

IM/DD is the whole game in one line: put data on optical power, read it with a square-law photodiode, and let FEC and equalization buy back the SNR that PAM4 spends. Everything else in this chapter is where each block in that chain lives and how the standards name it.

3.15.2 How it is measured

Name the reference plane before naming the metric. At TP2, use a calibrated power meter and digital communications analyzer (DCA) for average power, outer optical modulation amplitude (OMA), extinction ratio (ER), relative level mismatch (RLM), and transmitter and dispersion eye closure quaternary (TDECQ). At TP3, use a stressed source, attenuator, and bit-error-ratio tester (BERT) for receiver sensitivity. Use a vector network analyzer (VNA) for electrical and electro-optic *S*-parameters when bandwidth or reflections are suspect. Acceptance limits come from the optical physical-medium-dependent (PMD) specification and the program acceptance test plan (ATP), with the test planes in Section 3.9 and methods in Chapter 7 [8, 79].

3.15.3 How it fails

The common escapes are low ER from a bad bias point, low OMA from loss or limited drive, high TDECQ from bandwidth or nonlinearity, a receiver driven into saturation, and reflection-induced burst errors. Multi-lane links add power imbalance and electrical lane mismatch. A clean average-power reading does not clear the transmitter because ER, RLM, and noise can still close the eye.

Failure mode: Low extinction ratio

Symptoms. Average power passes, but OMA is low and receiver sensitivity degrades.
Likely causes. Incorrect laser or modulator bias, a compressed driver, a detuned ring, or an aged electro-absorption modulator.
Measurements. DCA level histograms, ER and RLM, bias sweeps, wavelength, and pre-FEC BER versus received power.
Mitigations. Restore the bias point, remove compression, retune the resonance, and add the failing corner to calibration and ATP.

3.15.4 How it is debugged

For degraded receiver sensitivity, first verify the power meter and reference plane. Then measure OMA, ER, RLM, and TDECQ at TP2. If the transmitter passes, inspect insertion loss and optical return loss (ORL), then run stressed sensitivity at TP3. A power sweep separates a noise-limited waterfall from saturation or a BER floor. A golden transmitter and receiver split the last ambiguity. Change one block at a time and preserve the failing state before cleaning, reseating, or retuning it.

Debug story

Observed. Receiver sensitivity was several dB worse than the qualification baseline.
Investigation. Average power passed, but the DCA showed low ER. A bias sweep restored both ER and the BER waterfall.
Finding. The receiver was healthy; the transmitted zero level was too high.
Root cause. A laser-bias calibration table selected the wrong temperature bin.
Resolution. The calibration was corrected, limits were added across temperature, and the escaped firmware revision was quarantined.

3.16 Interview takeaway

Key idea. IM/DD is intensity in, power out, with FEC and DSP closing the gap that PAM4's SNR penalty opens. Know OMA, ER, chirp and dispersion, pre-FEC BER, and the reference plane for every number. Then measure, bisect, and correct the failing block.

Junior mistake: treat average power as signal quality, or skip the named plane before arguing TDECQ (Section 4.8 and chapters 4 and 7).

Three questions to test yourself.

1. Why is PAM4 preferred over NRZ at 200G/lane despite its SNR penalty?
2. Average power is in spec but BER fails. What do you measure next?
3. What production escape leaves power in range while BER fails?

Chapter 4

Quantitative models: noise, RIN, and BER

The previous chapters argued mostly in architecture and measurement vocabulary. This one puts numbers behind the two questions every link engineer eventually asks: *what bit-error ratio (BER) will this receiver deliver?* and *what is the minimum signal it needs?* The answers follow a short chain of physics that has been stable for decades of IM/DD design: Gaussian statistics at the decision circuit, a handful of noise sources, and the way relative intensity noise (RIN) turns into an error floor. Understanding that chain is what lets you read a sensitivity number or a RIN floor without treating the datasheet as magic.

Everything here is backed by short, reproducible Python (in `sims/`), so the figures are computed curves rather than sketches. The models follow the treatment in Säckinger's *Analysis and Design of Transimpedance Amplifiers for Optical Receivers*,[1] and the sanity checks in the code reproduce that book's worked numbers.

[1] E. Säckinger, *Analysis and Design of Transimpedance Amplifiers for Optical Receivers*, Wiley, 2018, ch. 4.

4.1 The decision: from Q to BER

A binary receiver samples a noisy voltage and compares it to a threshold. If the noise is Gaussian and the threshold is optimally placed, the BER depends on a single quality factor Q, the separation between the one and zero levels measured in units of their combined noise:

$$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}, \qquad \mathrm{BER} = \tfrac{1}{2}\,\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right).$$

This binary *Q*-to-BER equation is an *equivalent-quality* approximation used throughout link budgeting. A full PAM4 treatment evaluates three decision boundaries, unequal level noise, nonlinearity, equalization, and Gray mapping;

do not treat the binary formula as a complete PAM4 model.[2] Two reference points anchor the binary curve: the classic uncoded target BER $= 10^{-12}$ needs Q $= 7.03$. For a named KP4-class optical PMD under its random-error assumption, a pre-FEC objective near 2.4×10^{-4} corresponds to Q ≈ 3.5 on this binary map (Chapter 3). That number is not a universal Reed–Solomon threshold; state the FEC architecture, error model, target PMD, and post-FEC metric.

[2] Q here is the electrical decision quality, not a laser-cavity or resonator Q.

```
from scipy.special import erfc, erfcinv
import numpy as np

def q_to_ber(q):
    return 0.5 * erfc(q / np.sqrt(2))

def ber_to_q(ber):
    return np.sqrt(2) * erfcinv(2 * ber)
```

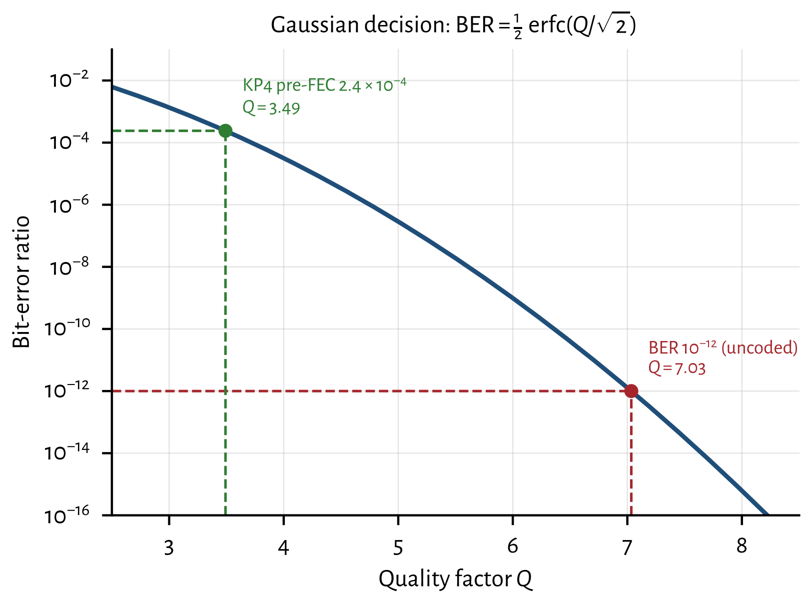

Figure 4.1: The Gaussian decision curve. Every dB of Q buys orders of magnitude of BER near the knee, which is why FEC (trading a modest Q for a huge BER improvement) is decisive.

Figure 4.1 shows why the curve is so steep near the operating point: a small change in Q (equivalently, in received power) moves the BER by orders of magnitude. This steepness is exactly what FEC exploits: nudging the required Q from 7.03 down to 3.5 relaxes the optical power budget by several dB.

4.2 The receiver noise budget

Q is only as good as our estimate of σ. Three noise sources dominate a short-reach IM/DD receiver, and they add in quadrature (they are independent):

Thermal / circuit noise the input-referred noise of the TIA and following stages, roughly white, so its variance scales with the noise bandwidth: $\sigma^2_{\text{th}} = i'^2_n \, \text{BW}$, where i'_n is a noise-current density (A/$\sqrt{\text{Hz}}$). *Signal-independent.*

Shot noise from the discreteness of photocurrent, $\sigma^2_{\text{shot}} = 2q\,I\,\text{BW}$. *Grows with signal*, so it is larger on the ones than the zeros.

RIN (relative intensity noise) the laser's own intensity fluctuations, $\sigma^2_{\text{RIN}} = \text{RIN}_{\text{lin}}\, I^2\, \text{BW}$, with $\text{RIN}_{\text{lin}} = 10^{\text{RIN[dB/Hz]}/10}$. *Grows with the square of signal*, the key fact of the next section.

Because shot and RIN noise are signal dependent, we evaluate σ_1 and σ_0 separately and form Q with their sum. The core of the model is just this:

```
def nrz_q(p_avg_w, responsivity, i_thermal_rms, bw,
          er_db=np.inf, rin_db_hz=-np.inf):
    p1, p0 = er_levels(p_avg_w, er_db)      # one / zero
        ↪ optical powers
    i1, i0 = responsivity * p1, responsivity * p0
    var1 = i_thermal_rms**2 + 2*Q_E*i1*bw   # thermal +
        ↪ shot
    var0 = i_thermal_rms**2 + 2*Q_E*i0*bw
    if np.isfinite(rin_db_hz):
        rin = 10**(rin_db_hz / 10)
        var1 += rin * i1**2 * bw            # RIN grows
            ↪ as I^2
        var0 += rin * i0**2 * bw
    return (i1 - i0) / (np.sqrt(var1) + np.sqrt(var0))
```

4.3 RIN and the BER floor

Here is the consequence that makes RIN worth its own section. Thermal noise is fixed, so pouring on more optical power raises $I_1 - I_0$ while σ_{th} stays put, so Q improves without limit. But RIN noise scales *with the signal itself*: $\sigma_{\text{RIN}} \propto I$. Once RIN dominates, the signal and its noise grow together and Q stops improving. Taking the high-power, high-extinction limit (thermal and shot negligible, $I_0 \to 0$):

$$Q_{\text{max}} = \frac{I_1}{\sigma_{\text{RIN},1}} = \frac{1}{\sqrt{\text{RIN}_{\text{lin}}\,\text{BW}}}.$$

This is a hard ceiling: no amount of transmit power or receiver sensitivity can push Q past it, so there is a BER floor set entirely by the laser and the bandwidth. Equivalently, the power penalty to hold a target Q is

$$\text{PP} = \frac{1}{\sqrt{1 - Q^2\,\text{RIN}_{\text{lin}}\,\text{BW}}},$$

which diverges as $Q \to Q_{\text{max}}$.

Engineering heuristic. If the waterfall floors, stop raising launch power as the primary fix. You are buying photons against a non-power-limited impairment.

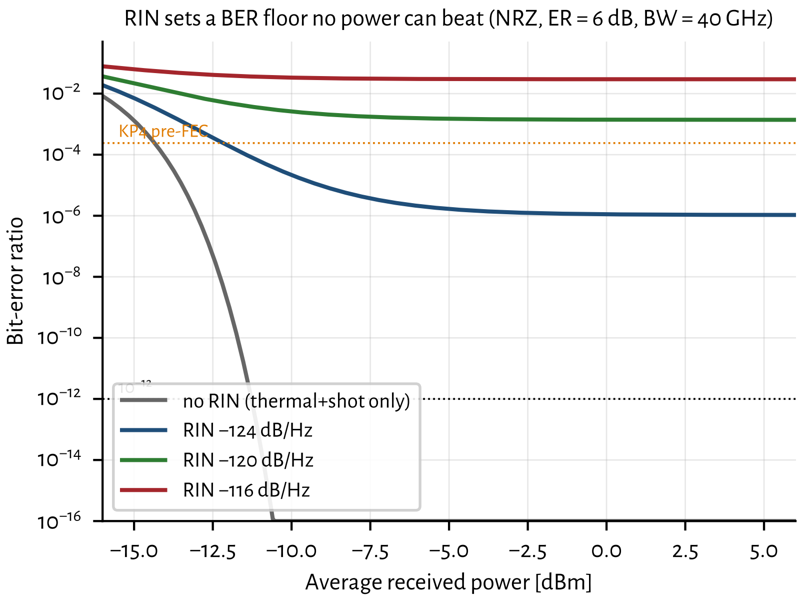

Figure 4.2: With RIN present, the BER stops falling no matter how much power is added. The thermal/shot-only curve dives; each RIN level flattens into a floor. (RIN values here are deliberately high to make the floor visible in-frame; good DFBs at < -150 dB/Hz have no floor at these rates; see text.)

Figure 4.2 shows the floor directly; Figure 4.3 plots the ceiling versus RIN for three lane rates. The bandwidth dependence matters: doubling the lane rate doubles the noise bandwidth and drops Q_{max} by $\sqrt{2}$ (≈ 1.5 dB of margin), so RIN that is harmless at 25G can bite at 200G. This is the quantitative reason the laser chapter (Chapter 5) lists RIN among the parameters that decide pass/fail.

4.3.1 Typical RIN values (2026)

How much RIN headroom do real sources have? Table 4.1 collects representative figures. Two cautions on reading them. First, standards quote RIN_xOMA, RIN referenced to the OMA and measured under a specified optical return loss (ORL) x, because back-reflections into the laser raise its noise, so a spec limit is a stressed, worst-case number, not the device's quiet intrinsic RIN.[3] Second, RIN degrades with optical feedback, so isolator-free and co-packaged designs care as much about *feedback tolerance* as about the isolated number, one reason quantum-dot lasers (near-zero linewidth-enhancement factor) are attractive for CPO [56].

A RIN number in dB/Hz is, by itself, incomplete: because RIN is *relative*, it only becomes an absolute noise current once the photocurrent $I = \mathcal{R}P$ is fixed. The intensity-noise current density is

$$i_{\text{RIN}} = \sqrt{\text{RIN}_{\text{lin}}}\, I \quad [\text{A}/\sqrt{\text{Hz}}], \qquad S_{\text{RIN}} = \text{RIN}_{\text{lin}}\, I^2 \quad [\text{A}^2/\text{Hz}],$$

[3] For example IEEE 802.3 / 100G Lambda MSA short-reach PAM4 links cap $\text{RIN}_{17.1}\text{OMA}$ at -136 dB/Hz with 17.1 dB ORL .

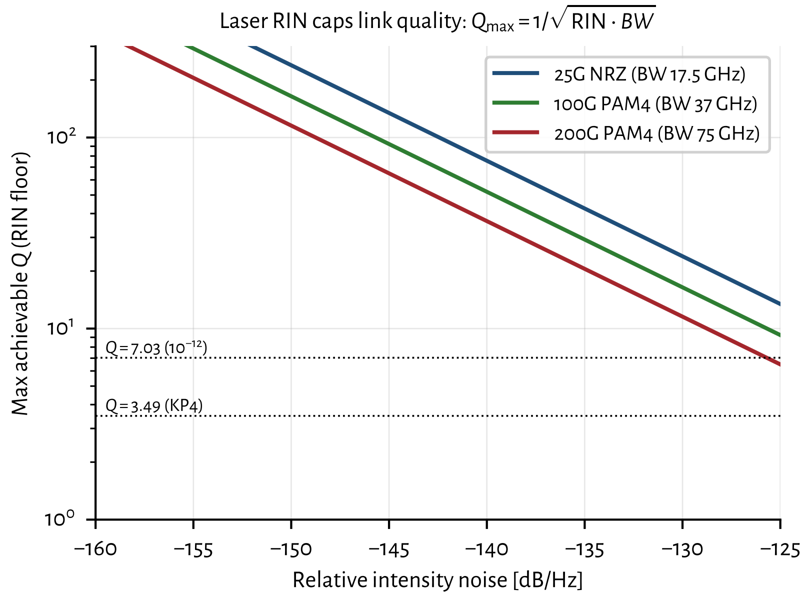

Figure 4.3: The RIN ceiling $Q_{max} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$. Wider receiver bandwidth (higher lane rate) integrates more RIN, lowering the ceiling. Where a curve dips below the dotted anchors, that link can no longer reach the corresponding BER.

so it scales linearly with received power. Table 4.1 therefore lists both the RIN and the current density it produces at a common reference operating point ($\mathcal{R} = 0.8$ A/W, $P_{rx} = 0$ dBm, i.e. $I = 0.8$ mA), the units a receiver designer actually compares against.

For scale, at that same 0.8 mA the *shot*-noise density is $\sqrt{2qI} = 16$ pA/$\sqrt{\text{Hz}}$ ($S = 2.6 \times 10^{-22}$ A^2/Hz) and a good high-speed TIA adds roughly 25 pA/$\sqrt{\text{Hz}}$ of thermal noise. So a VCSEL at -140 dB/Hz (80 pA/$\sqrt{\text{Hz}}$) already dominates both, while a heterogeneous source at -160 dB/Hz (8 pA/$\sqrt{\text{Hz}}$) is a minor term. The key asymmetry: thermal noise is fixed and shot grows only as $\sqrt{I}$, but RIN grows as I, so at low received power thermal wins and RIN is irrelevant, and only above a break-in power (Figure 4.4) does RIN take over. That is why quoting a RIN figure without an operating power says little.

Put these against the simplified dominant-RIN ceiling $Q_{max} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$. Assumptions: RIN dominates, noise bandwidth is defined, level structure is idealized, no stronger deterministic impairment is present, and the receiver is not compressed. At 200G-PAM4 bandwidths (BW ≈ 75 GHz), plugging the stressed -136 dB/Hz RIN$_x$OMA number into that intrinsic-style formula is illustrative only; the definitions do not automatically align. Even under that optimistic plug-in, $Q_{max} \approx 23$ is far above $Q = 7$ for 10^{-12}, so for well-behaved sources RIN is often not the limiter; thermal noise is. RIN becomes the story when feedback, aging, or a marginal source pushes the *effective* intensity noise toward -125 dB/Hz class. A third path is electrical: laser bias-driver current noise converts to equivalent RIN (Section 5.8) and must be budgeted separately from intrinsic laser RIN.

Source	RIN (dB/Hz)	i_{RIN} @ 0 dBm	Metric	Note
400G-FR4 / 100G Lambda class limit	≤ -136	≥ 127	stressed RIN$_x$OMA	named ORL (e.g. 17.1 dB)
Datacom VCSEL, 850 nm (MMF)	-135 to -145	45 to 142	intrinsic	quiet parts ≤ -145
Good datacom DFB / EML	-145 to -155	14 to 45	intrinsic	CPO ELS often target ≤ -145
Quantum-dot laser on Si	-140 to -150	25 to 80	intrinsic	feedback-tolerant class
Heterogeneous / injection-locked Si	-155 to -165	4.5 to 14	intrinsic	high-Q feedback
Lab record (QD + injection lock)	~ -168	~ 3.2	intrinsic	research

Table 4.1. Representative RIN by source (c. 2026) at $\mathcal{R} = 0.8$ A/W, $P_{\text{rx}} = 0$ dBm ($I = 0.8$ mA). Intrinsic RIN and stressed RIN$_x$OMA are different metrics; do not convert them through one equation unless the definitions align. Density $i_{\text{RIN}} = \sqrt{\text{RIN}_{\text{lin}}}\, I$.

Key idea. Thermal noise is beaten by power; dominant RIN is not. Under a simplified model, $\sigma_{\text{RIN}} \propto I$ imposes $Q_{\text{max}} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$. Keep intrinsic RIN and stressed RIN$_x$OMA separate. Feedback, aging, and bias-board noise erode the margin that quiet intrinsic sources appear to have.

4.4 Sensitivity and OMA

4.4.1 Common misconception: sensitivity does not guarantee a working link

Received power exceeding the datasheet sensitivity number does not guarantee low BER. Sensitivity is measured under specific, controlled assumptions: a clean transmitter, known extinction ratio, specified ORL, and a particular equalization state. In the field, BER depends on all of these simultaneously:

- received optical power;
- extinction ratio and OMA (not just average power);
- RIN under the actual ORL;

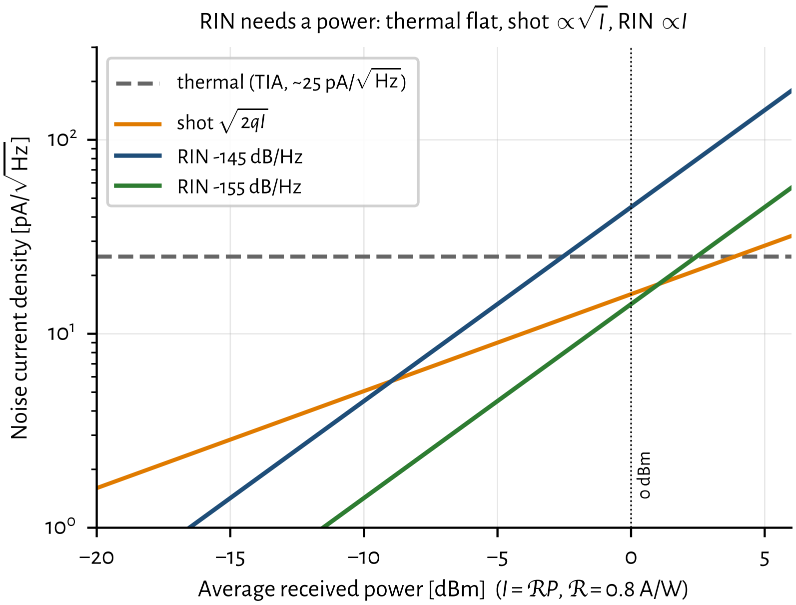

Figure 4.4: Noise current densities versus received power. Thermal is flat, shot $\propto \sqrt{I}$, RIN $\propto I$; the RIN curves cross the fixed thermal floor only above a break-in power, which is why RIN only "makes sense" once the optical power (hence $I = \mathcal{R}P$) is stated.

- jitter (random and deterministic);
- ISI from bandwidth limits and dispersion;
- receiver thermal noise and any TIA compression;
- equalization state and FEC margin.

Sensitivity is therefore a shorthand for receiver noise performance under reference conditions, not a complete description of whether the link closes. A link that "has enough power" can still fail if ER is poor, RIN is elevated, or the equalizer is saturated. Always close the full link budget (Section 7.7), not just the power line.

Tradeoff. Better receiver sensitivity vs receiver complexity

Improves: APD gain, stronger DSP, deeper equalization, or quieter optics can close a tight budget

Worsens: Power, latency, calibration burden, and reliability of the more complex path

When acceptable: When the sensitivity line is the true limiting ledger after Tx and plant are sound

Experienced decision: Improve the weakest link, not the most visible datasheet metric.

4.4.2 OMA versus average power

Two links may have identical average optical power but different BER. The reason is OMA and extinction ratio. At fixed average power P_{avg}, the OMA depends on ER through

$$\frac{\text{OMA}}{P_{\text{avg}}} = \frac{2(\text{ER} - 1)}{\text{ER} + 1},$$

where ER is the linear extinction ratio P_1/P_0. A link with 10 dB ER ($\text{ER}_{\text{lin}} = 10$) delivers $\text{OMA}/P_{\text{avg}} = 18/11 \approx 1.636$, while 6 dB ER ($\text{ER}_{\text{lin}} \approx 4$) delivers $6/5 = 1.200$. The difference is about 1.36 dB of OMA at the same average power. Whether that 1.36 dB matters depends on where the receiver sits on its BER waterfall: near the sensitivity cliff a 1 dB OMA change moves BER by orders of magnitude (Section 4.1 and figure 4.2), while well above sensitivity the same change is invisible. The impact also depends on RIN, bandwidth, equalization state, and TDECQ; ER alone does not set BER.

For PAM4, the relevant quantity is the outer OMA ($P_3 - P_0$), and level spacing (RLM) matters independently. A transmitter that launches adequate average power but has compressed inner eyes (poor RLM) will fail TDECQ even though a power meter reads in-spec. This is why TDECQ, OMA, and RLM are specified together, not average power alone (Section 7.4).

4.4.3 The sensitivity formula

Turning the question around (*what is the least power that meets a target BER?*) gives the sensitivity. Referring the receiver's input noise current i_n back to the optical input through the responsivity $\mathcal{R}$:

$$P_{\text{sens}} = \frac{Q\, i_n}{\mathcal{R}} \qquad \text{(average power)}, \qquad P_{\text{sens}}^{\text{OMA}} = \frac{2\, Q\, i_n}{\mathcal{R}}.$$

Modern short-reach standards specify *OMA* rather than average power. For binary NRZ, $\text{OMA} = P_1 - P_0$; for PAM4, outer OMA is $P_3 - P_0$. OMA decouples the sensitivity spec from extinction ratio. As a check, the textbook example ($i_n = 1\ \mu$A, $\mathcal{R} = 0.8$ A/W, BER $= 10^{-12}$) gives $P_{\text{sens}} = 7.03 \times 1\ \mu\text{A}/0.8 = 8.8\ \mu$W, or -20.6 dBm, which the code reproduces. A finite extinction ratio costs an idealized further $\text{PP}_{\text{dB}} = 10 \log_{10}[(\text{ER}_{\text{lin}} + 1)/(\text{ER}_{\text{lin}} - 1)]$: 0.87 dB at 10 dB ER, 2.2 dB at 6 dB ER. These penalties feed link budgets (Chapter 3) and TDECQ discussion (Chapter 7); do not double-count TDECQ as an independent penalty if the compliance method already includes it.

Worked example: DR4-class budget check. Take a 200G/lane DR-class estimate with Ge-on-Si PIN ($\mathcal{R} = 0.9$ A/W), TIA $i_n = 13$ pA/$\sqrt{\text{Hz}}$, noise bandwidth ≈ 60 GHz, and a KP4-class pre-FEC objective 2.4×10^{-4} mapped to binary $Q \approx 3.5$ (Sections 3.12 and 4.1). Reference plane: receiver optical input. Assume Gaussian noise, equalized eye, no compression, and RIN not yet dominant.

Integrated thermal-class noise: $i_n\sqrt{\mathsf{BW}} \approx 13 \times 10^{-12} \times \sqrt{60 \times 10^9} \approx 3.2\,\mu$A rms. Required OMA under the binary sensitivity formula:

$$P_{\mathsf{OMA,sens}} = \frac{2Qi_n\sqrt{\mathsf{BW}}}{\mathcal{R}} \approx \frac{2 \times 3.5 \times 3.2\,\mu\mathsf{A}}{0.9} \approx 25\,\mu\mathsf{W} \approx –16\,\mathsf{dBm}.$$

Illustrative allocations (not a normative budget): $\sim$3 dB TDECQ-class Tx impairment *or* remaining margin after a TDECQ-based OMA method (not both), $\sim$2 dB connector/fiber at stated planes, $\sim$2 dB system margin. That lands near –9 dBm class OMA at the receiver input, or roughly –6 to –4 dBm launched depending on reach. Parameter sensitivity: $\pm$1 dB in i_n or $\mathcal{R}$ moves the estimate about 1 dB; a RIN floor or MPI can invalidate the waterfall entirely.

Where the model stops: unequal PAM4 levels, level-dependent noise, strong ISI, compressed TIA, correlated RIN, or a different FEC/error model. If measured sensitivity is worse, bisect RIN, reflections, and ER (Section 7.7 and section 7.2.2).

Model-validity checklist. Before applying Q, Q_{max}, or P_{sens}, ask:

- Is the noise approximately Gaussian at the decision point?
- Is the noise bandwidth defined for this measurement?
- Are PAM4 levels equally spaced (or is RLM budgeted)?
- Is noise level-dependent (shot/RIN) accounted for?
- Is the receiver compressed or overloaded?
- Is residual ISI deterministic rather than noise-like?
- Is RIN intrinsic, stressed RIN_xOMA, or bias-board noise?
- Is the FEC threshold appropriate for this PMD and error model?
- Is the input metric average power or OMA, and at which reference plane?

4.5 Receiver technologies and their noise (2026)

The sensitivity formula $P_{sens} = Q\,i_n/\mathcal{R}$ has exactly two device inputs: the photodiode responsivity $\mathcal{R}$ and the amplifier's input-referred noise current i_n. So "receiver noise performance" is really a statement about the detector–TIA pair, and the winning short-reach recipe is the one the question anticipates: a *waveguide germanium-on-silicon PIN* feeding a *tightly integrated CMOS or SiGe-BiCMOS TIA*. The reasons are all in the two parameters above plus a parasitic (Section 4.5 details the TIA side).

Photodiodes and transimpedance amplifiers. The photodiode converts photons to photocurrent; the *TIA* (transimpedance amplifier) converts current to voltage with low input-referred noise. For PAM4 at 100–224G/lane:

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

- **PIN (Ge-on-Si):** no internal gain; lowest excess noise; common short-reach choice (Table 4.4). Capacitance at the TIA input dominates i_n.
- **APD**: internal multiplication. Any sensitivity benefit versus PIN depends on gain, excess-noise factor, bandwidth, receiver design, and BER target; published Ge/Si APD results often cite roughly 5–9 dB under specific conditions, not as a universal gain [61].
- **UTC/MUTC:** electron-only transport for > 200 GHz BW and high saturation; used when linearity and speed beat raw sensitivity [65].

PIN PIN photodiode: p-intrinsic-n structure; no internal gain. Ge-on-Si waveguide PIN is the mainstream short-reach detector.

APD Avalanche photodiode: internal gain improves sensitivity; excess noise factor and bias complexity vs PIN.

UTC Uni-traveling-carrier photodiode: electron-only transport; >200 GHz BW and high saturation current. Linear/LPO niche.

The TIA often embeds *CTLE* for LPO (Section 3.6 and section 10.5.1): a fixed high-frequency boost before the host SerDes ADC. Co-packaging PD and TIA (sub-mm interconnect) is the noise win repeated throughout this book (Chapter 2).

Why Ge-on-Si wins the mainstream. Germanium grown on silicon absorbs the O- and C-bands, is fully CMOS-process-compatible, and is built *on the same PIC* as the modulators and couplers. Modern devices reach $\mathcal{R} \approx 0.8$–1.0 A/W with dark currents of single-digit to tens of nA and, in 2025 research, –3 dB bandwidths beyond 100 GHz (e.g. a recessed Ge/Si PIN at 106 GHz, 0.93 A/W, < 10 nA; SiN-coupled lateral Ge > 110 GHz at 1 mA) [58]. Crucially, monolithic or 3D/flip-chip co-integration keeps the PD-to-TIA interconnect sub-millimetre, so the input node capacitance stays tens of fF, and since a TIA's input-referred noise rises with that capacitance ($i_n \propto C f^{3/2}$ in the front-end limit), *short is quiet*. This is the same capacitance argument that drives co-packaging in Chapter 2 (Miller's receiver point), now cashed out as noise current.

What the TIA must deliver. The TIA is the receive twin of the modulator driver (Section 3.14.3). At 224 GBd PAM4 you need bandwidth $\gtrsim$50–70 GHz for a 112 GBd Nyquist-class front-end (often less than the Tx driver BW because the optical channel and reference receiver already band-limit; LPO pushes for flatter, more linear TIAs), input-referred noise in the low teens of pA$/\sqrt{\text{Hz}}$ once co-packaged with a low-C PD, linearity / overload so large OMA and reflections do not crush PAM4 levels (RLM) or trip AGC into a bad corner, and optional CTLE for LPO/LRO so the host SerDes sees a usable eye without module DSP (Section 3.6 and section 10.5.1). Noise scales with input capacitance: $i_n \propto C f^{3/2}$ in the front-end limit. That is why co-packaging (or monolithic PD+TIA) is not optional at 200G+: every millimetre of bondwire is noise and BW you cannot recover with FEC.

Noise levels you actually budget. Table 4.2 puts the model numbers next to published front-ends. Shot noise at 0 dBm into $\mathcal{R} = 0.8$ A/W is $\sqrt{2qI} \approx 16$ pA/$\sqrt{\text{Hz}}$; a good TIA sits near that floor. Worse i_n or higher C burns sensitivity linearly via $P_{\text{sens}} = Q\, i_n/\mathcal{R}$ (Section 4.4).

Front-end	i_n (pA/$\sqrt{\text{Hz}}$)	BW / rate	Sensitivity note
Typical "good" short-reach TIA (book model)	$\sim$25	—	older rule-of-thumb floor
16-nm CMOS + co-pkg PD	16.9	32 GHz / 112G PAM4	–8.2 dBm class
55-nm SiGe $4\times$112 GBd linear TIA	13.2	65 GHz / 224G	$\sim$1.2 pJ/bit
Shot noise @ 0 dBm, $\mathcal{R} = 0.8$ A/W	$\approx$16	—	physics floor at that power

Table 4.2. Input-referred TIA noise densities used in link budgets (c. 2023–26 published front-ends). Integrate $i_n\sqrt{\text{BW}}$ for rms noise before applying Q (Section 4.4). Sources in text.

2026 linear PAM4 front-ends land around 10–17 pA/$\sqrt{\text{Hz}}$: e.g. 16.9 pA/$\sqrt{\text{Hz}}$ at 112G in 16-nm CMOS (–8.2 dBm sensitivity) [4][59] and 13.2 pA/$\sqrt{\text{Hz}}$ at 224G in 55-nm SiGe BiCMOS (65 GHz BW, 1.2 pJ/bit) [5][60]. Put these in the model: $i_n \approx 13$ pA/$\sqrt{\text{Hz}}$ integrated over ~ 60 GHz is ≈ 3.2 μA rms, giving an NRZ average sensitivity near -15 dBm; the PAM4 level penalty lands OMA sensitivities in the -8 to -10 dBm range these front-ends report.

[4] ??112G PAM4 linear TIA in 16-nm CMOS: 16.9 pA/$\sqrt{\text{Hz}}$; –8.2 dBm.

[5] ??224G SiGe TIA: 65 GHz BW, 13.2 pA/$\sqrt{\text{Hz}}$, $\sim$1.2 pJ/bit.

Industry snapshot — mid-2026 (receivers). Table 4.3 pairs detectors and TIAs with maturity labels. Commercial linear-optics TIAs (Semtech GN1834L/DL, GN1838DL) target LPO/LRO/CPO at 224G/lane with on-chip EQ (production / sampling class). Semtech has also shown 448G-class PMD ICs (TN14740 TIA) at OFC 2026 demos (vendor demonstration; not a volume datasheet claim) [37][6][64]. On the detector side, recessed Ge/Si PINs at 106 GHz / 0.93 A/W [58], Ge/Si UMC-APDs at 105 GHz with a published $\sim$9 dB sensitivity gain over PIN at 224/260G PAM4 under that paper's conditions [7][61], waveguide Ge/Si APDs toward 100 GHz at 2 A/W [8][62], and OFC 2026 Ge-on-Si APDs at 180 GBd PAM4 [9][63] mark the research edge. UTC/MUTC PDs remain the high-saturation / > 200 GHz niche [65].

[6] Semtech OFC 2026: 448G/lane TN14740 TIA + TN622 driver demos (provisional).

[7] Ge/Si UMC-APD: 105 GHz BW; 224/260G PAM4 at -10.9/-10.1 dBm.

[8] JLT 2026: waveguide Ge/Si APD >100 GHz; 2 A/W at 70 GHz.

[9] OFC 2026 Th3F.2: Ge-on-Si APD to 180 GBd PAM4 (O/C-band).

Reasonable alternatives to Ge PIN + quiet TIA. Table 4.4 lays the detector menu out. III-V InGaAs PINs (flip-chipped) trade monolithic integration for higher power handling and remain common in discrete modules. Avalanche photodiodes add internal gain that can improve sensitivity when gain, excess noise, bandwidth, and BER target cooperate (published short-reach results often land near 5–9 dB under specific conditions) at the cost of bias complexity and excess noise; device bandwidth above 100 GHz is no longer rare in research [61–63]. Uni-traveling-carrier (UTC/MUTC) PDs use electron-only transport for very high saturation current, linearity, and bandwidth (> 200 GHz) but modest responsivity, a fit for linear/LPO and

Part / paper	Type	BW / i_n or $\mathcal{R}$	Rate / sens.	Note
Semtech GN1834L/DL / GN1838DL	TIA	224G-class; linear+EQ	224 Gb/s/lane	Commercial LPO/CPO family
Semtech TN14740 (demo)	TIA	448G-class	448G/lane demo	OFC 2026 booth; provisional
SiGe 4×112 GBd TIA	TIA	65 GHz; 13.2 pA/$\sqrt{\text{Hz}}$	224G PAM4	Research / product-class paper
16-nm CMOS + co-pkg PD	TIA+PD	16.9 pA/$\sqrt{\text{Hz}}$	112G; -8.2 dBm	Co-packaged win
Recessed Ge/Si PIN	PD	106 GHz; 0.93 A/W	200 GBd-class	<10 nA dark
Ge/Si UMC-APD	APD	105 GHz @ $M\approx7$	224/260G; -10.9/-10.1 dBm	$\sim$9 dB over PIN
Ge/Si WG APD (300 mm)	APD	>100 GHz; 2 A/W @ 70 GHz	400G/lane target	7 V bias class
OFC 2026 Ge-on-Si APD	APD	70–100 GHz; 1.5–2 A/W	180 GBd PAM4	O- and C-band
UTC / MUTC-PD	PD	>110–200 GHz	high I_{sat}	Linear / LPO niche

Table 4.3. Receiver snapshot (c. 2025–26). Commercial TIA rows are vendor announcements; APD/PIN rows mix production-intent SiPh with research demos. Sensitivities are as published (FEC threshold varies).

> 200 GBd analog optics more than for raw sensitivity [65]. SOA-preamplified receivers bolt optical gain ahead of the PD for large effective responsivity and reach, but pay in ASE noise figure, power, and complexity.

SOA Semiconductor optical amplifier: inline optical gain (III-V); adds ASE noise. Used as preamplifier for sensitivity.

Outlook. Volume short-reach receive stays on PIN + SiGe/CMOS TIA: noise in the low teens of pA/$\sqrt{\text{Hz}}$ with >100 GHz Ge PINs is enough for DR/FR PAM4 when co-packaged, so the fight is capacitance and yield, not a new detector physics. Linear optics raises the bar: LPO/LRO need high linearity, on-chip EQ, and multi-lane density (Semtech's 224G family is the public commercial marker; 448G TIA demos are still provisional, Section 3.14.3). APDs are back in the 200G conversation when several dB of sensitivity gain changes a power-limited plant; UTC/MUTC matter when the impairment is saturation or >200 GBd analog fidelity rather than photons per bit. Bondwire and FAU still set C and BW, which is why CPO/NPO win on noise for the same reason they win on energy (Chapter 2).

Key idea. Receiver performance is $\mathcal{R}$ and i_n, with i_n set by PD+TIA capacitance. Budget 10–17 pA/$\sqrt{\text{Hz}}$ co-packaged TIAs and >100 GHz Ge PIN/APD detectors; use APD gain or UTC saturation when the link demands it. LPO makes TIA linearity as important as raw noise.

Detector	$\mathcal{R}$ (A/W)	-3 dB BW	Integration	Where it fits
Ge-on-Si waveguide PIN	0.8–1.0	60–>100 GHz	monolithic on SiPh	mainstream short-reach / CPO
III-V InGaAs PIN (flip-chip)	0.6–0.9	60–>100 GHz	hybrid / flip-chip	discrete modules, high power
APD (Ge/Si, InP, UMC)	effective $\uparrow$ (gain)	up to ~ 100 GHz+	hybrid / emerging	power-starved links; gain benefit condition-dependent
UTC / MUTC-PD	0.1–0.8	>110–200 GHz	III-V	linear/LPO, >200 GBd, high saturation
SOA-preamplified PD	effective $\gg 1$	~ 50 GHz	III-V PIC	tight power budgets; adds ASE noise

Table 4.4. Short-reach receiver detector options, c. 2026. Ranges span production to recent research; APD/UTC/SOA figures are effective (with gain) or device-record.

4.6 NRZ versus PAM4, at equal bit rate

The 224G-per-lane roadmap (Chapter 3) rides on PAM4, so it is worth seeing the trade quantitatively. PAM4 sends two bits per symbol using four levels, so at a fixed bit rate its symbol rate (and thus noise bandwidth) is halved, collecting less noise. But its three eyes each span only a third of the OMA, costing roughly $20 \log_{10} 3 \approx 9.5$ dB of vertical separation. Figure 4.5 pits the two at a common 100 Gb/s: PAM4's narrower bandwidth partly offsets its level penalty, but it still needs several dB more received power for the same BER, repaid by halving the electrical bandwidth the SerDes and optics must support. That balance is exactly why 100G/lane went NRZ and 200G/lane went PAM4.

4.7 Engineering lens

4.7.1 How it works

Binary link models often collapse receiver performance to a quality factor Q: signal separation divided by combined noise. That is a useful equivalent-quality map, not a complete PAM4 theory. Thermal noise yields to more power; dominant RIN does not, which is why a BER floor can exist. The rest of this chapter computes, measures, and defends Q under stated assumptions.

4.7.2 How it is measured

A BER model earns trust only when every term has a bench measurement. Use a BERT and calibrated optical attenuator for BER versus received OMA. Measure re-

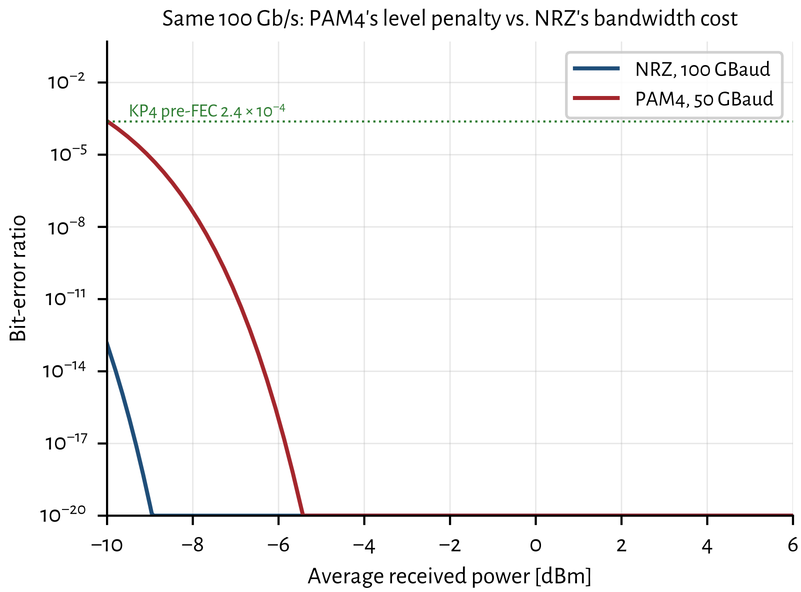

Figure 4.5: NRZ (100 GBaud) versus PAM4 (50 GBaud) at the same 100 Gb/s. PAM4 pays a level penalty but relaxes bandwidth; the crossover with the KP4 pre-FEC threshold sets the required operating power.

ceiver input-referred noise with the photodiode dark and illuminated. Use a DCA for OMA, ER, and eye quality, and a photodiode plus electrical spectrum analyzer for relative intensity noise (RIN). Repeat power sweeps at temperature and per lane. Acceptance is a curve, not one point: the required waterfall must cross the program's pre-FEC threshold with margin and without a floor (Sections 4.1 to 4.4).

4.7.3 How it fails

Thermal noise dominates at low optical power, shot noise grows with photocurrent, and RIN grows with signal power. Reflections, crosstalk, jitter, and compression break a simple Gaussian fit. The signatures differ: a horizontal BER floor points to signal-proportional noise or bursts; a uniform horizontal shift points to lost power or added receiver noise; one lane moving alone points to its optical or electrical path.

Failure mode: BER floor

Symptoms. BER improves with received power, then stops improving.
Likely causes. RIN, reflection-driven multipath interference, laser-bias noise, crosstalk, or periodic jitter.
Measurements. BER versus OMA, RF RIN spectrum (PD+ESA or RIN analyzer, not OSA), ORL sweep, FEC error distribution, and a quiet laser-bias source.
Mitigations. Remove the correlated noise source, improve ORL, isolate supplies, or reject the laser if its intrinsic RIN misses the ATP.

4.7.4 How it is debugged

When pre-FEC BER moves from well below the named FEC threshold into a 10^{-6}-class regime, save the failing condition and work in this order: received OMA, transmitter OMA and ER, receiver noise, RIN and ORL, electrical jitter, and lane crosstalk. Sweep power before changing settings. If the curve recovers with power, quantify the sensitivity shift. If it floors, split intrinsic laser noise from board noise and reflection feedback. If only temperature moves it, repeat the same terms hot and cold instead of assigning the change to “thermal margin.”

Debug story

Observed. One lane developed a BER floor after the product board replaced the bench supply.
Investigation. The optical power sweep flattened. Intrinsic RIN on the quiet source passed, but discrete tones appeared with the product rails active.
Finding. The laser was not the noisy element.
Root cause. A switching regulator coupled into the laser bias path.
Resolution. The bias filter and return path were changed, and a powered-board RIN check was added to validation.

4.8 The debugging fork: did received power change?

When BER degrades, ask one question first:

```
BER degraded
  |
Stable average received power?
  |-- NO  --> Power ledger / optical-path investigation
  |           laser / coupling / connector / fiber / MUX /
    ↪ monitor
  |-- YES --> Signal-quality path
              Tx / channel / Rx / DSP
              noise / timing / spectral / control
  |
Highest-value measurement
  |
Decision + recurrence control
```

Stable average received power deprioritizes gross optical loss but does not eliminate fast power fluctuations, clipping, monitor averaging, or reflection-dependent effects.

Did average received optical power change? If yes, investigate the power path:

- laser degradation (threshold rise, slope loss, COD);
- connector contamination or damage;

- coupling loss (fiber attach, FAU drift);
- fiber break or bend loss;
- MUX/de-MUX imbalance or grid misalignment.

Or did signal quality degrade at stable average power? If average received power is stable but BER worsened, isolate transmitter, channel, receiver, and DSP quality:

- noise: RIN (intrinsic, feedback-driven, or bias-driver), Rx noise, crosstalk, MPI;
- timing: jitter, CDR, skew, adaptation timing;
- spectral: wavelength, filtering, alignment;
- control: bias, APC, TEC/heaters, calibration, EQ authority.

This fork often narrows an investigation in minutes. Power-path failures show up on a meter; signal-quality failures need FEC timing, DCA, BERT, or spectrum analysis, depending on access. Apply it before opening the package, changing settings, or blaming a supplier (Section 7.12).

Why experienced engineers separate power from quality?

Because average optical power is easy to measure but tells very little about timing, noise, distortion, or adaptation. One meter reading prunes an entire branch of the tree.

What this usually means. Stable average power with rising BER

Usually: noise, timing, spectral alignment, control or calibration, intermittent contact

Not: gross attenuation as the primary story

4.9 Interview takeaway

Key idea. Four relations carry most of short-reach link design: the Gaussian BER(Q), the quadrature noise budget (thermal + shot + RIN), the RIN floor $Q_{\text{max}} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$, and the sensitivity $P_{\text{sens}} = Q\, i_n/\mathcal{R}$. Close each relation with a measured curve, then use the curve shape to debug the link.

Junior mistake: raise launch power into a BER floor, or quote sensitivity without the measurement conditions (Sections 4.3 and 4.8 and chapters 5 and 7).

Three questions to test yourself.

1. Why does RIN impose a BER floor that no amount of optical power can overcome?
2. Optical power is unchanged but pre-FEC BER worsened from well below the FEC threshold into the 10^{-6} class. Where do you start? (Section 4.8)
3. The BER curve flattens at high power. What mechanisms produce a floor?

Chapter 5

Choosing light sources and modulation

Read first: architecture paths; DML vs EML vs external modulation; LIV/SMSR/RIN; thermal versus aging; calibration.

Deep dive: bias-driver noise math; ELSFP pinout; optical-safety classes; CW-WDM source survey.

Reference: source/modulation matrix, laser PRD, measurement playbook, supplier ownership table.

Do not choose a laser by comparing data sheets in isolation. Start with reach, lane rate, fiber plant, power, cost, lifetime, manufacturing volume, and service policy. Those requirements select an architecture. The architecture then limits the useful source, modulation, detector, packaging, and validation choices.

5.1 System requirements and architecture paths

Freeze the system problem before asking a supplier for samples:

```
Reach / fiber plant
   |
Lane rate / bandwidth
   |
Laser choice
   |
Modulator path
   |
Receiver / detector
   |
Validation / qual burden (ATP, lock, thermal, life)
```

Reach and fiber decide whether multimode loss and modal bandwidth are acceptable or whether the link needs single-mode fiber.

Lane rate and bandwidth decide whether direct modulation closes the eye or whether the link needs an EML, MZM, or ring.

Power and cooling include laser wall-plug power, modulator loss, driver power, TEC power, and control overhead.

Cost and volume include fiber plant, assembly alignment, burn-in, test time, yield, and field replacement, not only die price.

Reliability and service decide whether the source can stay inside the package or must be a replaceable ELSFP.

The choices are coupled. A cost-driven, short multimode link points toward an 850 nm VCSEL, multimode fiber, silicon photodiodes, and direct modulation. A longer single-mode path points toward a 1310 nm DFB or EML and germanium or III-V detection. Dense WDM and co-packaged optics often move the CW source away from the modulator, which adds wavelength control, fiber attach, and service requirements. Section 5.6 and table 5.3 turns these paths into supplier specifications.

5.2 Light-source and modulation choices

The short-reach market uses a small set of source families. Each fits a different constraint set:

DFB (distributed feedback) a grating along the active region gives single-mode output; the workhorse continuous-wave (CW) or directly modulated source for CWDM and LAN-WDM (Section 5.4).

CW Continuous wave: unmodulated laser output. External modulators (MZM, ring, EAM) encode data onto a CW source.

DBR Distributed Bragg reflector: laser grating outside the gain region; enables tunable single-mode output.

DBR (distributed Bragg reflector) the grating sits outside the gain region. Choose it when tunability or separate control of gain and wavelength is worth added control and qualification work.

External-cavity laser a gain element and an external wavelength-selective cavity provide narrow linewidth and tunability. Choose it when spectral purity or lock range matters more than package size, cost, and control-loop simplicity. Most short-reach IM/DD links do not need it [RIO PLANEX]; the cited product is vendor orientation, not a datacenter source recommendation.

DML Directly modulated laser: modulate bias current; high chirp, low cost.

DML (directly modulated laser) modulate the bias current directly: cheap and low-power, but chirp-limited over dispersive fiber (Section 5.3).

EML (externally modulated laser) *EML*: a DFB integrated with an *EAM*. Low chirp and high bandwidth make it the dominant 100–200G/lane transmitter for single-mode links at DR (500 m) and shorter (Section 5.4 and section 3.14.3).

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

CW laser + TFLN MZM an external CW source feeds a thin-film lithium niobate Mach–Zehnder modulator on a separate chip. Very low chirp and $\gtrsim$100 GHz EO bandwidth make this the leading path to 400G/lane pluggables and high-baud FR links; see Section 3.14.3 and table 3.12.

CW laser + Si MZM an external CW source feeds a silicon Mach–Zehnder modulator on the same PIC (Section 3.14.3). Low chirp, flat passband, and CMOS fab integration make this the default for 100–200G/lane DR/FR SiPh modules; 400G/lane demos appeared in 2026.

CW laser + Si ring same laser architecture, but a microring or microdisk modulator on the PIC (Section 3.14.3). Smaller footprint and strong WDM/CPO fit; wavelength lock and thermal crosstalk dominate validation (Chapter 6).

CW-WDM / multi-wavelength sources high-power, multi-wavelength CW lasers (per the CW-WDM MSA) that feed comb-like WDM architectures (Sections 5.16 and 6.6).

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

VCSEL 850–940 nm multimode sources for short-reach links over multimode fiber; cheap but reach-limited and less relevant at 200G/lane.

External laser source (ELS/ELSFP) a pluggable laser module supplying CW light to a co-packaged switch, so a failed laser is field-replaceable (Section 5.14).

Attribute	VCSEL direct	DFB direct	DFB + EAM	CW DFB/DBR + MZM	CW DFB/DBR + ring	External cavity + modulator
Wavelength / fiber	850–940 nm / MMF	1310 nm / SMF	1310 nm / SMF	1310 or 1550 nm / SMF	WDM grid / SMF	Tunable grid / SMF
Modulation fit	Direct only	Direct	Integrated EML	Si or TFLN MZM	Resonant Si ring	External MZM or ring
Bandwidth / reach	Short, modal limit	Chirp-limited	High BW, low chirp	High BW, broad passband	High BW, lock-limited	High BW, architecture-specific
RIN / linewidth	RIN and modal noise	RIN and chirp	RIN plus EAM bias	RIN; linewidth usually secondary	RIN plus spectral alignment	Low linewidth; verify feedback response
Power / efficiency	Low Tx complexity	Low Tx complexity	Driver and EAM loss	Laser, driver, and MZM loss	Laser, heater, and lock power	Laser plus control overhead
Reliability	Junction and temperature wear	Facet and active-region wear	Laser plus EAM aging	Source, attach, and bias drift	Source, heater, and lock faults	Cavity, package, and lock faults
Manufacturing	Array-friendly, MMF plant	Simple Tx, SMF attach	Mature integrated Tx	Multi-die attach and RF match	Dense PIC, tight thermal control	Tight optical assembly and control
Evidence burden	Modal, temperature, aging	Chirp, LIV, RIN	LIV, RIN, EAM sweep, TDECQ	Source plus bias and RF path	Source plus resonance and crosstalk	Spectrum, lock, feedback, environment

Table 5.1. Decision matrix for common source and modulation paths. "Evidence burden" mixes characterization/ATP with life qualification; keep those jobs separate (Table 7.2 and chapter 8). Program limits come from Table 5.4.

5.2.1 Reading the source and modulation matrix

Table 5.1 is a decision map, not a datasheet. Architecture order is plant and reach first, then modulation class, then the burdens those choices create (RIN/linewidth, power split, life owner, factory skill, validation cost), and only then the specific source part. Freeze fiber and rate from the product; strike paths that fail bandwidth or plant class; score survivors on the remaining rows; treat the validation-burden row as ATP and FA cost you must fund. Fill Table 5.4 from the winning path and keep a one-line reject reason for each alternative. Later sections teach each family; do not let family enthusiasm override a row that already vetoes the path.

5.3 Directly modulated lasers and VCSELs

Before EMLs and silicon photonics took over single-mode datacenter ports, most volume optics were either a cheap *DML* on single-mode fiber or a *VCSEL* array into multimode fiber. Both still matter at the low-cost, short-reach edge of the market, and both show why chirp, modal bandwidth, and temperature push AI fabrics toward externally modulated single-mode sources.

A DML modulates laser bias current directly. The transmitter is simple and efficient, but the same carrier dynamics that make modulation easy also produce chirp: intensity changes drag the optical frequency along (Section 3.11). Over multimode or very short single-mode runs that is often acceptable. Over dispersive single-mode fiber at tens of GBd, the chirp turns into inter-symbol interference and closes the eye. Validation therefore focuses on extinction ratio, pattern-dependent chirp, and RIN, not just average power.

VCSELs took a different path. They emit from a vertical cavity at 850–940 nm straight into multimode fiber, so parallel arrays are easy to assemble and cheap to ship. That combination made VCSEL SR optics the default for early 40G/100G Ethernet inside the rack (100G-SR4 and its cousins): short ribbons of MMF, high lane count, low dollars per gigabit. The same physics that made them attractive also capped their future. Multimode fiber has modal bandwidth and modal noise limits; VCSEL bandwidth and reliability both degrade with temperature; and as lane rates climb toward 100 G and 200 G, those limits arrive sooner. The industry response has been incremental (better OM4/OM5 fiber, tighter specs, sometimes PAM4 on MMF) rather than a clean leap to 400G/lane SMF DR. In practice, MMF reach and modal dispersion keep VCSEL links in the SR box (Section 3.13), while hyperscale AI fabrics standardize on single-mode DR/FR and CPO.

Neither family is the path to 400G/lane SMF DR. EMLs and external modulators (Section 5.4, section 3.14.3, and table 3.12) own that space. Pattern-aware chirp linearization can stretch a DML a little farther, but it does not change the physics at FR distances: if you need low chirp and high EO bandwidth at fleet scale, you leave direct modulation behind.

5.4 DFB and EML: the workhorse transmitters

Once single-mode DR/FR became the hyperscale default, most short-reach ports started with an InP laser chip. Two configurations still dominate production: the CW or directly modulated DFB, and the EML that adds an electro-absorption modulator on the same die.

InP Indium phosphide: III-V semiconductor for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.

DFB. A distributed-feedback laser has a grating along the active region that selects one longitudinal mode. Spec-sheet metrics that matter in bring-up are threshold current, slope efficiency, SMSR (typically many tens of dB on a clean part), RIN, and wavelength vs. temperature/current. Used as a CW source for SiPh or TFLN modulators, or as a DML when chirp is acceptable (Section 5.3). Uncooled datacom DFBs ride case temperature with a known $d\lambda/dT$; cooled parts add a TEC and lock to a grid.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

EML. An electro-absorption modulated laser integrates a DFB with an *EAM* on one chip (Section 3.14.3). Reverse bias on the EAM sets absorption and extinction; chirp stays far below a DML. That combination, not marketing, is why EMLs became the volume answer for 100G/lane and then 200G/lane DR/FR pluggables: one chip, low chirp, mature supply chain. Validation adds EAM bias sweeps, aging of the absorption curve, and driver-match checks on top of the DFB LIV/SMSR/RIN suite (Sections 5.7 and 5.13).

When to pick which. Through 200G/lane DR, EML usually wins on cost and integration. A CW DFB (or ELSFP/CW-WDM bank) plus Si MZM, ring, or TFLN wins when the modulator must sit on silicon or needs $\gtrsim$100 GHz EO bandwidth (Table 3.12 and section 3.14.3). At CPO scale the laser often leaves the optical engine entirely so it can be replaced without pulling the ASIC package (Section 5.14). Looking forward, 400G/lane pluggables are pushing harder toward external CW plus TFLN or high-BW silicon modulators, while EMLs remain the workhorse of the installed 100–200G base.

Source	Typical use	Top risks
DML	short reach, cost-driven	chirp/dispersion, extinction ratio
EML	$\leq$DR, 100–200G/lane	EAM bias/aging, thermal
CW + TFLN MZM	400G/lane FR/DR, NPO	MZM bias drift, fiber attach, driver match
CW + Si MZM	DR/FR SiPh, 100–400G/lane	driver match, bias drift, fiber coupling
CW + Si ring	CPO, WDM transceivers	wavelength lock, thermal crosstalk, coupling
VCSEL	SR over MMF	modal noise, reach, temperature
ELS / ELSFP	co-packaged optics	connectorization, fleet serviceability

Table 5.2. When each source is used, and its top validation risks.

Use Table 5.2 as a short risk card once the path is roughly known. Use Table 5.1 when you still need to compare attribute rows across paths. Do not treat this card as a substitute for the full matrix or for Table 5.4.

5.5 Choosing the modulation path

The source decision and modulation decision must close together. Direct modulation minimizes parts and power but carries laser chirp into the link. An EML adds an EAM on the laser die and is a mature low-chirp path for 100–200G/lane. A silicon MZM uses more area and drive but gives a broad optical passband. A ring is compact and fits dense WDM, but adds resonance control and thermal-crosstalk tests. TFLN offers high bandwidth and low chirp with a separate material platform and assembly flow.

Table 5.1 compares the system consequences. The device operation, bandwidth, insertion loss, and driver interfaces live in Section 3.14.3 and table 3.12. Keep that physics in one place. Here the decision is whether the link can carry the added power, control, assembly, and validation burden.

5.6 Laser requirements: from roadmap to specs

Laser requirements only work when they are numbers a supplier can fail and a link budget can close. Start from the interconnect roadmap choice, then fill a short requirements slice; the ATP in Section 9.2 is how that slice is enforced on every lot.

Roadmap forks that set the laser. Each architecture decision forces a different requirements set (Table 5.3):

Roadmap choice	Laser implication	Specs you must freeze early
Pluggable EML vs CW+Si/TFLN	Integrated EAM vs external CW + modulator	EAM bias/aging and TDECQ vs CW power class, RIN, and modulator V_π match
On-package laser vs ELSFP/CW-WDM	Field replace vs FIT inside the package	Connector/ORL/mate cycles and hot-swap CMIS vs COD/aging inside ASIC thermal
Isolator vs isolator-free (CPO)	Feedback tolerance vs quiet RIN only	Stressed RIN_xOMA at stated ORL; monitor PD / lock policy
Single-λ vs CW-WDM / comb	One line vs N lines into rings/filters	Per-line power flatness, SMSR, grid, crosstalk (Section 5.16)
Retimed vs LPO	Module DSP hides Tx vs host sees raw eye	Laser+modulator TDECQ/RLM floor vs host COM budget (Sections 3.14.3 and 10.5.2)
Derate policy	Operating I, T, power below abs-max	Bias window, thermal class, FIT/E_a assumptions (Section 5.13)

Table 5.3. Architecture forks and the laser specs each one forces. Freeze these before DVT samples are built (Section 9.2).

Each "Specs you must freeze early" cell is the exit criterion for that fork. **Exit when** every active fork has numbers (or explicit N/A) before DVT samples are built.

Tradeoff. On-package laser vs field-replaceable ELS

Improves: Integration density and fewer optical connectors

Worsens: Package FIT, service downtime, and thermal risk next to the ASIC

When acceptable: When laser FIT times link count exceeds the service model

Experienced decision: Choose replaceability when fleet repair time dominates; choose on-package when connectors and mate cycles dominate.

One-page requirements slice. Table 5.4 is the PRD-sized list. Fill every row with a number (or an explicit "N/A for this architecture") before you negotiate ATP limits. Do not leave RIN without an ORL, or power without a case-temperature class.

Parameter	How to set the number	Measure / ATP	Reject if	Derate / ops note
Launch power / class	Link budget + connector loss + aging margin (Section 7.7)	Power meter; ELSFP class	Below min at rated T	Cap max power for COD
Wavelength / grid	PMD or ring FSR plan; $d\lambda/dT$ headroom (Chapter 6)	OSA / wavemeter	Off-grid at case T	TEC setpoints
SMSR floor	Datasheet + modal-noise budget	OSA	Below floor at T	Watch aging
RIN (quiet + stressed)	BER floor vs BW (Section 4.3); ORL from plant	PD+ESA; stated ORL	Above limit at ORL	Bias-driver noise budget (Section 5.8)
Bias window	LIV kink-free range at max case T	LIV	Kink in window	Run below abs-max I
EAM / MZM (if any)	ER, RLM, TDECQ at baud (Section 7.4)	DCA + bias sweep	TDECQ/RLM fail	Bias aging policy
ORL / isolator	Architecture: isolator-free needs tighter RIN	ORL meter; mate cycles	ORL out of range	Cleaning / ELS mate life
CMIS monitors	What fleet triage will read (Section 7.12)	CMIS dump	Missing alarms / bad state machine	Enable sequence (Section 7.9)
FIT / life	Fleet failures/day target (Section 5.13)	GR-468 + E_a	Screen escape	Burn-in depth; ELS replace

Table 5.4. Laser requirements one-pager. Every cell needs a program number; this table is the structure, not the limits.

Tradeoff. Higher optical power vs lifetime

Improves: Link margin and reach under loss and aging

Worsens: Thermal stress, COD risk, wall-plug efficiency, and cooling load

When acceptable: When power is the only remaining debit and the life model still closes at the new setpoint

Experienced decision: Do not solve every BER problem with more launch. Often noise, alignment, DSP, or coupling is the cheaper fix.

How to fill numbers (method, not invention). Work backward from the link, not forward from a marketing slide. The four steps below turn an architecture choice into ATP limits:

1. Close the optical ledger at target pre-FEC BER (Sections 3.12 and 7.7). That sets minimum launch OMA/power and maximum allowed penalties (transmitter and dispersion eye closure quaternary, TDECQ; ORL/RIN).
2. From receiver BW and the RIN ceiling $Q_{\max} = 1/\sqrt{\mathrm{RIN} \cdot \mathrm{BW}}$ (Section 4.3), set a stressed RIN limit with margin under the plant ORL you will actually see (not only a quiet bench).

3. From case-temperature and derating policy, set the LIV bias window and thermal class so the laser never sits on a kink or at abs-max in the fleet (Sections 5.13 and 7.9).

4. From service model, choose ELSFP mate-cycle / hot-swap requirements or accept on-package FIT and write COD/aging screens accordingly (Section 5.14).

Hand the filled slice to the supplier with the ATP checklist (Table 9.3). If a roadmap slide cannot point to a row in Table 5.4, the requirement is not real yet.

Exit when every cell in Table 5.4 is a program number or explicit N/A, with ORL stated wherever RIN appears and case-T class stated wherever power or bias appears. **Decision unlocked:** negotiate ATP limits, or reopen the architecture fork that left a cell empty.

Key idea. Laser leadership is a requirements sheet: architecture forks force specific specs (power, grid, RIN@ORL, SMSR, bias window, CMIS, FIT). Fill Table 5.4 from the link budget and fleet model, then enforce it with the ATP (Section 9.2).

5.7 LIV, SMSR, and RIN: the measurement playbook

Three questions decide whether a laser chip or module is usable. Can it make enough power in a kink-free bias window at the thermal class you claim (LIV)? Is it single-mode enough for the plant and WDM grid (SMSR)? Is intensity noise quiet enough at the ORL you will actually see (RIN)? The instruments are standard; the skill is knowing which failure each answer catches.

Bench order: LIV $\rightarrow$ SMSR $\rightarrow$ wavelength $\rightarrow$ RIN (quiet, then stressed ORL) $\rightarrow$ EAM or bias-driver checks as the path requires. Stop when distributions across temperature and units support the bias window, grid, and RIN policy in Table 5.4. If RIN fails, bisect electrical versus optical before you blame the diode: quiet SMU / high ORL first, then product bias rails, then an ORL sweep. Rise with reflection is laser feedback; rise independent of ORL points at the electrical path; discrete ESA spurs are supply or clock pickup (Section 5.8 and section 4.3.1).

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

LIV (light–current–voltage). The LIV curve plots optical power and forward voltage versus bias current. Read off threshold I_{th}, slope efficiency (mW/mA above threshold), kink-free operating range, and thermal rollover at high current or high case temperature. Figure 5.1 is a labeled schematic (not measured data).

High-temp LIV failures look like: I_{th} rise, slope collapse, early rollover, or a kink that moves into the bias window. Those map to aging, TEC saturation, or package thermal resistance (Section 5.13).

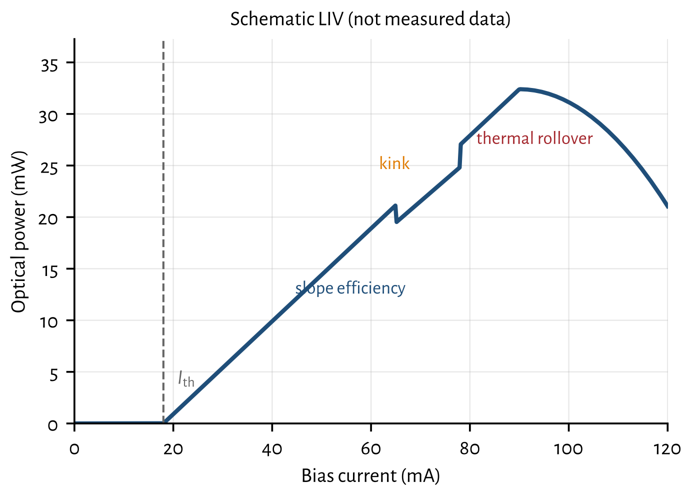

Figure 5.1: Schematic LIV curve with threshold, slope, kink, and thermal rollover labeled. Idealized for teaching; use measured LIV for pass/fail.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

SMSR (side-mode suppression ratio). SMSR is the power difference (dB) between the lasing mode and the strongest side mode on an optical spectrum analyzer (OSA). Datacom single-mode parts require high SMSR so side modes do not steal power or seed modal noise. Spec-sheet floors are part-specific; treat the datasheet or ATP limit as authoritative. SMSR collapse under temperature or aging is a reject: the laser is leaving single-mode operation.

RIN (relative intensity noise). Measure RIN with a calibrated photodetector and RF spectrum analyzer (or a dedicated RIN analyzer), under a controlled optical return loss. Distinguish *intrinsic* RIN (quiet bench, high ORL) from stressed RIN_xOMA used in Ethernet/MSA specs. IEEE 802.3 / 100G Lambda class links cap $\mathrm{RIN}_{17.1}$OMA at -136 dB/Hz with 17.1 dB ORL [1] [100G Lambda MSA / IEEE 802.3]. Quiet datacom DFB/EML parts typically sit well below that when feedback is controlled; CPO ELS designs care as much about feedback tolerance as about the quiet number (Section 4.3.1 and section 4.3).

[1] IEEE 802.3 / 100G Lambda MSA: $\mathrm{RIN}_{17.1}\mathrm{OMA} \leq -136$ dB/Hz.

See also Table 5.9. Do not keep measuring for its own sake once those exits close.

5.8 Laser drivers and the RIN budget

Modulator RF drivers (Section 3.14.3) deliver swing and bandwidth into an EAM or MZM. Laser *bias* drivers are a different circuit: they set a quiet constant current into the diode. Current noise on that path becomes optical intensity noise and adds in the RIN budget of Section 4.3. Confusing the two is a common debug miss: a great SiGe PAM4 driver can still ruin a CW laser if its supply or ground couples into

Parameter	Instrument	Pass/fail intent	Failure signature
LIV	SMU + power meter / integrating sphere	I_{th}, slope, kink-free bias window	high-temp rollover; kink in bias range
SMSR	OSA	single-mode purity vs. datasheet/ATP	side modes rise with T or age
RIN	PD + ESA / RIN analyzer	intrinsic and stressed RIN_xOMA	RIN rises with ORL; BER floor (Section 4.3)
Bias-driver noise	SMU vs. product bias board	RIN_{eq} from i_n (Section 5.8)	RIN rises with rails on, flat vs. ORL
Wavelength	OSA / wavemeter	grid placement, $d\lambda/dT$, $d\lambda/dI$	walk off ring or MSA grid
EAM bias (EML)	bias sweep + DCA/TDECQ	extinction, chirp, RLM	aging shifts absorption curve

Table 5.5. Laser measurement playbook: what to measure, with what, and what failure looks like.

the bias rail. The electrical-versus-optical bisection is in Section 5.7; use it before redesigning the diode.

Reference: bias-driver noise numerics. The conversion and table below are budget tools for quiet-CW sizing. Skip to path notes if you already have a measured product-board RIN.

From current noise to equivalent RIN. Above threshold, optical power tracks bias approximately as $P \propto (I - I_{\text{th}})$. Relative intensity fluctuations then track relative current fluctuations:

$$\text{RIN}_{\text{eq,lin}} \approx \left(\frac{i_n}{I - I_{\text{th}}}\right)^2, \qquad \text{RIN}_{\text{eq}}[\text{dB/Hz}] = 20\log_{10}\left(\frac{i_n}{I - I_{\text{th}}}\right),$$

where i_n is the one-sided current-noise density in $\text{A}/\sqrt{\text{Hz}}$ at the laser terminals (driver plus board pickup). The approximation assumes linear slope efficiency and ignores intrinsic laser dynamics; it is a budget tool, not a device model.

Worked numbers at $I - I_{\text{th}} = 50$ mA (typical CW DFB window): $i_n = 500$ pA/$\sqrt{\text{Hz}}$ maps to $\text{RIN}_{\text{eq}} \approx -160$ dB/Hz; 270 pA/$\sqrt{\text{Hz}}$ maps to about -165 dB/Hz. Commercial low-noise laser drivers quote roughly 50–500 pA/$\sqrt{\text{Hz}}$ at 1 kHz depending on current range (Table 5.6); the Koheron DRV200 family is a concrete example [2][Koheron DRV200]. Against a good datacom intrinsic RIN of -145 to -155 dB/Hz (Section 4.3.1), those 1 kHz densities look comfortable. The budget tightens when $(I - I_{\text{th}})$ is small (near threshold, derated CW, or low-current VCSELs), when you integrate broadband switching noise rather than a 1 kHz spot, or when SerDes/DSP rails dump discrete tones onto the bias network.

[2] Koheron DRV200: low-noise CW laser driver; i_n at 1 kHz by current range.

CW / ELSFP / CW-WDM paths. For external CW sources feeding Si or TFLN modulators, design the bias path as a low-noise current source with high supply rejection,

Driver class (example)	i_n @ 1 kHz	$\mathrm{RIN_{eq}}$ @ 50 mA	What it means
Ultra-low-noise CW (DRV200-A-40)	55 pA/$\sqrt{\mathrm{Hz}}$	≈ -179 dB/Hz	Bench / metrology floor
Low-noise CW (DRV200-A-200)	270 pA/$\sqrt{\mathrm{Hz}}$	≈ -165 dB/Hz	Typical quiet CW source
Higher-current CW (DRV200-A-400)	480 pA/$\sqrt{\mathrm{Hz}}$	≈ -160 dB/Hz	Still below -155 intrinsic
Shared digital LDO, poor PSRR	often $\gg$1 nA/$\sqrt{\mathrm{Hz}}$ + tones	can exceed -145	False "RIN" on ESA

Table 5.6. Bias-driver current noise converted to equivalent RIN at $I - I_{\mathrm{th}} = 50$ mA using $\mathrm{RIN_{eq}} = 20\log_{10}(i_n/(I - I_{\mathrm{th}}))$. Densities for the DRV200 rows are from the Koheron datasheet at 1 kHz; the last row is qualitative (board-dependent).

local decoupling at the diode, and a star ground that does not share return with SerDes switching currents. Automatic power control () loops that close through a monitor PD suppress slow drift; keep the loop bandwidth well below the RIN measurement band and quiet enough that the loop itself does not inject intensity noise. ELSFP and CW-WDM modules hide this circuitry inside the pluggable (Sections 5.14 and 5.16); acceptance still needs module-level RIN with the host bias and management rails connected, not only a quiet SMU on the bare die.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

DML and EML. A *DML* shares one diode for bias and RF: a bias tee (or on-chip bias network) combines a quiet DC source with the RF driver. Excess RF driver broadband noise, poor tee isolation, or supply ripple on the bias arm all raise measured RIN and chirp-related penalties. An *EML* splits the problem: keep the DFB bias as quiet as a CW source, and treat the EAM RF driver under Section 3.14.3. EAM drive amplitude sets extinction and chirp; DFB bias noise still lands in optical intensity before the modulator.

Key idea. Treat laser bias noise as a RIN term: $\mathrm{RIN_{eq}} \approx (i_n/(I - I_{\mathrm{th}}))^2$. Quiet CW drivers at tens to hundreds of pA/$\sqrt{\mathrm{Hz}}$ usually sit under a -145 dB/Hz intrinsic floor at 50 mA; digital supply pickup, near-threshold bias, and DML bias-tee leakage are what actually burn the budget.

5.9 How lasers fail

Six mechanisms account for most laser field returns. Each has a distinct telemetry signature, so classify before you open FA.

Threshold current increase I_{th} rises from its ship value at fixed temperature, usually with slope efficiency dropping in step. Points to active-region or facet degradation (Section 5.13).

Slope efficiency degradation Output power per unit bias current falls even when I_{th} is stable. A separate wear-out track from threshold rise; both show up on the same LIV sweep.

Wavelength drift The lasing line walks off its grid slot or ring resonance. Distinguish laser drift from TEC or ring drift by holding one actuator fixed and moving the other (Section 6.4 and chapter 6).

Aging (SMSR collapse, mode hopping) Side modes grow relative to the main mode, or the laser hops between modes under temperature or current. An OSA trend over time is the tell.

Thermal runaway A positive feedback loop where higher junction temperature raises threshold current and cuts slope efficiency, so more drive power turns to heat for the same optical output, raising temperature further until the TEC saturates and the laser rolls over. Triggered by a failed or saturated TEC, a blocked heat path, or operation above the rated thermal class. Distinct from ordinary wear-out because it is fast (minutes, not months) once it starts; the failure-analysis handbook has the full symptom-to-cause breakdown (Section 11.7).

Monitor photodiode failure The control loop's own sensor drifts or fails, so the laser looks unstable when the real fault is in the feedback path, not the gain medium (Section 5.20.3).

5.10 Separate thermal behavior from long-term aging

Thermal response is reversible on the time scale of a temperature sweep or cycle. It changes threshold current, slope efficiency, wavelength, EAM bias, TEC current, and ring alignment. Measure it with controlled case-temperature sweeps, loaded thermal corners, heater sweeps, and thermal cycling. Repeat the measurement after returning to the starting temperature. Recovery points toward an operating-point or control problem.

Long-term aging is cumulative. Threshold current rises, slope efficiency falls, contacts degrade, defects grow, and an absorption or spectral curve can move permanently. Measure those changes with HTOL, accelerated life testing, and periodic LIV, spectrum, and modulation readouts. A temperature cycle can expose a weak attach or calibration error, but it does not by itself establish a lifetime acceleration model.

Do not merge the data sets. A high-temperature BER failure that clears at room temperature needs thermal-margin work. A room-temperature baseline that keeps moving after each stress interval needs an aging or damage hypothesis.

5.11 Calibration: what drifts and what triggers retuning

Calibration corrects predictable unit-to-unit and temperature variation by storing operating points. It does not repair exhausted margin. A railed TEC, heater, or bias DAC, a permanently walked LIV or absorption curve, or a loop error that no longer converges is a design, life, or FA problem, not a "retune the table" task.

Calibration exists because no transmitter runs at a datasheet point. Every unit has its own threshold, slope, absorption curve, quadrature point, and resonance, and each of those moves with temperature and age. The operating points a product actually stores are:

Laser bias / APC target the bias current or monitor-PD power setpoint that holds launch power. Drifts as threshold rises and slope falls with age (Section 5.13); a drifting monitor PD corrupts it silently (Section 5.20.3).

EAM bias (EML) the reverse-bias point that sets extinction and chirp. Moves with case temperature and with absorption-curve aging, so production parts store bias versus temperature, not one number.

MZM quadrature the phase bias that holds the modulator at its linear point. Drifts with temperature, stress, and age; a bias-control loop tracks it, and a railed loop is a telemetry alarm, not a retune request.

Ring heater / lock point the tuner power that aligns resonance to the laser line. Consumes headroom as ambient rises and neighbors heat; a railed DAC means the tuning range is exhausted (Sections 6.4 and 6.5).

Tables are usually segmented by temperature. Segment boundaries are a real failure mode: the debug story in Section 5.20.4 is a healthy laser reading the wrong temperature segment after thermal cycling. Keep calibration tables under change control and record the table version with every test result, or failures cannot be replayed.

Recalibration should be triggered by evidence, not habit: a control-loop error residual that no longer converges, an actuator (TEC, heater, bias DAC) that approaches its rail, telemetry that disagrees with an external reference, a temperature excursion beyond the table range, or a repair, rework, or firmware change that invalidates stored coefficients. ATP must verify calibration at the temperature corners the fleet will see, not only at the station ambient (Section 7.9).

5.12 How lasers are qualified

Qualification projects the failure mechanisms of Section 5.9 forward from a short bench test to years of field life. Three stress classes do the work:

HTOL (high-temperature operating life) Run a sample lot at elevated temperature and bias for a fixed duration (often 1000 hours) and track LIV, SMSR, and wavelength drift. HTOL is the primary input to the Arrhenius life projection below.

Burn-in A shorter, sometimes 100%-screen stress that removes infant-mortality units before ship, rather than projecting life. Burn-in trades test time for escape rate (Section 9.4).

Environmental stress Temperature cycling, damp heat, vibration, and shock catch packaging, attach, and mechanical failure modes that HTOL does not. They qualify different risks and should not be treated as substitutes for long-term aging data (Section 8.3).

Together with the Arrhenius acceleration factor, these three stresses turn a qualification lot into a defensible FIT number.

Observable aging signatures. Watch LIV and spectrum over HTOL or field life:

- threshold rise and slope drop (active-region / facet degradation);
- SMSR collapse (mode competition);
- EAM bias creep on EMLs (absorption curve shift $\rightarrow$ TDECQ/RLM drift);
- RIN rise under feedback (ORL or isolator failure);
- COD (catastrophic optical damage) at the facet under overstress.

COD Catastrophic optical damage: sudden, irreversible facet failure in a laser under thermal or optical overstress.

Each signature should appear in the ATP and in field telemetry triage (Sections 7.12 and 8.3).

5.13 Aging curves, derating, and fleet FIT

Lasers wear out. At fleet scale that is not a footnote; it sets architecture (ELSFP vs. integrated laser) and operating policy (derating, burn-in).

Tradeoff. FIT claim vs sample humility

Improves: A crisp fleet failure-rate story for planning

Worsens: Overconfidence when sample-hours, lots, and sites are thin

When acceptable: When the bound, observation time, and population are stated with the claim

Experienced decision: Publish an upper bound with assumptions, or do not claim a FIT number.

Arrhenius life projection. Telcordia GR-468-CORE qualifies optoelectronic parts with accelerated stress (HTOL, temperature cycle, damp heat) and projects field life with Arrhenius acceleration [3] [80]:

$$\mathrm{AF} = \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{\text{use}}} - \frac{1}{T_{\text{stress}}}\right)\right],$$

where E_a is the activation energy for the wear-out mechanism under test, k_B is Boltzmann's constant, and temperatures are absolute. Document E_a, sample size, and confidence bounds when converting a 1000-hour HTOL lot into field-year FIT. Activation energies are mechanism-specific; use the value justified in the qual plan, not a generic number copied from another product.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.
HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.
[3] Telcordia GR-468-CORE: reliability assurance for optoelectronic devices.

Figure 5.2 is the mental model behind HTOL: same starting health, faster drift at higher temperature, a defined end-of-life threshold, and shorter time-to-failure as stress rises. For lasers the vertical axis is usually tied to LIV or power at constant current; the temperature axis must be junction temperature, not only chamber set point.

When the projection is valid. Acceleration assumes the stress speeds up the *same* physical mechanism the fleet will see. The projection fails in two ways: the stress activates a mechanism the field never sees (solder creep or moisture ingress at a stress temperature the product never reaches), or the field sees a mechanism the stress never exercises (connector wear, bias-rail transients, thermal cycling from traffic load). So a qual number is a hypothesis, not a fact: compare field-return Pareto and failure signatures against the qual projection, and treat divergence as evidence that E_a or the mechanism model is wrong, not that the fleet is unlucky (Section 7.12). Sudden fails (COD, ESD, cracked fiber attach) sit outside Figure 5.2; classify those separately before you fit Arrhenius parameters.

Derating. Run below absolute-max current, case temperature, and optical power. Derating extends wear-out life and reduces COD risk. Uncooled datacom parts already sit near thermal limits at high case temperature; cooled or faceplate ELSFP modules (Section 5.14) buy headroom by moving heat off the ASIC package.

Worked FIT example (assumptions labeled). FIT is failures per 10^9 device-hours. For illustration only, assume 50 FIT per laser (confirm against your supplier qual; do not treat 50 as a measured claim) and a fabric with 5×10^5 lasers (order-of-magnitude for a large AI cluster with several optical links per accelerator). Expected failures per day:

$$\frac{5 \times 10^5 \times 50 \times 24}{10^9} \approx 0.6 \text{ laser failures/day.}$$

That is why field-replaceable ELSFP modules, burn-in screens, and derating are design inputs, not afterthoughts (Chapter 8).

5.14 ELS and ELSFP: architecture, pinout, qual

ELSFP (External Laser Small Form-Factor Pluggable) is the OIF form factor for faceplate-pluggable CW laser modules that feed co-packaged optical engines [4] [49]. The lasers sit at the coolest part of the system (front panel), hot-swap when they fail, and keep thermal load off the ASIC and photonic engine.

ELSFP External Laser Small Form-Factor Pluggable (OIF): faceplate CW laser module for CPO; 24-pin card edge, CMIS, blind-mate MT.

[4] OIF ELSFP Implementation Agreement (OIF-ELSFP-02.0, 2023): 24-pin card edge, CMIS, blind-mate MT.

Mechanical and optical. The module uses a card-edge electrical interface and a blind-mate multi-fiber optical connector at the rear (MT-class ferrules), which improves eye safety for high CW power by keeping live fiber inside the chassis [49]. One ELSFP can feed more than one optical engine. OIF defines optical power classes, thermal classes, and wavelength assignments (e.g. DR-type 1311 nm and FR-type CWDM4 grids) so hosts and modules interoperate.

Management and hot-swap. ELSFP uses CMIS and the CMIS module state machine over TWI. On plug-in the module resets, initializes management, and stays in low-power mode with lasers *off* until the host transitions it to ModuleReady and explicitly enables lasers [49]. `ModPrsL` and `IntL` support presence detect and asynchronous alarms for safe hot-swap.

Reference: electrical pinout (OIF-ELSFP-02.0 Table 7). extitDeep dive / pin map. Skip unless you are wiring the host connector. Twenty-four contacts: multiple 3.3 V VCC and GND pins, module reset (`ResetL`), low-power mode (`LPModeL`), two-wire serial management (`SCL`/`SDA`), presence (`ModPrsL`), and interrupt (`IntL`), plus reserved pins for future power/ground [49]. Table 5.7 summarizes the published map.

Qual hooks for suppliers. Acceptance test plans should cover the checklist in Table 9.3 and section 9.2: laser LIV/SMSR/RIN inside the module; optical power-class compliance; connector mating cycles and contamination/ORL; burn-in before ship; CMIS register sanity; and thermal class at rated case temperature. Module bring-up must also prove the CMIS enable sequence and ModuleReady laser policy (Section 7.9). Field returns split between laser wear-out and connector/fiber-attach faults; keep both in the triage tree (Section 7.12).

5.15 Optical safety and laser classes

Reference / compliance deep dive. Read when you own APR/ALS ATP or multi-fiber hazard classification. Architecture chapters only need the rule that aggregate port power, not per-lane datasheet power, sets the class.

5.15.1 Hazard and laser classes

Laser safety for interconnects is governed by IEC 60825-1 (laser product classification) and IEC 60825-2 (optical-fiber communication systems, OFCS) [88]. Classes run from Class 1 (safe under normal use) through Class 1M (safe unless the beam is collected by optics), Class 3R/3B, and Class 4. At 1310 nm and 1550 nm the beam is invisible, which raises the operational risk: technicians cannot see exposure. The retinal-hazard band ends near 1400 nm, but corneal and skin hazards remain, and single-mode power confined to a $\sim$9 µm core is high radiance even at modest milliwatt levels.

Short-reach datacom modules are usually engineered so each fiber port stays Class 1 or Class 1M under rated launch power. That is a design constraint on EML/DFB bias and on how much power each lane launches, not a label you add after the fact.

5.15.2 Hazard level = aggregate, not per-lane

The safety case scales with *total* launched power at an accessible location, not with a single DFB data sheet. CW-WDM and ELS banks concentrate many lines on one MT or MPO ferrule (Section 5.14). A connector that breaks out eight or sixteen fibers can exceed a per-lane Class 1 budget even when each lane is modest. IEC 60825-2 assigns hazard levels (1 through 4) to each accessible port in the OFCS based on the radiant power that could escape during service [88]. That is why ELS architecture and fiber count drive classification, not the laser chip alone.

5.15.3 Open-fiber protection: APR and ALS

When fiber continuity is lost, open connectors and broken fiber can expose hazardous power. *APR* (automatic power reduction) holds output at or below Hazard Level 1M and probes for re-mate with safe low-power pulses. *ALS* (automatic laser shutdown) cuts power entirely and was common on older SDH links; for modern high-power systems APR with automatic restart is the preferred pattern because restart probes stay within the hazard limit [89]. ITU-T G.664 requires power reduction to Hazard Level 1M within about 3 s of a continuity break, a restart inhibit window, and restart only at safe power.

APR Automatic power reduction: holds laser output at safe level on fiber break; probes for re-mate at Hazard Level 1M.
ALS Automatic laser shutdown: cuts laser power on fiber break. APR with restart preferred in modern systems (ITU-T G.664).

These mechanisms tie directly to CMIS and bring-up policy: lasers enable only when the host commands ModuleReady (Sections 7.8 and 7.9). APR/ALS is what makes a live ELSFP hot-swap survivable in a running rack (Section 7.9).

5.15.4 What validation and ops owe

Optical safety is a validation deliverable, not a compliance sticker. ATP should verify APR/ALS trip threshold and timing on representative open-fiber faults; label modules and cages with the rated class; document max launched power per port and per MPO breakout; and write service procedures for multi-fiber connectors. At fleet scale, a hot-swap runbook that assumes ALS works but was never tested in ATP is a real hazard. Fold the APR/ALS check into the ELS hot-swap corner in Section 7.9 alongside mate-cycle and ORL tests.

5.16 CW-WDM source validation

Reference: multi-wavelength source survey. Architecture contract for CW-WDM locking lives in Chapter 6; this section lists per-channel acceptance checks and example products.

Multi-wavelength CW sources (CW-WDM MSA) feed dense ring or filter banks on a PIC (Section 6.6 and chapter 6). Validation is per-channel plus cross-channel:

- power flatness across λ (uneven OMA after the modulator bank);
- per-channel SMSR and wavelength grid placement;
- channel crosstalk and residual ASE between lines;
- lock to microring resonances under temperature and neighbor heating (Sections 6.4 and 6.5 and section 3.14.3);
- RIN and ORL sensitivity for each line (Section 5.7 and section 4.3.1).

Examples: Ayar Labs SuperNova (CW-WDM MSA-compliant, feeds TeraPHY) [5][50][6][48]; Broadcom ELSFP banks on Tomahawk CPO (Sections 5.14 and 10.10); quantum-dot comb lasers (Ranovus, Quintessent) aimed at many λ from one chip. Source tests live here; locking and on-chip MUX live in Chapter 6.

[5] Ayar Labs SuperNova CW-WDM source and TeraPHY optical I/O.

[6] CW-WDM MSA Rev 1.0 (2021): O-band 8/16/32-λ CW grids (9/18/36 nm spans); modular vs integrated ports; RIN/SMSR/linewidth methods.

5.17 Light-source supply strategy

The sourcing decision follows the same architecture fork as the optical design: buy a merchant source, buy a serviceable external module, or bind the source to the photonic package. Evaluate each path by qualification ownership, second-source portability, lot traceability, test access, field replacement, and change-control rights. A vendor list ages quickly and does not answer those questions.

Merchant DFB, EML, or CW die preserve module-level design freedom and can support a second source, but the integrator owns attach, driver match, screening, and package reliability.

External CW-WDM or ELSFP module moves source qualification and management into a replaceable unit. The system still owns connector, ORL, hot-swap, and host interoperability (Sections 5.14 and 5.16).

Multi-wavelength source reduces source count and can simplify WDM fan-out, but couples channel yield, power flatness, control, and replacement into one unit (Section 5.16).

Source integrated with the PIC reduces optical interfaces and can improve density, but makes laser yield and wear-out part of package yield and service life.

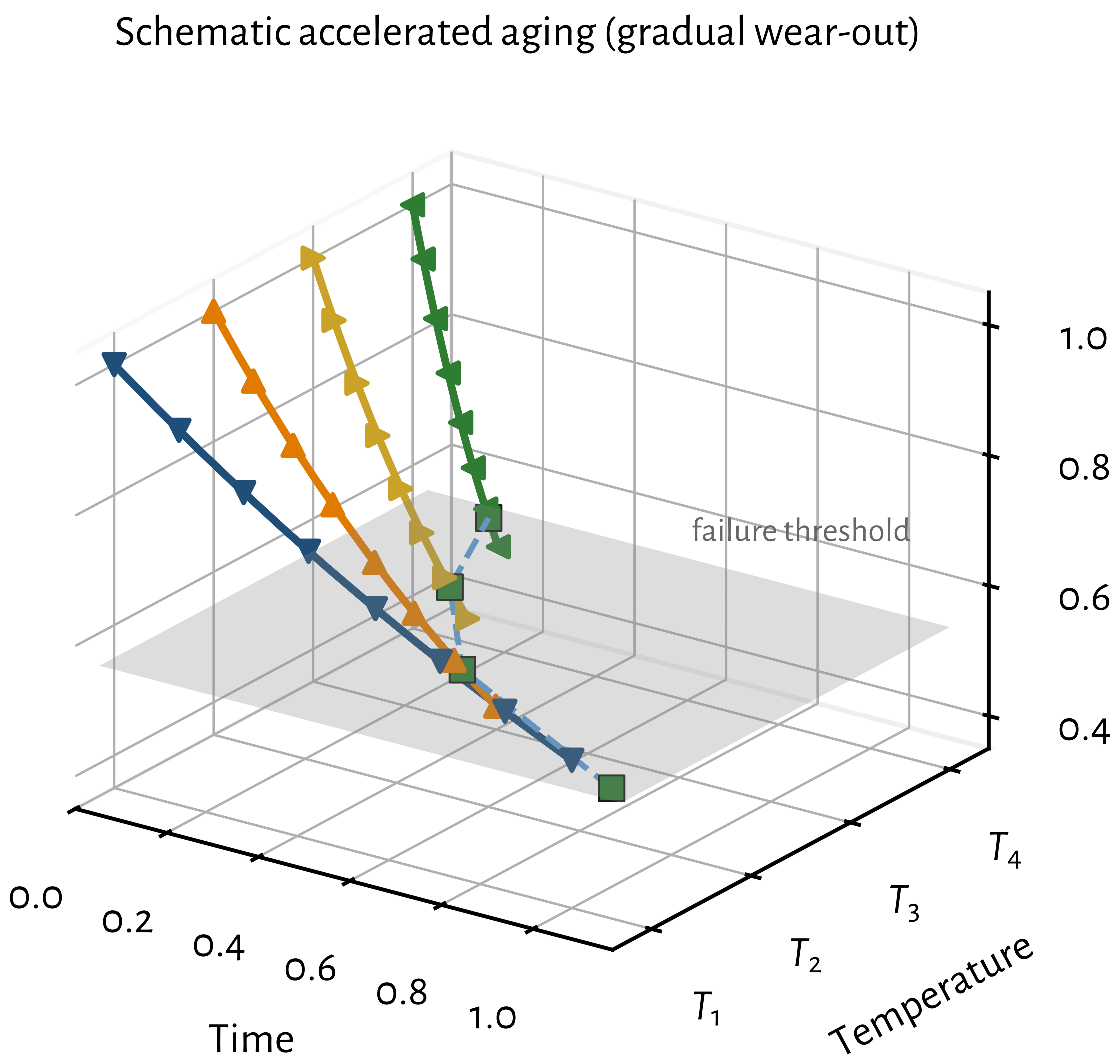

Figure 5.2: Schematic accelerated aging for gradual laser wear-out. A degradation parameter (for example optical power at fixed bias, or threshold current rise mapped to a falling health metric) is tracked versus time at several stress temperatures $T_1 < T_2 < T_3 < T_4$. Higher temperature reaches the failure threshold sooner. The dashed curve on the threshold plane is the time-to-failure trend that Arrhenius acceleration turns into a use-condition life estimate. Not measured data. Valid only for one thermally activated mechanism with junction temperature as the stress variable; sudden modes such as COD or ESD do not draw this surface.

Pin	Function	Requirements	Notes
1–3	VCC	1.5 A, 3.3 V	with noise filtering
4	TBD	reserved	future power
5	ResetL	pull-up 10 kΩ	reset module, LVTTL
6	LPModeL	MMC on only	low-power mode (low), LVTTL
7	TBD	reserved	future ground
8–10	GND	1.5 A, 3.3 V	with noise filtering
11	TBD	reserved	(reserved)
12	SCL	TWI clock	host 4.7 kΩ pull-up; module ≥10 kΩ
13	SDA	TWI data	same pull-ups as SCL
14	TBD	reserved	(reserved)
15–17	GND	1.5 A, 3.3 V	with noise filtering
18	TBD	reserved	future ground
19	ModPrsL	shorted to GND in module	presence (low), LVTTL
20	IntL	pull-up 10 kΩ	interrupt, LVTTL
21	TBD	reserved	future power
22–24	VCC	1.5 A, 3.3 V	with noise filtering

Table 5.7. ELSFP electrical pinout (adapted from OIF-ELSFP-02.0 Table 7). Lasers power only in ModuleReady after host command; default on plug-in is lasers off [49].

Approach	Qualification ownership and risk
Merchant DFB/EML/CW die	Integrator owns attach, driver match, screen, and module qual
External CW-WDM / ELSFP module	Supplier owns source module; system owns interface and service qual
Multi-wavelength source	Shared yield, power-flatness, and replacement risk across channels
Source integrated with PIC	Highest density; laser yield and life become package risks

Table 5.8. Light-source sourcing paths and the qualification ownership each one creates.

5.17.1 Reading the supplier ownership matrix

Table 5.8 is a qualification-ownership map. Choose the service model before you freeze optics. A merchant die leaves attach, driver match, screen, and package life with the integrator. An ELS or CW-WDM module moves source life into a replaceable unit while mate, ORL, hot-swap, and host interop stay system-owned. A multi-wavelength source couples channel yield, flatness, and replacement. An integrated PIC source makes laser yield and wear-out part of package life. Approve a path only when both ownership packs exist; do not treat a die cert as module qual, or a laser-module ATP as an optical-interface qual.

5.17.2 Why ownership order matters

Choose the service and ownership model before you freeze the optical architecture. Qualify the architecture you will service, not only the laser die.

5.18 Why lasers are the reliability bottleneck

At the scale of a large optical fleet the laser is usually the reliability-limiting component. It is an active device with wear-out physics that passive optics and even photodiodes largely lack:

- *Catastrophic optical damage* (COD) at the facet.
- Gradual facet and active-region degradation (accelerated by temperature, following Arrhenius kinetics; Section 5.13).
- EAM aging in EMLs; coupling and solder drift in packaged assemblies.

Because failures scale with the number of lasers, a fleet of $100{,}000$+ links turns a modest per-laser FIT rate into a steady stream of field failures (Sections 5.13 and 8.3). The mitigations shape architecture: field-replaceable external laser sources (ELSFP, CW-WDM), redundancy, burn-in screening to weed out infant mortality, and derating (running lasers below their maximum to extend life).

5.19 Margin erosion over temperature, lot, and life

A link rarely loses all margin in one event. The source can lose launch power as slope efficiency falls. Connector loss and ORL can rise after service. EAM or MZM bias can move. A ring can consume spectral headroom as its heater approaches range. Driver noise can raise the BER floor while none of these changes violates its stand-alone limit.

Track five ledgers:

Power margin launch power, coupling, connector and MUX loss, receiver sensitivity, and aging reserve.

Noise margin intrinsic and feedback-driven RIN, bias-rail noise, receiver noise, and crosstalk.

Timing margin source and modulator bandwidth, dispersion, driver and host jitter, and equalization reserve.

Spectral margin laser wavelength, SMSR, filter or ring passband, thermal drift, and lock range.

Control margin headroom in APC, TEC, heaters, ring lock, bias DACs, and calibration tables. A railed loop can fail the link while the diode is still healthy.

Recompute the link at combined production corners. A nominal part at nominal temperature says little about whether a slow loss in two ledgers will push a tail unit across the pre-FEC BER limit. Section 7.9 and table 7.6 carry the same ledgers into validation and fleet triage. The interview review compresses this checklist in Section A.8.4. The wall-chart form is Section D.10.

Why experienced engineers track five ledgers instead of one margin number?

Because links usually fail when several small spends add up. One room-temperature pass/fail hides which ledger is nearly empty.

Engineering heuristic. A railed heater, TEC, or bias DAC is often the failure before the diode is. Check control margin before you write a wear-out FIT story.

Tradeoff. More optical margin vs cost and power

Improves: Reach, temperature tolerance, aging headroom, and contamination tolerance

Worsens: Laser power, thermal load, wall-plug efficiency, and sometimes component lifetime

When acceptable: When a named uncertainty (aging, ORL, lot spread) still dominates the remaining risk

Experienced decision: Do not maximize margin. Allocate margin where uncertainty is highest. A predictable 1 dB connector loss often needs less attention than an unknown aging mechanism.

```
Nominal system margin
  |
Temperature debit
  |
```

```
Voltage / power-quality debit
   |
Channel / connector debit
   |
Manufacturing variation
   |
Aging / wear
   |
Interoperability variation
   |
Remaining margin
   |
Above deployment requirement?
   |-- YES --> proceed
   |-- NO  --> redesign / restrict / recalibrate / reject
```

Not every debit is naturally in decibels. Depending on the subsystem, remaining margin may be optical power, sensitivity, BER or FEC headroom, eye or TDECQ, jitter, control range, lifetime, or yield. Validation often measures the net externally visible result; do not double-count internal penalties the test cannot separate (Section 7.7).

5.20 Engineering lens

5.20.1 How it works

A laser is an active device with wear-out physics, which makes it both the first line of the link budget and the fleet's reliability bottleneck. The chapter's device families and LIV/SMSR/RIN measurements all serve one question: will this source stay in spec for years at temperature?

5.20.2 How it is measured

Qualify the laser as a set of curves across temperature, bias, ORL, and age, not a room-temperature data-sheet point. The measurement playbook (LIV, SMSR, RIN, wavelength, and EAM checks with their instruments and pass/fail intent) is in Section 5.7 and table 5.5; the stress classes that project field life are in Sections 5.12, 5.13 and 8.3 [80].

5.20.3 How it fails

The six field-return mechanisms are catalogued in Section 5.9: threshold rise, slope droop, wavelength drift, aging (SMSR collapse and mode hopping), thermal runaway, and monitor-photodiode failure. Manufacturing adds die, wafer, lot, and assembly spread to every one. The mechanism that most often misleads triage is a healthy laser behind a bad feedback sensor, so it gets the worked callout below.

Failure mode: Monitor photodiode drift

Symptoms. Reported power falls or the bias loop moves, but an external power meter does not show the same change.
Likely causes. Monitor-PD responsivity drift, transimpedance gain error, contamination in the monitor path, or a bad calibration coefficient.
Measurements. External power meter, monitor current, bias current, LIV, and loop error versus temperature.
Mitigations. Repair the monitor path or calibration, add disagreement alarms, and do not raise laser bias to compensate for a false reading.

5.20.4 How it is debugged

For power degradation, compare external optical power, monitor current, bias, and case temperature before changing the setpoint. Rerun LIV at the failing temperature and compare it with ship data. If LIV moved, inspect SMSR, wavelength, and RIN to classify active-region, facet, or modal change. If LIV is stable, move to coupling, connector, monitor-PD, and control-loop checks. For a wavelength excursion, inspect OSA data and TEC current together. For a bias anomaly, replace the product driver with a quiet source before blaming the diode.

Debug story

Observed. BER worsened after thermal cycling while average optical power stayed in range.
Investigation. The DCA showed that extinction ratio had collapsed. LIV and SMSR were unchanged, and an EAM bias sweep restored the eye.
Finding. The light source was healthy, but its modulator operating point was wrong.
Root cause. A calibration table used the wrong temperature segment after the cycle.
Resolution. The table and screening limits were fixed, and EAM bias sweep data became part of the thermal-cycle readout.

5.21 Engineering checklist

Decision or test	Question it answers	Evidence to retain
Architecture	Does the source and modulation path close reach, rate, power, cost, and service?	Requirement allocation and rejected alternatives
LIV	Is the operating window clear of threshold, kinks, and rollover?	Curves by unit, lot, temperature, and age
Spectrum	Does wavelength and SMSR stay inside the assigned grid and filter passband?	OSA or wavemeter data across corners
RIN and ORL	Does noise margin survive the reflection environment?	Quiet and stressed RIN with stated ORL and bandwidth
Modulation	Does bias, drive, chirp, and bandwidth close the eye?	Bias sweeps, TDECQ or equivalent, and driver conditions
Thermal behavior	Are reversible shifts within control and actuator range?	Temperature and heater sweeps, TEC current, recovery data
Long-term aging	Which parameters drift permanently, and at what rate?	HTOL intervals, LIV, spectrum, and modulation trends
Manufacturing	Can the ATP catch bad units and lot drift at useful test cost?	Limits, guard bands, GR&R, yield, and reaction plan
Fleet operation	Which monitors distinguish source, modulator, cooler, and optical path?	Telemetry map, alarm thresholds, and golden baselines

Table 5.9. Source and modulation engineering checklist. Each row ties a decision to evidence, not only a test name.

5.21.1 Reading the laser engineering checklist

Table 5.9 is the decision sequence for a laser program. Measurement methods for LIV, SMSR, RIN, and aging are taught earlier in this chapter; the notes below focus on what each row unlocks and when you may leave it.

Architecture. **Purpose.** Does the source and modulation path close reach, rate, power, cost, and service?

Uncertainty removed. Component enthusiasm does not allocate requirements. After architecture you have a chosen path, rejected alternatives, and owners for lock, attach, and life (Tables 5.1 and 5.8).

Exit criteria. Exit when requirement allocation and rejected alternatives are written and reviewed.

Decision unlocked. Freeze the path into Table 5.4, or reopen the matrix.

Risk if skipped. You validate a hero topology that cannot be serviced or powered in the rack.

LIV. **Purpose.** Is the operating window clear of threshold, kinks, and rollover across unit, lot, temperature, and age (Section 5.7)?

Exit criteria. Exit when kink-free bias windows and distributions support the bias policy.

Decision unlocked. Set bias and derate, or reject lots / redesign the window.

Risk if skipped. Soft BER and sudden dark fails appear without a ship baseline to compare.

Spectrum. **Purpose.** Do wavelength and SMSR stay inside the assigned grid and filter passband?

Exit criteria. Exit when OSA or wavemeter data across corners meet grid and SMSR floors.

Decision unlocked. Approve channel assignment, tighten temperature policy, or reject modal risk.

Risk if skipped. WDM unlock and modal noise show up as "random" BER.

RIN and ORL. **Purpose.** Does noise margin survive the reflection environment the plant will present (Section 4.3.1)?

Exit criteria. Exit when quiet and stressed RIN at stated ORL and bandwidth meet the BER-floor budget.

Decision unlocked. Approve isolator-free or isolator-required design; set plant cleaning rules.

Risk if skipped. Lab RIN looks fine; field ORL raises the floor.

Modulation. **Purpose.** Do bias, drive, chirp, and bandwidth close the eye at baud?

Exit criteria. Exit when bias sweeps and TDECQ (or equivalent) at named driver conditions meet Tx quality limits (Section 7.4).

Decision unlocked. Freeze EAM/MZM/ring bias policy, or reject the modulator class for this rate.

Risk if skipped. Average power passes while the eye fails under temperature.

Thermal behavior. **Purpose.** Are reversible shifts within control and actuator range?

Exit criteria. Exit when temperature and heater sweeps, TEC current, and recovery data show control headroom at loaded corners.

Decision unlocked. Approve thermal envelope, add heaters/TEC margin, or de-rate case T.

Risk if skipped. Lock and calibration faults are misread as permanent aging.

Long-term aging. **Purpose.** Which parameters drift permanently, and at what rate (Section 5.13)?

Exit criteria. Exit when HTOL intervals with LIV, spectrum, and modulation trends support the life claim or force derate.

Decision unlocked. Accept FIT/replacement plan, or hold ship for life risk.

Risk if skipped. Useful-life planning uses hope instead of a mechanism.

Manufacturing. **Purpose.** Can the ATP catch bad units and lot drift at useful test cost (Table 9.3)?

Exit criteria. Exit when limits, guardbands, GR&R, yield, and a reaction plan exist for the ship screens.

Decision unlocked. Open volume screens, or hold for process control.

Risk if skipped. Qualified engineering lots diverge from production without a catch point.

Fleet operation. **Purpose.** Which monitors distinguish source, modulator, cooler, and optical path (Section 7.12)?

Exit criteria. Exit when telemetry map, alarm thresholds, and golden baselines are named and owned.

Decision unlocked. Arm fleet triage; feed escapes back into ATP.

Risk if skipped. Field tickets cannot separate laser wear from connector or TEC faults.

5.21.2 Why the checklist order matters

Architecture first, or you characterize the wrong path. LIV and spectrum establish semiconductor and channel health before RIN and modulation argue about floors and eyes. Thermal separates reversible control from permanent drift before aging claims. Manufacturing and fleet close the loop so life and screens stay honest after ship. Later rows must not compensate for a missing architecture or ship baseline.

5.22 Interview takeaway

Key idea. Measure LIV, SMSR, wavelength, and RIN as distributions across temperature, lot, and age. Tie each requirement to an ATP row, each life claim to a physical mechanism, and each field alarm to a measurement that separates the laser from its driver, monitor, cooler, and optical path.

Junior mistake: declare wear-out from monitor telemetry alone, or raise launch power before naming which ledger spent (Section 5.19 and chapters 6 and 8).

Three questions to test yourself.

1. Why is the laser typically the reliability-limiting component in an optical link?
2. Optical power fell but the monitor photodiode reports no change. What do you check?
3. How would you qualify a second laser supplier without assuming the first supplier's failure distribution?

Chapter 6

WDM and wavelength-locked lasers

Read first: why WDM; capture versus hold; thermal crosstalk; laser-versus-ring isolation; one-lane versus all-lane signatures.

Deep dive: SOA noise figure and gain placement; polarization on the CW path; comb-source device classes.

Reference: WDM grid table, MUX defect map, CW-WDM MSA grid details.

Early datacenter optics mostly ran one wavelength per fiber. That worked while port counts were modest. At AI scale, fiber count itself becomes a first-order cost and cable-plant problem, so the industry packed more channels onto each strand. The price of that packing is control: once channel spacing tightens, or once the modulator is a wavelength-selective ring, someone must keep laser and filter locked together. Few phrases carry as much architectural information as "wavelength-locked laser," because locking only appears under those interconnect choices. Ring and MZM device physics stay in Section 3.14.3; per-λ laser ATP and aging stay in Chapter 5. This chapter covers grids, lock loops, thermal crosstalk, MUX budget, and CW-WDM architecture.

Tradeoff. Dense WDM vs lock complexity

Improves: Fiber count, faceplate density, and bandwidth per strand

Worsens: Heaters, TEC, lock firmware, thermal crosstalk, and ATP unlock corners

When acceptable: When fiber plant or connector count is the binding constraint

Experienced decision: Pay for locking only when multiplexing is worth the control burden.

6.1 Why multiplex wavelengths at all

At $100{,}000$+ accelerator scale, every extra fiber is another connector, another patch, and another failure mode. *Wavelength-division multiplexing* (WDM) puts many independent channels on a single fiber, each on its own wavelength, so bandwidth per fiber rises without adding fiber. Each wavelength can still be an ordinary IM/DD channel. WDM and IM/DD are orthogonal; you run IM/DD *per wavelength*.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Historically the industry climbed a ladder of spacing. Coarse CWDM4 used ≈20 nm slots and uncooled lasers. LAN-WDM tightened that for 2 km-class FR4. Dense grids and then CW-WDM O-band combs for CPO pushed spacing into the 100–800 GHz class and made active locking mandatory. Those spacings are standardized grids, not vendor choices: the 20 nm CWDM slots follow the ITU-T G.694.2 wavelength grid (18 channels, 1271–1611 nm), and the 50/100/200 GHz datacom DWDM spacings follow the ITU-T G.694.1 frequency grid anchored at 193.1 THz [90]. CWDM4 uses the four O-band lines of that CWDM grid; the CW-WDM combs in Section 6.6 define their own O-band grids for dense integration. Table 6.1 is that ladder as you will meet it in short-reach AI optics today.

Grid family	Spacing (class)	Channels / fiber	Cooling / lock	Typical short-reach use
CWDM4	≈20 nm	4	Uncooled; loose control	FR-class pluggables; faceplate WDM
LAN-WDM	≈800 GHz (≈4–5 nm @ 1310 nm)	4	Cooled or tight open-loop	2 km-class FR4 (edge of book scope)
Datacom DWDM	200/100/50 GHz	many	Locked to grid	Discrete DFB/EML DWDM modules
CW-WDM / CPO O-band	100–800 GHz class (MSA spans)	8 / 16 / 32	Locked; often ring-tuned	CPO engines, optical I/O chiplets

Table 6.1. WDM grids for short-reach AI interconnects. CW-WDM MSA normative grids sit in O-band with 9/18/36 nm spans and 8/16/32-line sets (Section 6.6); spacing is set by the chosen span and channel count, not by Ethernet CWDM4.

Exit when fiber count, lock burden, and cooling class pick one grid family for the product. **Decision unlocked:** accept uncooled CWDM4, or fund locked denser grids (LAN-WDM, DWDM, CW-WDM) with the control page they require.

6.2 Why "locked" is the operative word

WDM alone does not force active locking. CWDM4 packs four wavelengths with enough spacing that uncooled lasers can wander and still stay in their slots. Locking becomes the operative word only when either the channel spacing is tight or the modulator itself is wavelength-selective. Those two situations drive nearly every modern CPO and dense optical-I/O control loop.

6.2.1 Tight DWDM grids

To pack channels at 50–200 GHz class spacing, each laser must sit on its grid slot or adjacent channels collide. Emphasizing "locking" therefore hints at spacing tighter than CWDM. You do not stress locking for coarse, uncooled CWDM4; you do as soon as the grid looks like LAN-WDM, datacom DWDM, or a CW-WDM comb.

DWDM Dense wavelength division multiplexing: channel spacing $\leq$100 GHz; used in metro/long-haul and dense CPO WDM.

6.2.2 Microring modulators and WDM locking

Resonant ring and microdisk modulators are the dense-WDM workhorse (Section 3.14.3). Their resonance drifts by roughly 10 GHz/°C in silicon, so even a stable laser is not enough: the laser and the ring must be wavelength-locked. Either lock the laser to the ring, or thermally tune the ring onto the laser with a feedback loop. That laser–ring co-locking is the central control problem in ring-based WDM links and in co-packaged optics, and it is why neighbor heat and case-temperature ramps belong in validation, not only in the thermal section of a datasheet.

6.3 WDM filters, grids, and on-chip multiplexing

WDM is not only lasers and locking (Section 6.6): the PIC needs wavelength selective routing. The MUX/demux stage is a first-class link-budget line item (Section 7.7), not a packaging footnote.

PIC Photonic integrated circuit: waveguides, modulators, detectors, and couplers on one chip (typically SOI at 220 nm Si).

Signal journey through the MUX. Follow one wavelength from laser (or comb line) through the modulator, into the MUX, across the plant, and out the demux to the receiver. Stage insertion loss lowers OMA on every lane. Passband ripple and MUX imbalance make the weakest λ the budget limit even when the average looks fine. Adjacent-channel crosstalk closes eyes and raises TDECQ before average power looks dead. Grid misalignment clips the filter edge and can unlock a ring before the power meter alarms. Treat those signatures as plant and filter problems until per-λ power and isolation prove otherwise.

Hardware choices.

AWG / echelle gratings multiplex and demultiplex on silicon or glass. Insertion loss is often 2–5 dB per MUX stage; adjacent-channel crosstalk and passband ripple land in OMA and transmitter and dispersion eye closure quaternary (TDECQ).

Ring filter banks drop/port routing in microring banks sets how many λ share a bus waveguide. Thermal tuning per ring is common (Section 3.14.3 and section 6.5).

Hybrid some engines use a coarse AWG plus fine ring filters; count every stage in the ledger.

MZMs trade area for calm wavelength behavior. When each lane carries its own laser (DR/FR SiPh modules), silicon Mach–Zehnder modulators sidestep ring locking (Section 3.14.3). Rings remain the default when many λ share one PIC and area dominates (Section 3.14.3 and chapter 10).

Where MUX defects land. Table 6.2 maps common MUX faults to the measurement that catches them.

Fault	Optical symptom	Hits	Catch with
Stage insertion loss	Lower launch OMA on all λ	Link budget OMA	Power meter / OMA
Passband ripple / tilt	Uneven OMA across bank	Weakest λ	Per-λ OMA map
Adjacent-channel crosstalk	Closed eyes, RLM/TDECQ up	Tx quality / BER	Isolation sweep + DCA
MUX imbalance	One λ weak, neighbors OK	Single-lane BER	Per-lane power + BER
Grid misalignment	Filter edge clipping	TDECQ, unlock risk	OSA + lock status

Table 6.2. MUX/demux defects and where they appear in validation. Isolation and imbalance tests belong in ATP for any dense WDM engine (Sections 5.16 and 6.7).

Validation adds channel isolation sweeps, grid alignment across temperature, and MUX imbalance (uneven OMA per λ). Treat the weakest channel as the budget-limiting lane, not the average.

How to read the MUX budget. Table 6.2 is an FA map for the journey above. Approve the MUX for ATP when the weakest lane's OMA, isolation, and grid alignment close the budget; otherwise open packaging or PIC FA on the failing row.

6.4 Lock-loop mechanics

Wavelength locking closes the loop between source and filter. Pick a technique based on whether the laser, the ring, or both must be steered (Section 3.14.3 and chapter 7).

Error-signal sources.

Wavelength locker Etalon + dual-PD error signal that trims laser TEC or current onto a grid slot (common on discrete DFB/EML).

Etalon-based wavelength locker A fixed reference etalon plus a pair of photodiodes produces an error signal proportional to wavelength offset; feedback trims laser temperature or current onto the grid. Common on discrete DFB/EML modules.

Laser-to-ring thermal feedback Monitor the ring's through/drop power (or a dither tone) and heat the ring, or trim laser current, to park the carrier on resonance. Default for dense microring WDM banks in CPO and optical I/O.

Injection / external-cavity locking Stabilize a laser's wavelength and linewidth against an external reference cavity; higher performance, more parts. Rare in short-reach volume products.

Athermal design Engineer the device so its resonance barely moves with temperature, reducing the control burden. Athermal does not remove MUX grid alignment; it shrinks the loop authority you need.

Digital supervisory loop CMIS-exposed monitors and firmware on modern modules; link training at 224G/448G may iterate EQ and wavelength trim together (Section 10.3).

Capture versus hold. Decide the control job before you inventory actuators. *Capture* is acquiring lock from a cold or unlocked state: coarse scan of heater or TEC until the monitor error signal crosses zero, then close the loop. *Hold* is rejecting thermal drift and neighbor crosstalk once locked. Loop bandwidth must be fast enough to track case-temperature ramps and adjacent-heater steps, but slow enough not to fight the data path or inject RIN through bias modulation. Silicon rings at $\sim$10 GHz/°C set the disturbance scale: a 1 °C neighbor step is tens of GHz of resonance walk, which is a large fraction of a 100–200 GHz grid slot (Section 3.14.3).

Cold-start narrative. Power the module, identify each assigned λ, park the coarse TEC or heater until the error signal crosses zero, close the loop at operating optical power (not dark), then prove hold under neighbor heat and case-T ramp before you call lock validated (Section 6.7). Capture without hold is a demo, not a product.

What you trim. Three actuators show up repeatedly, and the bring-up order usually starts with the slowest, highest-authority knob. Laser TEC / temperature moves the whole comb or a single DFB on the frequency axis. Laser bias current is the fine wavelength trim (and also changes power), so watch RIN and SMSR when you use it as a locker (Section 5.8). Ring heaters park each microring onto its assigned λ and are the per-channel control in dense banks (Section 6.5).

Failure signatures. When a loop loses lock, bisect laser versus ring by toggling the TEC setpoint and the ring heater independently (Sections 6.7, 7.6 and 7.12). The one-lane and intermittent signatures this produces, and their causes, are catalogued in Section 6.9.3.

6.5 Thermal crosstalk and heater budget

Dense ring banks share a substrate. Heating one ring to stay on resonance shifts neighbors. That is , and it is why a single-lane lock test at room temperature is not a product test.

Thermal crosstalk Heater or self-heating on one ring shifts neighbors' resonances; worst case is full-bank traffic at max case *T*.

Where heat comes from. Self-heating from the ring's own heater (and absorbed optical power) shifts its resonance, so the lock loop must settle with the lane at operating optical power, not dark. Adjacent heaters on nearest-neighbor and next-nearest rings in a WDM bank are the next disturbance; the worst case is all neighbors at max heater power while you hold lock. Package and ASIC load add a common-mode walk: switch or XPU case-temperature ramps and local hotspots move the whole bank, and a shared TEC or cold plate sets how much of that walk the lock loop must reject (Section 7.9).

Design and validation implications. Budget heater range with headroom for crosstalk, not just for the coldest and hottest case alone. Layout (heater placement, thermal isolation trenches, ring pitch) is a reliability and yield problem as much as a control problem. In validation, simultaneous full-traffic on neighbors plus max case *T* is a *lock* test: unlock, BER walk, or TDECQ rise on one λ under neighbor load points at thermal design, not at a bad laser die (Sections 7.9 and 7.12).

6.6 External multi-wavelength sources (CW-WDM)

Dense WDM with ring modulators needs a source of many clean, stable wavelengths. The industry answer is a *disaggregated* external laser: a single multi-wavelength continuous-wave (CW) module supplies a comb of wavelengths over fiber to the photonic engine, where microrings imprint data onto each one. The *CW-WDM MSA* standardizes those sources for AI, HPC, and high-density optics [48]. Source-side measurement detail (per-channel power, SMSR, RIN, lock under neighbor heat) is in Section 5.16; this section is the architecture contract.

Architecture contract. The PIC needs many clean, stable lines with per-line power flatness, grid placement and SMSR, RIN at the plant ORL, and absolute wavelength stable enough that ring heaters stay in range across temperature with all ports active (Sections 5.16 and 6.5). Miss any one and a single λ looks like a modulator or lock failure when the source is the cause. External CW-WDM / ELSFP sources are field-replaceable: lasers dominate wear-out FIT (Chapters 5 and 8), so keeping them off the ASIC package turns COD or facet failure into a hot-swap.

Reference: CW-WDM MSA grid and measurement detail. *Reference.* Rev 1.0 (June 2021) defines O-band wavelength grids, port configurations, and measurement methods. It does *not* standardize mechanical form factors, management pins, or full link parameters; those stay application-specific or move to form-factor MSAs such as ELSFP [49][48].

Core normative content:

- **Grid sets:** 8+1 and 16+1 lines in a 9 nm span; 8+1 / 16+1 / 32+1 in 18 nm and 36 nm spans (shortest line optional in each set).
- **Spacings (class):** for the 18 nm span, channel spacing is 400 / 200 / 100 GHz for 8 / 16 / 32-line sets; the 9 nm and 36 nm spans scale spacing with span width (100–800 GHz class). Normative MSA grids are denser than 5 nm; coarser CWDM-like spacings are informative only.
- **Two physical configs:** *modular* (each fiber carries one λ) and *integrated* (each fiber carries the full comb).
- **Power classes and AS parameters:** output power classes span low to high launch; SMSR, RIN, and linewidth floors are defined with measurement methods, with many limits marked application-specific (AS) in the normative tables.

Informative appendix examples (not universal product guarantees) often quote ≈30 dB SMSR, ≈ -135 dB/Hz RIN, ≈20 MHz linewidth, $\pm$1 dB per-line power variation, and -20 dB ORL tolerance for 18 nm-span examples. Treat those as negotiation anchors; write the ATP to your link budget (Section 5.16 and table 5.4).

Exemplar: SuperNova + TeraPHY. Ayar Labs' optical-I/O stack is aimed at *scale-up* (XPU-to-XPU) rather than switch fabric [50]:

TeraPHY an optical-I/O chiplet co-packaged with the host XPU, carrying the microring modulators and receivers.

SuperNova the external CW light source, positioned as the first CW-WDM-MSA-compliant 16-wavelength source, delivering up to 16 wavelengths into each of 16 fibers. That is light for 256 data channels (vendor claim: about 16 Tb/s bidirectional), and roughly $64\times$ the wavelength count of CWDM4 pluggables.

Vendor performance claims versus pluggables plus electrical SerDes (5–10$\times$ bandwidth, 10$\times$ lower latency, 4–8$\times$ better power efficiency) are marketing numbers; use them as orientation, not as ATP limits.[1]

[1] Source: Ayar Labs, SuperNova light source, ayarlabs.com. Positioning: Ayar targets scale-up optical I/O between accelerators; Broadcom and NVIDIA CPO (Section 10.10) put the same building blocks on the *switch*.

6.6.1 Comb sources: one device, many lines

The SuperNova approach builds its comb from an array of discrete lasers, one distributed-feedback (DFB) die per wavelength, combined onto the output fiber. That is the shipping answer, and it scales cleanly to the 8 and 16 lines the MSA grids call for. Past a few dozen lines the die count, the combining loss, and the per-die wavelength trimming start to hurt, which is why a single device that emits a whole comb is attractive. Three device classes compete.

QUANTUM-DOT MODE-LOCKED LASERS (*QD-MLLs*) are the front-runner for a monolithic O-band comb. Mode locking in a single cavity produces evenly spaced lines at the cavity round-trip rate; quantum-dot gain adds low RIN, a near-zero linewidth-enhancement factor, and strong optical-feedback tolerance, the same properties that make quantum-dot lasers attractive for isolator-free co-packaging [2][102][56]. Reported O-band demos carry 14×100 Gb/s PAM4 over 10 km at ∼284 fJ/bit, and isolator-free variants target interconnect capacity beyond 3.2 Tb/s. These are research results, not qualified products, so treat the line counts and efficiencies as provisional.

[2] QD mode-locked laser combs: best single-device efficiency/size for DWDM; O-band demos to 14×100G PAM4 and beyond 3.2 Tb/s. Provisional.

KERR MICROCOMBS take the opposite route: pump one high-*Q* microresonator and let four-wave mixing fill in many evenly spaced lines on a chip [3][103]. The line count and the spacing uniformity are excellent, and a 2025 demonstration added a monolithic demultiplexer that autonomously locks to and tracks the comb lines. The catch is power. Pump-to-comb conversion efficiency is modest, so each line leaves the chip weak and usually needs a booster or per-line amplifier before it reaches the modulator bank (Section 6.6.2). Microcombs also need a clean pump laser and careful thermal control to hold the soliton state.

[3] Integrated Kerr/soliton microcomb source: hundreds of chip-scale lines with autonomous locking, but low per-line power. Provisional.

GAIN-SWITCHED AND QUANTUM-DASH COMBS sit between the two: a directly driven laser produces a flatter, lower-line-count comb with simple electronics. They have reached multi-terabit aggregate rates in the lab but see less datacom traction than QD-MLLs.

For any of them the contract from the MSA does not change: the source must hold per-line power flatness, SMSR, RIN, and grid placement across temperature with every port active (Section 5.16). A comb that delivers 32 lines but drops 6 dB across the band, or whose edge lines miss the grid, buys nothing over an array of DFBs the PIC already knows how to drive.

6.6.2 Gain and power distribution across the bank

Interview takeaway. An SOA rescues launch power or line flatness and spends noise figure and polarization-dependent gain. Prefer a higher-power source that already meets the budget; if you add gain, name whether it is a comb booster, per-line trim, or receiver preamp, and keep per-line flatness as a system ATP row.

Whatever generates the comb, the light still has to survive the trip to the modulators. One source feeds a multiplexer, a splitter tree, and per-line routing before it reaches a ring, and each stage takes its cut. When the source is a chip-scale comb with weak lines (Section 6.6.1), or when the fan-out is large, a *semiconductor optical amplifier* (SOA) restores the budget.

Where the gain sits. A booster SOA placed right after the comb lifts every line before the split, so one device pays for the whole fan-out. A per-line SOA after the demultiplexer instead corrects line-to-line imbalance, at the cost of one amplifier per wavelength. The receiver-side SOA preamplifier (Table 4.4) is a separate job: there the goal is sensitivity, here it is launch power.

The noise-figure cost. An SOA adds amplified spontaneous emission (*ASE*), and its noise figure sets how much. The quantum floor is 3 dB; commercial O-band SOAs land near 6–7 dB with roughly 15 dB of gain and about 1.5 dB polarization-dependent gain [4][104]. Every dB of noise figure eats into the signal-to-noise ratio the receiver eventually sees, so an SOA that rescues launch power can cost link margin if it runs deep into saturation or amplifies an already-noisy comb. Quantum-dot SOAs grown on silicon are attractive for the same reason QD lasers are: low noise, wide O-band gain, and CMOS-compatible integration.

[4] O-band SOA: $\sim$15 dB gain, NF $\leq$7 dB (3 dB quantum floor), $\leq$1.5 dB PDG; QD-SOA on Si suits CPO. Booster/distribution gain.

Holding the bank flat. Gain is not uniform across the comb span, and SOAs compress near saturation, so the line that starts strongest is not the line that ends strongest. Per-line power flatness is a system spec (Section 6.6), held with some mix of source-side pre-emphasis, gain-tilt control, and per-line trimming. The alternative to any distribution-side gain is a higher-power source, which is why array-of-DFB designs that already meet the launch budget often skip the SOA entirely.

6.6.3 Polarization on the CW distribution path

Interview takeaway. Receiver photodiodes do not need polarization control. The CW path into TFLN, grating couplers, and SOAs does. Hold that path on PM fiber to the preferred axis, or shorten it with co-packaging.

IM/DD is forgiving about polarization where it counts most. The photodiode is a square-law detector, so the received state of polarization does not affect the recovered bits (Section 3.4). A standard single-mode drop to the receiver therefore needs no polarization control. The external-source and CW-WDM architectures move the sensitive part upstream, onto the path between the laser and the modulator.

WHERE IT MATTERS. Three elements on the CW feed care about the launched state. A *TFLN* Mach–Zehnder drives the electro-optic effect through TE-polarized light on one crystal axis (Section 3.14.3); light on the wrong axis sees little modulation. Silicon grating couplers are polarization-selective by construction, so coupling loss swings with the input state, a *polarization-dependent loss* (PDL) that comes straight off the launch budget. And a booster or per-line SOA adds polarization-dependent gain (Section 6.6.2), so a drifting state becomes a drifting per-line power. None of these sit after the photodiode, so none show up in a receiver-side budget: they act on light that has not yet been modulated.

HOW IT IS HELD. Keep the CW feed on *polarization-maintaining* (PM) fiber and PM connectors from the external laser (Chapter 5) to the modulator input, launched on the coupler's preferred axis. On-chip the light is already single-polarization once it is in a TE waveguide, so the discipline is really about the fiber run and the mate at the package. A co-packaged design with the laser on the same substrate shortens that run to millimeters and sidesteps most of the problem; an external-laser design pays for a PM path and its alignment tolerance instead.

6.7 Lock validation playbook

Instruments and BER methods live in Chapter 7. What is special to WDM is the order: you cannot trust a BER number on a ring bank until the comb is identified, the resonances are parked, and the lock loops hold under neighbor heat. The sequence below is the usual bring-up; skip a step and you will debug the wrong domain.

Bring-up order.

1. **Grid ID:** confirm each CW line (or DFB) is on the assigned channel with an OSA / wavemeter (Section 5.16).
2. **Coarse align:** park rings near resonance with open-loop heater sweeps; check through/drop monitors.
3. **Close lock:** enable the feedback loop; verify capture on every λ at operating optical power.

4. **Stress neighbors / temperature:** max case *T*, neighbor heaters and traffic on (Sections 6.5 and 7.9). Confirm hold, not just capture.

5. **Close the link:** BER / TDECQ / sensitivity on the weakest lane first (Sections 7.4 and 7.7).

Exit when every assigned λ is grid-identified, captures at operating power, holds under neighbor heat and case *T*, and the weakest lane closes BER. **Decision unlocked:** proceed to loaded characterization, or stop and bisect laser versus ring before trusting any BER number.

Host-visible CMIS may report lock error, heater codes, or wavelength monitors; use those for fleet triage. Grid ID and absolute alignment still need an OSA or wavemeter when engineering access exists. Do not treat a "locked" flag alone as proof the comb is on the assigned grid.

Bisect laser versus ring. If one λ unlocks or walks, change one actuator at a time:

- change laser TEC / current with ring heater fixed: if the error follows the laser, the source or its locker is wrong;
- change ring heater with laser fixed: if the error follows the heater, the ring, monitor PD, or thermal crosstalk is wrong;
- if both look fine but OMA is low, inspect MUX imbalance and connector/ORL (Table 6.2 and section 7.2.2).

Fleet telltales. Slow BER creep with rising bias on one line is often laser wear-out (Section 8.5). Sudden unlock under neighbor load with healthy LIV is thermal crosstalk or lock firmware. One dark lane with neighbors up is COD, FAU, or a single ring/heater fail; classify with Section 7.12 before you open an 8D on the wrong supplier.

What this usually means. All lanes unlock after a temperature step

Usually: shared TEC, supply, firmware, or thermal control

Not: independent failures of every lane's laser in the same second

Engineering heuristic. One-lane unlock with healthy neighbors usually points at that lane's laser, ring, heater, or attach. All-lane unlock points at a shared resource first.

6.8 What wavelength locking implies about an architecture

Because locking is only worth its complexity under specific conditions, its presence narrows the design space considerably.

```
Locking present
   |
WDM (usually dense)
   |
Thermal / heater / TEC control
   |
ATP: lock range, unlock, crosstalk
   |
Fleet: unlock alarms, wavelength telemetry
```

Implication	Strength
Some form of WDM is in use	Near-certain: locking only matters with WDM.
Dense (D)WDM rather than coarse CWDM	Likely: locking implies tight spacing.
Microring-based silicon photonics with external multi-wavelength (CW-WDM) sources, often trending toward co-packaged optics	Common in modern AI interconnects (e.g. Broadcom and NVIDIA Quantum-X programs), driven by fiber-count pressure at scale.
Discrete DFB/EML DWDM (no rings)	Also possible; locking alone does not prove rings.

Table 6.3. What a wavelength-locking requirement implies.

The solid conclusion is that **WDM is present and wavelength control is central**; whether the implementation is ring-based silicon photonics with external multi-wavelength sources or discrete DFB/EML DWDM depends on the specific system.

6.9 Engineering lens

6.9.1 How it works

WDM buys fiber count back by stacking wavelengths, and the price is control: once the grid is tight or the modulator is resonant, laser and filter must stay locked. The chapter's lock loops, thermal budgets, and source specs all exist to hold that lock across temperature and neighbor load.

6.9.2 How it is measured

Start with an OSA or wavemeter to identify every source line, then sweep each ring or filter while recording through-port, drop-port, heater current, monitor-PD signal, and lock error. Measure insertion loss and adjacent-channel leakage per lane. Close the loop and repeat BER, OMA, and TDECQ under case-temperature ramps, neighbor heater activity, source-power spread, and restart. Capture range and hold range are different acceptance tests; Section 6.7 gives the order, and the source measurements follow the CW-WDM MSA methods [48].

6.9.3 How it fails

One degraded lane points toward its laser line, ring, filter channel, heater, monitor, or local fiber attach. All lanes moving together points toward the shared CW source, thermal controller, supply, polarization path, or common MUX. Capture failures occur during startup or a large temperature step. Hold failures occur after lock, often under neighbor heat, power droop, or control-loop interaction. Treat those as separate signatures.

One lane fails. Possible causes, roughly in order of how often each shows up in the field:

- **Wavelength locker failure:** the etalon-based error signal drifts or the locker photodiodes degrade, so the loop steers the laser to the wrong setpoint even though the laser itself is healthy (Section 6.4).
- **Ring drift:** the microring's own resonance walks off the assigned λ under local heating or process aging, independent of the laser (Section 3.14.3).
- **AWG issues:** an arrayed-waveguide grating channel shifts passband center or picks up excess adjacent-channel crosstalk, clipping one lane at the filter edge while neighboring channels stay clean (Section 6.3 and table 6.2). Distinguish from a laser or ring fault by sweeping the source across the passband and watching for a grid-alignment signature rather than a lock-error signature.
- **Thermal crosstalk:** an adjacent heater's disturbance exceeds this lane's hold-range budget while its own actuators read nominal (Section 6.5).

Intermittent failure. Failures that appear and clear on their own point to control-loop dynamics rather than a broken part:

- **Lock acquisition issues:** the loop fails to capture reliably from a cold or power-cycled state, so failures correlate with restarts or ELS hot-swaps rather than with steady-state operation (Section 6.7).

- **Heater control instability:** the thermal feedback loop oscillates or overshoots, often triggered by a control-loop bandwidth that fights the data path or a gain that was tuned for a different neighbor-load condition. Shows up as a heater current that hunts rather than settles.
- **Temperature excursions:** a transient case-temperature ramp (fan failure, load step, HVAC event) pushes the loop past its hold range momentarily; the lane recovers once the transient passes, which is the distinguishing signature versus a genuine hardware fault.

Failure mode: Wavelength drift

Symptoms. BER rises with temperature, one WDM lane moves, and lock error or heater demand grows.
Likely causes. Laser wavelength drift, ring resonance drift, thermal coupling, a saturated TEC or heater, or a weak monitor signal.
Measurements. OSA or wavemeter, heater and TEC current, lock error, neighbor activity, and per-lane BER.
Mitigations. Restore thermal headroom, widen capture only after noise is checked, improve calibration, and reduce shared thermal or supply coupling.

6.9.4 How it is debugged

First decide whether the fault affects one lane or all lanes. Apply the power-versus-quality fork, then ask which ledger moved: often spectral or control (Sections 4.8 and 5.19). Freeze heater and laser actuators long enough to measure the open-loop spectrum. Move the source while holding the ring fixed, then move the ring while holding the source fixed. If neither follows the failing BER, inspect MUX loss, polarization, connector ORL, and the electrical lane. For intermittent unlock, align lock-error, heater, temperature, supply, and neighbor-traffic traces on one time axis. A control loop cannot be debugged from final register values alone.

Debug story

Observed. One lane failed only after adjacent lanes began traffic.
Investigation. The source line stayed on grid, while the suspect ring heater railed and lock error grew with neighbor temperature.
Finding. The optical source and receiver passed; the resonance was being pushed out of hold range.
Root cause. Thermal coupling from the adjacent heater exceeded the control-loop budget.
Resolution. The heater map and feed-forward terms were corrected, and neighbor-load hold testing became an ATP corner.

6.10 Interview takeaway

Key idea. Wavelength locking is a control-system problem wrapped around an optical link. Measure the grid, capture, hold, shared thermal paths, and per-lane margin. Then use one-lane versus all-lane behavior to split the source, ring bank, MUX, thermal controller, and common supplies.

Junior mistake: blame every unlock on the laser, or debug lock from final register values without neighbor-load and time-aligned traces (Section 4.8 and chapters 7 and 11).

Three questions to test yourself.

1. Why does a microring modulator require wavelength locking while a Mach–Zehnder does not?
2. One WDM lane fails while neighbors are healthy. How do you determine whether the cause is the laser, ring, filter, heater, or connector?
3. All lanes unlock after a temperature step. What shared resource do you check first?

Chapter 7

Optical validation

Read first: validation lifecycle prose; bring-up; core measurements; CMIS sequence; production-like corners; fleet triage.

Deep dive: instrument theory of operation; Method A/B link-budget accounting; CMIS register map detail.

Reference: job definitions table, ladder Stage/Question/Evidence/Decision table, bring-up checklist, corner checklist.

A datasheet that closes on a quiet bench is not a product. A quiet-bench close proves one corner under one setup. It does not prove that the link meets its requirements across the temperatures, hosts, connectors, production spread, and lifetime the fleet will actually see.

Validation is the process of building enough justified confidence to make those product decisions. Passing tests is an output, not the purpose. The job is to remove the uncertainty that blocks the next call: continue, redesign, derate, qualify, open volume, or hold. Debugging asks which margin ledger was exhausted. Qualification asks how much margin remains after the expected stresses. Both are uncertainty reduction that ends in a decision (Chapter D and section D.16). This chapter sequences that work from a single device to a deployed fleet (Section 7.1), then covers bring-up under production-like corners and the hypothesis-driven debug method the work demands.

Companies overload EVT, DVT, PVT, and "verification" differently. Unless noted otherwise, this book freezes meanings by *job*, not by org chart. Abbreviations are collected in Chapter G.

How the jobs differ. *Characterization* maps how the design behaves. It discovers distributions, sensitivities, and cliffs. A characterization cliff improves understanding; it does not automatically fail the product. BER versus temperature, wavelength drift, receiver sensitivity, RIN, and bias trends belong here when the question is still "what happens when you change *X*?"

Verification asks a narrower question: does this implementation meet a stated requirement with a named method and reference plane? Transmit power, BER at a stated FEC objective, wavelength accuracy, and thermal operating range are verification when the requirement, plane, and method are already frozen.

Validation asks whether the product works for the intended customer and system use. Hosts, cable plant, temperatures, workloads, and install practice are in scope. A verified BER on a golden bench is not validation of the fleet claim.

Qualification asks whether you have evidence the design survives expected variation and life. The evidence is mechanism-based stress and sample humility, not a ritual checklist. HTOL, temperature cycling, humidity, and multi-lot process corners matter only when each stress maps to a named mechanism and an acceptance rule.

Production test asks whether you can repeatedly detect unacceptable units at volume. Factory and incoming screens catch escapes; they do not prove life. *ATP* (acceptance test) is one form of that process: a replayable accept/reject decision applied per unit, per lot, or by a documented sampling plan.

One module, five jobs. Take a temperature BER sweep on a new EML module. Mapping BER versus case temperature to find cliffs is characterization. Checking that BER stays below the frozen FEC objective at the named plane and method is verification. Running that check on the production host, cable plant, and install practice is validation. Running HTOL or humidity on a sample lot to bound life for a named mechanism is qualification. Running a fast lot screen for power, CMIS, and a sampled TDECQ row is production test. Same hardware, five different questions.

The split matters because the classic program failures are category errors: treating characterization cliffs as automatic product failure, treating ATP as life proof, or treating HTOL as host interoperability. The rest of this chapter is written to keep those jobs from collapsing into one another.

Term	Question	Decision it unlocks
Characterization	How does it behave?	Model / derate / redesign before loaded work
Verification	Does it meet a stated requirement?	Pass / fail that requirement at a named plane
Validation	Does it work for intended system use?	Approve / restrict system claim
Qualification	Does it survive expected variation and life?	Accept / derate / hold life risk
Production test	Can we detect unacceptable units at volume?	Screen / sample / hold lot
ATP	Replayable accept/reject process?	Ship / fail unit or lot

Table 7.1: Operating definitions used in this book. Decision unlocked: which job you are doing, and which decision that job can honestly support. Abbreviations: Chapter G.

The validation flow in Section 7.1 is the ordered way these jobs are sequenced so expensive evidence is not asked to answer the wrong question.

7.1 The validation flow

Table 7.2. Validation flow at a glance (Steps 1–11). Wall-chart: Section D.2. Interview drill: Section A.8.5. NPI names: Table 9.2.

The validation flow at a glance.

- **Step 1: Define requirements.** State what the product must do and clarify what later evidence must demonstrate.
- **Step 2: Review the architecture.** Close the important budgets on paper and clarify whether the proposed design is worth building.
- **Step 3: Bring up the hardware.** Establish reproducible basic operation and clarify whether later measurements will be interpretable.
- **Step 4: Characterize behavior.** Map distributions, sensitivities, and cliffs and clarify how the product normally responds.
- **Step 5: Validate margin and interoperability.** Challenge the product in realistic systems and clarify where the supported operating boundaries lie.
- **Step 6: Qualify reliability.** Stress credible failure mechanisms and clarify what lifetime and environmental claims the evidence supports.
- **Step 7: Validate manufacturing.** Measure process variation and detection capability and clarify whether the design can be built repeatedly.
- **Step 8: Run a controlled pilot.** Deploy a bounded, observable population and clarify whether laboratory and factory assumptions survive field use.
- **Step 9: Ramp mass production.** Sustain volume under ATP, SPC, supplier, and change controls and clarify whether production remains stable.
- **Step 10: Monitor the fleet.** Track cohorts, drift, and escapes and clarify whether the released population remains healthy.
- **Step 11: Feed learning into the next revision.** Convert field and manufacturing evidence into changed requirements, designs, qualification, or controls.

Each step answers a question that the previous step cannot. A later-stage success cannot compensate for missing evidence from an earlier stage.

7.1.1 One lifecycle, beginning before hardware

Validation does not begin when a module reaches the bench. It begins when the team decides what the product must accomplish and whether the proposed architecture can plausibly accomplish it. Requirements define success. Architecture work determines whether optical power, noise, thermal, electrical, reliability, manufacturing, and service assumptions can close together before tooling and hardware make changes expensive.

Once hardware exists, the nature of the uncertainty changes. Bring-up establishes that the unit is alive, configured correctly, and measurable. Characterization then builds the behavioral model: how performance moves with temperature, voltage, optical loss, reflections, lane, unit, and lot. These steps are deliberately separate. A module that links once is not characterized, and a detailed characterization sweep is difficult to interpret when basic operation is unstable.

Margin and interoperability place that behavioral model inside the intended system. The question is no longer merely how the module behaves in isolation, but whether enough headroom remains across supported hosts, peers, firmware, cable plants, thermal loading, and operating corners. This is where the team defines the real shipping envelope.

Reliability qualification asks a different question: whether time, stress, and environmental exposure cause permanent degradation. Manufacturing validation then asks whether the qualified result can be reproduced across lots, tools, suppliers, and production measurements. Qualification supports confidence in the design; manufacturing validation supports confidence in the process.

A controlled pilot tests the remaining assumptions in a bounded field population. Mass production scales only after the pilot exit is justified and the production controls are operating. Fleet monitoring continues because deployment can reveal mechanisms, interactions, and population effects that no laboratory campaign completely predicts. Those findings must flow back into the next requirements, architecture, qualification plan, ATP, supplier control, or service procedure.

This is one lifecycle. Simulation, bench measurement, qualification stress, and field evidence are not competing paths. They are different evidence sources selected within the appropriate step (Sections D.2 and D.16).

Why experienced engineers walk steps in this order?

Because each step removes a different uncertainty. A late-step pass cannot repair a missing requirement or unstable bring-up; treating HTOL as production readiness is the classic mix-up.

7.1.2 Step 1: Define the requirements

The first validation step is not a measurement. It is defining what success means so you do not validate the wrong product. Without a requirements slice you do not know which BER, envelope, lifetime, volume, or deployment environment later measurements must close.

Write performance, environment, reliability, manufacturing, and operational requirement classes with owners and, where applicable, named planes (Section 1.1 and table 5.4). Weak answer: "We test BER." Strong answer: first define target BER, operating envelope, lifetime, manufacturing volume, and deployment environment; then design validation around the risks. The evidence here is a signed contract, not a bench plot.

Representative evidence: Signed requirements slice with owners, planes, and

unambiguous pass/fail language.
Exit: The slice is specific enough that architecture and later steps have pass criteria, or hardware work is refused until it is.
Decision: Architecture target and validation scope, or hold until success is defined.
Interview trap: Jumping to instruments before success criteria exist.

Once success is defined, the team can determine whether any architecture can meet it.

7.1.3 Step 2: Review the architecture

Determine whether the architecture can meet the requirements before hardware exists. Validation starts on paper and in models (Section 1.1 and table 5.9). Even with clear requirements, optical power, thermal budget, electrical margin, reliability target, and manufacturing cost may not close together.

Close budgets under stated assumptions, or redesign before buying tooling. A quiet-bench prototype is not architecture proof. Simulation and paper budgets are Step 2 evidence when assumptions are named and later checkable; they do not replace later steps.

Representative evidence: Closed or explicitly open budget lines; named assumptions; redesign triggers.
Exit: Architecture is feasible under stated assumptions, or redesign is chosen before tooling.
Decision: Proceed to hardware bring-up, or redesign.
Interview trap: Skipping reliability and manufacturing cost until DVT.

Once the important budgets close under stated assumptions, hardware work has a defensible target.

7.1.4 Step 3: Bring up the hardware

Answer whether the hardware is alive and measurable. Bring-up is not qualification. Separate bench bring-up (trusted setup, interpretable measurements), system integration bring-up (target host init and traffic), and later margin/interop work. Before bring-up you do not know whether a failure is integration or product physics.

Confirm emit, receive, CDR lock, and usable pre-FEC BER at named planes before arguing about margin or life (Section 7.9 and table 7.4). The output is a known configuration that becomes the baseline for every later comparison, not a claim that the product is ready.

Tradeoff. Bring-up speed vs characterization depth

Improves: Faster path into margin work

Worsens: Missing baselines that make later BER tickets uninterpretable

When acceptable: When the unit already links and emits in class on the target host

Experienced decision: Do not skip characterization to “save time.” Without a baseline, margin and FA burn calendar later.

Representative evidence: CMIS ready state; first-light and Rx power; CDR lock; usable pre-FEC BER on golden or target host at named planes.
Exit: Unit reaches ready state, emits and receives light in class, holds lock, and shows usable pre-FEC BER with named reference planes.
Decision: Continue into characterization, or stop and debug integration.
Interview trap: Deep FA before basic operation, or treating loaded chassis corners as bring-up exit evidence.

Once operation is reproducible, measurements can be used to model behavior rather than debug basic integration.

7.1.5 Step 4: Characterize the behavior

Create the behavioral model. The output is not pass/fail. The output is how the system behaves. Characterization is exploratory: finding a cliff improves understanding; it does not automatically fail the product. Before this step you know one corner works. After it you know how distributions move with stress variables and which ledgers (power, noise, timing, spectral, control) are thin.

Map temperature, voltage, optical stress, and unit/lot distributions. On the transmitter path, measure TDECQ, OMA, ER, and level linearity; with component access, add LIV, RIN, and SMSR (Sections 5.7, 7.4 and 7.7). Characterization still studies the product largely as an isolated object.

Representative evidence: Response surfaces; distribution summaries; thin-ledger list; candidate guardbands.
Exit: Population behavior versus required corners is mapped, thin ledgers are named, and you decide whether derate or redesign is needed before loaded-fleet work.
Decision: Proceed to system validation, derate the envelope, or redesign.
Interview trap: Treating a hero sample at $25°$C as the fleet model, or parking life projection inside characterization.

Once normal behavior and thin margins are mapped, the product can be challenged in realistic system combinations.

7.1.6 Step 5: Validate margin and interoperability

Characterization tells how the product moves. Margin testing tells where it fails. Interoperability tells whether that boundary changes across hosts, peers, firmware, channels, and operating environments. These are one system question, not two lifecycle steps.

Establish remaining margin. A product does not fail because it reaches a nominal limit. It fails because all margins are consumed. Ask how much capability survives when temperature, voltage, loss, ORL, aging intent, and manufacturing variation stack under production-like stress (Sections 5.19 and 7.9). Compare pre-FEC BER, telemetry, retrain count, and control headroom against the characterization baseline. Track optical, electrical, thermal, reliability, and manufacturing margin categories separately, then stack once. Do not subtract the same physical effect twice under two names. Quiet-bench margin is not rack margin.

Margin waterfall (illustrative accounting, not a universal budget).

Initial optical margin
– temperature penalty
– connector / plant loss
– aging penalty
– contamination / ORL penalty
– manufacturing variation
= remaining field margin

Allocate margin where uncertainty is highest (Section 5.19).

Challenge the supported ecosystem. A component can pass isolated validation and still fail as a system. Prove supported host, peer, firmware, and channel combinations retain required headroom. Margin on a golden host does not prove the supported ecosystem. Exercise host ASIC / SerDes, peer module or second source, firmware and CMIS revision, faceplate temperature and airflow, and fiber/cable plant. Assign failures to host, peer, software, environment, or channel before changing laser bias tables.

Tradeoff. Golden-host speed vs interop risk

Improves: Fast bring-up and clean debug on a known good station

Worsens: Hidden host, firmware, or plant sensitivity until volume

When acceptable: When golden-host data are a stage gate, not the exit criteria for the ecosystem

Experienced decision: Use golden hosts for speed; require representative hosts before calling interop done.

Representative evidence: Cliff locations and remaining headroom by ledger; margin waterfall without double-counted penalties; loaded-corner results (Table 7.5); supported combination list and documented restrictions.
Exit: Failure cliffs and remaining headroom are known at the named plane and loaded corners, and supported host/peer/firmware/channel combinations retain required headroom or a documented restriction defines where the product may ship.
Decision: Approve the shipping envelope and supported ecosystem, restrict deployment or SKUs, or send the design back.
Interview trap: Calling golden-host margin the ecosystem exit, or subtracting the same impairment twice under two names.

Once the present-day shipping envelope is understood, qualification can ask whether time and exposure change it.

7.1.7 Step 6: Qualify reliability

Build mechanism-based evidence that the design survives expected variation and life. Temperature sweeps during characterization show how a healthy product behaves while it is hot or cold. Reliability stresses ask whether exposure causes permanent degradation. Keep operational environment test (works while exposed), reliability stress (exposure causes unacceptable permanent change), and life projection (justified field-life claim) distinct.

Do not run stresses only because a checklist names them. Follow failure mechanism $\rightarrow$ acceleration method $\rightarrow$ stress $\rightarrow$ pre/post observable $\rightarrow$ acceptance criterion $\rightarrow$ confidence (Sections D.3, 8.3 and 8.4). A stress without a failure mechanism is only exposure. Deep FIT/DPPM math lives in Chapter 8.

Representative evidence: Pre/post stress margins; mechanism notes; sample and confidence statement; production-proxy candidates.
Exit: Sample plan, mechanism, and projected life support the requirements slice, or ship is explicitly held for life risk.
Decision: Accept life risk for the envelope, derate life or use conditions, or hold.
Interview trap: Treating HTOL as host interoperability or as production readiness.

Reliability qualification is developed in Chapter 8, including mechanism selection, accelerated stress, observables, sample confidence, and acceptance decisions.

Once life risk is bounded for representative hardware, the next question is whether production can reproduce that result.

7.1.8 Step 7: Validate manufacturing

The factory is part of the design. Qualification proves the design. Manufacturing validation proves the process: build it repeatedly, measure it repeatedly, and detect bad units. Engineering samples often receive unusual attention; volume readiness requires manufacturing distributions, not only the best units.

This is the PVT question: whether yield, process control, and ATP coverage survive lot-to-lot variation (Sections 9.2 and 9.4 and table 9.2). DVT belongs earlier; do not park it inside this step. Prove ATP/sample/SPC coverage against known escapes with replay, separation, and production repeatability.

Representative evidence: Multi-lot yield; classified ATP/sample/SPC coverage; measurement capability; FAIR; escape-detection proof (Sections D.16 and 9.4).
Exit: Multi-lot yield, screen coverage, SPC stability, and FAIR evidence support opening volume, or shipment is held for process control.
Decision: Open volume toward pilot, hold shipment, or demand corrective action before ramp.
Interview trap: Calling two hand-selected lots "multi-lot" evidence.

Manufacturing validation is developed in Chapter 9, including production-intent builds, traceability, measurement-system analysis, yield, ATP, SPC, supplier control, and staged ramp.

Once the process and its screens are controlled, a bounded deployment can test the remaining assumptions.

7.1.9 Step 8: Run a controlled pilot

A pilot is not merely a small production shipment. It is a controlled experiment designed to determine whether validation and qualification assumptions remain true in deployment. Lab and factory evidence may still miss install practice, traffic mix, or environment.

Use known serial numbers and lots, representative hosts, enhanced telemetry, and explicit success and rollback criteria (Section 7.12). Telemetry should reveal whether BER, FEC behavior, retrains, temperature, optical power, lane behavior, and cohort rates match expected distributions.

Representative evidence: Pilot cohort database; field BER/FEC and telemetry trends versus lab baselines; exit metrics.
Exit: Pilot success/rollback criteria are met, or risk drives restrict or reject.
Decision: Expand deployment, restrict, pause a supplier or lot, or reopen an earlier step.
Interview trap: Calling open volume a pilot, or learning the escape from customer outage instead of a bounded trial.

Once pilot evidence matches the release model, production can expand under sustained controls.

7.1.10 Step 9: Ramp mass production

Sustain volume with process control after pilot exit. Depth lives in Table 9.2 and chapter 9. Pilot success does not prove ECO discipline, SPC stability, and RMA burn-down at volume. Hold or open volume based on control, not on hope that pilot luck continues.

Representative evidence: SPC trends; yield; RMA rates by code; ECO impact checks.
Exit: SPC, ECO, and RMA loops support sustained volume, or volume is held.
Decision: Open / hold volume; trigger supplier corrective action.
Interview trap: Treating pilot luck as proof of sustained control.

Once volume is open, fleet evidence becomes the continuing check on process and design assumptions.

7.1.11 Step 10: Monitor the fleet

Validation does not end at shipment. Keep escapes and drift detectable after release (Section 7.12 and table 7.6). Unknown failure modes, aging, supplier drift, and environmental effects appear only at scale. Every telemetry field must answer what decision it enables. Trend and disagreement alarms catch dying units; hard thresholds catch dead ones.

There is no terminal exit: ownership transfers into steady operations when schema, owners, and cohort baselines are in place.

Representative evidence: Schema-stable telemetry; cohort baselines; alarm history; RMA codes split by vendor and mechanism class.
Exit: Ownership transfer into operations with schema, owners, and cohort baselines in place (ongoing control, not silence).
Decision: Continue ship, restrict, pause a supplier or lot, or reopen an earlier step when cohort evidence falsifies a qual assumption.
Interview trap: Collecting telemetry that enables no decision.

Fleet observations have value only when they change the next product or its controls.

7.1.12 Step 11: Feed learning into the next revision

Close the loop into the next requirements and architecture revision. Fleet and FA evidence that does not change the product is wasted. Separate one-off install errors from systematic design or process debt (Section 1.1). Fleet tickets that never become requirements, or fixes without screen changes, leave the next escape open.

Representative evidence: Revision backlog with owners: requirement changes, architecture changes, ATP updates, derates.
Exit: Next-revision targets are written with owners, or residual risk is explicitly accepted without change.
Decision: Next-revision targets, ATP updates, or documented accept risk.
Interview trap: Closing RMAs without an owned control or requirements update.

The output becomes the input to Step 1 of the next revision.

7.1.13 Choosing evidence within a step

At any step, use the least expensive evidence that can responsibly answer the current question.

- Use a model when the assumptions are understood and the result can later be checked.
- Use a bench experiment when controlled measurement can separate the important hypotheses.
- Use qualification stress when the uncertainty concerns permanent degradation or lifetime.
- Use pilot or fleet evidence when the condition cannot be represented credibly in the laboratory.

The evidence source does not determine the lifecycle step. The engineering question does. Simulation may support Step 2 architecture review, while a validated model may also support a Step 5 margin decision. A chamber experiment may support characterization or qualification depending on whether it measures reversible operation or permanent degradation.

Before open volume

Step 1 requirements · Step 2 architecture · Step 3 bring-up · Step 4 characterization · Step 5 margin/interop · Step 6 qualification · Step 7 manufacturing/ATP · Step 8 pilot · Step 9–10 fleet control (Section D.17 and table 7.2).

For every metric at every step, state measurement, reference plane, condition, access level, and the decision unlocked (Sections A.2, 3.9 and D.16). Bad: "receiver sensitivity is -15 dBm." Good: sensitivity at the module optical input under the named BER target, temperature, wavelength, and FEC assumptions. A number without a plane and a method is not a measurement.

Engineering heuristic. Name the reference plane before you name the instrument. A pretty eye at the wrong plane is a wrong answer.

Key idea. Validation is a sequence of decisions, not a catalog of tests. Begin by defining success and closing the architecture on paper. Establish interpretable hardware, build the behavioral model, challenge system margin, qualify life, prove manufacturing control, and then compare the model with pilot and fleet reality. Each step should produce evidence for one explicit decision and identify the uncertainty that remains.

7.2 The core IM/DD measurements

Once the flow is clear, the measurement list is organized around isolation: transmitter, channel, and receiver. That split is older than PAM4. Long before TDECQ, field engineers learned that a dark link can be a dead laser, a dirty connector, or a dead TIA, and that guessing which one burns hours. Bisecting those three domains is still how you keep debug from turning into simultaneous retunes of everything.

7.2.1 Transmitter

Start with the light leaving the faceplate or the CPO fiber array. For PAM4, the headline metric is *TDECQ* (transmitter and dispersion eye closure quaternary): a reference equalizer is applied to the captured eye and the residual penalty is reported in dB (Section 7.4). Alongside it you read *OMA* (outer), extinction ratio, and *RLM* (level linearity), plus wavelength, spectral width, and RIN with a bias-driver versus feedback bisect (Sections 5.7 and 5.8 and section 4.3.1).

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

What else you add depends on the transmitter style. Laser-bearing modules need LIV, threshold, slope, SMSR, and chirp checks for DMLs (Sections 5.4 and 5.7). External MZMs (TFLN or silicon) need EO S_{21}, V_π, quadrature bias versus temperature, and driver-path eye symmetry at baud (Section 3.14.3 and section 7.4). Microring banks need resonance alignment, thermal tuning, neighbor crosstalk, and peaking-network EO S_{21} (Section 3.14.3 and chapter 6). The point of the list is not completeness for its own sake: it is knowing which instrument answers which hypothesis when the eye closes.

7.2.2 Channel

If the transmitter looks clean into a golden receiver and the link still fails, the channel is next. Insertion loss from fiber, connectors, MUX/de-MUX (Section 6.3), and on-chip coupling (Section 3.14.3) is the first ledger line. Use the specified maximum loss for the exact connector class, number of interfaces, cleanliness condition, and reference plane; do not treat "1–3 dB per mated pair" as a universal normal loss. Chromatic dispersion (Section 3.11) matters more on FR-class SMF sweeps than on short DR links. Optical return loss (ORL) is the quiet killer: reflections can create optical feedback noise, multipath interference, deterministic distortion, and power-independent error floors. That is why many DR/FR modules still carry isolators while some CPO engines rely on design margin and monitor photodiodes instead (Section 4.3.1 and chapter 5). Fiber attach (MPO/MTP, FAU, grating couplers) shows up as both yield and reliability (Section 8.7).

7.2.3 Receiver

Receiver work asks whether the front-end can still decide bits at the OMA that survives the channel. Measure sensitivity (minimum OMA for the named BER objective at a stated plane, pattern, and EQ) and stressed-receiver sensitivity with a calibrated stressor for margin (Section 7.5), plus overload before the TIA saturates. Underneath those system numbers sit the photodiode/TIA pair: responsivity, bandwidth, and input-referred noise (Section 4.5 and chapter 4).

7.2.4 Link level

Only after Tx, channel, and Rx each look sane do you trust a full-link verdict: pre-FEC BER against the KP4 threshold (Section 3.12), post-FEC BER, FEC symbol-error histograms (Section 7.3.1), and a signed link-budget ledger from transmitter OMA to receiver sensitivity with penalties and remaining margin. That ledger is the document you argue from in DVT; the BER alone is not.

7.3 Measurement mapping

The metrics above are scattered across Tx, channel, Rx, and link level because that is how you debug them. Table 7.3 collects the same metrics into one reference: what is measured, the instrument, why it matters, and the failure signature that points back to it. Use the chapter subsections for the debug logic; use this table to look up an instrument fast.

7.3.1 Reading the measurement map

Use the table for lookup. Use the notes below when a metric is new, or when you need the decision the measurement unlocks.

Metric	Instrument	Why it matters	Failure signature
OMA / TDECQ	DCA + reference equalizer	Scores transmitter quality against an ideal source; governs PAM4 acceptance (Section 7.4)	TDECQ rises with no average-power change; points to bandwidth, RLM, or bias
Extinction ratio / RLM	DCA level histograms	Sets OMA at fixed average power (Section 4.4); poor RLM inflates TDECQ	Compressed inner eyes with passing average power
Wavelength / SMSR	OSA or wavemeter	Confirms grid placement and single-mode purity (Section 5.7)	Side modes rise with T or age; line walks off grid
RIN	PD + ESA or dedicated RIN analyzer	Can create a power-independent BER floor when signal-proportional intensity noise dominates (Section 4.3)	BER improves with power then flattens (a floor); not every floor is RIN
Insertion loss / ORL	Power meter + ORL meter	First ledger line; reflections can cause feedback noise, MPI, distortion, or floors (Section 7.2.2)	Burst or patterned errors with stable average power; ORL dependence
Receiver sensitivity	BERT + calibrated attenuator	Minimum OMA at target BER, the budget's bottom line (Sections 4.4 and 7.5)	Waterfall shifts uniformly right without flooring
Pre-FEC BER / FEC histogram	BERT + FEC counters	The single number every other metric feeds; histogram shape reveals mechanism (Section 3.12)	Clustered errors point to bursts; sparse errors point to Gaussian noise margin
CMIS state / DDM	Host or CMIS tool	Confirms management layer before blaming optics (Section 7.8)	Module never reaches ModuleReady; DDM disagrees with bench truth

Table 7.3. Measurement mapping: metric, instrument, rationale, and failure signature in one reference. Row explanations follow; chapter subsections give the full treatment of each metric.

OMA / TDECQ. TDECQ asks how much worse this transmitter is than an ideal source after a reference equalizer. Outer OMA is the optical swing the receiver actually uses. Together they answer whether the Tx path still has signal-quality margin. **Exit when** TDECQ and OMA meet the PMD/ATP at the named pattern and temperature. **Decision:** continue, retune bias/equalization, or reject the transmitter path. **Risk if skipped:** you chase receiver noise while the eye was already out of budget.

Extinction ratio / RLM. Extinction ratio and level separation mismatch (RLM) set how much OMA you get at fixed average power and how linear the PAM4 levels are. Poor RLM inflates TDECQ even when average power looks fine. **Exit when** ER/RLM meet the mask at the failing corner. **Decision:** retune modulator bias or driver, or accept a derate. **Risk if skipped:** average-power APC hides a collapsing outer eye.

Wavelength / SMSR. Wavelength placement and side-mode suppression ask whether the spectral ledger still closes: on-grid for filters or rings, single-mode under temperature and age. **Exit when** the line sits in the allowed window with SMSR in spec. **Decision:** retune lock/thermal control, derate temperature, or replace the laser. **Risk if skipped:** BER failures get blamed on RIN when the line walked onto a filter edge.

RIN. Relative intensity noise sets how far Q can rise with power. Measure with a quiet bias path and under controlled ORL so you separate intrinsic laser noise from feedback. **Exit when** RIN at the stated ORL meets the budget. **Decision:** fix reflections/supply, replace the laser, or stop raising launch into a floor. **Risk if skipped:** you keep adding photons to a non-power-limited impairment (Section A.8.9).

Insertion loss / ORL. Insertion loss is the first power-ledger line. ORL asks whether reflections are seeding RIN or bursts. **Exit when** loss and ORL are inside the plant assumptions used in the link budget. **Decision:** clean/replace connectors, add isolation, or reopen the budget. **Risk if skipped:** burst tickets look like random laser death.

Receiver sensitivity. Sensitivity is the minimum OMA for the named BER objective at a stated plane, pattern, and EQ, the budget's bottom line. A parallel waterfall shift with no floor usually means the Rx path or channel loss changed. **Exit when** sensitivity meets the ledger with stated pattern and stress. **Decision:** golden-swap ownership, derate reach, or redesign Rx. **Risk if skipped:** Tx FA on an Rx-limited link.

Pre-FEC BER / FEC histogram. Pre-FEC BER is the system score every other metric feeds. An FEC *symbol-error histogram* is not a DCA eye histogram. KP4 is Reed–Solomon RS(544,514) on 10-bit symbols (Section 3.12). The decoder or host FEC counters report how many of those symbols were wrong before correction, usually as errors per codeword or as errors versus time / codeword index. That distribution is the histogram.

Average BER alone does not classify the failure. Two links can share the same pre-FEC BER and fail for different reasons:

- **Sparse / Poisson-like:** most codewords have zero or one error; rare twos and threes. Fits steady noise (thermal, steady RIN, Gaussian-ish margin).
- **Bursty / clustered:** long quiet stretches, then clumps of many errors in a short window (several bad symbols or consecutive bad codewords). Fits time-local events: MPI or reflections, connector intermittents, unlock, supply or clock glitches, vibration.

KP4 can correct up to 15 symbol errors per codeword. Sparse errors usually stay correctable. A burst can dump many errors into one codeword, so you see uncorrectables or flapping even when the long-run BER still looks acceptable. Shape separates sparse Gaussian-like errors from clustered bursts; a bursty histogram alone is not proof of MPI. Confirm with ORL, timing, swaps, and plant disturbance (Section 11.2 and section A.8.9). **Exit when** BER and histogram support the claimed mechanism class. **Decision:** contain, clean, retune, or open FA. **Risk if skipped:** average BER hides a bursty escape that ATP never stressed.

> **What this usually means.** BER waterfall floor that more launch power does not fix
>
> *Usually:* RIN, MPI, crosstalk, receiver saturation, or another non-power-limited impairment (Section A.8.9)
>
> *Not:* simple insertion loss that more photons will buy out

CMIS state / DDM. Management state and digital diagnostics confirm the control and monitor path before you blame photons. Disagreement between digital diagnostics monitoring (DDM) and an external meter is itself a finding (monitor-PD or calibration). **Exit when** state progression and DDM match bench truth at the named plane. **Decision:** fix firmware/seat/power, or proceed to optics. **Risk if skipped:** weeks of optical FA on a module that never reached ready.

7.3.2 Why this map is ordered by isolation

Transmitter metrics come before channel and receiver metrics because a bad Tx eye contaminates every downstream number. Channel loss and ORL come before Rx sensitivity arguments for the same reason. Link-level BER is last: it is the verdict, not the first bisect. If you start at BER alone, you still need this map to choose the next instrument.

7.4 Transmitter and dispersion eye closure quaternary (TDECQ)

TDECQ (transmitter and dispersion eye closure quaternary) deserves a closer look because it is the metric that governs many PAM4 transmitter acceptance methods. It answers a specific question: *how much worse is this transmitter than an ideal one, after a realistic receiver has done what it can to clean up the signal?*

The following is a representative IEEE-style procedure. Use the exact clause and PMD under qualification for the reference receiver, equalizer constraints, histogram locations, and target error ratio.

7.4.1 How it is measured

1. **Capture.** The optical waveform is acquired on a sampling oscilloscope (a DCA) through a standardized reference receiver (often a fourth-order Bessel–Thomson filter near half the baud rate under the named PMD) so every lab measures the same bandwidth.

2. **Equalize.** A defined *reference equalizer*, a *feed-forward equalizer* (FFE) with a small, bounded number of taps (commonly up to five in many PMDs), is applied. This models the modest equalization a real receiver would perform, so the transmitter is not penalized for *ISI* the system can remove anyway.

3. **Histogram.** Narrow vertical histogram windows are placed inside the symbol at the positions required by the clause. The noise distribution is evaluated at the three PAM4 decision thresholds.

4. **Compute.** The algorithm finds the RMS Gaussian noise σ that, added to the equalized signal, would just reach the clause's target symbol error ratio (often near 4.8×10^{-4} for KP4-class budgets). TDECQ is the ratio, in dB, of the noise an *ideal* transmitter could tolerate to the noise *this* transmitter can tolerate:

$$\text{TDECQ} = 10 \log_{10}\left(\frac{\sigma_{\text{ideal}}}{\sigma_{\text{measured}}}\right).$$

A worse transmitter tolerates less added noise before failing, so σ_{measured} shrinks and TDECQ rises. Lower is better; the numeric cap is PMD-specific.

7.4.2 Related quantities and failure signatures

SECQ Stressed eye closure quaternary: receiver-side stressed test, analog of TDECQ. Applies calibrated optical stress before BER test.

SECQ the stressed-eye counterpart used on the receiver side under a named PMD, adding a calibrated stressor to test margin rather than transmitter quality alone. Distinguish SECQ from a general stressed-receiver sensitivity test. See Section 7.5.

RLM (relative level mismatch) measures how evenly the four PAM4 levels are spaced; poor RLM (uneven levels) inflates TDECQ.

Because TDECQ folds several impairments into one number, the way it fails is diagnostic: uneven levels point to modulator or driver linearity (RLM); residual eye closure the equalizer cannot fix points to excess ISI or limited bandwidth; a noise-limited result points to low OMA, RIN, or reflections. For external MZMs (TFLN or silicon), also check EO S_{21} bandwidth, V_π and bias quadrature drift with temperature, and RF return loss on the driver-to-modulator path (Section 3.14.3). This is why *LPO*, which removes the module's own DSP, raises the stakes on transmitter quality: there is less downstream equalization to hide behind, so TDECQ-class metrics become even more central.

7.5 SECQ and stressed-receiver testing

SECQ (stressed eye closure quaternary) mirrors TDECQ on the *receiver* for a named PMD and clause: instead of scoring transmitter quality with a reference equalizer, the test applies a calibrated optical stressor (attenuation, ISI template, optional RIN) and asks how much margin remains before the receiver hits that clause's target pre-FEC metric.

Stressed-receiver sensitivity and overload tests (Section 4.4) use the same philosophy but are not automatically the same procedure as SECQ. Bracket the operating OMA range with impairments the link will see in the field, and name the PMD, FEC architecture, error model, metric, and test duration. For LPO, where the module DSP is gone, stressed Rx margin on the host-side receiver (Section 3.6 and section 10.5.1) is as important as TDECQ on the transmitter.

7.6 Instruments

A failing PAM4 link rarely announces which block is wrong. Ask the investigation question first, then pick the gear that can separate the leading hypotheses. Loopback topology tells you which side of the optical connector owns the fault (Section 7.2.2).

Where is the power? Power meter for average power; DCA for OMA. Walk planes before you change bias.

Did the eye or levels move? (digital communication analyzer) for PAM4 eyes, TDECQ, OMA, RLM (Section 7.4). Needs a reference receiver filter matched to the PHY under test.

Did BER shift or floor? for pre- and post-FEC BER and FEC symbol histograms (Section 3.12); / stressor for calibrated attenuation and optional ISI on sensitivity sweeps.

Is the spectrum or grid wrong? / wavemeter for wavelength, SMSR, side modes, and linewidth where supported (Chapter 5).

Is intensity noise the floor? PD + ESA or a dedicated RIN analyzer under a defined ORL (Section 4.3.1).

Does temperature own it? Thermal chamber + TEC controller for corners; essential for rings and laser grids (Section 3.14.3 and chapter 6).

Use electrical loopback (host SerDes), optical loopback (Tx→Rx on module), and golden-host/golden-module interop to bisect faults. If the fault follows the module under golden-host swap, stop blaming the SerDes; if it stays with the host, stop opening laser FA.

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

BERT Bit-error-ratio tester: instrument or ASIC function that generates PRBS and counts errors for pre-/post-FEC BER.

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

7.7 Building a link budget

A link budget is a signed dB (or power) ledger from transmitter to receiver. For IM/DD short reach, start from outer OMA at the Tx faceplate and subtract every loss and penalty until you compare against receiver sensitivity (with target BER and KP4 pre-FEC threshold, Sections 3.12 and 4.4).

```
Transmitter output (OMA)
  |
Coupling loss
  |
Connector loss
  |
Fiber / waveguide loss
  |
Penalties (Method A or B; see below)
  |
Receiver input
  |
Sensitivity requirement
  |
Remaining margin
```

Keep power budget, signal-quality penalties, timing, thermal, and control authority as separate ledgers when the impairment is not a pure optical-power number (Sections D.10 and 5.19).

Walkthrough before Method A or B. Name the reference plane (usually Tx faceplate OMA). Subtract each loss once: coupling, connectors, fiber or waveguide, and any plant allocation you have evidence for. Compare the remaining power to receiver sensitivity at the named BER / FEC objective. Choose Method A or Method B for transmitter quality; do not tax TDECQ twice by subtracting a compliance OMA/TDECQ limit and an independent TDECQ penalty.

Design allocation versus validation measurement. Distinguish margin allocation in design from margin verification in test. During design, engineers allocate transmitter output, receiver sensitivity, insertion loss, temperature degradation, aging, jitter, and manufacturing variation. During customer or system qualification, the integrator often measures net remaining margin across the operating envelope.

```
Design: allocate line items
  |
Build / integrate
  |
Test: measure net at named plane
  |
Room–T sensitivity margin
  |
Temperature / stress sweep
  |
Observed margin loss
```

```
   |
Remaining headroom
   |
Deployment decision
```

TDECQ accounting: two methods.

Method A: Composite compliance Use the PMD's specified OMA/TDECQ relationship. Do not subtract TDECQ again as an independent link-budget penalty.

Method B: Engineering decomposition Use a separate measured transmitter-quality penalty only when the accounting method is defined and does not duplicate the compliance limit.

Illustrative ledger (single-mode DR-class sketch). Start from Tx OMA on the DCA (or from average power and ER) at a named plane. Subtract connector/coupling loss using the specified maximum for the connector class, interface count, and cleanliness (an illustrative poor or multi-interface allocation can land near 1–3 dB per mated pair; that is not a universal normal loss). Subtract fiber loss ($\sim$0.3–0.4 dB/km at 1310 nm; often negligible at 500 m) and MUX/de-MUX if WDM (2–5 dB per stage, Section 6.3). Apply penalties with Method A or Method B above; add dispersion (Section 3.11) and reflection/MPI terms (Sections 4.3.1 and 7.2.2) only when not already absorbed. Compare the remainder to stressed sensitivity at the *named* PMD's pre-FEC objective (for a KP4-class optical PMD under its random-error model, often near 2.4×10^{-4}; state FEC, metric, and test duration). Keep production margin appropriate to fleet corners. Numbers here are an illustrative DR-class sketch, not universal limits. Electrical budgets parallel this for the host-to-module path: COM and pre-FEC BER (Section 10.5.2 and section 3.6). LPO requires *both* ledgers to close without module DSP help.

7.8 Module management: CMIS

Applicability header (fill for each program).

noitemsep,topsep=0pt Applicable CMIS revision:

noitemsep,topsep=0pt Applicable form factors:

noitemsep,topsep=0pt Applicable module states:

noitemsep,topsep=0pt Applicable data-path states:

noitemsep,topsep=0pt Optional diagnostics used:

noitemsep,topsep=0pt Vendor extensions permitted:

Do not imply that every CPO engine, ELSFP, pluggable, and future 448G implementation exposes identical CMIS behavior. Distinguish a standardized CMIS field, optional VDM, a vendor-specific diagnostic, and an inferred health metric.

7.8.1 What CMIS is, and why an optical engineer cares

CMIS (Common Management Interface Specification) is the vendor-neutral management layer between a host (switch ASIC, NIC, or test fixture) and a pluggable or on-board optical module that implements it. The host talks to the module over a two-wire bus (TWI, I2C-like) through a paged register map: identity, power mode, alarms, per-lane monitors, and (in later revisions) link-training and host signal-integrity tuning extensions [86]. Common form factors include QSFP-DD, OSFP, COBO, ELSFP, and some CPO engines that expose a CMIS contract.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

You touch CMIS on every bring-up and every field triage. It is how the host learns what module is seated, when lasers may turn on, what Tx/Rx power and temperature look like, and whether a link failed at the management layer or the optical layer. A module that passes BER on a bench with lasers forced on but cannot reach ModuleReady on a production host will fail in the fleet (Section 7.9).

7.8.2 The module state machine

CMIS defines a module state machine the host drives. After presence detect and power application, the module stays in low power until the host releases `LPModeL` (or the CMIS 5.x `LowPwr` equivalent). The host reads identifier pages, clears sticky interrupts, and steps the module toward ModuleReady. Only then should Tx lanes or ELS lasers enable. ELSFP modules that emit before ModuleReady are a reject: the host did not authorize light (Section 5.14).

Data paths have their own state machines in CMIS 5.x (data path states, and network path states for media-side links). For bring-up, map the sequence in Section 7.9 onto these transitions: presence and Vcc, CMIS init and ModuleReady, enable light, optical path check, electrical lock, traffic, snapshot. Skipping step 2 and jumping to BER is how interop failures hide until production.

7.8.3 The memory map: pages, monitors, control

The lower memory map holds module identity, status, interrupt flags, and alarm thresholds. Upper pages hold application descriptors, lane controls, tunable-laser support, versatile diagnostics (VDM), and command-data-block (CDB) firmware messaging [86]. Hosts select an application (lane count, host interface, media type) before bringing up traffic.

DDM (digital diagnostic monitoring) is the telemetry layer you read at scale:

DDM Digital diagnostic monitoring: per-lane telemetry in CMIS (Tx/Rx power, bias, temperature, LOS/LOL flags).

per-lane Tx and Rx optical power, laser bias current when exposed, module temperature, supply voltage, LOS/LOL flags, and alarm/warning bits. On WDM parts you also get wavelength or channel ID. This is exactly what Section 7.12 reads before anyone reaches for a DCA. On bring-up, dump the register map you will use in the field and treat that dump as the golden reference for later RMA comparisons.

7.8.4 CMIS as a validation deliverable

CMIS correctness is part of production readiness, not a firmware afterthought. ATP should prove the state machine reaches ModuleReady across voltage and thermal corners; DDM monitors track bench truth (CMIS Tx power versus DCA, module temperature versus case *T*); alarms fire at the right thresholds; and firmware revision is ECO-controlled like laser die revision (Section 9.2). Multi-source interop failures are often CMIS, media-type, or firmware mismatches, not marginal TDECQ (Section 7.9). At fleet scale the register map is the only eyes you have on a module in the rack. If CMIS is wrong, triage starts blind.

7.9 Module and system bring-up

Characterization proves a sample can meet metrics on a quiet bench. Bring-up proves a module (then a system) can be powered, managed, and linked the way production and the fleet will actually run it. Lab-to-production programs fail in the gap between those two if you only ever test golden hosts, clean fiber, and room-temperature faceplates.

Module bring-up sequence. Run this order on every new module (pluggable, ELSFP, or CPO engine with CMIS). Do not skip ahead to BER: a link that "works" with lasers forced on and CMIS ignored will fail the first host that enforces the state machine (Section 5.14).

1. **Presence and power.** Detect module (`ModPrsL` or equivalent). Apply rails in the host power sequence. Confirm Vcc and module temperature in CMIS. Stay in low power (`LPModeL` asserted or ModuleLowPwr) until management is sane.
2. **CMIS init.** Read identifier, vendor, firmware rev, supported media. Clear sticky interrupts. Confirm the state machine can reach ModuleReady (or the pluggable equivalent) under host command. Dump the register map you will use in the field; that dump is your bring-up golden reference.
3. **Enable light.** Exit low power; enable Tx lanes / ELS lasers only after ModuleReady. Confirm Tx optical power and laser bias (if exposed) against the power class. Lasers that come up before the host asks are a reject for ELSFP (Section 5.14).

4. **Optical path.** Mate fiber (clean first). Check Rx power and LOS. Optical loopback first if the host path is unproven.

5. **Electrical lock.** Bring host SerDes / module CDR. Confirm LOL clear, equalizer taps not pegged (Section 3.6). For LPO, this is the host eye and COM path (Sections 3.14.3 and 10.5.2).

6. **Traffic.** PRBS or live FEC traffic. Pre-FEC BER vs. KP4 threshold (Section 3.12); glance at FEC symbol-error histogram shape.

7. **Quality snapshot.** On a Tx-capable path: OMA/RLM/TDECQ or module diagnostics that proxy them (Section 7.4). Record CMIS + BER + case T together so later triage has a baseline (Section 7.12).

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

PRBS Pseudo-random binary sequence: deterministic test pattern that exercises all bit transitions for BER and eye measurements.

The numbered sequence above is primary. Table 7.4 is a wall-chart quick reference only; do not maintain a second conflicting procedure.

Step	Action	Pass signal	Fail → first look
1	Presence / Vcc / temp	CMIS alive, rails in range	cable, seat, PSU
2	CMIS state machine	ModuleReady (or equiv.)	firmware, TWI, LPMode
3	Enable Tx / ELS	Tx power in class; lasers on only when commanded	bias driver, enable pin, APC
4	Fiber / Rx power	Rx power up; LOS clear	dirty MT, polarity, break
5	CDR / SerDes lock	LOL clear; taps not saturated	host SI, LPO COM, retimer
6	Pre-FEC BER	below KP4 target with margin	Tx quality, ORL, Rx sensitivity
7	Snapshot	CMIS dump + BER + T logged	(needed for RMA later)

Table 7.4. Module bring-up checklist. LOS = loss of signal; LOL = loss of lock. Limits come from the ATP and PMD, not from this table.

Production-representative corners. Bench corners (T, V) are necessary and not sufficient. Chassis thermal, host rails, and ORL belong before Design Validation Test (DVT) exit on a representative unit. The full set in Table 7.5 belongs before Production Validation Test (PVT) exit (Table 9.2).

Corner	What to run	Why it catches	Points to
Chassis thermal	Module in target rack/sled at airflow and power load; not only a quiet chamber on a bench fixture	Faceplate T and TEC load differ from chamber setpoints	derate, TEC, ring unlock
Host rails live	Bias / CMIS powered from host supplies with SerDes traffic on	Switching noise into laser bias looks like RIN (Section 5.8)	PSRR, ground, APC
Dirty fiber / ORL	Controlled contamination or ORL stress on MT/FAU; clean vs dirty BER	Field installs are not lab-clean; ORL raises RIN and bursts	connector, isolator, feedback
Cable plant	Production fiber length, MPO count, and bend radius	Extra loss and reflections eat margin the ledger assumed	link budget (Section 7.7)
ELS hot-swap	Pull/replace ELSFP under traffic (or under controlled traffic stop per CMIS)	Service action the architecture promised (Section 5.14)	state machine, mate cycles
Neighbor load	Adjacent modules/lanes at full traffic and max case T	Crosstalk, shared supply droop, thermal crosstalk on rings	WDM lock, SI, PSU
LPO / linear path	Host COM and pre-FEC BER without module DSP crutch	LPO fails here first (Sections 3.14.2, 3.14.3 and 10.5.2)	host FIR, module linearity
Voltage corners	Host Vcc min/max with traffic	Brown-out and CMIS glitches	power design, ATP

Table 7.5. Production-representative corners. A quiet BERT at 25 °C with pristine fiber is characterization, not production readiness.

7.9.1 Reading the production-corner map

Quiet T/V characterization maps the part. Table 7.5 is the checklist. Read it in four groups before you invent new corners.

Thermal and loading. Chassis thermal and neighbor load ask whether sled airflow, faceplate gradient, shared heat, and tray traffic leave TEC, lock, and BER with margin. Chamber case-T alone does not answer that.

Electrical and host. Host rails live, voltage corners, and LPO / linear path ask whether SerDes traffic, supply noise, brown-out, and host FIR/COM still close the link when the module cannot hide behind a retimer.

Optical plant. Dirty fiber / ORL and cable plant ask whether production MPO count, bends, and field cleanliness match the signed loss and reflection assumptions.

Service and interaction. ELS hot-swap asks whether the replaceability story survives a live maintenance window: CMIS state, mate cycles, and recovery under the traffic policy you claim.

Chassis thermal, host rails, and ORL are the minimum before DVT exit on a representative unit. The full set belongs before PVT exit (Table 9.2). Later fleet monitoring must not invent coverage these corners never ran.

System bring-up.

Tradeoff. Best laboratory performance vs production yield

Improves: Hero samples that win bench demos

Worsens: Tighter tolerances, harder calibration, and escapes in volume

When acceptable: When the manufacturable design still meets the system requirement with guardband

Experienced decision: Optimize the system and the yield story, not the best component on a quiet bench.

A module that passes on a golden host can still fail in a real chassis:

- **Host path:** run the same sequence on the target NIC/switch ASIC SerDes, not only the lab BERT. LPO and half-retimed modules expose host FIR/CTLE mistakes that a retimed module hid (Section 10.5.1 and section 10.3).
- **Multi-lane / multi-module:** bring all lanes on a port, then neighbors in the same cage or tray. Watch thermal rise, supply droop, and CMIS temp alarms when the tray is loaded.
- **Golden swap:** known-good module in the suspect host slot, then suspect module in a known-good slot. That single swap splits host vs. module before you open FA (Section 7.12).
- **Interop:** at least one other vendor host or module if the program claims multi-source. Interop failures are usually CMIS, media type, or electrical eye, not laser physics.
- **ELS / CPO:** external laser modules add a second bring-up: ELSFP state machine and optical mate to the engine, then engine bring-up with light present (Sections 5.14 and 10.10). A dark engine with a healthy ELS is an optical connector or FAU problem until proven otherwise.

Exit criteria before "bring-up done." Call *bench bring-up* done when CMIS state machine and enable sequence are correct, the unit emits and receives light in class, CDR locks, pre-FEC BER is usable on a trusted setup at a named plane, and a CMIS+BER+T snapshot is filed. Call *system integration bring-up* done when the same sequence closes on the target host, golden-swap has split host vs. module issues, and multi-lane / neighbor load has not opened a new basic failure mode. Do *not* require loaded chassis-thermal / host-rail / ORL margin closure to declare bring-up done; that is Stage 3 margin and interop evidence (Section 7.1.6 and table 7.5). Everything after bring-up is characterization depth, margin/interop, supplier gates (Section 9.2), or fleet triage (Section 7.12).

Key idea. Bring-up is a sequence (presence → CMIS → light → lock → BER → snapshot), then a system proof on the real host. Production-representative corners prove remaining headroom; they do not redefine bench bring-up. A quiet bench pass is not DVT.

7.10 The debug mindset

Debug at this level is data-driven, not opinion-driven. The method is disciplined bisection: change one domain at a time, and let the measurement tell you whether the transmitter, the channel, or the receiver moved.

1. Isolate transmitter versus channel versus receiver, using loopbacks.
2. Sweep temperature and voltage to expose corner-dependent failures.
3. Correlate failures to DSP equalizer tap values (Section 3.6) and FEC symbol-error statistics (Section 3.12); these tell you *how* the link fails.

The third step is where modern PAM4 links differ from older eye-mask work. Tap saturation and FEC histograms often reveal the failure mode before a single waveform screenshot does. Treat those as primary evidence, not as afterthoughts logged once BER already fails.

[1]

[1] Worked example. "The link fails only at high temperature." Candidate causes: laser wavelength drift off the grid or off a ring resonance; TEC saturation; EAM bias shift; elevated receiver thermal noise; LIV rollover (Sections 5.7 and 5.13). You bisect with measurements (spectrum for wavelength, LIV for the laser, receiver sensitivity at temperature) rather than guessing.

7.11 The debugging fork in validation

Apply the debugging fork (Section 4.8) before sweeping parameters or changing firmware: check the power meter or CMIS Rx power monitor first. If power moved, the fault is in the optical path (laser, coupling, connector, fiber, MUX); if power held but BER or TDECQ worsened, it is signal quality (bandwidth, noise, jitter, bias, equalization, reflection). This one check prevents the most common validation mistake: retuning an equalizer or laser bias when the real cause is a dirty connector. Then check which margin ledger moved (Section 5.19) before descending to component physics.

Why experienced engineers separate power from quality first?

Because average optical power is cheap to measure and rules out gross attenuation, but it says almost nothing about timing, noise, distortion, spectral alignment, or adaptation.

What this usually means. Stable average power with rising BER

Usually: timing, adaptation, noise, spectral alignment, or intermittent control

Not: gross attenuation or a simple dirty connector as the whole story

Engineering heuristic. Never spend an hour on a DCA or spectrum sweep when a five-minute golden swap or attenuator step can eliminate half the tree.

```
Observation
  |
Possible ledgers (power / noise / timing / spectrum / control)
  |
Measurements (power first)
  |
Hypotheses removed
  |
Decision
  |
Recurrence control
```

Before debugging

Scope · time behavior · population · power or quality · highest-value measurement · decision · recurrence control (Section D.17).

Engineering heuristic. A passing BER on a golden host is not production readiness. Interop, margin, and manufacturing control still have their own questions.

7.12 Fleet and field triage

Lab debug asks: *what is broken on this unit?* Fleet triage asks: *which bucket does this failure belong in, and who owns the fix?* Optical programs at fleet scale own that split across performance, reliability, and manufacturability. Wrong bucket wastes weeks (sending a contaminated connector to laser FA, or rewriting a SerDes FIR when the laser is rolling over).

Engineering heuristic. Contain the population and clear the measurement system before you open supplier FA. A wrong ticket burns calendar time you cannot get back.

What this usually means. Temperature-only failures that recover cool

Usually: thermal margin, wavelength or lock drift, bias tables, receiver noise rise, or mechanics that move with case temperature

Not: a permanent wear-out mechanism already proven by ship LIV alone

Tradeoff. More telemetry vs operational complexity

Improves: Faster fleet debug, better cohort plots, earlier prediction

Worsens: Firmware cost, storage, and interpretation burden

When acceptable: When each new field answers a named decision

Experienced decision: Every telemetry field needs an owner and a decision it enables. Otherwise it is noise.

Three buckets. Classify every field issue before deep root-cause work:

Performance the design or operating point does not close the budget under the conditions seen in the fleet. Examples: TDECQ/RLM marginal at case temperature, host COM tight on LPO, ring unlock under thermal crosstalk, ORL-driven RIN that the architecture assumed away. Fix is usually retune, derate, firmware, or a design/spec change (Section 7.4 and sections 3.14.3 and 10.5.2).

Reliability the unit met spec at ship and later degraded. Examples: LIV threshold rise, SMSR collapse, EAM bias creep, COD, TEC wear, epoxy creep on fiber attach. Fix is Arrhenius-backed life projection, burn-in/screen, derating, or field-replaceable lasers (Sections 5.13, 5.14, 8.3 and 8.5).

Manufacturability a subpopulation fails early or never met the ATP; the issue tracks lot, date code, supplier site, or assembly step. Examples: FAU misalign yield cliff, solder void on a driver die attach, incoming DPPM spike, CMIS register map mismatch on one firmware rev. Fix is SPC, ATP tighten, first-article, DPA, and 8D/CAPA with the supplier (Sections 8.7 and 9.2).

A single symptom can sit in more than one bucket until you bisect. The tree below forces the split with telemetry first, then a short bench confirm, then an RMA label. Chapter 11 expands the same method into symptom-led bench and fleet procedures.

Telemetry you actually read. At scale you rarely start with a DCA. Start with what the host and module already report:

- *CMIS* monitors and alarms: module temperature, supply rails, Tx/Rx optical power, laser bias (when exposed), wavelength or channel ID on WDM parts, LOS/LOL flags, and interrupt history (`IntL` on ELSFP; Section 5.14).
- Host link state: CDR lock, pre-FEC BER, FEC symbol-error histogram shape (Section 3.12), equalizer tap saturation (Section 3.6).
- Fleet context: rack position, case temperature, time since install, date code / lot, neighbor-link correlation (one bad fiber vs whole tray).

Decision tree (symptom → bucket). Table 7.6 is the working map. Read left to right: observe, check telemetry, pick a provisional bucket, then run the named confirm measurement before you open an RMA or change a design rule.

```
Fleet symptom
  |
Scope analysis (how large?)
  |
Technical isolation
  |
Correlation analysis (which cohort?)
  |
Bucket: performance / reliability / manufacturability
  |
Contain / FA / ATP / telemetry
  |
Fleet monitoring
```

Scope sets severity and priors. Correlation after isolation unlocks contain, pause, replace, or supplier escalate (Section D.5).

7.12.1 Reading the fleet triage map

Each row is a provisional route, not a confirmed root cause. Capture telemetry first. Confirm with the smallest measurement that can falsify the bucket. Then assign an owner.

Link never comes up (fresh install). Ask whether the part is seated, powered, and managed before you open laser FA. CMIS presence, supply rails, Tx power flatline, and loss-of-signal (LOS) split install from product. Confirm with visual fiber/connector checks, a golden module swap, and a frozen CMIS dump. **Decision:** ops fix, or supplier ATP if the fail tracks a lot. **Risk if skipped:** manufacturing escapes get filed as design defects.

Intermittent LOS / burst errors. Ask whether the plant is reflecting or contaminating. Rx power dropouts and bursty FEC histograms point at connectors or ORL before intrinsic RIN. Confirm with inspect/clean, ORL meter, and RIN versus ORL. **Decision:** cleaning discipline or packaging FA if RMAs repeat. **Risk if skipped:** burst tickets become endless laser replacements.

Pre-FEC BER high, power OK. Power held, so leave the power ledger. Tap saturation, logged TDECQ/RLM, and case temperature point at signal quality or host SI. Confirm on a DCA and with host channel operating margin (COM) thinking on linear paths. **Decision:** host SI, module Tx design, or retune. **Risk if skipped:** you reseat fiber forever on a quality-path fail.

Symptom	First telemetry check	Bucket	Confirm on bench / FA	Typical fix owner
Link never comes up (fresh install)	CMIS presence, Vcc, Tx power flat-line, LOS	Mfg or install	Visual fiber/connector; golden module swap; CMIS dump	Ops install; supplier ATP if lot-correlated
Intermittent LOS / burst errors	Rx power dropouts; FEC bursts; ORL events	Perf (ORL) or mfg (contam.)	Clean/inspect MT; ORL meter; RIN vs ORL (Section 5.8 and section 4.3.1)	Ops cleaning; packaging if repeat RMA
Pre-FEC BER high, power OK	Tap saturation; RLM/TDECQ if logged; case T	Perf	DCA TDECQ/RLM; host COM; LPO vs retimed path (Section 7.4 and section 10.5.2)	Host SI / module Tx design
BER rises only at high case T	Module temp alarm; Tx power drop; λ walk	Perf or reliability	LIV at T; OSA grid; TEC current; EAM bias (Section 5.13)	Derate / TEC / laser supplier
Slow BER creep over weeks/months	Bias current up for same Tx power; SMSR if monitored	Reliability	LIV/SMSR vs ship ATP; Arrhenius lot history	Laser wear-out; ELS replace
Sudden hard fail, was healthy	Last good CMIS snapshot; neighbor links OK	Reliability (COD) or mfg (ESD)	Dark LIV; DPA on facet/solder; date-code cluster?	FA + supplier 8D
One date code / site fails early	Lot Pareto; burn-in escape rate	Mfg	Incoming SPC vs ATP; FA on sample of lot	Supplier CAPA; hold shipment
WDM / ring unlock, power OK	Channel ID; thermal of neighbors; lock-loop status	Perf	Resonance tune; crosstalk; CW-WDM line power (Sections 5.16, 6.5 and 6.7)	Lock firmware / thermal design
ELSFP swap restores link	Old module CMIS vs new; connector cycles	Reliability or mfg (connector)	Inspect MT; mating-cycle count; laser LIV in returned module (Section 5.14)	Laser vs connector split in FA

Table 7.6. Fleet triage map: symptom to provisional bucket to confirm measurement. Perf = performance (design/operating point); reliability = time-dependent wear; mfg = lot/process/install excursion. Row notes follow.

BER rises only at high case T. Ask whether the operating point or the device changed with heat. Telemetry for temp alarms, Tx sag, and wavelength walk comes first. Confirm with LIV at temperature, OSA, TEC/heater codes, and modulator bias. Cool-down recovery raises $P(\text{operating point})$; permanent shift raises $P(\text{aging})$. **Decision:** derate, thermal design, or supplier life action. **Risk if skipped:** ambient-only debug misses the spent control ledger.

Slow BER creep over weeks/months. Ask whether wear-out is spending the noise or power ledger. Bias current rising at fixed Tx power raises $P(\text{aging})$. Confirm against ship ATP baselines (LIV, SMSR, power, spectrum) and lot history; recovery after recalibration raises $P(\text{control/cal})$ instead. **Decision:** replace, derate, or update burn-in. **Risk if skipped:** FIT models stay optimistic until the fleet teaches you.

Sudden hard fail, was healthy. Ask whether the event is catastrophic optical damage, ESD, or a shared infrastructure hit. Neighbor links and the last good CMIS snapshot matter. Confirm with dark LIV and physical FA; check date-code clusters. **Decision:** FA plus supplier 8D, or infrastructure fix. **Risk if skipped:** one COD event becomes a false process CAPA.

One date code / site fails early. Ask whether this is a manufacturing subpopulation. Lot Pareto and burn-in escape rate are the first cuts. Confirm incoming SPC versus ATP and FA on a sample. **Decision:** quarantine, CAPA, hold shipment. **Risk if skipped:** a bad lot keeps deploying while FA studies one unit.

WDM / ring unlock, power OK. Ask whether the spectral or lock ledger was spent while average power looked fine. Channel ID, neighbor thermal, and lock-loop status are the telemetry. Confirm resonance tune, crosstalk, and line power. **Decision:** lock firmware or thermal design. **Risk if skipped:** unlocks get mislabeled as random BER.

ELSFP swap restores link. Ask whether the external laser, the connector, or the engine owned the fail. Compare old versus new CMIS and connector cycles. Confirm MT inspect and LIV on the returned module. **Decision:** split RMA codes for laser versus connector. **Risk if skipped:** FIT burns down the wrong wear-out mode (Section 5.14).

7.12.2 Why triage order matters

Scope before mechanism. Telemetry before destructive FA. Bucket before owner. Confirm before CAPA. Closing the loop into ATP is part of the incident, not optional paperwork. Reversing that order produces NFF piles and merged RMA codes that make life models dishonest.

How to walk an incident (order of operations).

1. **Stabilize and capture.** Freeze CMIS dump, host BER/FEC counters, rack T, and install age before anyone reseats the module. Reseating destroys connector evidence.
2. **Localize.** One link vs tray vs rack. Tray-wide points at power, cooling, or a shared ELS. Single-link points at that module, fiber, or host lane.
3. **Classify** with Table 7.6. Write the bucket on the ticket before FA starts.

4. **Confirm** with the smallest measurement that can falsify the bucket (golden swap, clean/inspect, LIV, TDECQ, ORL). Do not skip to DPA.
5. **Act.**
   - Performance: change operating policy (derate, FIR, lock loop) or open a design/spec defect.
   - Reliability: replace (ELSFP hot-swap when available), update FIT burndown, tighten burn-in or derate (Section 5.13).
   - Manufacturability: quarantine lot, incoming hold, supplier 8D with DPA photos and ATP deltas (Section 9.2).
6. **Close the loop.** Feed the signature back into ATP and CMIS alarm thresholds so the next incident trips earlier.

Worked paths (three common tickets). "*High temp only.*" CMIS shows module near thermal limit and Tx power sagging. Bucket starts as performance (thermal design / derate). A permanent LIV or spectrum shift at temperature that matches an aged lot raises $P(\text{aging})$ and justifies moving the ticket toward reliability; cool-down recovery without baseline shift keeps it in performance. Measure OSA wavelength before blaming the laser: a ring unlock is still performance (Section 3.14.3 and chapter 6).

"*Random burst errors, average power fine.*" Check FEC histogram for clustered errors and CMIS for Rx power dropouts. Clean and measure ORL. If RIN rises with ORL, treat feedback/ORL as the leading performance hypothesis until confirmed. If ORL is fine and bursts track a date code, treat intermittent fiber attach as the leading manufacturing hypothesis. If bursts grow over months at fixed ORL, suspect laser or driver aging (Sections 5.8 and 5.13).

"*ELSFP replace fixed it; returned module looks alive on the bench.*" Alive LIV with high ORL sensitivity or a dirty MT face supports connector/ORL over laser wear-out; confirm with IL/ORL and recurrence. Dead or kinked LIV supports a reliability path. Split those RMA codes or FIT math blames the wrong mode (Sections 5.14 and 8.7).

RMA labels that keep FIT honest. RMA codes should be distinct, not a single "optics fail":

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

- laser wear-out (LIV/SMSR/EAM baselines support aging; not proof alone);
- COD / sudden dark;
- connector / contamination / ORL;
- fiber attach / FAU;
- driver / bias electronics;

- host / SerDes / LPO eye (not module);
- NFF (no-fault-found; track these; high NFF means bad triage).

NFF No fault found: RMA unit passes all tests on return. High NFF rate points at triage or intermittent connector faults.

NFF rate and lot Pareto are as important as FIT. A rising NFF with clean LIV points at install practice or intermittent connectors, not Arrhenius.

7.13 Engineering lens

7.13.1 How it works

Validation is a chain of evidence, not a single pass: a number means nothing without its reference plane, its corner, and its method. The chapter's ladder, instruments, and triage tree are that evidence chain from bench to fleet.

7.13.2 How it is measured

Use the least complex instrument that can falsify the current hypothesis. Table 7.3 maps every key metric to its instrument, rationale, and failure signature in one lookup; the bring-up sequence (Section 7.9) orders those instruments into a workflow.

7.13.3 How it fails

Validation fails when the setup, sample, or acceptance rule does not match the product. Common misses are a stale calibration, the wrong reference plane, a golden host that hides interop risk, pristine fiber that hides ORL sensitivity, short BER dwell, one lane tested without neighbors, and chamber temperature used as a substitute for measured case temperature. These are test escapes even when the device physics is sound.

Failure mode: Low optical power

Symptoms. A lane is dark or below its launch-power limit.
Likely causes. A laser or enable fault, coupling loss, connector contamination, fiber polarity, calibration error, or a power-meter setup mistake.
Measurements. Known source and meter, inspection scope, CMIS state and bias, power at successive planes, and a golden fiber or module.
Mitigations. Correct the setup first, then repair the failing source, attach, connector, or control path. Add the signature at the earliest production test that can catch it.

7.13.4 How it is debugged

Preserve the failing state and record software, firmware, calibration, fixture, cables, temperature, and supply. Verify the meter with a known source. Walk from power to spectrum to waveform to BER, moving one reference plane at a time. Use a golden swap to split host, module, and fiber. Only then stress temperature, voltage, ORL, and neighbors. Every corrective action needs a repeated failing test, a repeated passing test, and a guard against recurrence in ATP or telemetry.

Debug story

Observed. A new module lot showed low optical power on one station.
Investigation. The same units passed on a second station. A known source exposed an offset in the first power-meter path.
Finding. The lot was good, and the station was reading low.
Root cause. A reference jumper had been replaced without updating the path-loss calibration.
Resolution. The station was recalibrated, jumper identity was placed under change control, and a start-of- shift source check was added.

7.14 Interview takeaway

Key idea. Validation is a chain of evidence. Start with calibrated power and management state, move through spectrum and waveform, then trust BER only after the blocks and reference planes are known. Run the target host, chassis, fiber, and neighbor corners before calling the product ready.

Junior mistake: call a golden-host BER pass "production ready," or open supplier FA before clearing the tester (Table 7.2 and chapters D and 9).

7.14.1 Interview Q&A: Optical Validation

Practice speaking these answers aloud. Prefer first-person reasoning over definitions. Detail lives earlier in this chapter (Section 7.1 and table 7.2). Score your answer using the chapter-end spoken-answer rubric (Section A.12.1).

Question 1. What is the purpose of optical validation? *Tests:* evidence-to-decision.

Spoken answer. "Validation builds enough evidence to decide whether the product is suitable for its intended system use. I treat it as a sequence of release decisions, not a catalog of tests."

Pressure follow-up. "What decision should validation enable?"
Answer pivot. "Continue, redesign, derate, qualify, open volume, or hold, with named remaining risk."

Trap: treating validation as "run BER and temperature."

Question 2. What is the difference between characterization, verification, validation, qualification, and production test? *Tests:* terminology discipline.

Spoken answer. "Characterization maps behavior and distributions. Verification checks a frozen requirement at a named plane. Validation asks whether the complete product works for intended system use. Qualification bounds permanent degradation from life and environment. Production test detects unacceptable units at volume. Same meters can serve all five; the question and decision differ" (Table 7.1 and chapters 8 and 9).

Pressure follow-up. "Where does burn-in fit?"
Answer pivot. "Burn-in is a production screen for early-life defects when justified. It does not replace life qualification."

Trap: calling them "levels of testing."

Question 3. Why is the validation sequence ordered, and which activities can overlap? *Tests:* order and overlap.

Spoken answer. "Earlier steps make later evidence interpretable. Requirements and architecture come before hardware spend. Bring-up before characterization so sweeps mean something. Characterization before margin so cliffs are not a surprise. Life and manufacturing evidence answer different residual uncertainties after the shipping envelope is clear. Some work can overlap in calendar time, for example architecture refresh while bring-up runs, or manufacturing validation while qualification samples age, but a later pass cannot substitute for missing earlier evidence" (Table 7.2).

Pressure follow-up. "Can you start qualification before characterization finishes?"
Answer pivot. "You can start planning, but without a behavioral model you risk stressing the wrong observables and accepting the wrong drift."

Trap: reciting Steps without saying why order matters.

Question 4. What happens during architecture review? *Tests:* budget and derating priors.

Spoken answer. "I ask whether the design can plausibly meet requirements before hardware makes changes expensive. I close optical, noise, thermal, electrical, reliability, and manufacturing budgets on stated assumptions, and I name the thin margins that validation must challenge. Architecture review is a path check, not life proof."

Pressure follow-up. "Where do derating rules come from?"
Answer pivot. "Vendor qual data, prior measurements, physics-of-failure models, field history, and design guidelines. They are priors. Qualification later checks whether this design and process support the claim" (Chapter 8).

Trap: "I simulated the link budget and it closed."

Question 5. What is the objective of hardware bring-up? *Tests:* reproducible baseline.

Spoken answer. "Bring-up establishes a known, reproducible operating state: identity, rails, firmware, CMIS, light, lock, alarms, basic power, and a simple link. I am not proving full margin yet. I am separating integration fails from product-performance questions" (Section 7.9).

Pressure follow-up. "What is your first action if ready state never appears?"
Answer pivot. "Freeze the setup, dump CMIS and rails, and isolate host versus module before any deep optical sweep."

Trap: "power on and check BER."

Question 6. What is characterization trying to produce? *Tests:* behavioral model.

Spoken answer. "A behavioral model: nominals, distributions, sensitivities, and recognizable signatures. The output is which margins are thin and which corners to challenge in system validation, not only pass or fail."

Pressure follow-up. "What belongs on every unit versus a sample?"
Answer pivot. "Cheap identity and power checks can be every-unit; expensive waterfalls and stressed RIN stay sample or audit unless escape data forces otherwise."

Trap: "measure over temperature."

Question 7. Give an illustrative temperature characterization sweep. *Tests:* conditions before causality.

Spoken answer. "Illustrative only. Sweep one lane at fixed attenuation, fixed pattern and dwell, pre-FEC BER, with the transmitter allowed to drift rather than held in constant-power control. Suppose case temperature goes 20 to 75°C: Tx power falls from about $+1.8$ to $+0.9$ dBm, bias rises from about 45 to 60 mA, RIN in a named band worsens from about -155 to -149 dB/Hz, and pre-FEC BER approaches 10^{-7}. That maps the complete-link response. It does not isolate the BER mechanism, because received power also moved. Next I would repeat BER at fixed received power to see whether quality degraded independently of average power. I would not claim RIN caused the BER change from the first sweep alone."

Pressure follow-up. "What else could move BER while power falls?"
Answer pivot. "Eye closure, ORL, control saturation, or a host EQ corner. Fixed-P_{rx} and ORL checks separate those."

Trap: blaming RIN from a coupled temperature sweep.

Question 8. What is the difference between margin validation and interoperability validation? *Tests:* cliffs versus ecosystem.

Spoken answer. "Margin finds failure boundaries by consuming optical, electrical, thermal, and control headroom. Interoperability asks whether those boundaries move across hosts, peers, firmware, and cable plants. Same lifecycle step, different conclusions" (Section 7.1.6).

Pressure follow-up. "Why not test only the reference host?"
Answer pivot. "Reference-host margin can look fine while a production host EQ or CMIS path moves the cliff."

Trap: "margin is internal; interop is another vendor."

Question 9. How would you run an optical-margin test? *Tests:* waterfall and planes.

Spoken answer. "Trusted baseline at a named plane and BER condition, then stepped attenuation with a full BER waterfall, not only the fail point. Repeat at relevant *T/V* corners and combine credible stresses. Report onset, boundary, uncertainty, and remaining margin without double-counting penalties" (Section 7.7).

Pressure follow-up. "How would you measure jitter margin?"
Answer pivot. "Inject controlled jitter on the electrical path with a qualified BERT, sweep amplitude and composition, and compare the BER or lock boundary to the allowed input jitter at documented EQ and CDR settings."

Trap: "add loss until it fails."

Question 10. How do you choose the next validation measurement? *Tests:* information per cost.

Spoken answer. "I pick the cheapest measurement that separates the strongest surviving hypotheses and changes the decision. Telemetry and controlled swaps before internal probing or DPA. I care what uncertainty it removes and what action each result unlocks."

Pressure follow-up. "When is destructive analysis justified?"
Answer pivot. "When non-destructive evidence cannot separate owners and the release or containment decision is blocked."

Trap: "run full optical characterization."

Question 11. Why is pilot deployment part of validation? Tests: bounded field experiment.

Spoken answer. "A pilot tests whether lab and factory models survive deployment. Bounded serials, enhanced telemetry, exit criteria, and rollback. Compare BER, FEC, retrains, temperature, power, and cohort rates to the release model. It is not a small shipment" (Section 7.1.9).

Pressure follow-up. "What would justify expanding the pilot?"
Answer pivot. "Exit metrics met, no unexplained cohort, and containment still reversible if the next stage fails."

Trap: "customers are happy."

Question 12. Give me a 60-second answer for validating a new optical module. Tests: time-boxed program.

Spoken answer. "Define measurable requirements and the release call. Review whether architecture budgets close. Bring up reproducibly, characterize distributions and cliffs, then validate margin and interoperability on production-like hosts and plants. Qualify named life mechanisms and validate manufacturing measurement, ATP, and yield. Run a controlled pilot, ramp under monitoring, and feed escapes back into requirements or controls."

Pressure follow-up. "Schedule is cut in half. What do you protect?"
Answer pivot. "Requirements, bring-up integrity, the thinnest margin corners, and a reversible pilot. I cut nice-to-have sweeps before I cut decision-critical evidence."

Trap: listing BER, temperature, reliability, and interop with no order.

Chapter 8

Reliability Qualification: Building the Lifetime Confidence Argument

Read first: FIT versus DPPM; qualification argument; sample confidence; burn-in versus HTOL; wear-out families.

Deep dive: Arrhenius life projection; GR-468 stress detail; connector mate-cycle physics.

Reference: qualification planning matrix; wear-out map.

Reliability qualification at a glance

- Start with the requirement. Define the life, environment, handling, and performance claim that needs support.
- Identify credible failure mechanisms. Ask what physical or electrical process could violate that claim.
- Select an acceleration method. Choose a stress that meaningfully accelerates the mechanism without creating an unrelated failure.
- Define observables. Decide what measurable change reveals degradation before running the stress.
- Define acceptance criteria. State how much change is allowable and why.
- Choose representative samples. Cover relevant lots, suppliers, sites, process corners, and product variation.
- Interpret the evidence. Connect observed degradation, confidence, and model assumptions to a bounded decision.
- Feed the result into production controls. Determine what must be controlled, screened, sampled, or monitored after qualification (Chapter 9).

> Qualification is not evidence because a standard test was performed. It is evidence when a named stress, observable, sample plan, and acceptance criterion support a specific reliability decision.

A five-year lifetime requirement cannot be verified by operating a product for five years before release. Reliability engineering therefore works indirectly. The team identifies the physical mechanisms that could cause the product to degrade, selects stresses that accelerate those mechanisms, measures signatures that reveal change, and determines whether the resulting evidence supports the intended use and lifetime.

The stress itself is not the argument. HTOL, humidity, temperature cycling, ESD, vibration, and connector cycling answer different questions. Running all of them does not automatically create confidence, and omitting one is not automatically irresponsible. The qualification plan must connect each activity to a credible threat.

Qualification also differs from characterization. A temperature sweep during characterization asks how a healthy product behaves while it is hot or cold. A qualification stress asks whether exposure causes permanent change. Reversible movement and lasting degradation require different evidence and different decisions.

Finally, qualification does not prove that every future production unit will be reliable. It establishes bounded confidence for the tested design, sample population, mechanisms, stresses, durations, observables, and assumptions. Chapter 9 addresses whether production can reproduce that qualified population consistently.

8.1 What qualification is, and is not

Keep these jobs distinct:

Characterization Maps behavior and distributions across corners.

Verification Checks measured results against frozen requirements.

Validation Demonstrates that the complete product is suitable for its intended system use (Chapter 7).

Reliability qualification Builds confidence that named lifetime and environmental mechanisms do not violate the intended claim.

Production screening Attempts to detect unacceptable units or early-life defects at manufacturing scale (Chapter 9).

Burn-in A possible production screen for specific early-life mechanisms. Burn-in is not automatically qualification and does not replace qualification HTOL.

Module example. Characterization maps LIV and BER versus temperature. Verification checks that ship power and BER meet the PRD. Validation shows the module works in the target host and plant. Qualification HTOL asks whether biased high-temperature exposure permanently shifts threshold or slope beyond the life claim. Burn-in may cull infant mortality on the line; it does not prove five-year wear-out life.

8.2 The language of scale: FIT and DPPM

Fleet arguments need two different numbers. One is about life in the field; the other is about quality at the factory door. A rate without population definition, observation time, and confidence is not a fleet claim (Section D.16).

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

FIT (failures in time) failures per 10^9 device-hours. State the population unit (laser die, module, or link), observed versus predicted FIT, confidence interval, useful-life / constant-hazard assumption, and whether the system is treated as repairable. Multiply a per-unit FIT by population and hours only after those definitions are fixed.

DPPM (defective parts per million) Distinguish incoming defects, outgoing defects, and escaped field defects. Always state the denominator, time window, and whether rework and retest are included.

Key idea. Zero failures do not prove a zero failure rate. 0 failures in 20 units is not the same evidence as 0 failures in 1,000 units. 0 failures after 10^3 device-hours is not the same evidence as 0 failures after 10^6 device-hours. Evidence strength depends on sample-hours, one-sided upper confidence bounds, censoring, and whether lots and sites are representative. Qualification and production SPC complement each other; neither alone is a fleet claim.

[1]

[1] An order-of-magnitude exercise: a $100{,}000$-GPU cluster with several optical links per GPU has on the order of 10^5–10^6 lasers. Even a small per-laser FIT then implies a nontrivial number of link failures per day, which is exactly why field-replaceable external laser sources and redundancy are attractive.

8.3 Qualification flows

Qualification is a life and variation argument, not a list of stresses. A requirement such as five-year operation is too abstract to test directly. You identify the mechanisms that could violate it, choose stresses that accelerate those mechanisms, monitor an observable signature, and decide whether the resulting evidence is sufficient for the claim (Section D.3 and section A.8.5).

Debugging is what you do when remaining margin hits zero. Qualification asks how much margin remains after the expected stresses. Place this work in

Section 7.1 Step 6; manufacturing validation and ATP live in Chapter 9. Keep the customer view and the vendor view distinct: the vendor designs internals; the customer characterizes externally visible behavior and decides deployment (Sections A.8.6 and A.8.7).

```
Requirement
   |
Budget (life / FIT / DPPM / power)
   |
Allocation (die / package / module / host)
   |
Verification (GR-468 / GR-1221 / JESD47 / ATP)
   |
Production + field monitoring
```

Margin budgeting

Every stress spends margin: temperature, voltage, ripple, contamination, insertion loss, connector wear, vibration, aging, process variation. Qual verifies what remains, not only that the part still links.

Customer view vs vendor view

Vendor: internals, device physics, implementation.
Customer: BER, sensitivity, FEC, telemetry, environmental sweeps, interop.
Engineering samples (Tx-only, Rx-only, breakout, PRBS) open isolation; otherwise stay on the external surface.

For a bookended product, begin with BER, FEC, sensitivity, telemetry, environment, and interop. Request engineering access only when that surface cannot decide (Section D.11).

Worked story: gradual laser degradation. Walk one mechanism end to end before opening the matrix. The requirement is a named life claim, for example five years at the use condition and thermal class in the PRD. The threat is gradual active-region wear: threshold rises and slope falls over life. That is not COD (sudden dark after a healthy ship LIV) and not ESD or latch-up (hard fail with supply signature and no optical aging trend).

The evidence strategy is representative lots and sites under justified HTOL: elevated temperature and bias chosen because they accelerate that mechanism, with sample size, stress hours, and confidence stated (Sections 8.2 and 8.3). HTOL is useful only when that acceleration argument holds. Observables are LIV and system proxies versus the ship baseline: I_{th}, slope efficiency, wavelength, launch power or BER drift, recorded before and during stress at named planes.

Acceptance is a bounded change after projection that still supports the life claim. Document E_a and confidence when converting HTOL hours to field years, and do not apply one E_a to mixed mechanisms. Production control is decided last and owned in Chapter 9: a room-temperature LIV/power ATP proxy, a sampled hot audit, SPC on I_{th} and slope, or an explicit statement that no cost-effective 100% screen separates the weak tail. A production burn-in may cull infant mortality; it does not replace this life argument.

Other mechanisms (solder fatigue, contamination, connector wear) follow the same template. Table 8.1 is the reference fill-in, not the primary explanation.

Qualification planning matrix. The matrix restates the laser story (and siblings) as mechanism $\rightarrow$ stress $\rightarrow$ observable $\rightarrow$ acceptance $\rightarrow$ production control. Fill cells for the product class and claimed life; do not treat blank rows as covered.

Failure mechanism	Stress	Observable	Acceptance	Production control
Laser degradation	Temperature / lifetime (HTOL)	I_{th}, slope, λ, BER	Named limit vs life claim	ATP / SPC / sampled audit / none
Solder fatigue	Temperature cycling	Resistance, BER, opens	Post-stress continuity / BER	Process control, FAIR
Contamination / corrosion	Humidity / damp heat	Loss, ORL, leakage	IL/ORL / functional limits	Handling, sealing, audit
Connector wear	Mate cycling	Insertion loss, ORL	Cycle-count IL budget	Supplier / hygiene control

Table 8.1. Qualification planning matrix (reference). Same template as the laser-degradation story above. Interview form: Sections C.15 and D.3. Decision unlocked: which mechanism-stress-observable row is missing before you claim life.

GR-468 in practice. Optoelectronics inherited a common qualification language from telecom: *Telcordia GR-468-CORE*[80]. Map each stress onto the qualification evidence path in Section D.3; do not invent a second sequence. Core stresses that still show up on every laser and module program: HTOL for life or mechanism evidence; burn-in as a production infant-mortality screen when justified; temperature cycling and damp heat; ESD and mechanical stress.

Keep the jobs distinct: **burn-in** screens infant mortality from a production population; **qualification HTOL** gathers life or mechanism evidence under accelerated operation. Do not imply that every GR-468-style HTOL is a per-unit screen. A 1,000-hour life test may justify a 100% room-temperature proxy, a sampled hot audit, a process monitor, or no direct production screen at all. Document E_a and confidence bounds when converting HTOL hours to field years, keep sample-size humility (Section 8.2 and chapter 9), and qualify the laser die, hermetic package, and module assembly separately when failures split across those boundaries (Sections 5.13, 5.14 and 8.7).

Arrhenius acceleration underpins life projection only when the named failure mechanism is temperature-accelerated in the assumed regime.

Sample strategy and confidence. Weak answer: "We test 20 units." Strong answer: the sample strategy depends on the failure-rate target, the confidence requirement, cost, and population variation. State lots, date codes, suppliers, manufacturing sites or lines, process corners, and whether the claim is zero-failure upper-bound or observed rate (Sections D.16 and 8.2). A rate without population, observation time, and confidence is not a fleet claim.

Engineering heuristic. A small sample that saw zero failures does not prove field life. It sets an upper bound. State that bound, or do not make the claim.

Tradeoff. More qualification vs faster release

Improves: Confidence and fewer late escapes

Worsens: Schedule, cost, and delayed learning in the field

When acceptable: When remaining risk is high and reversible controls cannot cover it

Experienced decision: Prioritize tests by risk $\times$ uncertainty $\times$ impact. Do not test everything equally.

GR-1221: the passive-component companion. GR-468 covers active optoelectronics. Its companion, *Telcordia GR-1221-CORE* (Generic Reliability Assurance Requirements for Passive Optical Components), covers the parts GR-468 does not: connectors, fiber couplers, WDM filters and MUX/DEMUX, splitters, and isolators [2][81]. It uses the same style of stress sequence (damp heat, temperature cycling, mechanical, and aging tests) but scores pass/fail on insertion loss and return loss rather than on LIV. A short-reach link that leans on an on-package or blind-mate MUX and on external multi-wavelength sources carries a passive reliability budget that lives in GR-1221, not GR-468 (Section 8.7 and chapter 6). Split the qual the same way you split the FIT: active laser die under GR-468, silicon under JESD47, passive optics under GR-1221.

[2] Telcordia GR-1221-CORE: reliability assurance for passive optical components (connectors, couplers, WDM filters, isolators).

Handoff to manufacturing. Qualification bounds life risk for representative hardware. Whether production can reproduce that result, screen escapes, and hold SPC is developed in Chapter 9. Failures that pass qual but fail field usually sit in derating policy, connector contamination, or a manufacturing coverage gap (Sections 5.13, 7.12 and 9.5).

8.4 Electronics reliability: driver, TIA, and DSP silicon

GR-468 covers the optoelectronic parts of the link: the laser die, the photodiode, and the hermetic or non-hermetic package around them. The modulator driver, TIA, retimer, and DSP (Section 3.14.3 and section 4.5) are ordinary CMOS or SiGe BiCMOS ICs, and they wear out and fail by a different, better-documented set of mechanisms. Treat them with the semiconductor industry's own qualification language, not with Arrhenius laser-aging math borrowed from Section 5.13.

JESD47: the silicon-side GR-468. JEDEC JESD47 is the baseline stress-test-driven qualification flow for a new IC, a device family, or a process change: temperature cycling, HTOL, HTSL (high-temperature storage life), autoclave or HAST (highly accelerated stress test) for moisture, and mechanical shock and vibration [3][93]. It plays the same role for driver and TIA silicon that GR-468 plays for the laser: a common list of stresses that a supplier runs once and a customer accepts instead of renegotiating a qual plan on every design win.

JESD47 JEDEC's baseline IC stress-test qualification standard: temperature cycle, HTOL, HTSL, autoclave/HAST, mechanical shock and vibration. The silicon-side counterpart to Telcordia GR-468 for optoelectronics.

[3] JEDEC JESD47M: baseline IC stress-test qualification flow (Aug 2025).

ESD and latch-up: failure modes GR-468 does not test. Two mechanisms are specific to ICs and absent from the laser-side wear-out map in Table 8.2:

ESD a discharge event during handling or assembly damages a gate oxide or junction. Component-level classification uses the human-body model (HBM) and charged-device model (CDM) test standards, *ANSI/ESDA/JEDEC JS-001* and *JS-002* [4][94]. A driver or TIA datasheet HBM/CDM rating is the number that protects the part on the factory floor, at fiber-attach and wire-bond stations where a laser die is also exposed.

[4] ANSI/ESDA/JEDEC JS-001 (HBM) and JS-002 (CDM) component ESD test standards.

Latch-up a parasitic thyristor structure in CMOS turns on under an overvoltage or current-injection event and holds a low-impedance path until power is cycled. *JESD78* defines the overvoltage and ± 100 mA current-injection test that classifies susceptibility by supply and signal pin [5][95]. A latched driver IC can look like a dead laser on the bench (no light, no LIV signature) until you check the supply current instead of the optical path.

JESD78 JEDEC IC latch-up test: overvoltage and ± 100 mA current-injection stress that classifies a CMOS/BiCMOS die's susceptibility to a parasitic-thyristor low-impedance latch.

[5] JEDEC JESD78F.02: IC latch-up test, overvoltage and ± 100 mA current injection.

Both mechanisms are 100%-screen or design-margin items, not something you project with an activation energy. If a driver fails ESD or latch-up in the field, that is a manufacturability or design-margin bucket item (Section 7.12), not a wear-out FIT argument.

AEC-Q100: a borrowed grade, not a requirement. *AEC-Q100* is the automotive industry's qualification standard for ICs, built on the same JEDEC JESD47/JESD22 stress methods with tighter ESD targets and named temperature grades from Grade 3 (–40 to 85°C) up to Grade 0 (–40 to 150°C) [6][96]. Datacenter optics does not require Q100; the fleet lives in a controlled data hall, not an engine bay. It is still a useful signal: a driver, TIA, or retimer die that also ships in an automotive part number typically carries a published Q100 grade, and that grade is a fast proxy for the

AEC-Q100 Automotive Electronics Council IC qualification standard: JEDEC stress methods plus tighter ESD targets and named temperature grades (Grade 3 to Grade 0). Not required for datacenter optics; useful as a borrowed signal when a driver/TIA die also ships an automotive part number.

[6] AEC-Q100 Rev-J: automotive IC qualification on JEDEC methods, tighter ESD and thermal grades.

ESD/latch-up margin and temperature-cycle depth behind the datasheet, without re-running the qual plan yourself.

Where this lands after qualification. Carry IC-level qual into the production acceptance and SPC structure in Chapter 9: require the supplier's JESD47 qual report and HBM/CDM/latch-up ratings for driver and TIA die at DVT, add an ESD handling audit to the incoming-QC checklist alongside laser LIV/SMSR sampling, and treat a driver/TIA silicon revision the same way you treat a laser die revision or a CMIS firmware rev: an ECO that needs first-article requalification, not a silent BOM swap.

8.5 Wear-out modes to know

Arrhenius math, derating, and the worked FIT example live in Section 5.13. This section is the mechanism catalog: how each failure shows up in ATP and telemetry, and which triage bucket owns it (Section 7.12). Do not run process CAPA on a wear-out part, and do not burn FIT math on a dirty connector.

Infant mortality versus wear-out versus packaging. Field failures come in three clocks, and mixing them up wastes CAPA. Infant mortality is early fails from latent defects; burn-in and HTOL screens remove them before ship (Section 8.3). Wear-out is gradual or sudden end-of-life in the laser or EAM under temperature, current, and optical power, projected with Arrhenius and derating (Section 5.13). Packaging and assembly faults (FAU align, epoxy creep, solder voids, connector wear) often dominate field returns once lasers are screened (Section 8.7). Destructive physical analysis (facet cross-section, EDX, FAU section) is required when the signature is ambiguous or when you need evidence for supplier 8D (Section 9.2).

Mechanism map. Table 8.2 is the working list for laser-bearing modules and CPO/ELS paths. Customize limits in the ATP; keep the classification discipline.

Mechanism	Observable	ATP / telemetry	Triage bucket
COD (facet)	Sudden dark or hard fail; was healthy	Dark LIV; DPA facet; date-code cluster?	Reliability (COD) or mfg (ESD)
Gradual facet / active region	I_{th} up, slope down over life	LIV trend vs ship ATP; HTOL lot history	Reliability (wear-out)
SMSR collapse	Side modes rise; modal noise / BER	OSA SMSR vs floor at T	Reliability; watch aging
EAM aging (EML)	TDECQ/RLM creep at fixed bias	EAM bias sweep + DCA; bias creep log	Reliability (EAM)
RIN rise	BER floor up; feedback sensitive	RIN @ ORL; isolator / connector check	Perf if ORL; reliability if isolator
TEC / thermal control	Unlock or λ walk; LIV may look fine	TEC current, case T, lock status	Perf (lock) or reliability (TEC)
Coupling / FAU / solder	Loss step, intermittent LOS, shock-related	ORL, mate cycles, DPA FAU/solder	Manufacturability / packaging
Driver/TIA latch-up (ESD)	Sudden hard fail; no light, no LIV signature; supply current spikes	Supply current vs bias; JESD78 rating; date-code cluster?	Mfg (ESD) or design margin
Connector wear / contamination	ORL creep after repeated mate cycles; RIN floor rise	Mate-cycle count vs IEC 61300-2-2 rating; endface grade (IEC 61300-3-35)	Manufacturability / packaging

Table 8.2. Wear-out and packaging mechanisms versus observables. Arrhenius projection and derating for the laser rows: Section 5.13. Electronics stress qualification: Section 8.4. Connector reliability: Section 8.7. Field classification workflow: Section 7.12.

8.5.1 Reading the wear-out map

Table 8.2 is a triage map, not a life calculator. Classify before corrective action. Reliability rows need life models and derate. Performance rows need plant and control fixes. Manufacturing and packaging rows need process and ATP screens. Mixing buckets wastes CAPA (Section 7.12). Arrhenius projection lives in Section 5.13; this section teaches how to read the map in three families.

Semiconductor wear. COD, gradual active-region degradation, SMSR collapse, and EAM aging change the diode or modulator physics. COD appears as sudden dark after a healthy ship LIV; gradual wear appears as rising I_{th} and falling slope; SMSR collapse raises modal noise while average power can still look fine; EAM aging creeps TDECQ/RLM at fixed laser bias. Separate them with dark LIV and facet DPA (COD versus ESD), LIV trends versus ship ATP (wear), OSA SMSR versus temperature and age (modal), and EAM bias sweeps with eye metrics (modulator). Controls are life review and derate for wear, SMSR/aging ATP for modal risk, and EAM life/ATP screens for modulator aging. Do not burn Arrhenius math on an ESD lot, and do not replace “good” DFBs while the EAM curve walks the eye closed.

Control and electronics. TEC/lock faults, driver/TIA latch-up or ESD, and ORL-driven RIN rise change the environment or the electronics path more often than the laser die. Unlock or λ walk with healthy LIV points at cooler or lock; sudden hard fail with supply-current spikes and no optical aging signature points at ESD/latch-up; a BER floor that tracks ORL points at reflections before isolator death. Separate them with TEC current, case T, and lock status; supply current and JESD ratings with date-code clusters; RIN at stated ORL versus clean plant (Section 4.3.1 and sections 6.7 and 8.4). Controls are lock/thermal policy, ESD handling screens, and connector/reflection budget, not laser FIT CAPA for every soft floor.

Packaging and optical interfaces. Coupling, FAU/solder, and connector wear or contamination produce step loss, intermittent LOS, shock correlation, or ORL creep after mates. Semiconductor metrics often stay clean. Separate them with ORL, mate-cycle history, endface grade (IEC 61300), and DPA of FAU/solder when needed (Section 8.7). Controls are assembly CAPA, attach screens, service hygiene, and connector ratings. Life models must not absorb packaging escapes.

Later evidence must not rewrite an earlier wrong bucket without new data. A clean facet DPA does not clear a connector if ORL was never measured. An HTOL pass does not clear a date-code ESD cluster.

8.6 The reliability bathtub: three failure regimes

Field failures follow three regimes with different clocks and different fixes. Mixing them wastes corrective action on the wrong mechanism.

Infant mortality (early life) Latent defects from manufacturing: weak solder joints, marginal laser die, contamination, firmware bugs that trip on first thermal cycle. Burns down rapidly with time. Fixed by burn-in, screening, tighter ATP, and first-article control (Section 8.3).

Useful life (constant rate) Random failures at a roughly steady FIT: cosmic-ray-induced single-event upsets, handling damage during unrelated service actions, and isolated material defects that passed all screens. Fixed by redundancy, field-replaceable modules, and design margin. Note: connector wear and contamination accumulate with mate cycles and are usage-driven degradation (Section 8.7), not constant-rate random failures; budget them separately from the flat-hazard FIT.

Wear-out (end of life) Physics-driven degradation: laser facet, active region, EAM absorption curve shift, TEC aging, epoxy creep. Onset depends on temperature, current, optical power, and time. Fixed by derating, Arrhenius-based life projection, and planned replacement intervals (Section 5.13).

A rising failure rate after years of service is wear-out and calls for replacement planning, not supplier 8D. A cluster of early failures on a new lot is infant mortality and calls for tighter screens. A steady trickle with no date-code correlation is useful-life random and calls for redundancy and fast repair. The triage tree in Section 7.12 forces this classification before corrective action starts.

8.7 Photonic packaging and module-level failures

Fleet FIT is not only laser wear-out. Once lasers are screened and derated, module and packaging failures often dominate field returns: the part that shipped with a clean LIV still loses light after shock, humidity, or a thousand ELSFP mate cycles. Fiber attach and FAU alignment fail from shock, humidity ingress, and epoxy creep; CPO fiber-array units add assembly steps that wafer test cannot catch (Section 10.10). Hybrid stacks (TFLN-on-Si, InP laser on Si, flip-chip drivers) introduce solder voids, underfill cracks, and RF return-loss drift (Section 3.14.3). Thermal paths matter too: uncooled datacom versus liquid-cooled XPO/CPO, and TEC failure that looks like wavelength drift off grid or off ring (Section 3.14.3 and chapter 6).

Connector reliability: MPO, mating cycles, and endface quality. Multi-fiber connectors are the highest-touch mechanical interface in the fleet: every ELSFP swap, every fiber-attach unit (FAU) rework, and every cable-plant install mates and unmates an MPO. The MPO/MT ferrule family (rectangular, 6.4 mm $\times$ 2.5 mm, guide-pin aligned, 8/12/16/24 fibers per row) is standardized in *IEC 61754-7*, split into one-fibre-row and two-fibre-row parts [7][97]. That standard fixes geometry, not lifetime; lifetime comes from two companion test methods. *IEC 61300-2-2* specifies the mate/unmate cycling test connector datasheets are rated against, and *IEC 61300-3-35* grades endface scratches, pits, and debris into pass/fail zones on the fiber core and cladding [8][98]. TIA-568.3 sets 500 cycles as the structured-cabling mating-durability floor; MPO/MTP-class connectors in practice are commonly rated well above 1000 cycles, but that headroom erodes fast with the wrong cleaning discipline (Section 7.2.2).

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

MPO Multi-fiber push-on: standard multi-fiber connector for parallel optics (8, 12, 16, 24, or 32 fibers per ferrule).

[7] IEC 61754-7-1/-2: MPO/MT mechanical interface, one and two fibre rows.

[8] IEC 61300-2-2 mating durability and 61300-3-35 endface inspection grading.

Three practical consequences follow for an ELSFP or CPO fiber-attach program. First, ORL creep is a mating-cycle and cleaning problem before it is a laser problem: a rising RIN floor after repeated ELS swaps (Table 8.2) is diagnosed with an IEC 61300-3-35-style endface inspection, not a laser FA request. Second, mate-cycle count belongs in the same telemetry you already read for CMIS and DDM (Section 7.8); track it per connector, not per module, since a connector can outlive several module swaps or vice versa. Third, write the mating-cycle and endface-grade limits into the ATP explicitly (Table 9.3) rather than inheriting a generic MPO datasheet number: an ELS bank that hot-swaps weekly reaches a 500-cycle floor in under ten years, and a CPO fiber array that is field-serviced more aggressively reaches it faster still.

ELSFP cycling adds connector wear and contamination that raise ORL (Section 7.2.2 and section 5.14); the mating-cycle and endface-grade limits above are exactly the numbers that turn "the connector feels loose" into an ATP line item instead of a guess.

Destructive physical analysis (cross-section, EDX) and structured 8D/CAPA with suppliers close the loop from RMA to design rule (Sections 7.12 and 9.2). Without that loop, packaging FIT gets mis-attributed to laser Arrhenius models and the wrong part gets redesigned.

8.8 From component FIT to fabric availability

The FIT arithmetic in Section 5.13 gives a rate: about 0.6 laser failures per day for a fleet of 5×10^5 lasers at 50 FIT. That number sizes the RMA pipeline and the ELS spares bin (Section 5.14), but it does not say what a failure costs or how a running job survives one. Two facts turn a per-component rate into a fabric problem.

First, a training or large inference job is synchronous. A collective (Section 10.7) waits for its slowest member, so a single dead or slow link stalls the whole group, not just one endpoint (Section 10.6). A link that flaps for a second is a stall for every accelerator in that collective. The optical FIT the earlier chapters budget therefore matters out of proportion to its share of the parts count.

Second, at cluster scale failures are continuous, not rare. Meta's published Llama 3 run is the clearest public data point: 16,384 H100 GPUs over 54 days logged 466 interruptions (419 unexpected), roughly one every three hours, while holding about 90% effective training time [9] [23]. GPU and HBM3 faults dominated at close to half; network switch and cable faults were 35 events, 8.4% of the total. The optical link is a minority of hard job stops, but 8.4% of a failure every three hours is still tens of network events per run, and the ELS, module, and connector FIT this chapter budgets (Sections 8.2 and 8.7) lands in exactly that bucket.

[9] Meta, *Llama 3 herd of models* (arXiv:2407.21783, 2024): 16,384 H100s, 54 days, 419 unexpected interruptions ($\sim$1 per 3 h, $\sim$90% uptime); network switch/cable 8.4%, GPU/HBM3 $\sim$47%.

So the design question shifts. It is no longer "how reliable is one link" but "how does a fabric of 10^5 links keep a job running through a failure every few hours." The answers are architectural, and the optical engineer feeds each one.

Redundancy and rails. Rail-optimized topologies (Section 10.2) already give parallel planes; dual-plane and dual-ToR designs let a lost link degrade bandwidth instead of dropping an endpoint. Redundancy multiplies the link and laser count, which feeds straight back into the FIT budget: more resilience is more parts that can fail.

Detection and reroute. Transient faults stay below the job. KP4 FEC (Section 3.12) absorbs the error bursts a marginal link throws; link-level retry and sub-second link-flap detection plus adaptive routing steer traffic off a degraded link before the scheduler notices. Vendor fabrics (NVIDIA Spectrum-X and Quantum, Broadcom Tomahawk) advertise adaptive or "cognitive" routing and link-level retry for this. Treat the specifics as vendor orientation, but the mechanism is why transient optical faults rarely reach the hard-stop bucket above.

Topology reconfiguration. When a link or rack dies for good, an optical circuit switch re-wires the topology around it in milliseconds, so the scheduler routes around the dead node instead of stalling the pod (Section 10.9) [21]. Component FIT still applies; the fabric survives each failure by re-wiring optically.

Sparing and field service. Hot spare nodes and lanes cover the interval between failure and repair. Field-replaceable external lasers (Section 5.14) make a dead laser a faceplate swap rather than a fabric outage, which is the architectural reason ELS decouples laser FIT from switch FIT. The connector mating-cycle and endface budget (Section 8.7) sets how many of those swaps the plant survives.

The cost of a failure closes the loop. A hard interruption is lost compute plus the time to detect, reroute or reschedule, and restart from the last checkpoint. Fast detection and reroute shrink that lost time, which is the fabric-level reason the module work in this chapter pays off: derating (Section 5.13), burn-in and screens (Section 8.3), and a tight ATP (Section 9.2) lower the failure rate, and a resilient fabric lowers the cost of each failure that slips through. The two multiply.

8.9 Engineering lens

8.9.1 How it works

At fleet scale, a modest per-part FIT times millions of parts is a steady stream of failures. Qualification, derating, and mechanism classification keep that stream small and classifiable. Manufacturing validation (Chapter 9) then asks whether the factory can reproduce the qualified population.

8.9.2 How it is measured

Reliability is measured as a distribution over stress and time. Qualification records failures and drift by mechanism, stress, lot, and sample history. Convert accelerated hours to field years only with a named mechanism, stated E_a, and confidence bounds (Sections 8.2 and 8.3). Fleet data add install age, temperature, firmware, supplier lot, and return code so wear-out and infant mortality are not mixed (Section 7.12).

8.9.3 How it fails

Programs fail at life through wrong mechanisms, weak samples, and overstated acceleration. A stress without a mechanism is only exposure. Zero fails in a small lot is an upper bound, not a FIT claim. Arrhenius math on an ESD cluster or dirty connector wastes CAPA.

8.9.4 How it is debugged

Classify the bathtub regime and wear-out map bucket before corrective action (Section 8.6 and table 8.2). Confirm the mechanism with LIV, supply current, ORL, and DPA as needed. Escalate supplier process CAPA only when the bucket is manufacturing; escalate life model and derate when the bucket is wear-out.

8.10 Interview takeaway

> **Key idea.** Reliability qualification is a mechanism-driven confidence argument. Begin with the lifetime or environmental claim, identify how it could be violated, select a credible acceleration method, define observable degradation, and choose representative samples and acceptance criteria before running the stress. The result supports a bounded decision; it does not guarantee every production unit. Chapter 9 explains how production reproduces and controls the qualified design.

Junior mistake: treat zero fails in a small HTOL lot as a FIT claim, or treat a temperature sweep as temperature qualification (Sections 8.1 and 8.3).

8.10.1 Interview Q&A: Reliability Qualification

Practice speaking these answers aloud. Prefer first-person reasoning over stress inventories. Detail lives in Sections 8.1 and 8.3 and table 8.1. Score your answer using the chapter-end spoken-answer rubric (Section A.12.1).

Question 1. What is reliability qualification, and how does it differ from validation? *Tests:* life claim versus system suitability.

Spoken answer. “Qualification is a bounded confidence argument that the design continues to meet requirements through intended life and environment. Validation asks whether the product is suitable for system use now. A temperature sweep can validate hot operation; HTOL or cycling asks whether exposure causes permanent degradation. Same gear, different question and acceptance rule" (Chapter 7 and section 8.1).

Pressure follow-up. "Does passing a standard qualify the product?"
Answer pivot. "Only if each stress maps to a named mechanism, observable, sample plan, and decision. A green checklist alone is not the argument."

Trap: listing HTOL, humidity, cycling, and vibration.

Question 2. How do you design a qualification plan? *Tests:* mechanism-driven planning.

Spoken answer. "I start with the life and environment claim. Then I list credible mechanisms, pick stresses that accelerate those mechanisms, define observables and acceptance before stress, and choose representative lots and sites. Each result must map to release, derating, redesign, production control, or more evidence" (Table 8.1).

Pressure follow-up. "The standard requires a low-risk test. What do you do?"
Answer pivot. "I run or waive it with a written mechanism rationale and owner, not silently. Customer or regulatory gates may still force the row."

Trap: "follow GR-468 and run the matrix."

Question 3. What is the difference between a failure mode and a failure mechanism? *Tests:* symptom versus physics.

Spoken answer. "Mode is the symptom: low power, high BER, dead lane. Mechanism is the physical process: active-region wear, solder fatigue, corrosion, contamination. Several mechanisms can share one mode, so qual and ATP must target mechanisms" (Table 8.2).

Pressure follow-up. "Name two mechanisms for low optical power."
Answer pivot. "Laser wear reducing slope, or coupling or connector loss. Different stresses and screens."

Trap: swapping mode and mechanism.

Question 4. Why can accelerated testing support a field-life claim? *Tests:* same-mechanism acceleration.

Spoken answer. "Only when the stress accelerates the same mechanism expected in use. For temperature-activated wear, Arrhenius may translate hours to field years if E_a and assumptions fit that mechanism. A severe unrelated fail is not a life prediction. I state physics, model, observables, and uncertainty" (Sections 5.13 and 8.3).

Pressure follow-up. “How do you know the activation energy?”
Answer pivot. “Mechanism-specific supplier data, literature, or fitted degradation on this process. I do not borrow one E_a across mixed failure mechanisms.”

Trap: “hotter means faster; convert hours to years.”

Question 5. Explain HTOL and what you would monitor. *Tests:* access-aware observables.

Spoken answer. “HTOL biases the part at elevated temperature to accelerate wear tied to powered operation. I need trusted baselines and intermediate reads, because units can stay functional while drifting. On engineering-access units I measure LIV, threshold, slope, SMSR, and sampled RIN. On bookended production modules I use external proxies: optical power, OMA, wavelength, BER, module current, telemetry, and control headroom. Supplier reports may carry die LIV when the customer only sees the module.”

Pressure follow-up. “Why is chamber temperature not enough?”
Answer pivot. “Junction or active-region temperature drives many mechanisms. Chamber air without bias or thermal path context can mis-rank stress.”

Trap: “run hot for a thousand hours and see if it still works.”

Question 6. What does temperature cycling qualify? *Tests:* package mechanics.

Spoken answer. “Repeated expansion mismatch: solder, bonds, adhesives, fiber attach, alignment. I watch continuity, intermittent lanes, power, sensitivity, BER, and permanent post-cycle shift versus reversible temperature behavior.”

Pressure follow-up. “Measure in situ or only after cycling?”
Answer pivot. “Periodic or in-situ catches intermittents that a final functional check can miss.”

Trap: treating cycling as hot/cold operation check.

Question 7. What is damp-heat testing trying to reveal? *Tests:* moisture mechanisms.

Spoken answer. “Corrosion, leakage, material or interface degradation, contamination movement. Biased or unbiased depending on mechanism. I look at leakage, loss, ORL, power, BER, and selected DPA, not only a post-test link pass.”

Pressure follow-up. “Why might bias matter?”
Answer pivot. “Bias can drive electrochemical paths that unbiased storage never sees.”

Trap: “checks if the package is waterproof.”

Question 8. Why do ESD qualification if units already pass functional test? *Tests:* latent damage.

Spoken answer. “ESD can kill, leave latent damage, or burn margin while the unit still links. Qual checks the protection network against the handling claim. Production controls cover handling. Final function alone can miss latent damage” (Section 8.4).

Pressure follow-up. “Can ATP screen latent ESD reliably?”
Answer pivot. “Often no. Leakage or sensitivity shifts may need targeted screens; many latent cases need process control, not 100% detection.”

Trap: “operators might touch the board.”

Question 9. How do you choose sample size for qualification? *Tests:* confidence, not folklore.

Spoken answer. “No universal count. I size to the rate or degradation claim, confidence, hours, acceleration, and lot diversity. Zero fails in twenty units is an upper bound, not zero field rate. I report bound, sample-hours, censoring, model assumptions, and remaining risk. Sufficiency means the release decision is supported, not that uncertainty is gone” (Sections 8.2 and 8.3).

Pressure follow-up. “Is lot diversity worth more than more units from one lot?”
Answer pivot. “Usually yes for process and supplier risk. Identical twins inflate false confidence.”

Trap: “we always use twenty or thirty units.”

Question 10. What should you do when a qualification test fails? *Tests:* contain, confirm, scoped requal.

Spoken answer. “Verify setup, preserve state, scope the population, separate mode from mechanism, and confirm with discriminating evidence. Contain, decide design, process, supplier, derating, or more data, then repeat enough qual to show the mechanism is addressed. I do not blindly restart the whole matrix.”

Pressure follow-up. “Failure is outside intended use. Still a fail?”
Answer pivot. “I still investigate relevance. If the stress is unrepresentative, I document that. I do not hide a real mechanism behind a claim of overstress.”

Trap: “FA the unit and rerun the test.”

Question 11. What is the difference between qualification, burn-in, and production screening? *Tests:* life evidence versus screens.

Spoken answer. “Qualification supports design-life claims on representative samples. Burn-in is an optional production screen for specific early-life defects. Broader screening is every-unit or sample ATP and process control. Burn-in does not replace qual; qual does not prove every shipped unit was built correctly" (Section 8.1 and chapter 9).

Pressure follow-up. "When is burn-in justified?"
Answer pivot. "When escape data and mechanism show it separates infant fails cheaper than alternatives, and the stress does not damage healthy units."

Trap: "burn-in is qualification on every unit."

Question 12. Give me a 60-second answer for qualifying a new optical transceiver. Tests: time-boxed qual argument.

Spoken answer. "Start from life, environment, handling, and performance claims. Name credible mechanisms, pick accelerations, define observables and acceptance, and sample across lots and sites. Monitor trends, interpret with confidence and model limits, confirm fails to a mechanism, and translate into release, derating, redesign, supplier control, or manufacturing monitors. State the supported claim and remaining risk."

Pressure follow-up. "Walk a laser-aging argument."
Answer pivot. "Illustrative five-year continuous claim. Threat: active-region wear raising threshold and cutting slope or OMA. On engineering-access samples I baseline LIV, SMSR, and sampled RIN; on bookended modules I use power, OMA, wavelength, BER, current, and headroom. Biased high-temperature stress with a justified acceleration model, intermediate reads, and drift limits tied to remaining system margin. Result supports release, derating, supplier limit, or redesign; production may sample LIV or bias SPC afterward, but that does not replace the qual argument."

Trap: "run the standard stress list."

Chapter 9

Manufacturing Validation and Production Readiness

Proving the factory can reproduce and protect the qualified design.

Read first: production reference freeze; gauge R&R; first-pass versus final yield; ATP versus SPC; NPI gates.

Deep dive: station correlation; escape classes; second-source evidence.

Reference: NPI gate table; ATP control checklist; yield-owner split.

Manufacturing validation at a glance

- Freeze the production reference. Define the exact design, BOM, firmware, calibration, tools, and processes being validated.
- Plan representative builds. Use production-intent materials, equipment, operators, suppliers, and stations.
- Create unit genealogy. Preserve the material, process, software, and test history of every unit.
- Validate the measurement system. Demonstrate accuracy, repeatability, reproducibility, stability, and station correlation.
- Understand yield and distributions. Determine normal production behavior and identify dominant variation.
- Validate ATP and screening. Prove that production tests detect relevant defects without excessive false rejects.
- Establish process controls. Use SPC, audits, supplier controls, and reaction plans.
- Ramp deliberately. Increase exposure only when evidence supports the next volume stage.
- Close the loop. Feed escapes and process movement back into design, qualification, ATP, and supplier controls.

> Manufacturing validation does not mean that one pilot lot passed. It means the production system is defined, measurable, capable, traceable, controlled, and prepared to react when it moves.

A qualified engineering design is not automatically a manufacturable product. Engineering units may be assembled by expert technicians, selected for favorable components, calibrated individually, reworked repeatedly, and measured with laboratory instruments that cannot support production takt time. A factory must reproduce the same required behavior across materials, operators, tools, shifts, sites, test stations, and time.

Manufacturing validation therefore begins by defining the production reference. The team must know exactly which design, firmware, calibration, materials, processes, fixtures, and test limits are being evaluated. It then builds representative units using the intended process, preserves their genealogy, and determines how much observed variation comes from the product versus the measurement system.

Only after the measurements are trusted can yield, capability, and process movement be interpreted. Production tests must then be shown to detect important defects efficiently. Statistical process controls and reaction plans keep the process stable as volume increases. Fleet and failure-analysis evidence complete the loop by revealing escapes that factory evidence did not predict (Chapter 11 and section 7.12).

The aim is not to prove that every future unit will be good. The aim is to establish a system that can repeatedly produce acceptable output, detect unacceptable output, identify affected populations, and correct drift before it becomes a fleet problem. The input is the design bounded by Chapter 8.

9.1 Manufacturing validation versus adjacent activities

Activity	Primary question	Typical evidence	Decision
Design validation	Does the product meet intended system requirements?	Characterization, margin, interop	Approve system envelope
Reliability qualification	Does the design survive named mechanisms and stresses?	Accelerated stress, degradation, confidence	Approve life/environment claim
Manufacturing validation	Can production reproduce and measure the result?	Production-intent builds, MSA, yield, capability	Approve controlled ramp
ATP	Can unacceptable units be detected economically?	Every-unit tests and validated proxies	Ship or reject unit
SPC	Is the process remaining stable?	Time-ordered process metrics	Continue, contain, investigate
Fleet monitoring	Does deployed behavior match the release model?	Telemetry, cohorts, returns	Expand, contain, improve

Table 9.1. Adjacent activities. Do not use qualification, manufacturing validation, ATP, SPC, and fleet monitoring interchangeably. Lifecycle placement: Section 7.1.

9.2 Supplier execution playbook

The supplier path is milestones, performance targets, quality, and manufacturability triage. That is not a soft skill. It is a concrete contract: requirements, gates, acceptance tests, process control, and corrective action when a lot goes wrong.

Why experienced engineers care about production lots?

Because manufacturing escapes almost always correlate with process history. Lot, date code, site, and firmware tags often beat another night on one returned unit.

Engineering heuristic. Ask for the process change list before you invent new physics. Most lot escapes sit next to a real change record.

Tradeoff. Second source vs qualification burden

Improves: Supply resilience and pricing leverage

Worsens: Validation, interop matrix, and manufacturing differences

When acceptable: When supply or concentration risk exceeds the qual cost

Experienced decision: Qualify second sources based on risk and evidence, not ideology.

Lab hero samples versus manufacturable yield: same tradeoff as in Chapter 7. Optimize the system you can build, not the best bench unit.

```
Design requirements
  |
Qualification
```

```
   |
ATP
   |
Production data
   |
Fleet data
   |
Failure analysis
   |
Updated limits or screens
   |
Next production cycle
```

Production validation is replayable and decision-oriented (Section D.13).

NPI as program maturity. New product introduction (*NPI*) is the manufacturing face of the validation Steps in Section 7.1. Write exit criteria a supplier can fail without ambiguity, not slogans. EVT/DVT/PVT/MP are stage names, not a calendar; dates and sample sizes belong in the program plan.

Why program maturity changes the question. Early in the program the question is whether the architecture can be made to work at all (EVT), then whether behavior and margin are understood across the intended corners (DVT). Once the part is characterized, the question shifts to whether named failure mechanisms threaten life (qualification), then whether production tooling and suppliers can reproduce that qualified result across lots (PVT). A pilot asks whether laboratory and factory assumptions survive a bounded, observable deployment. Mass production is not another validation experiment: it is sustained control with SPC, ECO, and RMA ownership (Sections 7.1.10 and 7.1.11).

Do not use an EVT hero sample as PVT evidence, and do not treat MP as fleet monitoring alone. Hold a gate if the exit data are missing. Table 9.2 retrieves the gate questions and release calls; it does not define a second lifecycle beside the Steps (Sections D.2 and 7.1).

Requirements and ATP are the contract. ATP and the requirements doc are the contract. Write both and keep them versioned together:

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

> **Engineering heuristic.** If a requirement has no ATP or sample line, it is a wish. If an ATP line has no requirement, it is cost without a decision.

1. **Requirements / PRD slice for the laser path:** fill Table 5.4 and section 5.6 (power class, grid, RIN@ORL, SMSR, derating, CMIS, FIT). Version it with the ATP.
2. **Acceptance test plan (ATP):** the measurable tests that prove those requirements on every ship lot (or on a defined sample). Map each ATP line to a GR-468 or design-validation stress where life is claimed (Section 8.3).

Gate	Question	Representative evidence	Release decision
EVT	Does it operate at all?	First light; CMIS bring-up; basic LIV/SMSR/RIN; one link closes BER	Continue / redesign integration
DVT	Does it meet spec across corners?	Full ATP at T/V; prod-rep corners; stress plan + FIT model frozen	Enter qual / PVT / hold
Qual	Env / reliability evidence ready?	Named mechanisms; sample plan; confidence (Section D.3)	Enter PVT / hold
PVT	Is it buildable at yield?	Multi-lot yield; SPC; burn-in escape; FAIR; production-host bring-up	Enter pilot / hold
Pilot	Do assumptions hold in a bounded field trial?	Known serials/lots; enhanced telemetry; exit/rollback criteria	Open MP / restrict
MP	Is quality sustained?	Steady DPPM; owned RMA Pareto; ECO control; fleet feedback	Keep shipping / CAPA / restrict

Table 9.2. NPI gates (reference). Decision unlocked: which release call the gate evidence supports. Pilot sits between PVT and MP; MP is sustained control, not fleet monitoring alone.

9.2.1 How production evidence is purchased

Every second of ATP costs line capacity. Every omitted screen leaves risk. Buy evidence in categories that cannot substitute for each other. Table 9.3 is a working checklist for an EML pluggable or an ELSFP CW module. Customize limits from the datasheet and the link budget; do not invent numbers in the ATP itself.

Every-unit fast screens. Power class, CMIS state and bring-up, basic lane operation, and selected electrical or optical checks that are cheap enough for 100% coverage. These catch dead units and firmware fails. They do not establish life.

Sampled expensive measurements. TDECQ, intrinsic and stressed RIN, detailed spectrum/SMSR, thermal corners, and longer BER dwell. Use lot sample or audit when the signature is expensive and correlated to escape risk.

Qualification-only stresses. HTOL, humidity, extensive cycling, and destructive analysis gather life or mechanism evidence. They are not per-unit ship screens (Sections D.3 and 8.3).

Process controls. SPC on LIV, SMSR, RIN, TDECQ, and mate-cycle yield; golden-unit tracking; gauge R&R; incoming quality; first-article / FAIR after tooling or site change (Section 9.4.3).

Fleet controls. Telemetry, lot traceability, RMA codes split by mechanism, and recurrence monitoring (Section 7.12). These catch what ATP never saw.

A fast production power check does not establish life. HTOL does not prove every shipped unit has correct firmware. SPC does not detect an unmeasured mechanism. LIV, SMSR, and wavelength protect semiconductor identity; RIN and ORL protect the reflection environment; EAM/DCA protects Tx quality on EML paths; CMIS protects field evidence; burn-in and ESD protect infant mortality and handling; thermal class protects the derate claim.

Item	Method	Control class	Pass intent	Ties to
LIV (I_{th}, slope, kink)	SMU + power meter	100% ATP (source) / module proxy	kink-free bias window	wear-out, derate
SMSR	OSA	100% ATP (source) / lot sample	single-mode vs. floor	modal noise
Intrinsic RIN	PD + ESA	Lot sample / FA	quiet source floor	BER floor budget
Stressed RIN_xOMA	PD + ESA @ ORL	Lot sample / 100% if escape	stressed Tx metric	named PMD
Wavelength / grid	OSA / wavemeter	100% ATP or sample	channel ID	WDM lock
Optical power class	power meter	100% ATP	class met	link budget
EAM / TDECQ (EML)	bias + DCA	Lot sample / audit	ER, RLM, TDECQ	Tx quality
CMIS / TWI bring-up	host / CMIS tool	100% ATP	state machine	telemetry
Connector / ORL	mate + ORL meter	Periodic audit / sample	cycles + endface	packaging
Burn-in (infant)	production screen	100% or lot sample	infant culled	not HTOL life
HTOL life evidence	accelerated life	Qualification only	mechanism + FIT claim	GR-468
Driver/TIA ESD	JESD47 report	Qualification / audit	rating on file	IC reliability
Thermal class	chamber	Lot sample / SPC	LIV/RIN/CMIS pass	derate
Process monitors	SPC charts	SPC / process	drift detection	escape prevention
Fleet cohort metrics	telemetry	Fleet telemetry	trend / alarm	recurrence

Table 9.3. Control checklist for laser-bearing modules (EML or ELSFP). Control class separates 100% ATP, lot sample, audit, qualification-only, SPC, and fleet telemetry. Intrinsic RIN and stressed RIN_xOMA are different metrics.

Exit for the ATP as a whole: every ship lot (or defined sample) has traceable pass data against versioned limits tied to the requirements slice. **Decision:** ship, hold, or reopen DVT limits.

Incoming QC and SPC. Qual lots are small. Production catches drift that qual missed.

- **Incoming:** sample or 100% screen against a subset of the ATP (at least power, CMIS, and a laser LIV/SMSR sample). Track DPPM by date code and site.
- **SPC**: control charts on I_{th}, slope, SMSR, Tx power, and burn-in fallout. A process shift is a hold, not a hope.
- **First-article / FAIR:** when tooling, epi, or assembly site changes, rerun a defined FAIR package before open PO volume. Treat CMIS firmware revs the same way as process changes.

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

Excursions: 8D / CAPA. When a lot fails ATP, incoming, or field triage lands in the manufacturability bucket (Section 7.12), run structured corrective action:

1. **Contain:** quarantine WIP and ship holds; identify suspect date codes in the fleet.
2. **Evidence pack:** failing ATP rows, CMIS dumps, LIV/SMSR/RIN plots, DPA photos (facet, solder, FAU cross-section) compared to a golden unit.
3. **8D / CAPA**: confirmed mechanism with the supplier (process step, material lot, firmware), corrective action, and preventive control (ATP tighten, SPC limit, poke-yoke).
4. **Verify closure:** containment confirmed effective; mechanism reproduced or physically confirmed; corrective action removes the failure; no unacceptable regression introduced; production control detects recurrence; next lots remain stable; field cohort trend improves. Re-run FAIR alone is not enough for environmental, intermittent, or fleet-specific escapes (Section D.16).

8D Eight disciplines (8D) corrective action: contain, root-cause, correct, and prevent with the supplier; close only with verification evidence.

Do not close 8D on "operator error" without a control that would have caught it at ATP or in process. If FA shows laser wear-out on a young unit, it may be a reliability screen gap, not a supplier process bug; reclassify with Section 7.12 before you argue FIT.

Milestone hygiene with partners. Align the partner calendar to gates, not slideware:

- freeze requirements before DVT samples are built;
- freeze ATP limits before PVT yield is claimed;
- freeze FIT/E_a assumptions before reliability marketing numbers ship;

- require ECO notice on laser die revision, TEC vendor, FAU epoxy, driver/TIA silicon revision (Section 8.4), and CMIS firmware.

Your job in those meetings is to name the measurement that would kill the gate. If nobody can point to an ATP row or a corner, the milestone is not real.

Key idea. Manufacturing readiness is a frozen reference, a trusted measurement system, multi-lot yield, ATP coverage, and owned SPC with reaction plans. Classify escapes with the wear-out map (Table 8.2) before FIT or 8D. Gate suppliers on ATP, multi-lot SPC, and FAIR. Do not run process CAPA on wear-out, or FIT math on a dirty connector.

9.3 Yield analysis

Yield is not one number. It splits by stage and by failure mode, and each split points at a different owner.

Wafer / die yield Process-limited: waveguide loss, ring resonance spread, heater shorts, photodiode dark current. Caught at wafer probe. Owner: foundry SPC.

Assembly yield Packaging-limited: fiber-array attach alignment, solder voids, wirebond pull, epoxy placement. Caught at module ATP. Owner: assembly supplier.

Test yield (first-pass) ATP-limited: units that fail one or more acceptance criteria on first pass. May include measurement-system false rejects (gauge RR). Owner: test engineering.

Escaped DPPM (post-screen field failures) Units that passed all screens but fail in the fleet. Each escape is either a preventable coverage gap (the screen exists but missed the defect) or a residual latent failure that no cost-effective screen separates from good units. Owner: quality and reliability engineering.

Yield stage	Main limit	First catch	Owner
Wafer / die	Waveguide, resonance, heater, PD dark	Wafer probe	Foundry SPC
Assembly	FAU align, solder, wirebond, epoxy	Module ATP	Assembly supplier
Test (first-pass)	ATP fails; may include false rejects	ATP station + gauge RR	Test engineering
Escaped DPPM	Passed screens; failed in fleet	Field RMA / triage	Quality / reliability

Table 9.4. Yield stages, first catch, and owner. Split escapes further in Table 9.5.

Track yield by ATP row, lot, supplier site, tester, and date code. A yield drop that correlates with one tester is likely a measurement problem. A yield drop that correlates with one supplier lot is likely a process problem. A yield drop with no observed correlation requires further investigation: verify gauge repeatability, expand stratification (shift, fixture, material lot, firmware), and test guardband or specification mismatch as hypotheses before concluding. Do not open supplier corrective action until the measurement system is cleared (Section 9.4).

Engineering heuristic. Clear the tester with a golden unit before you escalate a supplier. Station drift masquerades as a process excursion more often than engineers admit.

What this usually means. A golden unit fails on only one production station

Usually: fixture, calibration, cable, software limit, or operator path on that station

Not: a sudden die-level failure of every good unit that station has ever seen

9.4 Production test at volume

Before production

ATP · SPC · telemetry · supplier gates · monitoring owners · RMA-to-ATP feedback (Section D.17).

9.4.1 Test time is a cost, coverage is a risk

Every second in the acceptance test plan (ATP) times millions of units is line capacity and real money. Every skipped measurement creates uncontrolled escape risk; it is not automatically a field DPPM event (Section 8.2). The core tension in high-volume manufacturing is how much coverage you buy per second. The expensive optical steps are thermal soak and corner runs, TDECQ on a sampling scope, BER dwell long enough to trust a low pre-FEC target, laser burn-in, and mate-cycle stress on ELSFP connectors. Some screens are statistical (sample burn-in from a lot, audit TDECQ on a subset). Safety and enable-sequence faults usually require 100% coverage. For source-level production where LIV or SMSR are directly available and correlated to escape risk, they may be 100% screens. For closed modules, use the validated module-level proxy or a documented sampling plan (Sections 5.15 and 7.8 and table 9.3).

Tradeoff. More production screening vs cost

Improves: Escape detection and earlier catch

Worsens: Cycle time, tester cost, and false rejects that burn good units

When acceptable: When a named mechanism has a cheap, reliable detection signature

Experienced decision: Choose the cheapest control that reliably detects the failure mode: 100% ATP, sample audit, SPC, or supplier process control.

Tradeoff. Burn-in vs cycle time

Improves: Infant-mortality removal when the screen separates

Worsens: Line capacity, cost, and stress on healthy units

When acceptable: When escape data and mechanism justify the screen on this population

Experienced decision: Keep burn-in only while it buys escapes you cannot catch cheaper elsewhere.

9.4.2 Where the test happens: wafer, die, module, system

Push defect detection as far upstream as correlation allows. Wafer-level or PIC probe catches process shifts (waveguide loss, ring resonance drift, bad heaters) before fiber attach and packaging spend. Killing a bad die at probe is orders of magnitude cheaper than an RMA (Section 8.7). Module ATP is the full functional test: optical power class, TDECQ or proxy, sensitivity spot-check, CMIS bring-up, and connector/ORL on ELS parts. System or golden-host bring-up catches interop: media type, firmware rev, equalizer defaults, and the corners in Section 7.9. Wafer test cannot catch fiber attach, FAU alignment, epoxy creep, or connector wear. Those failures must survive to module ATP and, for some signatures, to fleet telemetry (Section 7.12).

9.4.3 ATE-to-bench correlation and gauge R&R

Production testers are built for speed and cost, not lab fidelity. Correlation asks whether ATE TDECQ, OMA, or sensitivity tracks the DCA/BERT reference within a known offset and spread. Teach the measurement system explicitly:

Repeatability Same station, operator, and unit.

Reproducibility Across stations, shifts, and operators.

Bias Production station versus reference lab.

False reject / false accept Guardband from measurement uncertainty.

Golden-unit stability and station drift Catch a stale golden or drifting ATE before it becomes a yield cliff or DPPM escape.

Keep a golden module (and golden laser subassembly for ELS), run gauge R&R across testers and shifts, and correlate CMIS monitors to bench instruments the same way you correlate TDECQ (Section 7.8). If the ATE and the DCA disagree, fix the correlation before you argue with the supplier about spec.

9.4.4 Screens, guardbanding, and SPC

Burn-in (infant-mortality screen) and HTOL (life/mechanism evidence) trade different risks against test time and cost (Sections 8.3 and 8.5). Test limits are usually guardbanded tighter than the customer spec so field DPPM stays inside target under drift. SPC control charts on LIV, SMSR, RIN, TDECQ, and mate-cycle yield by lot, site, and date code catch a process shift before it becomes an 8D (Section 9.2). Production test is a yield, DPPM, and cost trade under a fixed reliability target. It is not a pass/fail checkbox after the optics already work on a golden bench.

9.5 Escaped defect analysis

```
Supplier escape containment flow
   |
Escape detected
   |
Provisional containment
   |
Scope analysis
   |
Refine contained population
   |
Investigate / confirm mechanism
   |
Corrective action
   |
Recurrence control
```

Provisional containment, scope refinement, mechanism confirmation, and recurrence control are different actions. Apply Sections D.9, D.12 and D.16 to separate immediate hold from FA and from the ATP, sample, SPC, or telemetry change that closes the loop. Organize the spent margin with the five ledgers before naming a component (Sections 4.8 and 5.19).

An escaped defect is a unit that passed every production screen and failed in the field. Post-screen field failures split into two categories with different corrective actions:

Class	Meaning	Typical action	Lands in
Preventable coverage	Screen could have caught it	Add/tighten ATP or SPC	Escape DPPM, CAPA
Residual latent	No cost-effective screen	FIT / redundancy / replace	Residual FIT model

Table 9.5. Escape classes. Preventable rows change production; residual rows change the life model.

Preventable coverage escapes. A defect that a cost-effective screen could have caught but did not, because the screen was absent, miscalibrated, or insufficiently stressed. Each one requires an ATP or process-control change. Common mechanisms in optical modules:

- **Contamination:** particles trapped during assembly that only move under thermal cycling or vibration, shifting coupling or raising ORL intermittently.
- **Marginal alignment:** fiber-attach or FAU position within spec at room temperature but outside at the thermal corner, creating a temperature-dependent loss that the ATP chamber did not run.
- **Weak solder or wirebond:** passes pull test but fatigues under repeated thermal cycling, creating intermittent opens or increased resistance.
- **Firmware corner:** a state-machine path or calibration table entry that production never exercises but the fleet hits on a specific boot sequence, temperature bin, or host interaction.
- **Packaging stress:** residual mechanical stress from underfill cure or lid attach that relaxes over time, shifting alignment or birefringence.

For each preventable escape, trace the failure signature back to the earliest point in the production flow where it could have been caught, and add or tighten the screen there.

> **Engineering heuristic.** An escape without an ATP or SPC change is unfinished work. Containment stops the bleed; the control stops the next lot.

Residual latent failures. A defect that no cost-effective screen can separate from good units at the time of test. Examples include cosmic-ray-induced single-event upsets, rare material inclusions below detection limits, and process-marginal units that sit inside all guardbands but fail under a specific field combination of temperature, ORL, and neighbor load. These go into the residual FIT model, not into ATP. Document why no reasonable screen exists so the decision is auditable and revisitable as test technology improves.

9.6 Worked case study: yield loss blamed on the laser supplier

Illustrative numbers only. A 240-unit production-intent build shows 90% first-pass yield. Low OMA and high laser bias dominate the Pareto. Failures appear to cluster on one laser date code. Two ATP stations disagree by about 0.4 dB on the same units. Retest recovers many fails.

1. **Verify measurement first.** Golden units and station-to-station correlation show station B reads low by ~ 0.4 dB. Do not open laser supplier CAPA yet (Section 9.4.3).
2. **Stratify the population.** Split by station, fixture, operator, shift, and laser lot. Lot and station are confounded: the suspect date code ran mostly on station B.
3. **Controlled swap.** Move the same units and fixture across stations. The OMA offset follows the station, not the laser lot.
4. **Find the mechanism.** Fixture insertion loss has drifted. Repair and recertify the station; remeasure the held population.
5. **Quantify residual supplier difference.** After station repair, one laser lot still sits slightly high in bias. Contain that lot, tighten incoming LIV sample, and document the residual as a material signal rather than the original yield cliff.
6. **Update controls.** Golden-unit cadence, gauge R&R, and SPC on station offset resume. Restricted ramp continues only after a confirmation lot clears first-pass yield and station agreement.

A yield problem can belong to the product, material, process, measurement system, software, or data. Verify measurement before redesigning the product or blaming the supplier.

9.7 Engineering lens

9.7.1 How it works

Manufacturing validation proves that the qualified design (Chapter 8) can be reproduced and protected at scale: frozen reference, representative builds, genealogy, trusted measurement, yield and ATP coverage, SPC with reaction plans, and staged ramp.

9.7.2 How it is measured

Production records first-pass and final yield, retest and rework rates, fallout by ATP row, measurement distributions, gauge repeatability, station correlation, and escaped defects per million. Keep the chain from incoming material through module ATP so a drift can be traced to its first observable point (Sections 7.12, 9.2 and 9.4).

9.7.3 How it fails

Programs fail at scale through variation and escapes. Variation produces a weak tail across wafer, lot, site, or assembly line. An escape is a defect the current screen cannot see or a control that was not run. Yield can drop because the product changed, the process moved, incoming material moved, calibration changed, or the tester moved. Do not open supplier corrective action until the measurement system is cleared.

Failure mode: Yield drop

Symptoms. First-pass yield falls from its stable baseline, often on one ATP row, tester, lot, or shift.
Likely causes. Process drift, supplier-lot variation, assembly change, calibration or fixture drift, software revision, or a changed guardband.
Measurements. Pareto by test and lot, golden-unit history, gauge repeatability, station correlation, incoming data, and destructive analysis on selected failures.
Mitigations. Contain suspect material, clear the tester, identify the first changed input, correct it, and verify with a controlled lot before release.

9.7.4 How it is debugged

For a yield fall, freeze software, limits, and suspect material. Split by tester, shift, lot, supplier site, and ATP row. Golden unit across stations; failing unit on a reference bench. Station-follows means repair the measurement system; unit-follows means upstream process and mechanism FA. Contain first, confirm second, change the process third, verify on fresh data.

Debug story

Observed. Module yield fell sharply after a supplier lot change.
Investigation. The failure Pareto pointed to one-lane TDECQ. Golden units passed all stations, and failed units kept the bad lane on the reference bench. Cross-sections showed a shifted fiber-array attach.
Finding. The electrical path and testers were stable.
Root cause. An assembly fixture change moved one fiber row outside its coupling window.
Resolution. The lot was held, the fixture was restored, first-article coupling checks were tightened, and the supplier control plan was revised.

9.8 Interview takeaway

Concise interview answers.

How would you validate a factory? Freeze the production reference; build representative lots with genealogy; clear MSA; read first-pass yield and distributions; prove ATP coverage; install SPC with owners and reaction plans; ramp only as evidence supports exposure.

How many units would you build? Enough to cover the question: lots, sites, operators, tools, stations, and variants. A balanced design beats a large but confounded build. State what decision the count unlocks.

What does gauge R&R tell you? How much observed variation is measurement versus product. It does not prove the product meets life claims (Section 9.4.3).

First-pass versus final yield? First-pass is before retest/rework. Final can hide instability if retest-until-pass or undocumented rework recovers units (Section 9.3).

Sudden yield drop? Contain material; clear the tester with golden units; stratify by station, lot, and ATP row; then change process or supplier controls (Section 9.6).

ATP versus SPC? ATP ships or rejects a unit. SPC asks whether the process remains stable over time. Neither replaces qualification (Table 9.1).

Second source? Evidence beyond form/fit/function: margin, interop, manufacturing readiness, and reliability re-entry where mechanisms may change (Chapter 8 and section 9.2).

When may an ATP limit change? When the requirement, measurement capability, and escape risk are re-argued with data, not to recover yield alone. Preserve genealogy of the limit change.

Key idea. Manufacturing validation proves that a qualified design can be reproduced and protected at scale. Freeze the production reference, build representative lots, preserve genealogy, validate the measurement system, understand distributions and first-pass yield, prove ATP coverage, and establish SPC with reaction plans. Increase volume only as evidence supports greater exposure. The goal is not one successful lot; it is a production system that remains capable, traceable, measurable, and correctable.

Junior mistake: escalate a supplier before the measurement system is cleared, or treat two hand-selected lots as multi-lot evidence (Section 9.4.3 and chapters B and 11).

Three cold questions.

1. Why must the measurement system be validated before drawing process conclusions?
2. Why can a high final yield conceal substantial production risk?
3. What evidence is required before relaxing a production-test limit?

Chapter 10

AI datacenter networking

Optics only make sense once you see the fabric they sit in. Training and inference clusters are not one network; they are several overlapping networks with different bandwidth, latency, and reach needs. This chapter places the devices and validation methods from earlier chapters into that system context: how AI clusters are wired, why the interconnect sits on the inference critical path, and where optics dominates cost and power.

10.1 Scale-up versus scale-out

The industry borrowed two words from computer architecture and overloaded them for AI fabrics. Keeping them straight matters, because they buy different optics.

Scale-up tight coupling inside a node or rack: GPU-to-GPU over a memory-semantic fabric (for example NVLink). Very high bandwidth, very low latency. Increasingly optical as bandwidth outgrows copper reach.

Scale-out coupling across racks and pods over a switched network (Ethernet or InfiniBand). This is where pluggable and co-packaged optics have long lived.

10.1.1 Three networks, two that set the optics budget

The OIF CEI-448G framework names *three* distinct networks in a large AI cluster [14]. Figure 10.1 shows two; the third is easy to overlook:

Scale-up accelerator-to-accelerator inside a pod or rack. Highest bandwidth per link, lowest latency, often lossless. Dominates compute time if under- provisioned.

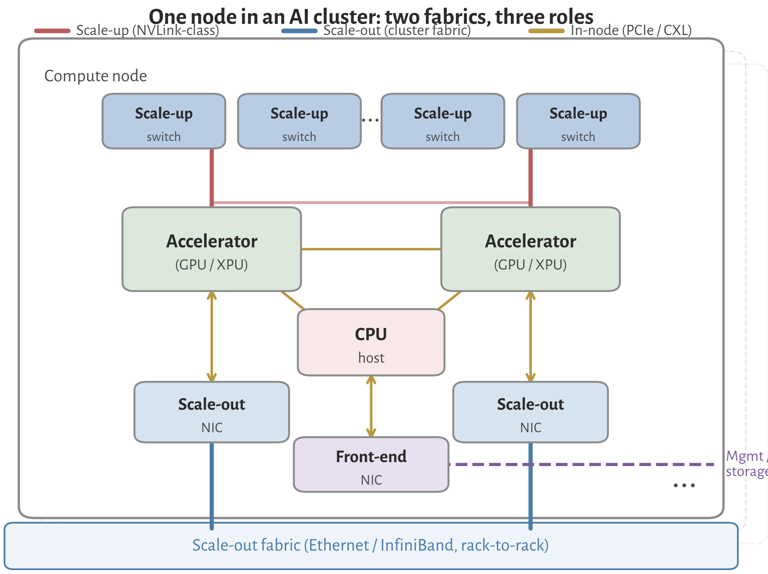

Figure 10.1: Schematic AI compute node (one tray in a larger cluster). *Accelerators* are the heavy compute engines (typically GPUs; the book also uses XPU for GPUs and custom ASICs). **Scale-up** links (red) tie accelerators through a high-bandwidth, low-latency switch fabric inside the node or rack (NVLink-class). **Scale-out** links (blue) leave via a dedicated NIC per accelerator into the datacenter Ethernet or InfiniBand fabric. The CPU and front-end NIC handle host, storage, and management traffic.

Scale-out accelerator-to-accelerator (or node-to-node) across the cluster over a switched fabric (InfiniBand, *UEC*/Ethernet with RoCE or Ultra Ethernet Transport). Lower per-link bandwidth than scale-up, but must reach 10^5+ endpoints.

UEC Ultra Ethernet Consortium: open Ethernet stack for AI/HPC scale-out (UET transport, PHY extensions).

Front-end CPU-attached traffic: management, control plane, checkpointing, storage, and training-data ingest. Analogous to a conventional cloud datacenter network; not the IM/DD bottleneck this book focuses on, but it shares the same rack power and cabling budget.

Scale-up carries the majority of *interconnect* bandwidth inside a training job; scale-out multiplies link count across the building. That is why both push 448G-class lane rates and why optics shows up first on scale-out, then on scale-up as copper reach shrinks [14].

10.1.2 Scale-up versus scale-out at a glance

Table 10.1 condenses OIF Table 1 (CEI-448G framework, §2.2): order-of- magnitude targets for node count, physical extent, and media type [14]. Numbers are industry snapshots, not hard limits, but they explain why "optics inside the rack" and "optics between racks" arrive on different timelines.

	Scale-up		Scale-out	
Metric	Today	Next gen	Today	Next gen
Accelerator nodes in domain	$\sim$100	$\sim$1 k	100 k+	$\gg$100 k
Physical extent	Rack	Rack to row	Datacenter	Datacenter
Network properties	Lossless, low latency		Large scale; multi-tier switching	
Media within rack	Electrical: passive PCB / twinax backplane		Electrical: twinax backplane	
Media between racks	AEC (adjacent racks)	Optical (within row)	Optical (pluggable or CPO on switch/NIC)	

Table 10.1. Scale-up vs. scale-out snapshots (adapted from OIF CEI-448G framework Table 1, 2025). Scale-up stays on copper as long as rack densification keeps channels short; scale-out is already optical at datacenter scale.

10.1.3 Standards bodies: who owns what

448G/lane signaling is not owned by one standards body, and that is by design. Electrical reach, Ethernet naming, connectors, and rack packaging evolved in different rooms; AI fabrics forced them to meet at the same lane rate. OIF's CEI-448G framework (§2.3) lists the groups that must align [14]. Table 10.2 maps each body to the layer an optical engineer actually touches. The short version: OIF sets the electrical baud and reach classes; IEEE names the Ethernet optical PMD; UALink and UEC own scale-up and scale-out protocol stacks; SNIA and OCP decide connectors and where the optics physically live. The *LPO MSA* is not a standards body, but it publishes the only open end-to-end spec that stitches CEI Linear electrical limits to IEEE optical PMD limits for DSP-less modules (Section 10.3.1). Prose below expands each row, OIF and non-OIF.

OIF. Common Electrical I/O (*CEI*) Implementation Agreements are the modular electrical PHY recipes: die-to-die (*XSR*), chip-to-module (*VSR*), chip-to-chip (*MR*), and backplane (*LR*). CEI-448G is the current AI-driven push beyond 224G/lane [14]. Related OIF tracks include energy-efficient interfaces (EEI), CMIS management extensions at 448G, and coherent DCI (out of scope for this book). For optics, CEI tells you the *baud and reach* the module or CPO engine must meet; IEEE 802.3 tells you how Ethernet names the optical PMD.

UALink Consortium. *UALink* specifies an open *scale-up* stack (transaction, data link, and physical layers) optimized for direct accelerator memory access, not IP/Ethernet [1][15][14]. UALink 1.0 targets 200G/lane and pods up to roughly 1 k accelerators; the physical-layer working group is gathering requirements for $\sim$400G/lane. Optical engineers care because higher lane rate reduces port count on the accelerator package, and because scale-up eventually hits the same copper

UALink Ultra Accelerator Link: open scale-up fabric for accelerator load/store (200G/lane class).

[1] UALink Consortium, UALink 200G 1.0 scale-up specification (April 2025).

Body	Fabric	What matters for short-reach optics
OIF CEI	All reaches	Electrical PHY: XSR/VSR/MR/LR, 448G PAM4/6/8, LPO/LRO; CMIS for module management
UALink	Scale-up	Open load/store pod fabric; 200G/lane today, ∼400G/lane next; drives XSR-class PHY needs
UEC	Scale-out	Ultra Ethernet Transport, congestion control; PHY group surveying 400G/lane enhancements
SNIA SFF	Host / backplane	Cables, connectors, PCIe/EDSFF form factors; SFF 448G COM for backplane (complements CEI front-panel work)
OCP	Rack / system	Open rack designs, CPO/XPO placement, optical circuit switching (OCS) for AI clusters
IEEE 802.3	Scale-out (+ optics PMD)	Ethernet MAC rates (802.3dj, 400G/lane SG), KP4 FEC, IM/DD PMD clauses (DR/FR, TDECQ)
100G Lambda MSA	Scale-out optics	Originated the 100G/λ single-mode PMDs (100G-FR/LR, 400G-DR4/FR4/LR4) later folded into IEEE 802.3; $\mathrm{RIN}_x\mathrm{OMA}$ transmitter method the DR/FR and LPO specs inherit
IBTA	Scale-out	InfiniBand Architecture; NDR 400G, XDR 800G/port at 200G/lane, 1.6T switch links; reuses QSFP/OSFP + MPO optics with Ethernet
IEEE 802.1	All fabrics	Link-layer security: MACsec (802.1AE) line-rate encryption; 802.1X port access control
PCI-SIG / CXL	Scale-up, in-node	PCIe 7.0 (128 GT/s PAM4) and CXL 4.0 memory fabric; PCI-SIG Optical Workgroup for optical PCIe
DMTF / OpenConfig / SONiC	Management	Box and fleet telemetry and config (Redfish, OpenConfig, SONiC); complement OIF CMIS module management

Table 10.2. SDO and consortium roles at 448G/lane. The top six rows follow OIF CEI-448G §2.3; the 100G Lambda MSA, IBTA, IEEE 802.1, PCI-SIG/CXL, and the management stack are added for coverage beyond OIF's own list.

reach wall as CEI XSR (Section 10.5).

Ultra Ethernet Consortium. *UEC* builds a complete Ethernet-based stack for AI/HPC *scale-out*: transport (UET), link layer, and PHY enhancements [2][16][14]. UEC 1.0 (June 2025) targets cluster-scale congestion control and RDMA-class performance over standard Ethernet hardware, including NICs, switches, optics, and cables. The PHY working group is surveying 400G/lane improvements. This is the protocol layer above the transceiver; IM/DD module specs still come from IEEE and MSAs.

[2] Ultra Ethernet Consortium, UEC Specification 1.0 (June 2025).

RDMA Remote direct memory access: zero-copy network transfer between accelerator memories; RoCE (Ethernet) or native IB.

NIC Network interface card: host adapter connecting an accelerator or CPU to the scale-out fabric (Ethernet or IB).

SNIA (SFF Technology Affiliate). SNIA's SFF working group defines connectors, cables, and form factors for storage and compute backplanes, not front-panel Ethernet [3][18][14]. The SFF 448G project (SFF-TA-1043) parallels CEI-448G but focuses on backplane COM, package insertion loss, and PAM choice on copper channels inside the box. Pair OIF VSR (module connector) with SFF (host PCB and PCIe cable) when debugging a link budget.

[3] SNIA SFF TA-1043 / OIF 448G workshop: backplane 448G scope (2025).

Open Compute Project. OCP publishes open rack, server, and networking designs for hyperscale [4][19][14]. Relevant work includes Open Systems for AI, short-reach optical interconnect guidelines, and the Optical Circuit Switching (OCS) subproject (2025) for reconfigurable cluster fabrics. OCP rarely specifies baud rate; it specifies *where* the optical engine lives (faceplate pluggable vs. CPO vs. XPO) and how it is cooled and serviced.

[4] OCP Open Systems for AI; OCS subproject for AI cluster fabrics (2025).

IEEE 802.3. Ethernet standards define MAC rates, FEC (KP4 in Clause 91), and interoperable optical PMDs [8][10][14]. 802.3dj (200G/lane) and the 400G/lane study group name the product generations that consume 224G and 448G-class optics. One naming trap: IEEE quotes **400 Gb/s per lane** (MAC/info rate); CEI quotes $\sim$**448 Gb/s** (line rate with coding overhead). Same physics, different accounting (Chapter 3 and table 3.6).

100G Lambda MSA. The single-mode PMDs this book leans on did not start at IEEE. The *100G Lambda MSA*, formed in 2017 by a broad group of suppliers and hyperscalers, wrote the first interoperable optical specifications built on one wavelength carrying 100 Gb/s PAM4 ($\approx$53 GBd): 100G-FR and 100G-LR for single-wavelength 100 GbE, and 400G-DR4/FR4/LR4 for four-lane 400 GbE over duplex single-mode fiber [5][54]. IEEE 802.3 then adopted the same 100 Gb/s-per-λ approach into its DR/FR/LR PMD clauses, and the LPO MSA 100G-DR-LPO profile (Section 10.3.1) inherits both that modulation and the RIN_xOMA transmitter method the MSA defined (Section 4.3). For a short-reach engineer this is the body behind the reach-class names on almost every single-mode datasheet: when a module is called "DR4" or "FR4," the per-wavelength recipe traces to this MSA even where the compliance point is now quoted against an IEEE clause.

[5] 100G Lambda MSA (2017): 100G/λ PAM4 single-mode PMDs (100G-FR/LR, 400G-DR4/FR4/LR4); basis for IEEE DR/FR/LR clauses and 100G-DR-LPO.

InfiniBand Trade Association. The *IBTA* maintains the InfiniBand Architecture Specification, the other dominant scale-out fabric for AI and HPC alongside Ethernet [6][17]. Data rates follow the same per-lane SerDes ladder: NDR reaches 400 Gb/s per port over four 100 Gb/s lanes, and XDR reaches 800 Gb/s per port over four 200 Gb/s lanes, with switch-to-switch links at 1.6 Tb/s (XDR figures are from the 2023 spec announcement and remain provisional). For the optical engineer the physical layer is close to the Ethernet case: the same QSFP and OSFP form factors, the same MPO fiber, and DR/FR-class IM/DD optics at matching lane rates. InfiniBand changes the transport and congestion model (credit-based flow control, RDMA), not the transceiver. It is absent from the OIF §2.3 list because IBTA runs its own specification, but it occupies the same scale-out slot as UEC.

IBTA InfiniBand Trade Association: owns the InfiniBand Architecture spec (NDR 400G, XDR 800G/port); reuses QSFP/OSFP optics.

[6] IBTA InfiniBand Architecture: NDR 400G, XDR 800G/port at 200G/lane; reuses QSFP/OSFP + MPO optics with Ethernet.

IEEE 802.1 (link-layer security). Encryption is a link-layer function, and at 800G and 1.6T it has to run at line rate. *MACsec* (IEEE 802.1AE) provides connectionless confidentiality, frame integrity, and data origin authenticity at the MAC layer, transparent to the MAC client; IEEE 802.1X handles port access control and the key agreement that establishes MACsec associations [7][12]. This lands on optics indirectly: line-rate MACsec costs latency and gate count in the switch ASIC or NIC, not in the transceiver, and an LPO or DSP-less module carries the ciphertext transparently. Encryption is therefore orthogonal to the optical PMD choice (Section 10.3.1), but it belongs in the standards map because a fabric-security requirement can still gate a module program at qualification.

[7] IEEE 802.1AE MACsec (line-rate L2 encryption) and 802.1X port access control.

PCI-SIG and CXL. Inside the node, the load-store fabric is PCI Express and *CXL*. PCI-SIG released PCIe 7.0 in 2025 at 128 GT/s per lane using PAM4 and flit encoding, and the CXL Consortium built CXL 4.0 on that same PCIe 7.0 electrical base for coherent memory pooling [8][25][9][26]. Both are 2025 releases, so treat the exact rates and the optical-interface scope as provisional. Both are copper-first and short today, but PCI-SIG runs an Optical Workgroup defining a technology-agnostic optical PCIe interface, and CXL memory disaggregation across a rack is exactly the reach that pulls optics into a bus that was never optical before. For a short-reach optics team this is the emerging in-node and in-rack case to watch: PAM4 at PCIe rates over fiber, with the same modulator and detector problems as an Ethernet lane.

CXL Compute Express Link: coherent memory interconnect built on PCIe PHY (CXL 4.0 on PCIe 7.0). Potential optical reach.

[8] PCI-SIG PCIe 7.0 (2025): 128 GT/s PAM4, flit encoding; Optical Workgroup for optical PCIe.

[9] CXL 4.0 (Nov 2025) on PCIe 7.0: coherent memory-pooling fabric; relevant to optical CXL.

Management and telemetry. Module management is OIF CMIS (Section 7.8), but the box and fleet layers above it are not OIF. *DMTF Redfish* is the RESTful standard for server and switch management and telemetry; *OpenConfig* defines vendor-neutral data models and streaming telemetry (gNMI) for network devices; and *SONiC* is the open network operating system many hyperscalers run, with its own dataplane telemetry hooks [10][87]. These are where per-module CMIS monitors surface as fleet signals: optical power, case temperature, FEC symbol-error counts, and pre-FEC BER aggregated across 10^5 links (Chapter 8 and section 8.8). A module program that ships without a telemetry contract into one of these systems is effectively invisible at fleet scale.

[10] DMTF Redfish, OpenConfig (gNMI), and SONiC: box/fleet telemetry and config above OIF CMIS.

The frontier is optics entering the scale-up domain (optical NVLink-class links, co-packaged switches), because copper reach at 200G/lane is only about a meter.

10.2 Topologies and why optics count explodes

Cluster topology is where optical count stops being "a few modules per server" and becomes a fleet problem. Large AI fabrics mostly use fat-tree / Clos, rail-optimized, or dragonfly layouts. In a classic k-ary fat-tree, link count scales as $O(k^3)$ while endpoints scale as $O(k^2)$: optics multiply faster than compute. Rail-optimized designs (one NIC rail per accelerator row, all-to-all within a rail) rose with collective-heavy training and inference because they cut oversubscription on all-reduce paths, at the price of more parallel optical planes. Dragonfly and other hierarchical topologies trade some global bisection bandwidth for fewer long links.

The optical engineer cares because every topology choice sets link count, which sets laser count, module count, and FIT budget (Chapter 8); rail layouts drive fan-out from leaf to spine and push denser 800G/1.6T ports toward CPO/XPO (Sections 10.10 and 10.11); and scale-up inside the rack stays electrical longer while scale-out between racks is already optical (Table 10.1). That explosion in link count is the economic reason a hyperscaler builds an in-house optical engineering team.

10.3 Pluggable form factors and module styles

Faceplate modules differ in aggregate rate, lane count, electrical reach, and where the DSP lives. The form factor sets the passive channel the SerDes must survive, and the last decade of AI networking has mostly been a fight over that choice: keep a retimer in the module for interop, delete it for watts (LPO/LRO), or move the optics onto the package (CPO) and service the laser from the faceplate (ELSFP).

QSFP-DD / OSFP the incumbent datacom pluggables. QSFP-DD carries 800G/1.6T class products (eight lanes at 100G or 200G per lane). OSFP targets higher power and density (1.6T today, 3.2T roadmaps) with a larger cage and better thermal path. Both impose a VSR-class electrical channel from host PCB through the connector (Section 10.5 and section 10.5.1).

Retimed module DSP/CDR inside the module (Section 3.6). Default for interoperability; ∼15–20 W class at 800G/1.6T.

LPO / LRO linear or lightly retimed: delete or slim module DSP (Sections 3.14.3 and 10.5.1 and figures 3.2 and 3.6). Power drops to ∼7–9 W but host EQ and transmitter and dispersion eye closure quaternary (TDECQ) margin tighten.

ELSFP / external laser field-replaceable CW source for CPO (Section 10.10 and chapter 5): decouples laser FIT from switch FIT.

XPO liquid-cooled mega-pluggable (Section 10.11): 12.8 Tb/s per module, front-panel serviceability with CPO-class density.

CMIS (Section 7.8) is the management layer for module identity, monitors, and (at 224G/448G) link-training and host-side signal-integrity tuning extensions. Optical engineers touch it when debugging lock, FEC, and equalizer settings across vendors.

Half-retimed and asymmetric modules. Not every pluggable is fully retimed or fully linear. Common 2025–26 variants sit between those poles:

LPO linear drive and linear receive: no module DSP/CDR (Section 3.6 and section 3.14.3).

LRO / RTLR retimed transmit, linear receive (OIF RTLR): DSP on electrical $\rightarrow$ optical only; eases host TX while keeping a simpler RX path [72].

TRO / half-retimed symmetric opposite or partial retiming variants appear in vendor roadmaps; always check which direction carries the DSP.

Fully retimed DSP both directions; default for interoperability at 800G/1.6T.

Power scales with DSP content: retimed $\sim$15–20 W, LRO $\sim$9 W, LPO $\sim$7–9 W at 800G class (Section 10.13). The architecture choice is a trade between host margin (COM, TDECQ) and module watts.

10.3.1 The LPO MSA: stitching IEEE optics to CEI Linear

OIF CEI tells you the electrical recipe at the module cage. IEEE 802.3 tells you how Ethernet names the optical PMD and its TDECQ/OMA limits. Neither document, by itself, is a complete product spec for a linear pluggable that deletes the module DSP. The *LPO MSA* (Linear Pluggable Optics Multi-Source Agreement) fills that gap: one open specification for both sides of the module, with normative test points and host responsibilities spelled out [79][72].

The first published revision, *100G-DR-LPO* (v1.0, March 2025), targets 100 Gb/s per lane at 53.125 GBd PAM4 on single-mode fiber from 0.5 m to 500 m. It is explicitly a data-center profile: low power, low latency, high port density, RS(544,514) FEC on the host, and form-factor agnostic (QSFP, QSFP-DD, OSFP are examples, not the spec). The naming pattern generalizes to `n00G–DRn–LPO` for $n \in \{1, 2, 4, 8\}$ lanes. Optical reach and modulation track IEEE DR-class PMDs; the electrical interface tracks OIF CEI-112G-LINEAR-PAM4. That split is the template 224G linear modules follow: CEI-224G-Linear on the AUI, 802.3dj optical PMD limits on the fiber (Section 3.14.2 and section 3.13).

Who owns which block. In an LPO link the host is not a passive cable driver. It runs KP4 FEC (Section 3.12), full SerDes equalization (CTLE, FFE, DFE, CDR; Sections 3.6 and 3.7), and optional nonlinear compensation (NLC) and startup protocol functions. The module is analog: linear driver, modulator or laser, photodiode, TIA, and at most a fixed CTLE. No retiming, no FEC, no heavy DSP. CMIS (Section 7.8) is the management contract. SFF hardware specs (QSFP-DD, OSFP cages) set the mechanical envelope. The MSA's job is to define what must pass at each interface between those blocks so modules from different vendors close on the same host.

Tradeoff. Retimed module vs LPO host burden

Improves: Module watts and faceplate density when DSP leaves the module

Worsens: Host SerDes margin, equalization depth, and interop risk on every platform

When acceptable: When host channels and thermal budgets are proven for the linear path

Experienced decision: Pay for retimers when host margin is the scarce resource; delete them when watts dominate and the host can own EQ and FEC.

The test-point ladder. LPO MSA normative compliance is organized around six electrical/optical test points (Table 10.3), the concrete instance of the general TP0-to-TP5 planes in Section 3.9. Think of them as the validation script: host TX at TP1a, module optical TX at TP2, stressed optical RX at TP3, module electrical RX at TP4, and stressed host RX at TP4a. Section 10 of the MSA adds a host-to-host end-to-end BER test with FEC-encoded traffic, which is how you prove interop after the point tests pass.

Point	Location	Principal measurements
TP1a	Host SerDes output	EECQ (electrical eye closure), host TX quality
TP1	Module electrical input	Module input stressor calibration
TP2	Optical TX (2–5 m patch cord)	TDECQ, TECQ, $\text{OMA}_{\text{outer}}$, RIN_{xOMA}, ER
TP3	Optical RX input	Stressed receiver calibration (SECQ), sensitivity masks
TP4	Module electrical output	Module RX linear output (EECQ)
TP4a	Stressed host input	Host RX under worst-case module output

Table 10.3: LPO MSA test points (100G-DR-LPO).

Optical limits that matter. The MSA optical tables (Section 7.4 and chapter 7) inherit IEEE 802.3 measurement methods with LPO-specific reference equalizers. The numbers you will quote in a datasheet review:

- TDECQ and TECQ capped at 3.4 dB per lane, measured with a 9-tap T-spaced reference FFE and SER target 4.0×10^{-4}.
- $\text{OMA}_{\text{outer}}$ coupled to transmitter quality: launch power in OMA rises with max(TECQ, TDECQ) along a piecewise mask (for example $-3.2 + \max(\text{TECQ}, \text{TDECQ})$ dBm in the mid-TECQ region).
- $\text{RIN}_{\text{xOMA}} \leq -138$ dB/Hz at 17.2 dB ORL (the “x” subscript tracks return loss, same convention as IEEE DR PMDs).

- Illustrative link budget 6.7 dB total at 500 m (3 dB channel loss plus 0.3 dB MPI penalty plus 3.4 dB TDECQ allocation). Do not stack that TDECQ allocation on a SECQ-stressed sensitivity that already embeds Tx quality.

- Stressed receiver sensitivity -3.1 dBm OMA_{outer} at TP3 with SECQ = 3.4 dB and aggressor lanes at 4.2 dBm OMA.

TECQ is TDECQ without the dispersion test fiber. The MSA uses both because outer OMA limits are written against max(TECQ, TDECQ), and a module can be fiber-limited even when the electrical eye looks clean.

Electrical limits and host COM. On the electrical side the MSA points to CEI-112G-LINEAR-PAM4 for the host-to-module interface and defines EECQ (electrical eye closure quaternary) at TP1a and TP4a. That is the electrical analogue of TDECQ: a host that passes optical TDECQ at TP2 but fails EECQ at TP1a still will not interoperate. Host PCB insertion loss, connector, and module input return loss sit in the channel reference model (Section 7). For 224G deployment, swap CEI-112G-LINEAR for CEI-224G-Linear and run the same two-ledger program (Sections 3.14.2 and 10.5.2).

What to read first. For a bring-up engineer the reading order is: Section 5 (system overview and host FEC requirements), Section 7 (electrical TPs and channel model), Section 8 (optical TX/RX tables), Section 9 (parameter definitions, especially 9.5 TDECQ/TECQ and 9.10 stressed RX), then Section 10 (host-to-host FEC BER). Cross-check every optical definition against the cited IEEE 802.3 clause; the MSA adds reference equalizer taps and SER targets, not new physics.

Key idea. OIF owns the electrical baud at the cage. IEEE owns the optical PMD metrics on the fiber. The LPO MSA is the product contract that binds them for DSP-less modules: normative test points, host FEC/EQ duties, and TDECQ/OMA masks you can test without a module retimer safety net.

10.3.2 The LPO supplier base and the adoption question

The MSA defines what a linear module must pass; it does not settle whether the market buys one. By 2025–26 the supplier base had formed around the analog parts that replace the DSP: high-linearity TIAs and linear drivers. Macom, Semtech, and Maxlinear are the named component proponents, and once the DSP is gone the TIA and driver become the make-or-break blocks [72].

The demonstrations track a fast rate climb. Eoptolink showed a 200G/λ four-channel LPO link with no DSP or CDR at OFC 2024 and moved a second-generation

100G/lane 800G and 400G single-mode line into volume, claiming full TP2 compliance at the transmit interface [75]. Macom exhibited its PURE DRIVE 200 Gb/s LPO parts at OFC 2024 and extended them toward 212 Gb/s/lane for a 1.6T module, with the TIA and driver as the headline [76]. Marvell, a DSP house, announced a 200G/lane TIA and laser-driver chipset for 800G and 1.6T LPO aimed at scale-up XPU fabrics [77]. Macom and Eoptolink are founding members of the LPO MSA (Section 10.3.1).

LPO REACHES INTO PCIE SCALE-UP. Alphawave and Innolight demonstrated a 64 Gb/s/lane PCIe 6.0 subsystem (controller plus PHY) over Innolight's LPO OSFP optics at OFC 2024, then a 128 Gb/s/lane platform pairing a PCIe 7.0-ready SerDes PHY with the same optics [78]. The pull is the switch-fabric pull again: larger, faster AI nodes need more PCIe reach than copper gives, and the linear module skips the DSP latency and power a retimed module would add (Sections 10.1 and 10.12).

WHETHER LPO WINS IS A SEPARATE QUESTION. Host support exists: Juniper's Broadcom-based QFX switches take LPO optics without hardware changes, and Arista has shown Broadcom TH5 compatibility for over two years [72]. The market read is still cautious. Cignal AI argues LPO stays a small share, at least at 800G, because the installed fabric was already designed around DSP-based modules; 100G/lane (800GbE) LPO looks late and likely to hold only a small slice long term[11] [74]. At 200G/lane and 1.6T the balance tips toward LRO: early 1.6T LPO parts draw more than 30 W, a thermal problem at faceplate density (Section 10.13.1), while a transmit-DSP LRO design promises under 20 W with easier integration and interop [72]. Cignal notes that every vendor showing 1.6T LPO at OFC also showed an LRO part [74]. That matches the book's technical read: LPO and LRO win where the host electrical channel and TDECQ close without the module DSP, and lose where they do not (Sections 3.14.2 and 10.5.1).

[11] Cignal AI: LPO likely a small share (esp. at 800G) given DSP-based installed base; LRO favored at 1.6T. Analyst report, provisional.

10.4 Emerging link styles

The form-factor list above is the present. The industry argument for 2025–26 is where the DSP and the laser should live next. Retimed pluggables remain the interop default: a module DSP cleans the electrical channel, at the cost of watts. LPO and LRO delete or slim that DSP so the host SerDes carries more of the EQ burden, which is attractive for AI power budgets and painful for margin and interop. CPO moves the optical engine onto the switch or XPU substrate to cut electrical reach, and usually keeps the laser external and field-replaceable (ELSFP/CW-WDM). XPO appeared in 2026 as a middle path: a much larger, liquid-cooled pluggable that keeps front-panel serviceability while pushing density toward CPO territory (Section 10.11). The rest of this chapter treats electrical reach, eye budgets, and vendor

CPO programs as consequences of that argument.

10.5 The electrical link: reach, conditioning, and the eye budget

Every form factor above is really an answer to one physical question: *how far can the electrical signal travel before it is cheaper (in power and dB) to convert to light?* As the per-lane rate climbs, that distance collapses, and it is what pushes the optics from the faceplate toward the die.

TRACE LOSS SCALES WITH FREQUENCY. A PCB stripline's insertion loss grows roughly with $\sqrt{f}$ (skin effect) plus a dielectric term linear in f, so it is quoted per inch *at the Nyquist frequency*. Going from 112G to 224G PAM4 moves Nyquist from 28 GHz to 56 GHz, and the loss per inch roughly doubles. Recent 224G board studies measure ≈ 2.8 dB/inch for regular stripline and ≈ 1.9 dB/inch with skip-layer routing at 56 GHz, against a next-generation *target* of 1 dB/inch that demands ultra-low-loss dielectric, HVLP copper, and short via stubs [68]. Figure 10.2 shows the consequence: at a fixed PCB-trace budget, doubling the baud rate roughly halves the copper reach.

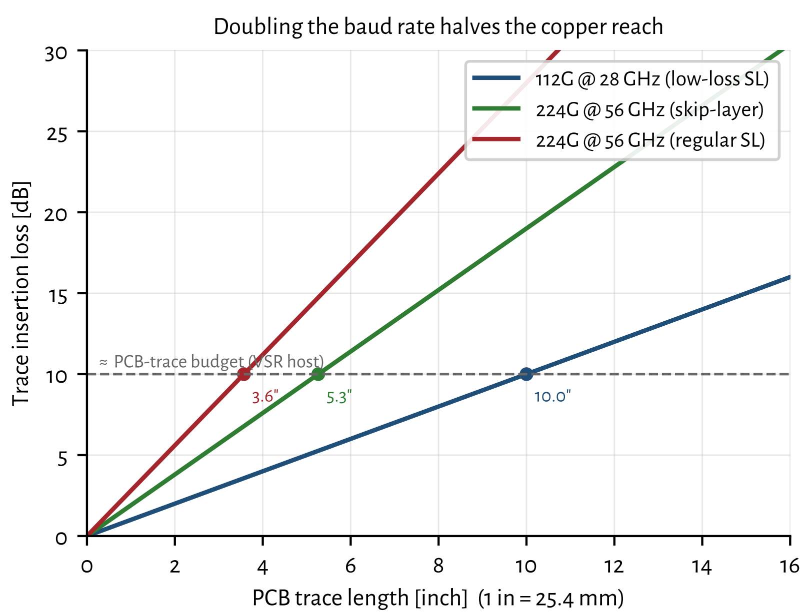

Figure 10.2: PCB trace insertion loss versus length at each rate's Nyquist. The reach to a fixed budget shrinks from ~ 10 inches (112G) to a few inches (224G), which is why the optical conversion must move closer to the ASIC.

THE CEI CHANNEL CLASSES NAME THE REACHES. OIF's Common Electrical I/O project defines the electrical link budgets the whole industry designs to. Table 3.8 is the CEI-224G lookup card (XSR / VSR / MR / LR, plus Linear for DSP-less modules); the reach map is Figure 3.5. At 56 GHz Nyquist the same names mean much shorter copper than at 112G [66]:

XSR *XSR*: die-to-die / die-to-engine: the shortest, lowest-power tier, the one CPO and chiplet optics live in ($\lesssim$50 mm package).

VSR *VSR*: chip-to-module: $\approx$200 mm of host plus 20 mm of module and one connector: the classic pluggable channel.

MR *MR*: chip-to-chip across a board: $\approx$500 mm, one connector, $\sim$32–34 dB die-to-die.

LR *LR*: backplane or copper cable: $\approx$1000 mm of host and daughter cards, two connectors, up to $\sim$40 dB die-to-die (including a 1 m cable).

Linear *CEI-224G-Linear*: same faceplate-class ports without a module DSP/CDR; the electrical foundation for LPO (Section 3.14.2 and section 10.3).

Active copper: ACC, AEC, and DAC. Passive direct-attach copper (DAC) survives only where the reach is short: at 224G a passive DAC is good for roughly 0.5–1 m (strong 224G PHYs have demonstrated 2 m). *ACC* adds redrivers (CTLE + VGA) in the cable; *AEC* adds retimers (EQ + CDR) and stretches twinax to $\sim$2.5 m at higher power (Section 10.5.1 and section 3.6) [70]. Beyond that, and increasingly *within* the rack for scale-up, the answer is optics (Table 10.1).

ACC Active copper cable: redriver (CTLE+VGA) inside cable; extends DAC reach.
AEC Active electrical cable: retimer (EQ+CDR) inside cable; longer twinax.

Co-packaged and near-package copper: copper moves inward too. Optics is not the only thing the reach wall pushes toward the package. *CPC* (co-packaged copper) mates a copper cable or connector directly onto the ASIC package substrate, and *NPC* (near-package copper) places that connector just outside the package on the socket. Both leave the host at the package edge and skip the lossy PCB run to the faceplate, where most of the trace budget in Figure 10.2 is spent. The move mirrors CPO: shorten the electrical path until the channel closes, but keep the signal in copper instead of converting it to light.

On-substrate copper has been validated at 224G-PAM4 with a stated roadmap to 448G, and compression substrate connectors now target 224 Gb/s PAM4 and beyond [12] [105]. Inside its reach, copper keeps the properties that make it the default: a long-reach 224G SerDes into a CPC cable runs near 4 pJ/bit with no electro-optic conversion, adds less latency than a retimed optical hop, and costs less per lane. Passive co-packaged copper reaches about a meter at 448G-PAM4 under optimistic connector assumptions (Section 3.14.3), enough to cover many scale-up links inside a rack or between adjacent racks.

[12] Co-packaged copper (Molex): on-substrate routing validated at 224G-PAM4, roadmap to 448G, for short scale-up runs. Marvell 224G LR SerDes $\sim$4 pJ/bit. Provisional.

Past that wall the conversion to light pays for itself. When the reach exceeds what a clean substrate channel can carry, or the port count makes copper bulk and cabling weight unmanageable, optics takes the link (Table 10.1). CPC and NPC do not remove that crossover; they move it, buying copper one more rate generation before the optics win. Read the placement ladder in Table 10.4 from the copper side and this is the same trade seen in reverse.

SO THE OPTICS MARCH INWARD. Shortening the electrical path is exactly what trades power and reach for serviceability, the through-line of this chapter. Table 10.4 lays out the ladder from faceplate to interposer.

Placement	Host electrical reach	Energy/bit	Serviceability
Pluggable (faceplate)	full VSR host run (∼200 mm) + connector	highest (>30 pJ/bit w/ DSP; less for LPO)	hot-swap at faceplate (best)
On-board optics (OBO / COBO)	shorter mid-board trace	lower	board-level (solder/socket); largely bypassed for CPO
Near-packaged (NPO)	engine beside the ASIC on substrate	lower still	module on substrate; limited
Co-packaged (CPO)	mm-scale die-to-engine (XSR)	<5 pJ/bit	soldered; ELSFP lasers field-replaceable
Optical I/O on interposer	on-die / interposer	<2 pJ/bit	not field-serviceable (emerging)

Table 10.4. The optics-placement ladder. Moving the electro-optic conversion inward cuts trace loss and energy per bit but erodes field serviceability, the central tension behind pluggables, OBO/NPO, CPO (Section 10.10), and the XPO middle ground (Section 10.11). A mid-board fiber connector can shift breakage risk from the costly engine to a cheap jumper.

ON-BOARD OPTICS: THE STEP THE INDUSTRY MOSTLY SKIPPED. OBO (standardized by the *Consortium for On-Board Optics*, COBO) moves the optical engine off the faceplate and onto the main PCB, cutting the copper run without abandoning silicon photonics [COBO / OBO–NPO–CPO]. It works, but it gives up hot-plug serviceability while only partly closing the power gap, so with CPO maturing, most hyperscalers are leapfrogging OBO/NPO straight to co-packaging, keeping serviceability via field-replaceable lasers (Section 10.10) and the XPO pluggable hedge (Section 10.11).

THE DIE-TO-DIE INTERFACE IS A STANDARD TOO. At the innermost tier (Table 10.4), the electrical hand-off between chiplets is increasingly set by *UCIe* (Universal Chiplet Interconnect Express), an open die-to-die interconnect whose 2.0 revision (August 2024) added 3D packaging and in-field manageability, with a

UCIe Universal Chiplet Interconnect Express: open die-to-die interconnect for chiplets on a package or interposer; parallel counterpart to CEI XSR.

3.0 revision roughly doubling bandwidth in 2025 [13][24]. UCIe is the parallel counterpart to CEI XSR: XSR is a serial die-to-OE link, while UCIe carries wide parallel lanes across a package or interposer. It reaches optics because co-packaged engines and optical-I/O chiplets attach to the compute die over a UCIe port, and optical-UCIe proposals aim to carry that traffic over fiber instead of a few millimeters of substrate. The chiplet-protocol and packaging detail sits outside this book's scope; the point for short reach is that the die-to-die interface, copper or optical, sets the shortest reach an optical engine must beat.

[13] UCIe Consortium: open die-to-die chiplet interconnect; 2.0 (Aug 2024) adds 3D packaging + manageability; 3.0 (2025) doubles bandwidth.

10.5.1 Reshaping, retiming, and where the DSP lives

To survive that lossy channel the signal is conditioned at several points (Section 3.6), and it helps to keep two device classes distinct.[14]

[14] Rule of thumb: a redriver reshapes, a retimer rebuilds timing. Neither fixes reflections or a bad return path.

Redriver an analog *reshaper/amplifier*: a *CTLE* plus a VGA. It sharpens and boosts the eye but has *no* clock recovery, so it cannot reset accumulated jitter; it adds almost no latency. This is the active element in active copper cables (ACC) and on host boards used to stretch a marginal trace.

Retimer adaptive EQ *plus* a *CDR* that re-samples and regenerates the data. It breaks the channel into independent jitter/loss segments, at the cost of power and per-hop latency. Retimers sit in active electrical cables (AEC) and in fully retimed modules (Broadcom's Agera is a host retimer).

Inside the optical module the traditional "cleanup" is a full DSP (retimer + FEC engine). The retimed-to-linear spectrum trades exactly cleanup for power and latency:

Fully retimed DSPs in both directions: least sensitive to *ISI*/dispersion, but highest power (∼14–18 W/module at 800G) and most latency (∼8–10 ns/hop).

LPO (linear) *LPO*: *no* DSP or CDR in either direction: only a CTLE in the TIA and a linear driver, so the host SerDes must do all the *FFE*/CTLE equalization *and* carry the FEC. Lowest power (∼7–9 W), lowest latency (<3 ns), but most ISI/dispersion-sensitive and shortest reach (<2 km) [72].

LRO / TRO (retimed TX, linear RX) DSP only on the electrical→optical path, roughly half the power, sensitivity, and latency, with easier interop.

So yes: reshapers/amplifiers (redrivers) and retimers are routinely placed on the host path, and the module's own DSP is the last line of cleanup. The entire LPO bet (Section 10.13 and section 3.14.2) is to *delete* those active stages and let the host SerDes' CTLE/FFE carry the channel, trading electrical margin for power and latency.

10.5.2 The electrical eye: acceptable voltage and noise

What must the signal actually look like where it hands off to the module? The CEI-224G reference and draft numbers set the envelope [66]. Differential swing sits roughly in the 0.36–1.05 Vppd band (reference TX near 1 V peak-to-peak differential; lossier channels use the larger swing). Transmitter *SNDR* ($\mathrm{SNR_{TX}}$) is about 33 dB: the transmitter's own noise-plus-distortion ceiling. PAM4 level uniformity needs RLM ≥ 0.95 (1.0 = perfectly even eyes). Jitter budgets are tight: random ≈0.01 *UI* rms and bounded/uncorrelated ≈0.02 UI pk. The go/no-go metric that folds those pieces together is *COM* (channel operating margin) ≥ 3 dB: a statistical SNR of the fully equalized link (Section 10.5.2).

SNDR Signal-to-noise-and-distortion ratio: transmitter quality ceiling including nonlinearity.

UI Unit interval: one symbol period ($= 1/R_{\mathrm{sym}}$). Jitter specs are in UI.

Jitter and level uniformity. At 112 GBd, 1 UI ≈8.9 ps; 0.01 UI rms jitter is ∼90 fs rms. Random jitter adds vertical eye closure; deterministic jitter (ISI, crosstalk) shows up in bathtub curves and COM. *RLM* ≥ 0.95 keeps the three PAM4 eyes even; poor RLM wastes vertical margin the same way excess TDECQ does on the optical side (Section 7.4).

Retimers reset jitter per segment; redrivers do not. That is why long ACC chains accumulate timing budget stress while AEC segments stay independent (Section 10.5 and section 10.5.1).

Here is the number that reframes "acceptable noise." After the reference receiver's equalizer (an 8-tap *DFE*), the eye at the slicer is only about 4–10 mV tall and ∼0.06 UI wide, with ∼11–14 dB of vertical eye closure, at a *pre-FEC* error rate near 10^{-4}. The channel is deliberately driven deep into ISI and closed nearly shut; *KP4* FEC (pre-FEC 2.4×10^{-4} to post-FEC 10^{-15}, Chapter 4) is what turns that into a working link. "Acceptable," then, is not a clean open eye; it is whatever keeps COM ≥ 3 dB and the pre-FEC BER under the KP4 threshold. It is also why the optics must present a clean, highly linear interface, especially for LPO: with only a few mV of margin, any added noise or nonlinearity from the driver or TIA comes straight off COM.

Channel operating margin (COM) in one page. *COM* is the electrical go/no-go statistic for CEI-class channels: after the reference transmitter, channel, and receiver (CTLE + 8-tap DFE) are applied, COM is the SNR margin at the slicer, in dB. COM ≥ 3 dB is the usual pass line (Section 10.5.2 and section 3.6).

COM connects directly to optics because the module is the last analog segment before FEC. A retimed module can absorb some host-channel sin; LPO cannot (Section 10.3 and section 3.14.3). When COM is tight, debug in this order: host TX SNDR and RLM, connector return loss, equalizer tap saturation, then module input swing and TIA noise (Section 4.5 and chapter 4).

Optical-side analogs are not identical: TDECQ scores the transmitter with a reference equalizer; SECQ stresses the receiver (Sections 7.4 and 7.5). Think of COM as the *electrical* counterpart to those optical margin tests.

Key idea. Electrical reach is the hidden clock behind form factors. Trace loss per inch roughly doubles from 112G (28 GHz) to 224G (56 GHz), collapsing copper reach to a few inches on-board and ∼1 m in cable. Each step inward (pluggable, OBO, NPO, CPO, on-interposer optical I/O) buys power and reach and spends serviceability. The signal is held together by redrivers (reshape), retimers (reclock), and DSPs (both), which LPO deletes, inside a brutal budget: ∼1 Vppd in, 33 dB TX SNDR, COM ≥ 3 dB, and a post-equalizer eye of only a few mV rescued by FEC.

10.6 The two phases of inference, revisited

Chapter 1 introduced prefill and decode; here is why they matter for the network.

Prefill compute-bound and highly parallel: crunches the whole prompt through every layer at once.

Decode memory-bandwidth-bound and autoregressive: one token at a time, streaming all weights each step. The *KV cache* (scaling with batch × context length) is a major and growing memory consumer.

Why decode is memory-bandwidth-bound. It comes down to *arithmetic intensity*, FLOPs performed per byte read from memory. Each weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ costs $d_{\text{out}} d_{\text{in}}$ parameters to read. In decode you generate one token, so every layer is a matrix–*vector* product (GEMV): about $2\, d_{\text{out}} d_{\text{in}}$ FLOPs against $2\, d_{\text{out}} d_{\text{in}}$ bytes of FP16 weights, i.e. each weight is used once and discarded:

$$\text{AI}_{\text{decode}} \approx \frac{2\, d_{\text{out}} d_{\text{in}}}{2\, d_{\text{out}} d_{\text{in}}} = 1 \text{ FLOP/byte.}$$

Prefill instead pushes *N* prompt tokens through the *same* weights at once, a matrix–matrix product (GEMM) that reuses each loaded weight *N* times, so $\text{AI}_{\text{prefill}} \approx N$ FLOP/byte. On a roofline this is decisive: an H100 delivers ∼1000 TFLOP/s FP16 on ∼3.35 TB/s of HBM, so its balance point sits at

HBM High-bandwidth memory: stacked DRAM on interposer beside the accelerator die. Memory BW sets the decode roofline.

$$\frac{1000 \times 10^{12}}{3.35 \times 10^{12}} \approx 300 \text{ FLOP/byte.}$$

Decode at ∼1 FLOP/byte runs ∼300× *below* that ridge (under 1% of peak FLOPs) so token rate is set by HBM bandwidth / model bytes, not by compute [71]. Two effects compound it: the full weight set must be streamed from HBM *per token*, and attention re-reads the KV cache every step, with traffic that grows with context length.

BATCHING COUPLES LATENCY AND THROUGHPUT. Larger batches amortize the weight-streaming cost (better throughput and energy per token) but make each individual response wait (worse latency). Squeezing both at once is a central goal of an inference platform.

SHARDING PUTS THE NETWORK ON THE CRITICAL PATH. Because frontier models (especially mixture-of-experts) are sharded across many accelerators, each token triggers collectives: all-reduce for tensor parallelism, all-to-all for expert routing, point-to-point for pipeline stages. For MoE all-to-all in particular, interconnect bandwidth and tail latency can dominate. This is the concrete reason hyperscale inference platforms balance "compute, memory, and networking."

MoE Mixture of experts: transformer architecture that routes tokens to specialist subnetworks; all-to-all collective demand.

Phase	Bottleneck	Network role
Prefill	compute	moderate (parallel)
Decode	memory bandwidth	high for sharded models (collectives)
MoE routing	interconnect	dominant (all-to-all)

Table 10.5: Where optics fits the inference bottlenecks.

10.7 Collective communication and optics demand

Once a model is sharded across many accelerators, the network stops carrying only point-to-point streams. Training and large inference jobs spend a large fraction of wall time in *collective* patterns: all-reduce for tensor-parallel layers, all-to-all for MoE expert routing, and steadier point-to-point streams for pipeline stages. Those patterns set optical requirements even when the PHY still looks like ordinary Ethernet or InfiniBand.

All-reduce tensor-parallel layers sum partial results across a group; needs high bisection bandwidth and low latency; latency sensitive at small message sizes (decode).

All-to-all MoE expert routing sends tokens to remote experts; bandwidth dominates; tail latency spikes if any link in the pod is slow (Table 10.5).

Point-to-point pipeline parallelism moves activations between stages; steady streams rather than global sync.

Optical engineering maps to these patterns indirectly. More rails and higher per-lane rate cut time spent in collectives; CPO/XPO raise faceplate bandwidth so fewer hops are needed (Sections 10.2, 10.10 and 10.11). Protocol choice (UEC vs IB) sets lossless delivery and congestion behavior (Section 10.8), but the PHY job remains the same: deliver pre-FEC BER below KP4 at the lowest pJ/bit (Sections 3.12 and 10.13). The next section names the fabric stacks that carry those collectives; the sections after that show where the optics physically sit.

Why experienced engineers chase one slow rail before rewriting the fabric?

Because synchronous collectives amplify a single straggler. Average utilization can look healthy while one optical lane sets the job tail.

Engineering heuristic. Judge the fabric by delivered step time and collective tail, not by aggregate port rate on a clean bench.

Telemetry versus complexity: same decision criteria as Sections C.14 and 7.12. Instrument ledgers that unlock contain or repair; skip vanity counters.

10.8 Fabric options

InfiniBand, Ethernet, and the Ultra Ethernet Consortium's AI-tuned Ethernet are the scale-out contenders. Momentum in 2025–26 favors open Ethernet for AI; large vertical AI stacks commonly use Broadcom Tomahawk switch silicon (Chapter 1 and section 10.10).

InfiniBand / NVLink fabric lossless, RDMA-native; NVIDIA Quantum switches; strong in closed NVIDIA stacks; CPO photonics on Quantum-X (Section 10.10).

RoCEv2 Ethernet RDMA over converged Ethernet; widely deployed; competes on congestion control and tail latency versus IB.

Ultra Ethernet (UEC) UET transport, enhanced PHY, congestion control aimed at AI collectives (Table 10.2). PHY work tracks 400G/lane class electrical and optical I/O.

Collectives (all-reduce, all-to-all for MoE) sit on this fabric (Section 10.6). The optical job is raw bandwidth and predictable latency at $\sim$200–400G per lane, not long-haul reach.

10.9 Optical circuit switching

The fabric options above are all *packet* switches: every link terminates in a switch ASIC that turns light into electrons, reads the header, and turns it back into light for the next hop. That O-E-O conversion is where much of a fabric's power and latency goes (Section 10.13), and at 10^5-plus endpoints it repeats at every tier. An *optical circuit switch* does a different job. It is a Layer-1 switch that steers light from an input fiber to an output fiber with no O-E-O conversion, so it is transparent to bit rate, modulation format, and wavelength. The same switch that passes 200 Gb/s PAM4 today passes 448G or a full WDM comb tomorrow with no change (Chapter 6 and section 3.14).

OCS Optical circuit switch: Layer-1 switch steering light with no O-E-O conversion; transparent to rate/format/wavelength. MEMS OCS: ms switching, $\sim$2 dB loss, topology reconfiguration.

Transparency is the point, and also the limit. Because an OCS never looks at a packet, it cannot buffer, arbitrate, or route per packet. It sets up a *circuit*: a fixed light path held open as long as the topology needs it. The mirrors or waveguides that steer the light reconfigure in milliseconds, far slower than a packet time, so an OCS reshapes the *topology* between jobs or around failures, not the traffic inside a job. In return it deletes a whole tier of packet switches and their pluggable optics, along with the power, cost, and FIT that tier carried (Section 10.2 and chapter 8).

The device and its parameters. Most production OCS today is free-space: a fiber-collimator array launches beams onto a two-axis MEMS mirror array, and each mirror tilts to aim its beam at the chosen output collimator. Table 10.6 lists the main technologies. For an optics engineer the parameters that matter are the ones that land in the link budget and the fleet model.

MEMS Micro-electro-mechanical systems: tilting mirror arrays in OCS; millisecond switching, $\sim$2 dB loss, 136$\times$136 radix.

Insertion loss a hop through the switch costs roughly 2 dB, straight off the optical budget (Section 7.7). Higher launch power or better receiver sensitivity has to cover it (Chapter 5 and section 4.4).

Radix how many fibers the switch connects. Production MEMS OCS runs about 136×136; circulators reuse each port in both directions and so double the effective radix (below).

Reconfiguration time milliseconds for MEMS, which fixes OCS as a topology switch. Silicon-photonic and SOA switches reach nanoseconds, but at low radix and higher loss (Table 10.6).

Crosstalk, return loss, polarization a mirror that leaks light into the wrong port is crosstalk; a reflective interface raises ORL and feeds laser RIN (Section 7.2.2 and section 4.3). Free-space paths are largely polarization-insensitive, which suits IM/DD.

What it buys at fleet scale. Google's Jupiter datacenter fabric replaced a patch-panel Clos interconnect with a layer of MEMS OCS under software-defined control, and reported roughly 30% lower capex, 41% lower power, and 3x faster fabric reconfiguration while a direct-connect topology carried the same production traffic[15] [20]. The switch, Palomar, is a 136×136 MEMS OCS with about 2 dB insertion loss and millisecond switching, and circulators realize bidirectional links through it to double the effective radix [20]. The same building block reshapes AI pods: a

[15] Poutievski *et al.*, "Jupiter Evolving," ACM SIGCOMM 2022: MEMS OCS + SDN; Palomar 136×136, $\sim$2 dB, ms switching; $\sim$30% capex, $\sim$41% power, 3x reconfiguration.

Technology	Switching	Insertion loss	Radix	Where it fits
MEMS mirror array (free-space)	ms	$\sim$1–3 dB	100s (136×136 shipping)	DC fabric and AI-pod topology reconfiguration
Piezoelectric beam steering	ms	low–moderate	10s–100s	free-space alternative to MEMS
Liquid crystal (LCoS)	ms	moderate	wavelength-selective	wavelength add/drop, WSS roles
3D robotic fiber	seconds–minutes	$\sim$0.5 dB	1000s	automated patch and provisioning, not per-job
Silicon photonic (MZI / SOA)	ns–μs	higher (integrated)	10s	fast, low-radix; research and niche

Table 10.6. OCS technologies. Free-space MEMS switches dominate AI deployments today; robotic-fiber switches trade speed for radix and very low loss; integrated photonic switches trade radix and loss for nanosecond speed.

TPU v4 pod wires 4096 accelerators through 48 OCS into a 3D torus that reconfigures per job and routes around failed racks, so a dead node becomes a topology the scheduler works around instead of a pod-wide outage [16][21]. That reconfiguration is the fabric-reliability lever Chapter 8 points at: component FIT still applies, but the fabric survives each failure by re-wiring optically rather than stalling the job.

[16] Jouppi *et al.*, TPU v4 optically reconfigurable supercomputer, ISCA 2023: 4096 chips, 48 OCS, reconfigurable 3D torus, route around failures.

What OCS asks of the transceivers. An OCS layer changes the module spec in ways this book cares about. Because the switch adds a fixed loss and is wavelength-transparent, single-fiber duplex reaches (FR, one fiber each way) fit an OCS better than parallel-fiber reaches (DR, many fibers), so an OCS deployment pulls the plant toward FR optics and toward circulators for bidirectional operation on one fiber [17][22]. The insertion loss argues for higher launch power and tighter ORL budgets, since every mated interface and mirror is a reflection the laser sees (Sections 4.3.1 and 7.2.2). Wavelength transparency means a WDM link passes through the switch unchanged, so CW-WDM and ring-based engines (Chapter 6 and section 6.6) compose with an OCS without a translation layer. None of this is exotic optics; it is the same DR/FR PMD and laser work from earlier chapters (Section 3.13 and chapter 5), specified against a channel that now includes a switch.

[17] Cignal AI, optical circuit switching market (2025): OCS technologies and transceiver impact (FR over DR, circulators). Analyst report, provisional.

Status and where it sits next to CPO. By 2025–26 OCS moved from a Google-specific technique to an industry theme, a headline topic at OFC 2026 with MEMS, piezoelectric, liquid-crystal, robotic-fiber, and silicon-photonic approaches competing on radix, loss, speed, and reliability [22]. It complements co-packaged optics rather than competing with it: CPO shortens the electrical path at the switch package (Section 10.10), while OCS removes packet-switch hops between packages and racks. A fabric can use both, CPO optics feeding an OCS layer, and the reliability question for each is the one Chapters 7 and 8 keeps returning to.

Key idea. An optical circuit switch reroutes light at Layer 1 with no O-E-O conversion, so it is transparent to rate, format, and wavelength and adds only insertion loss to the link budget. Millisecond MEMS switching makes it a topology and failure-reroute switch, not a packet switch. It buys fabric power, cost, and resilience (Google Jupiter and the TPU pods are the production proof), and it pulls the transceiver plant toward FR optics, circulators, and higher launch power. OCS and CPO are complementary bets on the same power and reliability problem.

10.10 Co-packaged optics: 2025–26 status

By 2025–26, co-packaged optics crossed from demonstrations into shipping products, pushed by the power and reliability limits of pluggables in AI scale-out. The programs converge on a common recipe: a photonic engine co-packaged with the switch ASIC, 200 Gb/s per channel, microring modulators (Section 3.14.3), *field-replaceable* lasers pulled out of the package, and TSMC COUPE packaging underneath.

10.10.1 Broadcom Tomahawk and CPO

Broadcom entered CPO earlier than most switch vendors and treated it as a product line, not a one-off demo. The current flagship is *Tomahawk 6*, a 102.4 Tb/s single-chip Ethernet switch (shipping 2025) offered with either copper or co-packaged optics, on 100G/200G SerDes.[18] The CPO variant, *TH6-Davisson*, began shipping in October 2025 as Broadcom's *third-generation* CPO switch. The public numbers sketch the architecture:

[18] Broadcom's broader AI-Ethernet stack: Tomahawk and Jericho switches, Thor NICs, Agera retimers, Sian optical DSPs, and CPO. Large hyperscale inference fabrics draw on this family (Chapter 1).

- 102.4 Tb/s optically enabled, built from sixteen 6.4 Tb/s "Davisson DR" optical engines at 200 Gb/s per channel.
- Photonic engines fabricated with *TSMC COUPE*; 64 Condor 3 nm SerDes cores (eight 212.5 Gb/s PAM4 lanes each).
- About 70% lower optical-interconnect power than pluggables.
- *Field-replaceable ELSFP laser modules*: lasers, the highest-failure component, made serviceable in the field.
- Scale-up to 512 XPUs; 100,000+ XPUs in a two-tier fabric at 200 Gb/s/link.

The lineage matters: CPO shipped on Tomahawk 4 and the second-generation *TH5-Bailly* (51.2 Tb/s), which logged "millions of hours" of reliability testing before Davisson. A fourth generation at 400 Gb/s per channel is already in development.

10.10.2 NVIDIA

NVIDIA's CPO story is the scale-up and scale-out fabric vendor converging on the same TSMC COUPE + microring recipe as Broadcom, announced as product families at GTC 2025: *Quantum-X* (InfiniBand) and *Spectrum-X* (Ethernet) Photonics switches, 200G SerDes, 1.6 Tb/s ports. Headline marketing claims include 3.5× power efficiency, 4× *fewer lasers*, and large resiliency gains versus pluggables; treat those as vendor orientation, and validate against your own FIT and power model.

- *Quantum-X InfiniBand*: 144 ports of 800G (≈115 Tb/s), liquid-cooled; available late 2025. Each package integrates eighteen silicon-photonics engines fed by 36 laser inputs through six *detachable* optical sub-assemblies.
- *Spectrum-X Ethernet*: up to 409.6 Tb/s; available in 2H 2026.
- Ecosystem: lasers from Lumentum, silicon photonics with Coherent, packaging with TSMC.

10.10.3 TSMC COUPE (the shared foundation)

COUPE (Compact Universal Photonic Engine) stacks an electronic IC on a photonic IC via SoIC-X hybrid bonding (a 6 nm EIC on a 65 nm SOI PIC), giving a low-impedance die-to-die interface. The roadmap: pluggable qualification in 2025, CoWoS-based CPO integration and *mass production in 2026*, with 800G/1.6T engines now and 3.2T/6.4T (toward 12.8 Tb/s on-package) to follow. TSMC cites the energy-per-bit trajectory from >30 pJ/bit for copper toward <5 pJ/bit for CPO on substrate and <2 pJ/bit once optical I/O moves onto the interposer (Section 10.13). The hard problems it names (wafer-level test, fiber-array-unit integration, and high-speed optical packaging assembly) are exactly the validation and manufacturing challenges of Chapters 7 to 9.

COUPE Compact Universal Photonic Engine: TSMC's SoIC-X hybrid-bonded EIC-on-PIC packaging for CPO. Mass production 2026.

> **Key idea.** The 2025–26 CPO wave shares one architecture: 200G/lane microring engines on TSMC COUPE, with field-replaceable lasers because lasers fail first. Tomahawk-class CPO at 200G/lane makes IM/DD validation and ELSFP laser reliability direct gates on how many accelerators a pod can wire together dependably.

10.11 The serviceable-density middle ground: XPO (OFC 2026)

CPO buys density and power at a cost the operator feels: the optics are soldered to the switch, so a single failed engine can mean pulling the whole line card. That tension (pluggable serviceability versus co-packaged density) defined much of *OFC 2026*, where a third path drew the most attention.

Program	Technology	Status
Broadcom TH6-Davisson	102.4 Tb/s, 200G/ch, COUPE, ELSFP lasers	3rd-gen CPO, shipping Oct 2025
Broadcom TH5-Bailly	51.2 Tb/s CPO	2nd-gen, extensively field-tested
NVIDIA Quantum-X (IB)	144×800G, MRM, COUPE, detachable lasers	available late 2025
NVIDIA Spectrum-X (Enet)	up to 409.6 Tb/s, MRM, COUPE	2H 2026
TSMC COUPE	SoIC-X EIC-on-PIC packaging	mass production 2026
Samsung (foundry)	optical engines / turnkey CPO	OE 2027, CPO 2029
Ayar Labs	TeraPHY optical I/O + SuperNova CW-WDM source	merchant scale-up optical I/O

Table 10.7. CPO programs, 2025–26.

Arista, with Coherent, Marvell, Lightmatter, and a broad partner list, launched the *XPO* (eXtra-dense Pluggable Optics) multi-source agreement. The bet is to keep the front-panel pluggable form factor (slide a module out, snap a new one in) while closing most of the density and power gap to CPO:

XPO eXtra-dense Pluggable Optics: liquid-cooled 12.8 Tb/s pluggable (Arista MSA, OFC 2026); front-panel serviceable at CPO density.

- **12.8 Tb/s per module**: 64 electrical lanes at 200 Gb/s PAM4 (with a roadmap to 400 Gb/s lanes for 25.6 Tb/s), roughly $4\times$ the density of a 1.6T-OSFP pluggable.
- **204.8 Tb/s per OCP rack unit**, from up to sixteen modules, front-panel density approaching co-packaged designs.
- **Integrated liquid-cooled cold plate** rated for 400 W+ per module, with blind-mate dripless quick-disconnects; this, not the connector, is what makes the high per-module power serviceable.
- **Universal reach and interface**: SR/DR/FR/LR plus ZR/ZR+, and fully-retimed, half-retimed, or linear (LPO/LRO) optics in one form factor.

[19]

THE BROADER OFC 2026 PICTURE. XPO landed inside a clear consensus: 1.6T transceivers went mainstream and 3.2T (400G/lane) previews appeared, with initial demos expected around 2027; CPO moved from demo to imminent, with

[19] XPO is best read as a hedge against CPO's serviceability problem. If field-replaceable ELSFP lasers (Section 10.10) prove insufficient and CPO repair economics stay painful, a liquid-cooled pluggable that reaches CPO-class density lets operators keep the failure model they already trust. The reliability and validation questions of Chapters 7 to 9 are exactly what decides which path wins.

Attribute	Retimed / LPO pluggable	XPO	CPO
Capacity	0.8–1.6 Tb/s/module	12.8 Tb/s/module	100+ Tb/s on-package
Density	baseline	$\sim$4$\times$ (204.8 Tb/s per RU)	highest
Power path	full electrical run to faceplate	short run to dense faceplate	shortest (on substrate)
Cooling	air (or LPO savings)	integrated cold plate, 400 W+	switch-package liquid cooling
Serviceability	field-replaceable (best)	field-replaceable (slide-out)	soldered; ELSFP lasers replaceable
Energy/bit	highest	intermediate	lowest (<5, then <2 pJ/bit)
Maturity	shipping	MSA launched OFC 2026	shipping (Broadcom, NVIDIA)

Table 10.8. Where XPO sits between pluggables and co-packaged optics.

new MSAs (Open CPX, "socketed CPO") blurring the pluggable/co-packaged line; and hollow-core fiber (record loss now $\sim$0.091 dB/km) advanced toward low-latency intra-datacenter use (Section 10.12.1). The through-line is the one this book opened with: rising per-lane rate forcing optics closer to the silicon and squeezing every last pJ/bit.

10.12 The latency budget

The book has a link budget in dB (Section 7.7) and, next, an energy budget in pJ/bit. It needs a third ledger. Inference puts the optical link on the critical path (Section 10.6), so you should be able to add up a link's latency the way you add up its loss and its power. The contributors fall into two groups: fixed digital costs that do not care about distance, and a propagation term that does.

THE FIXED DIGITAL COST DOMINATES A SHORT LINK. FEC is the largest single term. KP4 RS(544,514) (Section 3.12) costs roughly 20 to 100 ns to encode and again to decode, set by the codeword length and the implementation, not the fiber [20][99]. The module DSP adds another 8 to 10 ns per hop when the link is fully retimed. The analog stages, driver, modulator, photodiode, and *TIA*, together contribute well under a nanosecond of group delay and are almost never the problem. On a 10 m in-rack link the fiber itself is about 50 ns, smaller than one pass through the FEC.

[20] KP4 RS(544,514) decode latency ≈20–100 ns in practice (arXiv:2402.09364; vendor notes). Provisional.

PROPAGATION DOMINATES ONLY ONCE THE FABRIC IS LARGE. Light in standard single-mode fiber travels at roughly $c/1.47$, about 4.9 ns/m, because the glass has a group index near 1.47. A 100 m row-scale run is then about 490 ns,

Contributor	Typical latency
PCS framing / serialization	a few ns
FEC encode + decode (KP4)	∼20–100 ns each
Module DSP / retimer (per hop)	∼8–10 ns (fully retimed)
LPO (no DSP)	<3 ns
Driver, modulator, PD, TIA	sub-ns
Fiber propagation (silica)	∼4.9 ns/m
hollow-core fiber	∼3.3 ns/m
Switch hop (O-E-O)	∼100–560 ns

Table 10.9: Approximate one-way latency contributors, 200G/lane class.

comparable to a switch hop and larger than the digital terms. Each switched tier adds an O-E-O conversion: a cut-through Ethernet switch forwards in a few hundred nanoseconds (∼560 ns for an 800GbE class device), while InfiniBand reaches under 100 ns per hop [21] [100]. Across a multi-tier scale-out fabric, the switch hops and their conversions, not the fiber, set the tail latency that stalls a collective (Section 10.6). This is the latency argument for optical circuit switching (Section 10.9) and for co-packaging: both remove conversions and electrical runs from the path.

[21] Per-hop switch latency: ∼560 ns for an 800GbE SONiC switch; sub-100 ns for InfiniBand XDR. Vendor figures, provisional.

LATENCY IS WHERE THE MODULE ARCHITECTURE SHOWS UP AGAIN. The retimed-to-linear choice you met in Section 10.13 for power repeats here. A fully retimed module spends ∼8–10 ns per hop; *LRO* roughly halves that; *LPO* deletes the module DSP and lands under 3 ns [72]. On a link with a few hops, deleting the DSP saves more time than shortening the fiber does. Latency and energy point the same way, which is why LPO and CPO are attractive for the scale-up domain where both matter most.

10.12.1 Hollow-core fiber

The one distance term you can attack is the group index. Hollow-core fiber (*HCF*) guides light mostly through air rather than glass, so its group index sits near 1.0 and light travels close to vacuum speed. A double nested antiresonant nodeless

HCF Hollow-core fiber: guides light in air ($n \approx 1$); 33% lower latency than silica SMF. Loss record 0.091 dB/km (2025).

fiber (*DNANF*) design reported 0.091 dB/km at 1550 nm, below the $\sim$0.14 dB/km floor that silica has held since the 1980s, and about 0.2 dB/km across a 66 THz window [22] [101]. The latency payoff is the headline for short reach: propagation drops from $\sim$4.9 to $\sim$3.3 ns/m, roughly 45% faster or about a third lower latency, which on a 100 m run saves $\sim$150 ns, the size of a switch hop. Microsoft reports deploying cabled HCF in the Azure network.

[22] Chen *et al.*, DNANF hollow-core fiber, Nature Photonics 2025: 0.091 dB/km at 1550 nm, $\sim$45% faster than silica. Record, provisional.

For short-reach IM/DD the low nonlinearity and near-zero dispersion of air guiding matter less than they do for coherent long-haul; the draw here is purely latency and, secondarily, the wider low-loss window for more wavelengths. The open problems are practical: low-loss splicing and connectorization to solid-core plant, yield at volume, and cost. These figures are recent records, so treat them as provisional.

10.13 Energy per bit and the power wall

Frontier AI is *power-limited*: a site's usable megawatts, not just capital, caps how much compute it can host. Every watt spent moving bits is a watt not spent on compute, and interconnect energy multiplies across a fabric with millions of links, which is why energy per bit (*pJ/bit*) has become a headline design metric.

The trajectory that CPO is chasing, per TSMC's COUPE disclosures:

Table 10.10: Approximate interconnect energy per bit.

Link style	Energy per bit
Conventional copper / retimed	>30 pJ/bit
Co-packaged optics on substrate	<5 pJ/bit
Optical I/O on the interposer	<2 pJ/bit

This is the quantitative reason CPO exists: removing the power-hungry electrical run between a switch ASIC and a front-panel pluggable (and the module's retiming DSP) is roughly a 70% cut in optical-interconnect power in Broadcom's Davisson (Section 10.10). *LPO/LRO* attacks the same target from the pluggable side by deleting the DSP; CPO attacks it by shortening the electrical path.

WHY IT COMPOUNDS. Multiply even a few pJ/bit by aggregate fabric bandwidth and link count and interconnect becomes a meaningful slice of cluster power. At a fabric moving petabits per second, a 5 pJ/bit versus 30 pJ/bit choice is megawatts, directly setting how many accelerators fit under a fixed power envelope. Laser wall-plug efficiency feeds the same budget: fewer, more efficient lasers (NVIDIA claims 4$\times$ fewer) cut both power and failure count at once.

THE MACRO TREND BEHIND THE METRIC. The pJ/bit fight has grid-scale stakes. US data centers drew about 176 TWh in 2023, roughly 4.4% of national electricity, up from 58 TWh in 2014; the Lawrence Berkeley National Laboratory projects 325 to 580 TWh, or 6.7 to 12% of US electricity, by 2028 [23][73]. Interconnect is a growing share of that draw as lane rates climb [72]. Every pJ/bit an LPO or CPO link removes is multiplied across that base, so interconnect power reads as an infrastructure problem, not only a per-module one.

[23] LBNL (2024): US data centers 176 TWh in 2023 (∼4.4% of US electricity), up from 58 TWh (2014), projected 325–580 TWh (6.7–12%) by 2028.

10.13.1 The thermal envelope

Every watt from the budget above turns into heat that has to leave the box, so interconnect power is also a cooling problem, and cooling sets a second ceiling beside the power wall. On a faceplate switch the optics are a large part of that heat: a 32-port 800G switch dissipates on the order of a kilowatt, and the pluggable modules account for roughly half of it [24][106]. Air cooling loses headroom as per-module power climbs past the mid-teens of watts, which is why dense high-rate switches are moving to liquid and immersion cooling, and why the LPO power cut (from ∼14–18 W to ∼8 W, Section 10.13) reads as a thermal cut as much as an electrical one [72].

[24] FiberMall (2026): 32-port 800G switch >1 kW, optics ∼50% of thermal load; power 15–25% over rated. Vendor, provisional.

CO-PACKAGING CHANGES THE SHAPE OF THE PROBLEM. Moving the optics onto the switch substrate (Section 10.10) puts heat-sensitive optical engines a few millimeters from a high-power ASIC. Absolute temperature still matters, but the steep on-package gradient becomes the performance-limiting term: rings drift off resonance and lock loops fight neighbor heaters (Section 6.5), and the laser is the least tolerant part of all [25][107]. This is the thermal half of the argument for external lasers (Section 5.14): holding the laser off the hot interposer at a controlled temperature protects both its wavelength and its life.

[25] Cao *et al.*, Front. Optoelectron. (2025): liquid-cooling thermal management for high-bandwidth CPO; power density and thermal crosstalk limit reliability.

COOLING IS A RELIABILITY LEVER, NOT ONLY A POWER ONE. Laser wear-out follows Arrhenius kinetics (Section 5.13): the acceleration factor is exponential in inverse junction temperature, so a few degrees of cooling buys a measurable drop in FIT and, across 10^5-plus lasers, fewer failures per day (Section 5.13). Power, cooling, and reliability are one constraint seen three ways. The link that fits under a fixed power and cooling envelope, and stays cool enough to last, is the one that scales.

Key idea. Energy per bit is a first-order lever on cluster size under a fixed power budget. The industry path, retimed pluggable (>30 pJ/bit) to LPO to CPO (<5, then <2 pJ/bit), is why "balance compute, memory, and networking" (Chapter 1) is a power statement as much as a performance one.

10.14 A first-order cost model

The book quantifies two of its three themes. Power has a ledger in pJ/bit (Section 10.13); reliability has one in FIT (Section 5.13). Cost is invoked everywhere but never counted. It deserves the same first-order treatment, kept deliberately relative and order-of-magnitude. What follows is an illustrative model, not a price sheet: absolute module prices move too fast and vary too much by volume to write down usefully, so the numbers here are assumptions you should replace with your own.

Total cost of ownership (TCO) for an optical link splits into three buckets. The first is acquisition, the *bill of materials* (BOM) and the yield of optical assembly and test: laser count, whether a DSP die is present, and the packaging and coupling steps that dominate transceiver cost. The second is lifetime energy, the module's power drawn over years and multiplied by a cooling overhead. The third is service: the failures per day implied by the fleet FIT (Section 5.13), each one costing a replacement part and a hands-on visit. Acquisition is capital; the other two are recurring.

TCO Total cost of ownership: acquisition (BOM, yield) plus lifetime energy plus service (FIT $\times$ replacement cost).

BOM Bill of materials: component and assembly cost of a module or system. DSP presence is a large BOM driver.

Energy is the bucket you can compute. Take an 800G module drawing $\sim$15 W fully retimed against $\sim$8 W for LPO (Section 10.13) [72]. Over a five-year life at $0.10/kWh, with a *power usage effectiveness* (PUE) of $\sim$1.3 to cover cooling, the retimed module burns about 850 kWh, near $85 of electricity; the LPO module about 460 kWh, near $46. The $\sim$$40 gap per module is a meaningful fraction of what the module itself costs, and it recurs every life cycle. Scaled up, the number stops being small: a vendor estimate puts the LPO saving on a 500,000-accelerator cluster on the order of 100 MW and roughly $100 million a year in electricity [26] [108]. Treat that figure as vendor orientation, but the order of magnitude is the point.

[26] Semtech (2025): LPO $\sim$50–60% less power than retimed; 500,000-GPU cluster $\sim$100 MW and $\sim$$100M/year electricity saving. Vendor, provisional.

Recurring cost is a large share of the total. A gigawatt-class AI site has been quoted near $38 billion of up-front capital and roughly $0.9 billion a year to run [27] [109]. Over a multi-year life the running cost is a real fraction of the capital, and interconnect power is a growing slice of it, so a pJ/bit cut reads directly as dollars saved. That estimate is an analyst breakdown, provisional, but it sets the scale: power and cost are the same argument seen through two units.

[27] Epoch AI (2025): 1 GW AI site $\sim$$38B up-front capex, $\sim$$0.9B/year opex. Analyst breakdown, provisional.

The architecture choice moves all three buckets at once. Per delivered Gb/s, a fully retimed pluggable carries the highest BOM (a DSP die and its packaging) and the highest power. LPO and LRO delete or relax that DSP, cutting both the BOM and $\sim$40–50% of the power [72], and move the cost that remains into host validation risk rather than the module. CPO and XPO remove the faceplate connector and the long electrical run and push energy under 5 pJ/bit (Table 10.10), but raise assembly and test cost and couple the optics to an expensive switch ASIC, which is why the lasers are field-replaceable: a laser failure must not scrap the package (Section 10.10). Co-packaged and near-package copper is cheaper still per Gb/s wherever it reaches, carrying only the connector and the host SerDes energy (Section 10.5); the crossover to optics is set by the reach wall, not by cost. At the fabric level, optical circuit switching is itself a cost lever: replacing a tier of O-E-O packet switches with a MEMS OCS cut capex ∼30% and power ∼41% in Google's Jupiter (Section 10.9) [20].

Key idea. Cost tracks power and reliability, not against them. The cheapest link per Gb/s is the one you do not light (copper within reach), then the one with the fewest active stages (LPO, then CPO), weighed against the validation risk and the failure blast radius each one adds. A pJ/bit saved is dollars saved every year the fabric runs, which is why the energy budget and the cost model point the same way.

10.15 Synthesis: from workload to fleet

This chapter sits at the end of the book because it joins the decisions from every earlier chapter into one system. The path below is not a waterfall; teams iterate. But each step produces evidence that either advances the design or sends it back.

```
Workload / collectives
  |
Reach / topology
  |
Placement (pluggable / LPO / CPO / CPC)
  |
Laser / modulator / WDM
  |
Budgets (optical / electrical / energy / thermal)
  |
Validation ladder
  |
Supplier / ATP / SPC
  |
Fleet telemetry + corrective action
```

1. **Workload and collective requirements.** What traffic pattern, latency target, and tail tolerance does the job demand? (Chapter 1 and sections 10.6 and 10.7)

2. **Reach and topology.** What physical extent, link count, and oversubscription does the cluster need? Does copper close, or does optics take the link? (Sections 10.2 and 10.5 and table 10.1)

3. **Electrical versus optical placement.** Pluggable, LPO, CPO, XPO, or co-packaged copper? The answer follows from reach, power envelope, and serviceability. (Table 10.4 and sections 10.5, 10.10 and 10.11)

4. **Laser, modulation, and WDM selection.** DFB, EML, CW+Si MZM, CW+ring, or CW+TFLN? Single-wavelength DR or dense WDM with locking? The choice sets

the ATP, the supplier base, and the fleet FIT model. (Chapters 5 and 6 and tables 3.12 and 5.1)

5. **Link, power, and thermal budgets.** Close the optical ledger (OMA to sensitivity with penalties), the electrical ledger (COM), the energy ledger (pJ/bit), and the thermal envelope together. (Sections 7.7 and 10.13 and sections 10.5.2 and 10.13.1)

6. **Noise, sensitivity, and receiver design.** Match photodiode, TIA, and equalization to the power the laser and channel deliver. Budget RIN, shot, and thermal noise against the pre-FEC BER target. (Chapter 4 and sections 4.3 to 4.5)

7. **Validation evidence.** Walk the ladder: bring-up, characterization, margin corners, stress qualification, and production readiness. Name the instrument and reference plane for every number. (Chapter 7 and sections 7.1 and 7.9)

8. **Yield and supplier readiness.** Multi-lot yield, SPC, ATP coverage, NPI gates, first-article, and 8D discipline. Prove the part can be built at volume before committing the fleet. (Chapter 9 and sections 9.2 and 9.3)

9. **Fleet telemetry and corrective action.** CMIS monitors, FEC histograms, triage buckets, RMA codes, and the feedback loop from field to ATP. The system is not done until the fleet can detect, classify, and correct a failure without the design team. (Sections 7.12 and 8.8)

A design review should be able to point at evidence for every step. Where evidence is missing, the step is not done. Where two steps conflict (power budget versus serviceability, yield versus guardband), the trade must be stated and owned, not hidden behind a single-number spec.

10.16 Engineering lens

10.16.1 How it works

An AI fabric is judged by delivered workload time, not port rate, and the optical layer sits on that critical path. The chapter places every earlier device into scale-up and scale-out networks and shows where copper ends and optics wins the link.

10.16.2 How it is measured

Measure the network at workload, fabric, link, and module layers. At workload level, record step time, collective latency, accelerator idle time, and tail behavior. At fabric level, record delivered bandwidth, queue occupancy, route balance, retries, and failed links. At link level, record pre-FEC BER, FEC error distributions, equalizer state, and flaps. At module level, record optical power, temperature, wavelength or lock state, and electrical power. A peak-rate test does not replace an all-reduce or all-to-all run on the intended topology (Sections 7.12 and 10.7).

10.16.3 How it fails

Scaling can fail through oversubscription, poor route balance, head-of-line blocking, a straggling link, excess retries, switch-radix limits, fiber-count and service errors, or a power and cooling ceiling. Architecture adds distinct escape paths: pluggables can run out of faceplate power, linear optics can expose host margin, and co-packaged optics can turn package thermals, lock control, or fiber service into the limiting risk. A working link can still be a bad system choice if it cannot be monitored, replaced, or operated at fleet scale.

Failure mode: Collective slowdown

Symptoms. Accelerator count rises, but job throughput flattens or falls and tail step time grows.
Likely causes. Fabric oversubscription, uneven routing, one degraded rail, retransmits, switch congestion, or synchronization amplifying a straggler.
Measurements. Collective traces, route and queue telemetry, per-link FEC and retry counters, topology map, and a scale sweep.
Mitigations. Repair weak links, rebalance routes, remove oversubscription, change the collective or topology, and gate the next scale point on delivered workload performance.

10.16.4 How it is debugged

Start with the workload symptom and identify the slow collective, rail, or time window. Compare topology and route data with link counters. A single lane with rising FEC points to the optical path; many clean links with full queues point to fabric capacity or scheduling. On the optical path, use the power-versus-quality fork and five ledgers (Sections D.4, 4.8 and 5.19). Remove one rail, reroute one group, or replace one suspect module to test causality. Keep optics, switch, and workload timestamps aligned. Otherwise a link flap and a collective stall cannot be ordered reliably.

Engineering heuristic. Align timestamps across optics, switch, and workload before arguing causality. Without a common clock story, every flap looks like every stall.

Debug story

Observed. All-reduce tail latency rose after a rack expansion, while average link utilization looked normal.

Investigation. Per-rail traces showed one path with FEC bursts and retries. A module swap moved the symptom with the module.

Finding. The fabric had capacity, but one marginal optical lane set the collective tail.

Root cause. A contaminated connector raised ORL and produced burst errors without a large average-power change.

Resolution. The connector was replaced, inspection was added to the expansion runbook, and collective-tail alarms were tied to link-level error bursts.

10.17 Interview takeaway

Key idea. An AI fabric is judged by delivered workload time, not aggregate port rate. Connect collective traces to queue, route, FEC, optical, thermal, and service data. Choose pluggables, linear optics, or co-packaging by the measured system constraint and by how the fleet detects, contains, and repairs each failure.

Junior mistake: rewrite the topology before scoping one weak rail, or ignore optics FA when collectives stall (Section 4.8 and chapters D, F and 11).

Three questions to test yourself.

1. What is the difference between scale-up and scale-out, and why do they use different optics?
2. A single lane has rising FEC errors but its Rx power is stable. Is this a power problem or a signal-quality problem?
3. How do pluggable, linear pluggable, and co-packaged choices trade host margin, power, cooling, fiber count, and repair time?

Chapter 11

Failure analysis handbook

Read first: incident workflow; power loss; BER waterfall shift versus floor; eye/TDECQ; lane imbalance; thermal versus aging; yield drop.

Deep dive: BER-floor RIN/MPI separation; contamination inspect-before-clean procedure; aging-versus-thermal router.

Reference: FA checklist table; FA output categories.

This chapter is a symptom-first field guide. Start with what the bench, production line, or fleet reports, then run one incident path every time.

Preserve the failing state before reseat, reboot, or clean. Those actions often destroy the only evidence that separates contact, firmware, and true wear (Section D.16). Next name the population: unit, lane, lot, vendor, site, or fleet. Contain when the population can grow; a perfect mechanism story does not unship yesterday's lot. Classify time behavior (sudden, gradual, intermittent) and which margin ledger moved first (power, noise, timing, spectrum, or control). Choose one measurement that can kill or promote the leading hypotheses at the access you have (black-box versus engineering; Sections A.2 and D.11). Confirm ownership with a controlled swap, stress, or physical evidence before you call a mechanism confirmed. Do not say "data proves" for the last surviving hypothesis alone.

Recurrence-control closure. An incident is not closed when the unit recovers. Close only when a production or fleet control catches the same signature next time: requirement, qualification plan, ATP, SPC, telemetry, or supplier process (Section D.3 and section 7.1.12). Each symptom section ends with a short *Recurrence control* line for that signature. Name the FA output category when you file the case (Section 11.13).

Why experienced engineers preserve state before reseating?

Because reseat, reboot, and clean often destroy the only evidence that separates contact, firmware, and true wear. Scope without a snapshot is theater.

Engineering heuristic. If two explanations fit equally well, prefer the one that requires the fewest independent failures.

```
Preserve failing state
   |
Scope (unit / lot / vendor / site / fleet)
   |
Time behavior + first-moving ledger
   |
Measure / isolate (name access)
   |
Contain / confirm
   |
Recurrence control + FA output category
```

The debugging pyramid in Section 1.16, the power-versus-signal fork in Section 4.8, the fleet router in Table 7.6, and the wall-chart trees in Chapter D are the same method at different scales. Earlier chapters own mechanism physics. This chapter owns order of operations. Symptom routes:

Power loss Split launch-power loss from coupling, connector, MUX, fiber, monitor, and receiver-plane errors (Section 11.1).

BER increase Named-plane waterfall first; classify shift versus floor, then power, eye, noise, timing, and spectrum (Section 11.2 and section 11.2.1).

Eye closure or low ER Split bandwidth, drive, bias, reflection, dispersion, and resonance alignment (Sections 11.3 and 11.6).

Lane imbalance Split source, modulator or ring, filter or MUX, fiber attach, driver, and receiver (Section 11.4).

Temperature sensitivity Track power, wavelength, lock error, TEC or heater headroom, receiver margin, and package stress (Section 11.12).

Wavelength drift Split source movement from filter, ring, TEC, and control-loop movement (Section 11.5).

Intermittent bursts Preserve counters before reseating. Check connector contamination, ORL, supply noise, lock state, and weak attach (Section 11.8).

Yield drop Clear the tester, then split lot, site, process step, assembly, firmware, and calibration (Section 11.10).

11.1 Power loss

```
Power loss
   |
External meter vs CMIS
   |
Plane walk (source -> coupling -> MUX -> connector -> Rx)
   |
LIV at failing T
   |
Decision: source / path / monitor
   |
Recurrence: earliest ATP power check
```

Observed behavior. Received power or OMA falls at one or more lanes. BER may remain stable at first, then rise as receiver margin is consumed. The module monitor and an external power meter may disagree.

Likely hypotheses. Launch power can fall because the laser is disabled, thermally rolled over, or aged. Power can disappear after the source through modulator loss, coupling shift, MUX loss, fiber bend, connector contamination, or a wrong reference-plane calibration. A drifting monitor photodiode can report loss that does not exist.

Measurements, mechanism isolation, and confirmation.

1. Compare CMIS Tx and Rx power, external power at the faceplate, bias current, and case temperature. Disagreement identifies the first suspect plane.
2. Walk optical power plane by plane with a known source and calibrated meter. Do not change bias or equalization while locating the loss.
3. Rerun LIV at the failing temperature. A moved LIV raises $P(\text{source})$; a stable LIV raises $P(\text{path or monitor})$ until confirmed. Do not treat either as a confirmed mechanism before controlled confirmation.
4. Preserve telemetry, then inspect connectors before cleaning (Section 11.9). Measure insertion loss and ORL. Use a golden fiber and module swap to separate field plant from module.
5. For a weak lane, compare sibling lanes and per-lane coupling. Lot or lane clustering points toward assembly or MUX variation.

Corrective action and recurrence control. Repair the first plane where power diverges. Correct calibration or monitor coefficients before changing source bias. Add a power check at the earliest production plane that can catch the signature and retain golden-path baselines for fleet comparison. *Recurrence control:* ATP or sample power at that plane; golden-path baselines.

11.2 BER increase: waterfall shift or floor

Interview path: name the optical plane, sweep a waterfall at that plane, then classify shift versus floor before picking instruments. One operating-point BER is not a classification.

```
BER up
  |
Name plane
  |
Waterfall (BER vs power)
  |-- power not held --> power path (\Cref{sec:fm-power-loss})
  |-- shift --> sensitivity / OMA / IL / eye / timing
  |-- floor --> \Cref{sec:fm-ber-floor}
  |
Decision + ATP / telemetry control
```

Observed behavior. Pre-FEC BER rises. A waterfall is BER versus received power from a VOA sweep at a named reference plane. The curve either moves to higher power (shift), stops improving at a horizontal asymptote (floor), or shows both (soft floor on a shift).

Likely hypotheses. A shifted waterfall points toward lost power, receiver sensitivity, eye closure, timing, or dispersion: the link still responds to more photons, but the power needed for a given BER has moved. A floor points toward signal-proportional noise, reflection, crosstalk, or bursty impairments: past a point, more power does not help. The distinction prevents a power fix on a noise-limited link. FEC error timing further splits the cases: randomly sprinkled errors fit Gaussian or steady RIN; clustered bursts fit MPI, connector intermittents, or shared supply and clock events.

Measurements, mechanism isolation, and confirmation.

1. Name the plane, then sweep received power and plot BER. Confirm shift, floor, or both.
2. If the waterfall shifted, compare received OMA, TDECQ, receiver sensitivity, equalizer taps, wavelength, and temperature with the golden baseline. Golden-swap Tx and Rx to assign ownership.
3. If the curve floors, continue in Section 11.2.1 and split intrinsic RIN, electrical noise, ORL, MPI, and crosstalk.

4. Use FEC error timing and lane correlation to separate random noise from bursts and shared disturbances. Preserve counters before reseating if the pattern is intermittent.

Corrective action and recurrence control. Restore the margin ledger that moved, then repeat the full BER sweep at loaded corners. Store waterfall shape, not only pass/fail BER, so later fleet changes can be classified without guessing. Interview study treatment: Section A.8.9. *Recurrence control:* waterfall or sensitivity sample in ATP/telemetry, not pass/fail BER alone.

11.2.1 BER floor

Observed behavior. Pre-FEC BER improves as you increase transmit or received power, then stops improving and flattens at a constant floor regardless of how much more power you add. The FEC histogram may still look random (steady RIN) or bursty (MPI, intermittents); the floor shape alone does not decide that split.

Likely hypotheses. A BER floor means additional received power no longer removes the dominant impairment. That is a diagnostic pattern, not one mechanism. RIN can create a floor when signal-proportional intensity noise dominates the receiver budget: $\sigma_{\mathrm{RIN}} \propto I$, so Q can saturate at $Q_{\mathrm{max}} = 1/\sqrt{\mathrm{RIN}_{\mathrm{lin}} \cdot \mathrm{BW}}$ under a dominant-RIN model (Section 4.3). Do not define every floor as RIN-limited. Other leading mechanisms include multipath interference (MPI), bias-rail noise that converts to equivalent intensity noise (Section 5.8), pattern-dependent distortion or residual ISI, crosstalk, timing or CDR limits, and DSP or equalization limits.

MPI may produce deterministic interference, power-independent floors, pattern sensitivity, environmental sensitivity, or time-correlated errors depending on coherence, path delay, motion, and modulation. Useful evidence includes ORL dependence, delay-related structure on an ESA, aggressor dependence, pattern dependence, thermal or mechanical sensitivity, and FEC error timing. Do not treat "bursty FEC histogram" alone as proof of MPI.

Required access.

- **Black-box / bookended:** attenuation sweep, pre-FEC BER and FEC timing, Tx/Rx power telemetry, temperature, host/module swap, lane remap.
- **Engineering access:** intrinsic RIN, product-board RIN, $\mathrm{RIN}_x\mathrm{OMA}$ at a named ORL, optical eye, controlled reflector, ESA, optical breakout.

Measurements and isolation.

1. Confirm the floor exists: sweep received power (or Tx OMA) at a named reference plane and plot BER vs. power. A floor appears as a horizontal asymptote.

2. Where engineering access exists, bisect optical vs. electrical RIN: quiet SMU (intrinsic) versus product bias board. If the floor moves, the electrical path is injecting noise (Section 5.8).

3. Sweep ORL with a controlled reflector. If the floor worsens with lower ORL, the path is feedback-sensitive. Check isolator, connector, and fiber-attach cleanliness.

4. Gather MPI evidence (ORL, delay structure, pattern, thermal/mechanical, FEC timing). Treat MPI as a leading mechanism until confirmed.

5. Compare measured RIN_xOMA at the stated ORL against the named PMD revision and ATP limit (for example -136 dB/Hz at 17.1 dB ORL for a cited DR-class clause [100G Lambda MSA / IEEE 802.3]). State the plane and condition.

Corrective action and recurrence control. If intrinsic RIN is confirmed as the limiter, replace or derate. If electrical RIN, fix the bias supply. If ORL-driven, clean or replace connectors and verify isolator function. If MPI is confirmed from multiple reflections, reduce mated interfaces or improve their ORL. Update the earliest economical control (ATP, sampled audit, SPC, or telemetry), not every deep FA measurement. *Recurrence control:* RIN@ORL sample or ORL audit at the stated plane.

11.3 Low extinction ratio

Observed behavior. Transmitter OMA looks low on the DCA even though average power is in range. TDECQ may or may not fail depending on how the reference equalizer compensates.

Likely hypotheses. Linear extinction ratio is $\text{ER}_{\text{lin}} = P_1/P_0$ (for PAM4 outer levels use P_3/P_0). In decibels, $\text{ER}_{\text{dB}} = 10\log_{10}(\text{ER}_{\text{lin}})$ and $\text{ER}_{\text{lin}} = 10^{\text{ER}_{\text{dB}}/10}$. Low ER means the off level is too high or the on level is too low. In an EML, ER is set by EAM reverse bias; in a DML, by modulation depth relative to threshold.

An idealized receiver OMA penalty for finite ER is

$$\text{PP}_{\text{dB}} = 10\log_{10}\left(\frac{\text{ER}_{\text{lin}}+1}{\text{ER}_{\text{lin}}-1}\right).$$

At 10 dB ER the penalty is $\sim$0.87 dB; at 6 dB ER it rises to $\sim$2.2 dB (Section 4.4). This is an idealized receiver-penalty model, not a measured compliance quantity such as TDECQ.

Measurements, mechanism isolation, and confirmation.

1. Measure ER on the DCA (outer OMA / average power, or directly from the histogram levels). Compare against the PMD limit.
2. **EML:** Sweep EAM bias. ER should peak at the optimal bias point; if the curve has shifted (aged EAM), the operating point needs recalibration. Check EAM bias DAC code vs. datasheet.
3. **DML:** Check bias current vs. LIV. If bias is close to threshold, modulation depth is limited. Increase bias (but watch thermal rollover and RIN).
4. **MZM:** Check quadrature bias. If the MZM has drifted off quadrature ($V_\pi/2$ point), extinction degrades. Log the bias-control loop error signal; a saturated loop indicates drift beyond correction range.
5. **Ring:** Check resonance alignment. If the ring is detuned from the laser wavelength, extinction drops. Monitor the thermal tuner current and wavelength-lock error.

Corrective action and recurrence control. Recalibrate the modulator operating point. For EML aging, update the EAM bias setpoint in firmware or flag the module for replacement if the absorption curve has shifted beyond the correctable range. For MZM drift, verify the bias controller and its monitor PD. For rings, retune or check for neighbor thermal crosstalk (Section 6.5). *Recurrence control:* ER or modulator-bias check in ATP or cal audit.

11.4 Lane imbalance

Observed behavior. In a multi-lane module (DR4, FR4, DR8), one or more lanes show significantly different OMA, TDECQ, or pre-FEC BER compared with siblings in the same module. The weak lane may be marginal or failing while others are healthy.

What this usually means. One lane only, siblings healthy

Usually: lane-specific optics, attach, driver or TIA channel, or local thermal gradient

Not: a shared firmware image bug as the first explanation, unless remapping proves otherwise

Likely hypotheses. Multi-lane modules share a substrate, laser array (or CW-WDM source), and thermal environment. Lane-to-lane variation comes from: (1) die-level non-uniformity in the laser or modulator array (threshold, slope, V_π, coupling), (2) packaging variation in fiber-array alignment (one channel of the FAU slightly misaligned), (3) driver or TIA channel mismatch on the electronic IC, or (4) thermal gradient across the die (edge lanes hotter or cooler than center lanes).

Measurements, mechanism isolation, and confirmation.

1. Measure all lanes: OMA, ER, TDECQ, RIN, wavelength. Identify the outlier.
2. **Is it optical or electrical?** Swap the electrical lane assignment (if the host supports lane remapping) or test the suspect optical lane with a known-good driver channel. If the problem follows the optical path, it is laser/modulator/fiber-attach. If it follows the electrical channel, it is driver/TIA.
3. **Check coupling.** Measure per-lane fiber-coupled power with an integrating sphere or power meter array. A single weak lane with normal LIV suggests FAU misalignment.
4. **Thermal map.** Use an IR camera or CMIS per-lane monitors (if available) to check for hot spots. Edge lanes near the package wall or near a TEC boundary may run hotter.
5. **Time-growing imbalance.** A lane that worsens over time raises the probability of lane-specific source, modulator, receiver, coupling, packaging, or control degradation. Do not immediately call it laser aging. Where engineering access exists, compare LIV, bias, lock error, and coupling at t_0 and now before naming the mechanism.

Required access. Black-box path: per-lane power/BER/FEC, host lane remap, sibling comparison, temperature. Engineering-access path: LIV, optical eye, TDECQ, per-lane coupling, bias sweeps, FAU inspection.

Corrective action and recurrence control. FAU misalignment is a manufacturing escape; tighten incoming inspection or first-article coupling specs. Thermal gradient: redesign TEC zoning or derate the hot lane. Confirmed source wear-out: flag the lot and check burn-in screening effectiveness. Driver mismatch: work with the IC supplier on channel-to-channel gain flatness. *Recurrence control:* per-lane coupling or TDECQ first-article / SPC.

11.5 Wavelength drift

Observed behavior. In a WDM system, one or more channels walk off the ITU grid or the ring/filter passband. BER degrades as the channel moves off the receiver filter or MUX/DEMUX passband. In ring-modulator CPO, this manifests as sudden unlock of the wavelength control loop.

Likely hypotheses. Laser wavelength moves with temperature and bias current. If the TEC or wavelength-locker servo cannot track, the channel walks off its assigned slot. In microring systems, resonance also moves strongly with temperature, and neighbor heaters create thermal crosstalk that pushes adjacent channels (Sections 6.4 and 6.5).

Measurements, mechanism isolation, and confirmation.

1. Measure wavelength on an OSA or wavemeter. Compare to the target grid.
2. **Laser-side:** Check TEC current. If saturated (at max or min drive), the thermal load exceeds TEC capacity. Check case temperature and airflow.
3. **Locker-side:** Read the wavelength-locker error signal. A healthy loop holds near zero; drift means the servo is losing lock. Check locker etalon alignment and PD balance.
4. **Ring CPO:** Check the ring thermal tuner DAC code. If it has railed (max heater power), the ring cannot reach the target wavelength. Check for neighbor heating (all adjacent lanes at full traffic and max case T).
5. **Aging:** If drift is progressive over weeks/months, suspect laser mode hop or gradual TEC degradation (reduced ΔT capacity).

Corrective action and recurrence control. TEC saturation: reduce case temperature (improve airflow or liquid cooling) or derate the laser operating current. Ring unlock: increase heater headroom in the design, reduce thermal crosstalk with layout changes, or shift the CW-WDM source grid to re-center the ring tuning range. Aging: schedule preventive replacement (ELSFP hot-swap, Section 5.14). *Recurrence control:* TEC headroom and lock-error telemetry alarms.

11.6 Eye closure (high TDECQ)

Observed behavior. TDECQ exceeds the PMD limit even though average power and ER look acceptable. The DCA eye appears compressed or distorted after the reference equalizer is applied.

Likely hypotheses. TDECQ measures how much noise the transmitter can tolerate before BER exceeds the FEC threshold, relative to an ideal transmitter (Section 7.4). High TDECQ means the equalized eye is poor. Common causes: (1) insufficient EO bandwidth (modulator or driver roll-off), (2) poor level linearity (RLM < 0.95; driver or modulator compression), (3) pattern-dependent effects (ISI from bandwidth limit, reflections, or impedance mismatch), (4) chromatic dispersion on FR-class fiber eating into the margin.

Measurements, mechanism isolation, and confirmation.

1. Inspect the raw (unequalized) eye on the DCA. Is it bandwidth-limited (rounded transitions), compressed (uneven levels), or noisy (RIN/jitter)?
2. **Bandwidth:** Measure EO S_{21} of the modulator (if accessible) or the combined Tx path. Compare to the Nyquist frequency. If 3-dB BW is below Nyquist, bandwidth is the limiter.
3. **RLM:** Compute relative level mismatch from the DCA histogram. If RLM < 0.95, the PAM4 levels are unevenly spaced. Check driver linearity (swept DAC code) and modulator transfer function (bias sweep).
4. **Reflections:** Check S_{11} of the RF path (driver to modulator) and the optical return loss. Reflections cause post-cursor ISI the FFE cannot fully cancel.
5. **Dispersion:** Measure TDECQ with and without the test fiber (TECQ vs. TDECQ). If TDECQ is significantly worse than TECQ, dispersion is the marginal contributor. Check Tx wavelength and chirp against the fiber length and dispersion coefficient.

Corrective action and recurrence control. Bandwidth: upgrade driver or modulator (higher-BW die or peaking network). RLM: tune driver pre-emphasis (DAC levels for each PAM4 symbol). Reflections: fix wirebond, impedance discontinuity, or connector. Dispersion: tighten wavelength tolerance or shorten the fiber (reroute). *Recurrence control:* TDECQ/RLM sample tied to the failing corner.

11.7 Thermal runaway

Observed behavior. Module temperature rises continuously until shutdown (CMIS over-temperature alarm) or until the laser degrades catastrophically. May start subtly: BER creeps up as case T rises during traffic ramp or with neighbor-lane loading.

Likely hypotheses. In a faceplate pluggable, double-digit-watt module power must leave through the cage and heatsink [106]. If airflow is blocked, the cage is overloaded, or the TEC inside the module is fighting a losing battle against junction temperature, the thermal loop runs away. In CPO, the optical engine sits on the switch substrate beside the ASIC; cooling-path faults concentrate heat on the source and ring controls [107]. Both sources are vendor or research orientation, so the product thermal model remains authoritative.

Measurements, mechanism isolation, and confirmation.

1. Read CMIS module temperature and compare to the module's rated case T range. If case T exceeds the max, the system cooling is inadequate.
2. **Check TEC current.** A TEC at max drive current is saturated; it cannot pump more heat. The junction temperature is higher than the case T suggests.
3. **Measure LIV at temperature.** If threshold rises and slope drops steeply with T, the laser is near thermal rollover (Section 5.13). The operating point may be marginal.
4. **Neighbor loading.** Bring all lanes and neighbor modules to full traffic simultaneously. If the problem only appears under full-cage load, the thermal design margin is insufficient.
5. **Airflow audit.** Inspect the faceplate for blocked vents, incorrect fan speed, or missing blanking panels (bypass airflow). For liquid-cooled systems (CPO, XPO), check coolant flow rate and inlet temperature.

Corrective action and recurrence control. System-level: improve airflow, lower ambient, or reduce module count per cage. Module-level: derate the laser (lower bias current reduces self-heating) or switch to a lower-power module style (LPO instead of retimed, Section 10.5.1). CPO: ensure the cold-plate thermal interface material (TIM) is intact and the liquid loop meets flow-rate spec. Long-term: specify a tighter thermal class in the laser requirements (Section 5.6). *Recurrence control:* loaded-cage thermal class in ATP or DV plan.

11.8 Intermittent failures

```
Intermittent / burst
   |
Preserve state (do not reseat yet)
   |
Scope time + change history
   |
ORL / connector / supply / lock / attach
   |
Decision: contain / clean / redesign / ATP dwell
   |
Recurrence: dwell + FEC histograms
```

Observed behavior. Links flap, lose lock, or show bursts of FEC errors while average power and a short bench BER test look normal. The symptom may clear after reseating, cooling, or restarting firmware.

What this usually means. Intermittent bursts that clear on reseat or cool-down

Usually: connector or mate stress, ORL or MPI, supply noise, lock loss, weak attach, or firmware state

Not: a confirmed wear-out FIT story from a short room-temperature BER pass

Likely hypotheses. Intermittent faults come from contacts, reflections, weak optical or electrical attach, supply noise, wavelength-lock loss, firmware state, or environmental stress. Reseating can remove the evidence by cleaning a contact, changing fiber stress, or resetting a state machine.

Engineering heuristic. Treat cool-down recovery as a clue, not a fix. Capture the failing corner before the unit returns to room temperature forever.

Measurements, mechanism isolation, and confirmation.

1. Freeze CMIS state, FEC error timing, LOS or LOL history, temperatures, rails, lock error, and neighbor activity before touching hardware.
2. Scope the pattern across lane, module, tray, rack, lot, and firmware revision. Shared timing points toward power, cooling, firmware, or a shared source.
3. Correlate bursts with Rx power, ORL, supply and clock spurs, lock-loop state, vibration, and temperature. Use trigger capture rather than averages.
4. Run controlled disturbance tests one at a time: connector motion, thermal ramp, neighbor load, rail load, and firmware state transition.
5. Repeat on a golden unit and fixture. If the fault follows the unit, inspect fiber attach, solder, contacts, and control logs before destructive analysis.

Corrective action and recurrence control. Fix the confirmed contact, attach, supply, lock, or firmware cause. Add event-triggered telemetry and a production stress that reproduces the fault. Keep intermittent and no-fault-found RMA codes separate from laser wear-out.

Engineering heuristic. A rising NFF rate is often a triage and evidence problem, not proof that the field is healthy. Separate intermittent codes from wear-out before you trust the FIT.

Recurrence control: event-triggered FEC telemetry plus a dwell stress.

11.9 Connector contamination

Observed behavior. Intermittent link failures, burst errors, or elevated BER that clears after reseating or cleaning a connector. Rx power may fluctuate or show sudden drops. Often affects one direction of a duplex link.

Likely hypotheses. A particle of dust on an MT, LC, or MPO ferrule endface scatters and absorbs light, raising insertion loss and back-reflection (lowering ORL). Debris in the core zone can cause large loss even when most of the ferrule looks clean. Elevated ORL feeds back into the laser and raises RIN, causing burst errors even when average power looks acceptable. In high-power CW-WDM and ELSFP systems, trapped particles can burn onto the fiber endface and cause permanent damage.

Measurements and isolation. Preserve evidence before you disturb the mating:

```
Preserve telemetry and failure history
   |
Photograph and inspect before disturbance
   |
Record contamination and damage
   |
Clean and re-inspect
   |
Measure insertion loss and ORL
   |
Retest BER and sensitivity
```

1. Capture CMIS/host counters, timestamps, Tx/Rx power, alarms, and which port failed before reseating or cleaning.
2. Photograph and inspect with a fiber-endface scope (200–400×) before disturbance. Record particles in the core zone, scratches, pits, residue, and burn marks.
3. Clean and re-inspect (dry-click cleaner or lint-free wipe with IPA). If the endface still fails IEC 61300-3-35 zone criteria, replace the jumper [98].
4. Measure insertion loss and ORL across the mated pair at the named plane. Compare to the link-budget allocation (Section 7.7).
5. Retest BER and sensitivity. Clearing after cleaning supports contamination as the leading mechanism; confirm with IL/ORL and watch for recurrence. Log connector location and date code.

Corrective action and recurrence control. After evidence is preserved: clean, re-inspect, and verify IL/ORL and BER. Preventive: dust caps on unused ports, “inspect before connect" in the service runbook, sealed cassettes or trunk cables that minimize open-ferrule exposure. For high-power paths (ELSFP, CW-WDM), burn damage requires replacement, not re-cleaning. Track contamination RMAs as a distinct failure code (not "laser failure") so FIT accounting stays honest (Section 7.12).

Engineering heuristic. Inspect before you clean, and photograph before you disturb. Cleaning first can erase the only evidence that the mate was dirty.

Recurrence control: inspect-before-connect runbook; contamination RMA code.

11.10 Yield drop

Observed behavior. First-pass yield falls below its stable baseline. The loss may cluster on one ATP row, lane, tester, shift, supplier lot, assembly site, or firmware revision.

Engineering heuristic. Calibration drift is usually more likely than simultaneous hardware failure across a previously healthy population.

Likely hypotheses. Process drift, incoming material variation, fiber-array alignment, die-attach or solder change, tester or fixture drift, stale calibration, a software limit change, or a guardband that no longer matches measurement spread.

Measurements, mechanism isolation, and confirmation.

1. Contain suspect work in process and freeze tester software, limits, fixtures, and calibration records.
2. Build a Pareto by ATP row, lot, date code, site, tester, and shift. Do not average away a one-lane or one-station signature.
3. Run a golden unit across stations and the same failed unit on the reference bench. This clears the measurement system before supplier action begins.
4. Compare upstream wafer, die, assembly, and incoming data at the first point where good and bad populations separate.
5. Select failure analysis that distinguishes the remaining mechanisms, then verify the correction on a controlled lot and watch the field code.

Corrective action and recurrence control. Restore the changed process or measurement input, add a statistical control at the first observable point, revise ATP only with correlation data, and require a new first-article check before releasing volume (Section 9.2). *Recurrence control:* SPC at the first separating observable; FAIR after change.

11.11 Aging, thermal response, and margin erosion

Use time scale and recovery to route the incident. A reversible shift during a temperature or neighbor-load sweep is a thermal operating-point problem. A baseline that moves over stress hours or field months is aging. A sudden permanent step after a cycle points toward damage or an assembly defect, not ordinary thermal response.

1. Return the unit to its starting temperature and operating point. Record whether power, wavelength, bias, TDECQ, and BER recover.
2. Compare with ship and pre-stress data. Permanent LIV, spectrum, or bias-curve movement supports aging or damage.
3. Repeat the temperature sweep with source, wavelength-selective element, receiver, and neighbors isolated in turn.
4. Update the power, noise, timing, spectral, and control ledgers (Section 5.19). Several small shifts can explain a BER failure even when each component remains inside its stand-alone limit.
5. Route reversible thermal loss to cooling, control, calibration, or derating. Route cumulative change to HTOL and life-model review. Route lot-clustered permanent steps to manufacturing failure analysis.

On margin versus power, see Section 5.19: add margin only on the ledger that is empty; do not raise launch power by habit.

The corrective action must restore margin at combined corners. A room-temperature retest does not close a high-temperature incident, and one clean HTOL readout does not explain a reversible lock failure.

11.12 Temperature sensitivity

Observed behavior. BER, TDECQ, optical power, or lock stability degrades during a case-temperature ramp or under loaded-neighbor conditions. The unit may recover when cooled.

Likely hypotheses. Laser threshold and slope drift, wavelength movement, ring-resonance drift, TEC or heater saturation, EAM or MZM bias error, receiver-noise rise, package stress, or a thermal gradient that affects one lane.

Measurements, mechanism isolation, and confirmation.

1. Record case temperature, external optical power, wavelength, bias, TEC or heater current, lock error, OMA, ER, TDECQ, and pre-FEC BER on one time axis.
2. Repeat with neighbors off and on. A shared shift points toward thermal or supply coupling; a lone lane points toward its local path.
3. Hold the source fixed and move the wavelength-selective element, then reverse the test. This splits laser drift from ring or filter drift.
4. Rerun LIV and receiver sensitivity hot and cold. Separate lost launch power from lost receiver margin.
5. Inspect package and cooling interfaces if optical and electrical blocks pass alone but fail in the assembled system.

Corrective action and recurrence control. Restore thermal headroom, correct calibration and control limits, reduce coupling, or derate the operating point. Add the loaded-neighbor temperature ramp to the ATP or design-validation plan that missed it. *Recurrence control:* loaded-neighbor temperature ramp in ATP or DV.

11.13 Failure-analysis checklist

Failure-analysis output categories. Every FA result should land in one bucket before the case closes:

Design issue Fix architecture or derate.

Process issue Fix manufacturing or assembly.

Supplier issue Fix incoming quality or supplier process.

Test escape Improve detection (ATP, sample, SPC, telemetry).

Unknown mechanism Continue investigation with an owner and interim control.

The checklist in Table 11.1 is a lifecycle, not a suggestion list. Each step removes a class of uncertainty before the next step spends lab time. Skip Preserve and you often destroy the only evidence that separated host, plant, and product.

Step	Question	Required record
Preserve	What evidence will a reseat, reboot, clean, or retest destroy?	CMIS, BER and FEC history, rails, temperature, firmware, fixture, and time
Scope	One unit, lane, lot, vendor, site, or fleet?	Population and correlation plot
Classify	Sudden or gradual, constant or intermittent, thermal or cumulative?	Timeline and recovery test
Locate margin	Did power, noise, timing, spectrum, or control move first?	Golden comparison and margin ledger
Falsify	Which measurement best separates the leading hypotheses?	Expected result for each hypothesis before the test
Confirm	Does the fault follow the suspected block under a controlled swap or stress?	Repeated failure and passing control
Correct	Does the fix restore the original failing condition with margin?	Before and after data at loaded corners
Prevent	Where can production or fleet monitoring catch recurrence earliest?	ATP change, control limit, alarm, owner, and due date

Table 11.1. Failure-analysis checklist. The incident is not closed until the cause is reproduced, corrected, and covered by a recurrence control.

11.13.1 Reading the failure-analysis checklist

Preserve. What volatile evidence will a reseat, reboot, clean, or retest destroy? Snapshot CMIS, BER/FEC history, rails, temperature, firmware, fixture, and time before any invasive action. Photograph endfaces before cleaning if contamination is plausible. **Exit when** the failing-state pack is filed. **Decision:** proceed to scope, or stop ops from "just reseating." **Risk if skipped:** the fail clears and you learn nothing.

Scope / classify / locate (short). Name the population (unit, lot, vendor, site, fleet) before instruments (Section 7.12). Record sudden versus gradual and whether cool-down recovers. Name which ledger moved first: power, noise, timing, spectrum, or control (Section A.8.4). **Exit when** population, time class, and first-moving ledger are evidenced. **Risk if skipped:** wrong containment width or the wrong instrument tour.

Falsify. Which single measurement best separates the leading hypotheses? Write expected results before you run the test. **Exit when** one measurement kills or promotes the leading set. **Decision:** move survivors to Confirm, or reopen Locate margin. **Risk if skipped:** confirmation bias.

Confirm / correct (short). Confirm the fault follows the suspected block under swap or stress, then show before/after margin at the original failing corner. **Exit when** ownership is assigned and the fix restores that corner. **Risk if skipped:** you “fix” the wrong block or ship a lab-only cure.

Prevent. Where can production or fleet monitoring catch recurrence earliest? Version an ATP row, SPC limit, or alarm with owner and due date. **Exit when** recurrence control is owned and the cohort is under watch. **Decision:** close the incident, or monitor-only if rate and impact justify it (Table A.1). **Risk if skipped:** you solve one RMA and train the factory to recreate it.

11.13.2 Why the steps occur in this order

Preserve first because reseat and reboot destroy state. Scope next so containment width matches the population. Classify and locate margin before falsify so you pick the cheap separating test. Confirm before correct so you do not ship a story. Prevent last so the factory and fleet catch the next escape. Later steps must not compensate for a missing preserve pack or a wrong scope.

11.14 Interview takeaway

Key idea. A useful failure analysis starts with a symptom and ends with a new control. Preserve the failing state, split shared from local behavior, clear the measurement system, and choose one measurement that can falsify the leading hypothesis. The corrective action is incomplete until production or fleet data show that the same signature no longer escapes.

Junior mistake: reseat first, or close without a recurrence control in ATP or telemetry (Section 7.12 and chapters B, C and F).

Three questions to test yourself.

1. BER rose on one rack after a software rollout. What data do you preserve before reseating modules?
2. Received power is unchanged but BER worsened. Which measurement do you make next, and which hypotheses does it separate?
3. A golden unit fails only on one production station. What must be cleared before opening supplier failure analysis?

Chapter A

One-week optical systems interview review

Use this appendix as a standalone drill for about a week of focused prep. Its purpose is not to cover the most optics, and it is not a memorization guide. It teaches how a Staff engineer approaches ambiguous problems until the process is automatic under time pressure.

Read order when time is short: the Staff answer pattern here, then one or two cases in Chapter B, then the "30-second answer" callouts in Chapter C, then the wall-chart trees in Chapter D. Do not re-read every worked answer unless a topic is weak. Compression is the point: recover the method in minutes, not hours. When a claim needs a release or ship decision, use the shared evidence block in Section D.16.

Key idea. Every answer should end with the engineering decision. Interviewers remember decisions more than measurements. Close with "Therefore I would continue validation," "stop shipment," "contain Supplier B," or "update the ATP."

A.1 Staff engineer answer pattern

Strong engineering answers follow this sequence:

```
Staff answer pattern
  |
Define success
  |
Scope the problem
  |
Identify possible mechanisms
  |
Choose highest-value measurement
  |
```

```
Interpret evidence
  |
Make a decision
  |
Prevent recurrence
```

A senior engineer is not expected to know the failure immediately. They are expected to know how to reduce uncertainty. The same loop drives validation, qualification, failure analysis, and supplier decisions:

```
Central engineering loop
  |
Requirement
  |
Risk
  |
Failure mechanism
  |
Measurement
  |
Evidence
  |
Decision
  |
Control
  |
Learning
```

Do not invent a second framework. Use this loop inside the validation lifecycle (Table 7.2), the qualification evidence path (Section D.3), and the failure-analysis handbook (Chapter 11).

Why experienced engineers force this loop before naming a part?

Because the loop names the decision and the control. A part name without a measurement and a containment plan is a guess dressed as confidence.

Practice the same loop on full narratives in Chapter B.

Answer anatomy under time pressure.

Situation What is observed, at what plane and condition?

Initial hypothesis space Transmitter, channel, receiver, DSP, control loop, environment. Do not jump to one cause.

Highest-value next measurement The test that separates the largest number of hypotheses at the lowest cost and access level.

Decision Continue, contain, request FA, update qual, change ATP, derate, or monitor-only.

Recurrence prevention What prevents this from happening again?

A.2 Measurement access hierarchy

Every diagnostic answer should name the access level before asking for deeper hardware:

Level 0 Existing telemetry and logs.

Level 1 System experiments (swaps, attenuate, remap, dwell, reload).

Level 2 External module measurements at a named faceplate or optical plane.

Level 3 Engineering samples, breakout, or internal monitors.

Level 4 Destructive FA / DPA.

A good engineer does not immediately request destructive analysis. Maximize information gained per cost. Prefer Level 0–2 until they stop reordering beliefs (Sections D.11 and D.16).

Why experienced engineers climb access levels slowly?

Because Level 0–2 often reorders beliefs at low cost. Destructive FA early spends the sample and the calendar before you know what question the tear-down must answer.

Engineering heuristic. Pick the next measurement by information value per cost, not by instrument prestige (Section B.1).

A.2.1 Deciding under uncertainty

Experienced engineers often decide before certainty exists. That is not recklessness. It is matching evidence strength to impact, reversibility, delay cost, and confidence. Use the same Staff loop (Section A.1); do not invent a second process.

Ask four questions before you act:

- What is the impact if we are wrong?
- How reversible is the action?
- What does delay cost (escapes, schedule, learning)?
- How strong is the evidence now (observation, correlation, hypothesis, confirmation)?

Stopping shipment is high impact and hard to reverse, so it needs stronger evidence. Adding a telemetry field or a dwell is low impact and easy to reverse, so weaker evidence can still justify it. Practice narratives live in Chapter B.

Decision example. Should we stop deployment of a suspect lot?

Evidence: One manufacturing lot shows rising pre-FEC BER; average power is stable; sibling lots on the same hosts look healthy; the cohort is growing

Decision: Pause shipment and field install of the affected lot; keep good lots moving

Why: Containment is reversible. Waiting for perfect FA while the population grows is not.

Decision example. Should we redesign the module?

Evidence: One corner failure; mechanism still unknown; no lot or host correlation yet

Decision: Do not redesign yet. Increase evidence: scope, power-versus-quality fork, and the cheapest separating measurement

Why: A redesign is expensive and slow. Unknown mechanism plus one unit is not enough to move the architecture.

Decision example. Should we add more telemetry before the next pilot?

Evidence: Fleet triage is slow; lot and actuator fields are missing; storage budget is available

Decision: Add the minimum fields that unlock contain versus monitor decisions, with owners

Why: The action is reversible and cheap compared with another week of blind RMAs (Section C.14).

A.3 Four principles

Principle 1: Engineering reduces uncertainty.

Key idea. The purpose of engineering is not to find certainty. It is to reduce uncertainty enough to make the next decision.

Key idea. The job is not to wait for complete knowledge before making a responsible and appropriately reversible decision.

Validation, measurement, debugging, qualification, supplier choices, and production are the same work under different names. Ask the scale of the problem (device, module, rack, fleet) before you chase a confirmed mechanism: scale picks the owner.

Principle 2: Measurements unlock decisions.
Instruments do not exist to produce plots. They exist to unlock an action. A power meter asks whether you should chase the optical path or signal integrity. An OSA asks whether spectral alignment is still plausible. An LIV asks whether the device itself changed. A DCA or TDECQ asks whether the eye is still inside budget. A BER waterfall asks whether you have a sensitivity shift or a noise floor. On every answer, name the decision unlocked (ship, derate, second-source, ATP change, partner action) as well as the instrument (Table A.1).

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

Principle 3: Every measurement updates your beliefs.
Treat engineering as hypothesis testing:

$$\text{observation} \longrightarrow \text{hypotheses} \longrightarrow \text{measurement}$$
$$\longrightarrow \text{belief updated / hypothesis eliminated.}$$

Key idea. Inside the Staff loop (Section A.1), each measurement reorders competing hypotheses.

Speak the update out loud. “Calibration drift is my leading hypothesis, wavelength walk is second, and receiver noise is lower probability; the next measurement should sharply reorder those beliefs.” A measurement that leaves the ranking unchanged was the wrong measurement, or the wrong reference plane.

Principle 4: Measurements characterize margin.
Engineering is not only proving that a product works. It determines how much uncertainty and margin remain before failure. Debugging identifies exhausted margin. Qualification verifies that remaining margin is still acceptable after expected stresses. The same ledgers (power, noise, timing, spectral, control) appear in both jobs.

Engineering priors. Before touching the bench, assign higher probability to common failure modes than to exotic ones. Calibration drift is more likely than simultaneous laser aging and receiver degradation. A supplier-specific hot-corner escape is more likely than a new physics mechanism. Choose the first measurements to test those higher-probability hypotheses while eliminating as many alternatives as possible. Priors are not prejudice; they are how you spend lab hours.

Key idea. Engineering is decision making. Decision making is uncertainty reduction. Measurements reduce uncertainty. Therefore measurements exist to improve decisions.

This role owns laser direction inside an IM/DD interconnect effort. Lab measurement is how you decide, not the whole job. Hands-on fluency still matters:

LIV, RIN, ORL, TDECQ, and a BER waterfall. Senior people lose the level if they stop at plots and never close on a decision and a control.

Work the day plan at the end of this appendix. Memorize the one-page cheat sheet before Day 7. Night-before drill is Chapter C: open the matching thirty-second framework. Use each LLM practice box the day it is scheduled. Do not add new topics after Day 6 beyond that drill.

A.4 Engineering decision trees

Interview questions are usually solved by walking a sequence of uncertainty-reduction decisions. The exact branches differ. The philosophy never changes. Debugging asks what broke and which margin ledger is spent. Qualification asks what uncertainty remains before shipment. The playbooks in Chapter C are specialized trees under these two universal ones. The same trees, plus supplier, escape, unknown-failure, and margin-budget variants, live as a wall chart in Chapter D.

Memorize the two shapes, not the ASCII: Debugging = scope → power/quality → isolation → decision → control. Lifecycle = requirements → architecture → bring-up → characterization → margin → interop → qualification → manufacturing → pilot → MP → fleet → feedback. Full wall charts, including power fork, scope versus correlation, black-box access, and measurement selection, are in Chapter D. Night-before drill opens the matching playbook in Chapter C.

A.5 The answer spine

Move every answer through the same sequence. Memorize four phases, not nine nodes:

```
Answer spine
   |
Define success / scope
   |
Hypothesis space
   |
Highest-value measurement (name access level)
   |
Interpret evidence (observation / correlation / hypothesis /
     ↪ confirmation)
   |
Decision + owner
   |
Recurrence control
```

The diagram is a memory aid, not the answer. Speak one clear paragraph per phase under time pressure, and expand a node only when asked. Do not jump from a symptom to a component. End every debug answer with the decision (Table A.1). The systems loop in Section 1.1, the debugging pyramid in Section 1.16, and the failure-analysis method in Chapter 11 are the full versions of this spine.

Requirements. Start by stating what the system must do before you name a part. Reach, lane rate, aggregate capacity, power envelope, lifetime, cost, and manufacturing volume are the constraints that decide everything downstream. A 500 m DR link at 100 G/lane with a tight power budget is a different problem from a 2 km FR link or a dense WDM co-packaged engine. Say the requirement out loud even when the interviewer hands you a failed module, because the requirement tells you which margins matter and which architectures were even eligible. If you cannot name the requirement, you cannot tell whether a measurement is relevant.

DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.

FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Architecture. Translate the requirement into an architecture path: fiber plant, wavelength, source, modulator, receiver, and digital stack. A VCSEL path commits you to 850 nm multimode fiber and direct modulation. A DFB or EML path at 1310 nm commits you to single-mode fiber and leaves a choice among direct modulation, electro- absorption, a silicon MZM, or a ring. Each path then sets detector material, thermal control, test coverage, and service policy. Architecture is where you show that component choices are not preferences; they are consequences of the requirement (Table 5.1).

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

DFB Distributed-feedback laser: grating in the active region for single-mode output. Dominant 1310 nm source for DR/FR and EML gain sections.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

Measurements. Name the instruments you would use and the uncertainty each one removes. Always state the reference plane, pattern, temperature, and pass criterion with the instrument. A number without a plane is not a measurement. Ask out loud: what decision does this measurement unlock?

A power meter does not "measure power" as an end in itself. It answers whether the optical path still owns the failure, or whether you should move to signal integrity. An LIV setup answers whether the laser itself changed threshold or slope. An OSA answers whether spectral alignment is still plausible (wavelength and SMSR).

A DCA answers whether the eye is still inside budget: OMA, ER, RLM, TDECQ, and eye shape. A BERT and FEC counters answer whether you have a sensitivity shift, a floor, or a burst pattern.

A VNA answers electrical or electro-optic bandwidth. An ORL meter and inspection scope answer whether the plant is reflecting or dirty.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

Observations. Report what the instruments actually showed, not what you hoped they would show. Separate facts from interpretation. "Received power held at

-4 dBm while pre-FEC BER rose from 10^{-8} to 10^{-5} at $70°$C" is an observation. "The laser is dying" is not. First scope the failure (unit → lot → vendor → fleet), and note sudden versus gradual and recoverable versus permanent. Preserve telemetry before you reseat, clean, reboot, or rewrite a calibration table. Observations that disappear under debugging are observations you never had.

Hypotheses. Turn observations into a short ranked list. Start from engineering priors (common modes first), then apply the power-versus-quality fork (Section A.8.3). Keep the list short enough that the next measurement can kill more than one item. A hypothesis you cannot falsify with a bench step does not belong on the list yet.

Isolation. Good engineers perform measurements. Do not perform two experiments when one can separate the hypotheses.

Key idea. Great engineers perform the minimum measurement that eliminates the largest number of hypotheses.

Optimize uncertainty removed per hour of lab time, not uncertainty removed in the abstract. A fast sequence often beats a complete one:

$$\text{golden swap} \longrightarrow \text{power meter} \longrightarrow \text{bias sweep}$$
$$\longrightarrow \text{then DCA / OSA / RIN as needed.}$$

A golden swap of transmitter versus receiver splits Tx from Rx. An optical-versus-electrical lane swap in a multi-lane module splits the optical path from the driver or TIA. A bias sweep at the failing temperature asks whether the optimum moved away from the stored table. An LIV compared with ship data asks whether the device aged or only the setpoint drifted. A BER-versus-power waterfall asks whether the curve shifted or floored.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

Confirmed mechanism. State the mechanism that survived isolation, with the evidence that killed the alternatives. "EAM bias table segment wrong above $60°$C" is a confirmed mechanism. "Bad eye at high temperature" is still a symptom. Name the physical or process mechanism when you can: facet wear, monitor-PD corruption, FAU misalignment, MPI from a dirty connector pair, control ledger exhausted (TEC or heater at rail), supplier lot with high threshold. If the evidence only reaches "calibration drift" and not a deeper mechanism, say so. Overclaiming a confirmed mechanism is worse than stopping one layer early with honesty. Sometimes the product decision is due before the mechanism is known; that case is Section A.6.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

Corrective action. Fix the mechanism you named, under the condition that failed. Retune the table, replace the lot, change the thermal design, clean and inspect the plant, filter the supply, or rewrite the ATP limit. Containment comes first when the fleet is already exposed: stop ship, quarantine lots, or derate while the permanent fix is built. Then walk the verification ladder: reproduce the failure, verify the fix, regression-test neighbors, requalify if the change touches life or safety, release to production, and watch fleet telemetry. Candidates often stop at “I fixed it.” Staff engineers do not.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

Recurrence control and the decision. Close with the control that stops the same escape next month, then state the product decision in one sentence. The control is usually a new or tightened test, an SPC chart, a telemetry alarm, a process-control limit, a supplier screen, or a firmware guard that refuses to boot with a railed actuator. Name the owner and the measurable closure criterion. Then say the action: continue validation, stop shipment, contain the lot, update ATP, escalate the supplier, or monitor only. A confirmed mechanism without recurrence control is a story. Recurrence control without a decision is incomplete. In the interview, ending on the decision is what separates a debug narrative from a product-engineering answer.

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

Key idea. I first want to understand the scope of the problem, then determine which margin ledger is being spent, choose the measurement that eliminates the largest number of hypotheses, make the product decision, and finally add the control that prevents the next escape.

LLM practice

Summary. Four principles: uncertainty reduction; measurements unlock decisions; every measurement updates belief; measurements characterize margin. Chunk the spine as Understand / Investigate / Resolve / Prevent. Every answer ends with a decision. Name the ledger that moved.

Paste to an LLM and answer aloud. Quiz me on the four principles, priors, and the four spine phases. Give me a failed module at high temperature with stable average power. Stop me if I skip scope, the power fork, the control ledger, the product decision, or recurrence control.

A.6 Staff judgment under uncertainty

The spine above assumes you can reach a mechanism. Senior work often cannot. The product decision still has a deadline. Staff engineers make the decision with the evidence they have, name residual risk, and keep the FA path open.

A.6.1 Decide before the mechanism is known

Evidence can be enough even when the physical story is not. Example:

Evidence BER degrades. Only Supplier B. Only the hot corner. About 3% of the population.

Mechanism Unknown.

Decision Stop shipment of Supplier B. Continue Supplier A. Expand the ATP hot-corner screen. Begin FA and request DPA on fails. Review telemetry and RMA daily until the mechanism closes.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

FA Failure analysis: structured investigation from symptoms and telemetry to mechanism, corrective action, and recurrence control.

DPA Destructive physical analysis: cross-section / EDX on failed units to confirm facet, solder, or FAU failure modes.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

That is real engineering. Waiting for SEM photos while bad lots keep shipping is not.

A.6.2 Measurements unlock actions

Measurements do not unlock understanding as an end in itself. They unlock actions. Keep this vocabulary ready and use it in answers (Table A.1).

Action	Typical unlock
Ship / don't ship	Population meets ATP and life model, or it does not
Continue validation	Uncertainty still blocks a ship or architecture call
Escalate supplier	Lot, site, or vendor signature after scope
Derate	Margin too thin; product can ship under tighter use
Second source	Single-vendor risk exceeds fleet tolerance
Contain lot	Date-code or lot escape; stop further exposure
Modify ATP	Escape path found; production must catch it next
Open RMA / request FA	Field or partner unit needs mechanism work
Perform DPA	Need physical confirmation of facet, solder, FAU, die
Change firmware	Control loop, table, or guard is wrong
Retune calibration	Device healthy; setpoint or table segment wrong
Monitor only	Rate tiny, flat, no customer impact; watch trends

Table A.1. Decision vocabulary for interview answers. Name the action, the owner, and the residual risk when the mechanism is still open.

A.6.3 Reading the decision vocabulary

Measurements unlock actions (Table A.1). Expand ship, contain, and modify-ATP under time pressure; retrieve the rest from the table and the summary below.

Ship / don't ship. Hero samples do not answer ship. **Exit when** population data meet versioned ATP and life claims, or explicitly fail them. **Decision:** ship, hold, or restrict. **Risk if too early:** you learn the escape from customer outage.

Contain lot. Date-code or lot escapes need a hold today. **Exit when** suspect lots are identified and ship/fleet holds are in place. **Decision:** quarantine while FA continues. **Risk if too late:** bad lots keep shipping while SEM photos are pending.

Modify ATP. An escape path must become a production catch. **Exit when** the new row has limits, guardband, and GR&R. **Decision:** version the ATP; hold failing lots. **Risk if skipped:** FA closes; the factory ships the same escape tomorrow.

Other actions (expand on ask).

Continue validation Name the missing corner; do not delay with open-ended testing.

Escalate supplier Needs lot/site/vendor correlation, not one dirty connector.

Derate / second source Product decisions with written envelope or dual-qual evidence.

RMA / FA / DPA Climb only as far as the decision requires; preserve first (Section 11.13).

Firmware / calibration Allowed only when hardware baselines clear.

Monitor only Tiny, flat, no impact; armed alarms; escalate on growth.

A.6.4 Learning summary

Ship decisions Population ATP and life, or hold / derate / dual source.

Containment Lot and supplier actions follow scope, not a single ticket.

Learning loop Escape → ATP or alarm change with owner and date.

Escalation ladder Telemetry → bench → swap → FA/DPA as needed.

Monitor only Tiny, flat, no impact; armed alarms; review on a clock.

A.6.5 Time is a resource

Every measurement costs hours. Optimize uncertainty removed per hour of lab time. Prefer the cheap cut that kills many hypotheses before the slow characterization. Telemetry and a golden swap often beat an immediate RIN setup. SEM and DPA come last, not first.

A.6.6 Measurement hierarchy

Not every failure deserves destructive analysis. Climb only as far as the decision requires:

```
telemetry -> simple bench -> swap -> characterization -> FA /
    ↪ DPA
```

A.6.7 Fleet economics

Product engineering is economics as well as physics. A tiny, flat, no-impact fail rate can stay on a monitor-only plan. A growing, supplier-specific rate demands containment the same day.

Monitor only Failure rate ∼0.003%, no customer impact, no trend, no growth. Keep telemetry alarms and review weekly.

Contain immediately Failure rate ∼2%, growing, supplier-specific. Stop ship, quarantine lots, notify the partner, open FA, tighten ATP.

Intentionally not fixing a confirmed mechanism is sometimes correct. Leaving a growing escape unowned is never correct.

A.6.8 Ownership language

When the interviewer asks "What would you do?", answer as the owner:

> As owner I would stop shipment of the affected lots, notify the supplier, request FA with DPA on a sample, update the ATP hot-corner screen, review fleet telemetry and RMA codes daily, and schedule a qualification rerun before open volume resumes.

Ownership is the difference between a debug narrative and a Staff answer.

LLM practice

Summary. Staff judgment: decide with incomplete mechanism evidence; name an action from the decision table; spend lab hours for uncertainty removed; climb the measurement hierarchy only as far as needed; apply fleet economics; speak as the owner.

Paste to an LLM and answer aloud. Give me three incomplete-evidence scenarios. For each I must state evidence, confidence weights, the action I take today, residual risk, and the FA path. Fail me if I wait for a perfect confirmed mechanism before containing a growing lot.

A.7 Common interview traps

Interviewers listen for these mistakes. Avoid them on purpose.

Trap 1: Naming a component first. Start with scope, not a part number.

Trap 2: Calling symptoms confirmed mechanisms. "Bad eye" is a symptom. "EAM bias table wrong above $60°$C" is a confirmed mechanism.

Trap 3: Stopping after the fix. Always end with recurrence control and the product decision.

Trap 4: Listing instruments. Name the decision each measurement unlocks, not a gear catalog.

Trap 5: One unit as the fleet. Scope to lot, vendor, rack, and trend before you generalize.

Key idea. Before any debugging answer, ask: Scope? Which ledger moved? Power or quality? Fastest measurement? Decision? Control?

A.8 Concepts to know cold

Split the load. Memorize the core five cold; keep the supporting five as retrieval hooks you can expand under follow-up.

Core five. Scope; power versus quality; Five Ledgers; measurement $\rightarrow$ uncertainty $\rightarrow$ decision; containment $\rightarrow$ correction $\rightarrow$ recurrence control.

Supporting five. Lifecycle; BER signatures; thermal versus aging; production statistics; supplier qualification.

The cards below expand those ten ideas.

A.8.1 Start at the system

For a new link, walk downward from capacity to the service model. Capacity sets lane rate. Lane rate and reach together set the fiber plant and the source class. Power and cost then decide whether you can afford a TEC, a retimer, or a dense WDM engine. Only after those constraints are named do you choose the modulator and the receiver, then the DSP and FEC stack, then the validation and field-service model.

$$\begin{aligned}\text{capacity} &\longrightarrow \text{lane rate} \longrightarrow \text{reach} \longrightarrow \text{power}\\ &\longrightarrow \text{fiber plant} \longrightarrow \text{source} \longrightarrow \text{modulator}\\ &\longrightarrow \text{receiver} \longrightarrow \text{digital signal processing and FEC}\\ &\longrightarrow \text{validation and service model.}\end{aligned}$$

A good answer explains why each choice constrains the next one. A VCSEL path points toward 850 nm, multimode fiber, silicon detection, and direct modulation with a short reach. A DFB path at 1310 nm points toward single-mode fiber, germa-

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\lesssim$100 GHz).

DSP SerDes DSP-based SerDes: ADC on the Rx lane, digital FFE/DFE and timing recovery, DAC on Tx. Standard above $\sim$50 GBd; the "DSP" that cancels ISI at 224G lives inside the SerDes.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

MMF Multimode fiber: 850–940 nm VCSEL links (OM3/OM4/OM5); reach and modal BW limit to SR distances.

DFB Distributed-feedback laser: grating in the active region for single-mode output. Dominant 1310 nm source for DR/FR and EML gain sections.

SMF Single-mode fiber: 9 μm core; O-band (1310 nm) or C-band (1550 nm). G.652.D and G.657 for datacenter plant.

nium detection, and a choice among a directly modulated laser, an EML, or an external silicon-photonic modulator. Neither path is "better." Each closes a different reach, power, cost, and manufacturing problem. The decision matrix is in Table 5.1.

A.8.2 Scope before mechanism chase

Before you open an instrument, walk the failure up the scope ladder (unit → lot → vendor → site → fleet). Each rung changes the owner and the next action (Sections D.5 and D.9). Also ask time and change: sudden versus gradual, intermittent versus constant, and what changed just before the symptom. Scope often removes more hypotheses than the first bench measurement. A fleet-wide gradual drift cannot be a single dirty connector. A vendor-lot signature points to supplier containment before you redesign the module.

DML Directly modulated laser: modulate bias current; high chirp, low cost.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

Preserve the failing state and its telemetry before you reseat, clean, reboot, or change calibration. Capture CMIS monitors, pre-FEC BER, bias currents, temperatures, LOS and LOL flags, and firmware versions. An intermittent that disappears under debugging is still a real failure; you destroyed the evidence (Section 7.12 and table 11.1).

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

> **Why experienced engineers ask about scope first?**
>
> Because scope eliminates enormous parts of the hypothesis space before you touch the lab. Unit, lot, vendor, site, and fleet point at different owners.

> **What this usually means.** Fleet-wide sudden failure after a rollout
>
> *Usually:* software, configuration, shared environment, or shared infrastructure
>
> *Not:* independent wear-out of thousands of unrelated lasers in the same hour

A.8.3 Use the power-versus-signal-quality fork

First ask whether received optical power changed. That one question splits the debug tree. Full instrument paths and worked examples live in Sections 4.8 and 7.11.

If power changed, stay on the power ledger: source enable, coupling, connectors, ORL, plant loss, and monitor calibration. Confirm with an external meter at a named plane before retuning eyes or equalizers.

If power held but BER worsened, leave the power ledger. Signal quality, receiver sensitivity, wavelength or lock, and calibration tables are next. A sensitivity shift can look like transmitter degradation until you golden-swap.

> **Engineering heuristic.** Confirm power at a named external plane before you trust a monitor that may share the fault under investigation.

A.8.4 Track five margin ledgers

Links rarely fail from one dramatic excursion. They fail when several small shifts spend different ledgers at once. The five-ledger map is a teaching and debug framework: name which ledger moved, what spent it, and which decision that update unlocks. Full device treatment is in Section 5.19.

```
Power · Noise · Timing · Spectral · Control
```

Power. **Question:** Is there enough light at the decision point? Launch, insertion loss, coupling, connectors, multiplexers, and receiver sensitivity live here. Track average power and OMA separately: average power can look healthy while OMA collapses. **Evidence:** external meter at a named plane, monitor-versus-meter agreement, sensitivity. **Decision:** clean/replace plant, retune APC, derate reach, or leave the power ledger. **Risk if ignored:** quality-path debug on a loss problem.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

Noise. **Question:** Is the error rate limited by impairments that power cannot buy out? RIN, receiver thermal noise, shot noise, crosstalk, supply noise through weak PSRR, and MPI from reflections live here. Signal-dependent noise and other non-power-limited impairments make BER floors (Section A.8.9). **Evidence:** waterfall shape, RIN under ORL, FEC histogram timing. **Decision:** remove the impairment, do not raise launch into a floor. **Risk if ignored:** endless OMA increases.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

Timing. **Question:** Is there still equalization and jitter reserve? Bandwidth, dispersion, ISI, jitter, and SerDes FFE/DFE tap use live here. An eye that still opens on a DCA can hide a SerDes with no remaining taps. **Evidence:** TDECQ/RLM, tap saturation, COM on linear paths. **Decision:** retune EQ, shorten channel, or redesign SI. **Risk if ignored:** “good eye” tickets that fail only in the host.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

Spectral. **Question:** Is the line still inside the filter or lock window? Wavelength, SMSR, passband, thermal drift, and lock range live here. A heater near its DAC rail spends spectral margin while BER still passes. **Evidence:** OSA/wavemeter, lock-loop status, actuator codes. **Decision:** retune lock/thermal design, derate temperature, or replace the source. **Risk if ignored:** unlocks mislabeled as random BER.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

Control. **Question:** Do the loops still have authority to hold the operating point? APC, TEC, heaters, ring lock, bias DACs, and calibration tables live here. Prefer product language: “control ledger exhausted,” not only “TEC current hit max.” A railed actuator or bad table can fail the link while the diode is healthy. **Evidence:** actuator codes, cool-down recovery, table reload trials. **Decision:** retune tables, fix thermal design, or route to aging FA. **Risk if ignored:** healthy silicon sent to reliability for a firmware bug.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.
TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

Why ledger language comes before component names. Name the spent ledger before naming a laser, TIA, or connector. The ledger picks the measurement. The measurement updates belief. The belief unlocks contain, retune, derate, RMA, or monitor-only. Component names without a ledger are guesses.

Why experienced engineers name the ledger before the part?

Because the ledger picks the measurement and the owner. "Bad laser" without a spent ledger skips the cheap tests that would falsify it.

Engineering heuristic. An actuator near its rail is often the story before the diode is dead. Check control authority before you declare wear-out.

A.8.5 Use the validation ladder

Key idea. Validation is staged uncertainty reduction. Engineering is decision making; decision making is uncertainty reduction; measurements reduce uncertainty; therefore measurements exist to improve decisions.

For every stage, name the question the stage answers and the uncertainty it removes. A test that answers no question is cost, not confidence. Do not re-list the lifecycle here; memorize the order in Table 7.2 and the exit criteria in Section 7.1.

Night-before path: validation playbook in Section C.8. Practice prose: Section A.10.2.

A.8.6 Margin budgeting

Every environmental or use stress consumes part of the system margin: temperature, voltage variation, supply ripple, fiber contamination, insertion loss, connector wear, aging, mechanical vibration, and process variation. Qualification is not a tour of each mechanism for its own sake. It verifies that after the expected stresses, remaining margin is still acceptable. Stress consumes margin. Qualification measures remaining margin. Debugging finds which ledger is exhausted when that remaining margin hits zero (Section A.8.4).

A.8.7 Customer view versus vendor view

The vendor designs internals. The customer characterizes externally observable behavior. As a customer you often do not need laser threshold, driver architecture, or TIA topology. You measure BER, sensitivity, FEC statistics, launch and receive power, telemetry, and environmental response; eye metrics when engineering access exists. If the product is a black box, qualification focuses on that external surface. If engineering samples are available, request transmitter-only, receiver-only, breakout, or diagnostic hardware to isolate Tx and Rx margins independently. Keep the view explicit in second-source and qualification answers (Section D.11 and section A.10.5).

Tradeoff. Customer realism vs diagnostic visibility

Improves: Bookended tests match the deployed product and ownership model

Worsens: Limited root-cause visibility without engineering samples

When acceptable: When black-box evidence already unlocks contain, derate, or supplier action

Experienced decision: Use black-box evidence first. Request deeper access only when it changes the decision.

A.8.8 Know what each instrument answers

Do not recite instrument names. Use Measurement → uncertainty removed → decision unlocked (Section D.14). Fast map: power meter (power ledger), LIV (device vs setpoint), OSA (spectral), RIN/ORL (floor), DCA (eye), BERT/FEC (waterfall shape), VNA (electrical plant), thermal chamber (reversible vs aging), bias sweep (control ledger). Details and reference planes live in Section 7.6 and table 7.3.

A.8.9 Read a BER waterfall: shift, floor, and burst pattern

These three words show up in almost every debug answer. Know what each one looks like on the bench, what it rules in, and what it rules out. The operational procedures are in Section 11.2 and section 11.2.1.

What a BER waterfall is. A BER waterfall is a plot of bit error ratio versus received optical power. You build it by sweeping a calibrated VOA in the path, holding pattern, temperature, and host fixed, and counting errors long enough at each power that the BER is statistically meaningful. Plot received power on the horizontal axis (name the reference plane: usually TP3 at the receiver connector) and pre-FEC BER on a log vertical axis. A healthy link falls steeply as power rises: more photons, better signal-to-noise ratio, fewer errors. That falling curve is the waterfall. A single BER point at the operating power is not a waterfall. Without the sweep you cannot tell whether you are short on power or limited by noise that scales with the signal.

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

What a shifted waterfall means. A parallel shift means the whole curve moved left or right while keeping a similar slope. The link still improves when you add power; it just needs a different power to hit the same BER. A rightward shift (worse sensitivity) means you now need more received power for the same pre-FEC BER. Common causes are lost launch or coupling (power ledger), a quieter or noisier receiver (TIA noise, responsivity), eye closure that lowers effective OMA at constant average power (ER, RLM, TDECQ), wavelength walking onto a filter edge, or equalizer misadaptation. A leftward shift means the link got healthier. The diagnostic move after you see a shift is a golden swap: known-good transmitter, then known-good receiver, to decide which side owns the sensitivity change. Raising launch power can still help a shifted link, because the waterfall has not stopped responding to power.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

What a BER floor means. A floor is a horizontal asymptote: as you raise received power, BER improves for a while and then stops. Additional received power no longer removes the dominant impairment. That is a diagnostic pattern, not a single mechanism. Relative intensity noise is a common case (Q saturates when noise scales with the signal), but floors also arise from multipath interference, reflections, pattern-dependent distortion, residual ISI, crosstalk, timing or CDR limits, supply noise, DSP or equalization limits, error propagation, and nonlinear behavior. The interview trap is to keep raising launch power or cleaning for loss when the curve has already floored, or to name RIN before bisecting. Confirm the floor with a full sweep, then choose the next measurement from the leading hypothesis: quiet laser versus product bias board, ORL reflector sweep, pattern change, FEC error timing, or equalizer diagnostics.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN$_x$OMA under fixed ORL.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

What a burst pattern means. Average BER alone hides how errors arrive in time. A burst pattern means errors cluster: many errored symbols in a short window, then quiet intervals, rather than a steady sprinkle of random bit flips. FEC histograms and pre-FEC error counters with timestamps make this visible. Gaussian thermal noise and well-behaved RIN tend to spread errors. MPI from a pair of reflective interfaces, connector intermittents, ESD events, supply glitches, and unlocked CDR intervals tend to cluster them. Lane- correlated bursts across a module point at a shared supply, clock, or thermal event. A single-lane burst pattern points at that lane's optical path or connector. In the fleet, bursty pre-FEC counters with stable average power are often intermittents: preserve the counters and CMIS event log before you reseat anything, or the evidence disappears (Section 11.8).

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

How to say the three together. "I sweep BER versus received power. A parallel shift is a sensitivity or power-margin problem; more power can still help. A floor means power is no longer the limiting variable; more power cannot help until the dominant impairment is removed. A bursty FEC histogram means time-correlated events such as MPI, retrains, or intermittents, not plain Gaussian noise." Offer to draw the shifted curve next to the floored curve on the whiteboard.

A.8.10 Separate thermal response from aging

Thermal effects are usually reversible when you return to the starting temperature: wavelength shift, ring detuning, EAM or MZM bias movement, TEC loading, and receiver-noise increase. Aging is cumulative and does not reverse on cool-down: threshold rise, slope loss, contact degradation, defect growth, and permanent absorption or spectral change.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

The practical test is simple. Return to the starting temperature and compare with pre-stress data at the same junction temperature. Full recovery supports a thermal or control hypothesis, often a calibration-table segment or an actuator that ran out of range. A permanently moved LIV baseline supports aging, damage, or assembly change. Mixing the two owners wastes weeks: a reliability team cannot fix a wrong table, and a firmware team cannot fix a worn facet (Section 5.10).

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

A.8.11 Accelerated life tests need a mechanism

HTOL predicts field life only if elevated temperature and bias accelerate the same physical mechanism the fleet will see. The stress must not introduce a different mechanism, such as solder creep at a temperature the product never reaches, and it must not omit a dominant field stress, such as thermal cycling or connector wear.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

In the interview, state the assumed mechanism, the activation energy and why it applies to this process, the sample size and confidence bound, the stress and use temperatures, the measured drift parameter (threshold, slope, wavelength), and the field-data check. Treat the projected FIT result as a hypothesis that field returns must confirm or falsify. A FIT number without those pieces is a spreadsheet, not a prediction. The life acceleration model is Arrhenius (Section 5.13).

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

A.8.12 Calibration is part of the product

The operating points a product actually stores are as much the product as the die. Know them by name: laser bias or APC target; EAM bias versus temperature; MZM quadrature; ring heater or wavelength-lock point; monitor-photodiode calibration; and receiver and power-monitor offsets. Tables are usually segmented by temperature, and segment boundaries are a real failure mode.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

Recalibrate when evidence demands it, not by habit: loop residual stops converging, an actuator approaches its rail, telemetry disagrees with an external reference, temperature leaves the table range, or repair, rework, or firmware changes invalidate coefficients. ATP must verify calibration at the temperature corners the fleet will see, not only at station ambient (Section 5.11).

PD Photodiode: converts photons to photocurrent. Ge-on-Si PIN is mainstream; APD adds gain; UTC for high saturation.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

A.8.13 Corrective action must prevent recurrence

A confirmed mechanism is not the end of the answer. Close with immediate containment; the design, process, calibration, firmware, or supplier correction; verification under the original failing condition; an acceptance test, process control, alarm, or telemetry control that catches recurrence; and an owner with a measurable closure criterion. An interview answer that stops at “we replaced the laser” sounds like repair. An answer that ends on the new ATP corner or the new telemetry alarm sounds like product engineering.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

LLM practice

Summary. Ten cold concepts: start at the system; scope before mechanism chase; power versus signal-quality fork; five margin ledgers; validation ladder; instruments named by uncertainty removed; waterfall shift versus floor versus burst; thermal response versus aging; HTOL needs a mechanism; calibration is part of the product; corrective action must prevent recurrence.

Paste to an LLM and answer aloud. Act as my interviewer for optical systems. Ask me one question from each cold concept, in random order. Push for a named reference plane, a next measurement that kills hypotheses, and a recurrence control. Grade me on process, not on jargon volume.

A.9 How to present a technical experience

Each story is a spoken version of the answer spine. Prepare one component or bench story and one system, production, or fleet story. Use only work you personally performed, and separate your contribution from the team’s. Walk the same eight beats every time:

1. **Situation.** What failed or needed improvement?
2. **Scope.** How large was the impact (unit / lot / fleet)?
3. **Responsibility.** What did you own versus the team?
4. **Hypotheses.** What possibilities did you consider after the power-versus-quality fork?
5. **Measurements.** What evidence did you collect, at what plane and access level?
6. **Decision.** What action did you recommend?
7. **Impact.** What changed after the action?
8. **Prevention.** How did you stop recurrence, and what metric confirmed the control?

LLM practice

Summary. Prepare two true stories, one bench or component and one system, production, or fleet. Each story must hit the eight beats and end on a measured recurrence control. Separate your work from the team's.

Paste to an LLM and answer aloud. I will tell you my two story titles only. Interview me on each for five minutes. Force the eight beats in order. Cut me off if I invent metrics, skip scope, or end without a control that would catch the next escape.

A.10 Questions to rehearse aloud, with model answers

Rehearse these until the structure is automatic. For night-before review use the matching playbook in Chapter C; the prose below is practice depth. Each answer starts with scope or requirements, names measurements and the hypotheses they separate, and ends with a decision and a control. Do not recite; adapt the skeleton to the follow-up questions.

How to use the worked answers. Each long question opens with a **30-second answer** callout. Prefer the canonical wording in Chapter C; the expansions below are for practice. The 10-minute section is read-only reference. Expand only where the interviewer asks:

$$\text{30-second answer} \longrightarrow \text{interviewer asks} \longrightarrow \text{expand only there.}$$

A.10.1 How would you set laser requirements for a new IM/DD link?

This is the ownership question. Start from the system, not from a laser datasheet. The output is a requirements slice a supplier and an ATP can both test against (Tables 5.1 and 5.4).

30-second answer (memorize). Freeze reach, lane rate, fiber, power, lifetime, and volume. Those choose the source path. Then write OMA and RIN at a named plane and ORL, thermal and control headroom, a named HTOL mechanism, and an ATP with supplier reaction plan. Therefore I would freeze requirements that let us ship and second-source, not pick the laser that looks best on a bench.

3-minute answer (practice). Walk four steps: (1) system constraints choose the architecture path before any part number; (2) optical budget at named planes (OMA, RIN at ORL, SMSR, chirp); (3) thermal, life, and control headroom with a named HTOL mechanism; (4) ATP methods, FAIR triggers, and RMA codes split by supplier.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

Name one hard number you would fight for (RIN under ORL, or APC headroom at hot) and what fails if it is missing.

10-minute reference (read only). Open Section C.8 only if the interviewer expands into the ladder; otherwise expand one constraint into the budget table and ATP/FAIR landing. Architecture forks: Section 5.1 and table 5.1.

A.10.2 How would you validate a new optical transmitter from bring-up through production?

This is the question most likely to open the interview. The ladder itself is in Section A.8.5 and table 7.2. Frame first: validation is staged uncertainty reduction. Each stage answers a question the previous stage could not.

> **30-second answer (memorize).** See Section C.8 for the canonical 30-second answer. Deliver that first; expand below only if asked.

3-minute answer (practice). Walk the ladder in order. For each stage, name one instrument, the uncertainty removed, and the decision unlocked (continue, redesign, tighten ATP, stop ship). End on which ledger the telemetry must watch.

10-minute reference (read only). Open Section C.8 for the thirty-second playbook. Expand only the stage the interviewer picks using Table 7.2: entry condition, key uncertainty, exit criteria, decision unlocked. Body detail is in Chapters 7 and 8. Prefer customer-visible measurements unless engineering access is available (Section D.11).

A.10.3 BER worsens at high temperature but average power is stable. What do you do?

Classic fork question (Section 11.12 and section A.8.3).

> **30-second answer (memorize).** See Section C.4 for the canonical 30-second answer. Deliver that first; expand below only if asked.

3-minute answer (practice). **Situation:** BER worsens at high temperature while average received power is stable.
Hypothesis space: ER collapse, wrong modulator bias, wavelength walk, railed TEC/heater, hotter receiver noise; not gross optical loss.
Highest-value measurement: remaining margin at the failing temperature

(Level 0–1), then Level 2 external eye / Level 3 bias and OSA if access exists.
Decision: retune table, fix thermal design, or restrict the corner.
Recurrence: put the loaded corner in ATP and watch the control ledger in telemetry.

10-minute reference (read only). Playbook: Section C.4. Offer the calibration-table segment-boundary story or the railed-heater story if asked. Aging does not reverse on cool-down; recoverable failures are operating-point problems.

Practice case: Section B.7.

A.10.4 How do you distinguish laser aging from calibration drift?

Separates device physics from control-loop bookkeeping (Sections 5.10 and 5.11).

30-second answer (memorize). See Section C.5 for the canonical 30-second answer. Deliver that first; expand below only if asked.

3-minute answer (practice). Black-box first: BER/FEC, telemetry, recal trial. Then, with engineering access, remeasure LIV and other physical baselines against ship data at fixed junction temperature. Compare monitor-PD to an external power meter. Aging: life model, derating, burn-in, or replacement. Drift: table version control, temperature-segment verification, monitor integrity, ATP corner under load. Do not mix owners (Section D.11).

10-minute reference (read only). Playbook: Section C.5. Monitor-PD corruption is the silent drift mode: APC holds the wrong launch while telemetry looks fine. Recalibration recovery raises $P(\text{drift})$; it does not confirm the device is unchanged. Confirm with external baselines when access exists.

A.10.5 How would you qualify a second laser or photonic-integrated-circuit supplier?

This question tests supplier judgment, not vendor names. Night-before playbook: Section C.6. The frame: the first supplier's failure distribution does not transfer. Qualify against the requirements slice, not against the incumbent's datasheet (Table 5.4 and section 9.2). Prefer customer-visible remaining margin; request engineering access only when black-box evidence is insufficient (Section D.11).

30-second answer (memorize). See Section C.6 for the canonical 30-second answer (component / PIC path). For a finished module or cable second source, use Section C.7. Deliver that first; expand below only if asked.

Step 1: freeze the requirements, not the part number. Write what the new part must close: link budget, RIN at the stated ORL, bias window, wavelength class, thermal class, and lifetime FIT target. Those are the acceptance criteria. The incumbent's datasheet is evidence that one process can meet them, not a template the second source must copy. A second source that matches the datasheet but fails the link-budget corners is not qualified.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN$_x$OMA under fixed ORL.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Step 2: characterize representative multi-lot distributions, not hero units. Measure threshold, slope, wavelength, SMSR, and RIN across wafers, lots, and temperature. Compare spreads against the incumbent, not only means. A hero sample from a new supplier demonstrates nothing about the process. Ask for wafer maps and lot genealogy so edge-of-wafer outliers are visible before they enter your module line. Record the same reference planes and fixtures you use on the incumbent, or the comparison is fiction.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

Step 3: run the full qualification on this process. HTOL with the supplier's own activation-energy justification for the named mechanism, temperature cycling, damp heat, and burn-in. Wear- out mechanisms and infant-mortality rates are process-specific: epi, facet coat, attach, and hermeticity all differ. Borrowing the incumbent's E_a is the most common quiet mistake. State the projection's fine print: sample size, confidence bounds, and evidence that the stress accelerates the field mechanism without inventing a new one.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Step 4: correlate ATPs and plan the ramp. Run the same units on the supplier's line and yours so an ATP limit means the same thing at both sites. Size guard bands from the combined repeatability. Ramp with lot traceability, a FAIR, and an agreed reaction plan for excursions. Keep fleet RMA codes split by supplier so field data can falsify the qualification. A second source that cannot be traced in the fleet is not a second source; it is a blind risk.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

FAIR First-article inspection report: re-qualify after tooling, site, epi, or CMIS firmware change before open volume.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

How to say it aloud. "Requirements first, distributions second, process-specific qual third, ATP correlation and split field codes fourth." Offer to go deep on HTOL validity or on what you would put in the reaction plan.

Practice case: Section B.6.

A.10.6 A single lane is weak in a multi-lane module. How do you isolate optical, electrical, thermal, and assembly causes?

Multi-lane modules give you a free control group: the sibling lanes. The frame is pattern recognition across lanes before any single-lane deep dive (Section 11.4).

Step 1: confirm one outlier, not a gradient. Measure every lane for OMA, ER, TDECQ, and wavelength at the same temperature and pattern. A single-lane cliff is a local defect. A smooth edge-to-center gradient is thermal or FAU alignment. A checkerboard pattern often points at electrical routing or a shared supply. Write the pattern down before changing anything. The siblings tell you what "normal" is for this unit.

Step 2: bisect optical versus electrical. Swap the electrical lane assignment, or drive the suspect optical path with a known-good driver channel. If the weakness follows the optical path, the candidates are the laser or EML element, the modulator, the attach, or the fiber. If it follows the electrical channel, the candidates are the driver, the TIA, the package routing, or the host SerDes lane. This one swap often cuts the tree in half.

Step 3: separate assembly from device and thermal. Per-lane coupled power with a normal LIV on the weak lane points at fiber-array alignment or a contaminated ferrule: the device is fine, the light is not leaving. A thermal map or per-lane temperature monitors separate an edge-lane thermal gradient from a device defect. If the imbalance grows over HTOL or burn-in time, one array element is aging faster, which is a lot or screening question, not a design question.

Step 4: pick the corrective action from the pattern. Across lanes, units, and lots, the pattern chooses the fix: a rework instruction for alignment, a coupling-spec change for the FAU, a thermal- design change for edge lanes, a driver or TIA lot screen, or a supplier corrective action on one array element. Do not rewrite the whole module spec for a single-lane assembly escape.

How to say it aloud. "Siblings first, then optical-versus-electrical swap, then assembly versus device versus thermal, then the pattern picks the control." Offer the FAU-alignment story as the most common production escape.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

A.10.7 Received power is unchanged but required receiver power increased. What hypotheses remain?

Apply the power-versus-quality fork (Section A.8.3): power ledger intact, so eye quality or the receiver (Section 11.2).

> **30-second answer (memorize).** See Section C.1 for the canonical 30-second answer. Deliver that first; expand below only if asked.

3-minute answer (practice). List location-tagged hypotheses (Tx ER/RIN/jitter, channel MPI or filter walk, Rx TIA/PD/equalizer). Golden-swap to own the side. Waterfall with a VOA: parallel shift means sensitivity moved; a floor means power is no longer limiting. Stressed-eye if Rx is indicted. Control follows the mechanism: table, hygiene, supply filter, or lot screen.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

SECQ Stressed eye closure quaternary: receiver-side stressed test, analog of TDECQ. Applies calibrated optical stress before BER test.

A.10.8 Why can a link show a BER floor that more launch power does not fix?

This question has a short physics answer and a longer debug answer. Give both (Section 11.2.1 and section 4.3).

Step 1: state the pattern in one sentence. A BER floor means additional received power no longer removes the dominant impairment. Raising launch into a floor does not buy margin. That is a diagnostic pattern, not a confirmed mechanism.

Step 2: name common non-power-limited causes. Signal-dependent noise such as laser RIN (often feedback-driven through poor ORL); multipath interference from reflective interfaces; residual ISI or pattern-dependent distortion; crosstalk; timing or CDR limits; supply noise through weak PSRR; DSP or equalization limits; error propagation; nonlinear behavior. For RIN specifically, $Q_{\text{max}} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$ is the ceiling when that term dominates. Each candidate is fixed by removing the impairment, not by raising OMA.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

PSRR Power-supply rejection ratio: how well a bias board or driver rejects rail noise. Weak PSRR converts electrical noise into optical intensity noise through the laser.

Step 3: diagnose by waterfall shape and bisect. Sweep received power and plot the BER waterfall. Confirm it floors rather than shifts. Then choose the next measurement from the leading hypothesis: RIN with a quiet lab source versus the product bias board; ORL reflector sweep; pattern change; FEC error histogram (MPI and bursts cluster; Gaussian noise spreads); equalizer or CDR diagnostics.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

Step 4: fix the mechanism, then prevent recurrence. Isolator or connector hygiene for feedback. Supply filtering for electrical noise. Aggressor isolation for crosstalk. Equalizer or pattern fixes when ISI or DSP limits. Replace the laser only if intrinsic RIN remains out of spec after board and ORL are cleared. Add the missed stress to qualification and ATP.

How to say it aloud. "A floor means power is no longer the limiting variable. I confirm with a waterfall, then bisect the leading non-power-limited hypothesis." Offer the Q_{max} line when RIN is the candidate.

A.10.9 What makes an HTOL projection credible?

A FIT number without a mechanism is a spreadsheet. The frame is four conditions, said in order (Section 5.13). HTOL is only as credible as those four.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Condition 1: a named failure mechanism. The projection applies to a specific drift parameter: threshold current, slope efficiency, wavelength, or a defined hard failure such as COD. "The laser" is not a mechanism. Name the parameter, the end-of-life criterion, and the distribution you are projecting.

COD Catastrophic optical damage: sudden, irreversible facet failure in a laser under thermal or optical overstress.

Condition 2: a justified activation energy. E_a must be measured or justified for that mechanism on that process, with sample size and confidence bounds. Copying an E_a from another product, another vendor, or a textbook table is the most common way a projection lies. State the Arrhenius form and the temperatures used, and be ready to say what happens if E_a is wrong by 0.1 eV.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

Condition 3: the stress matches the field mechanism. The stress must accelerate the mechanism the fleet will see, without adding a new one. A stress temperature that triggers solder creep the field never sees produces a pessimistic fiction. A bench HTOL that omits thermal cycling, connector wear, or bias-rail transients produces an optimistic fiction. Say which field mechanisms the HTOL does and does not cover.

Condition 4: a field feedback loop. Compare the RMA Pareto against the projection. Divergence is evidence the model is wrong, not that the fleet is unlucky. Update E_a, the mechanism list, or the use condition when the field disagrees. A projection that never meets field data is theater.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

How to say it aloud. "Named mechanism, justified E_a, stress matches field, field feedback. Without those four, FIT is a spreadsheet." Offer one example of a stress that invents a false mechanism.

A.10.10 Which data must an automated test save so a failure can be replayed?

This question is about whether you have ever had to defend a ship decision months later. The frame: enough data that a unit failing today can be re- analyzed without the station or the person.

What identity and setup must be saved. Sample identity down to serial, lot, and date code. Instrument identity and calibration state, including the path-loss table version. Fixture, cable, and reference-plane description, because a power number without a plane is not a measurement. Software, firmware, and calibration-table versions on the device under test. Test conditions: temperature, supplies, pattern, dwell, and timestamps that let results correlate with environmental logs.

What measurement content must be saved. Raw measured data, not only pass/fail. Eyes, LIV points, spectra, and BER sweeps that can be re-plotted. Station-to-station golden-unit correlation results, so measurement drift can be separated from product drift. Without the raw data, a later engineer can only argue about the limit, not about the unit.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

The replay test of sufficiency. Given a field return, can you reproduce the exact shipping measurement and decide whether the unit changed or the measurement did? If the answer is no, the archive is incomplete. That question is the acceptance criterion for the data system, not a nice-to-have.

How to say it aloud. "Identity, instrument state, reference plane, raw data, and versions. The test is whether I can replay ship data on a return." Offer one war story about a missing path-loss table if you have one.

A.10.11 When would you choose an EML, silicon MZM, or ring modulator?

Choose by constraint, not preference. State the requirement slice first, then name what each path buys and costs (Tables 3.12 and 5.1).

When an EML wins. An EML wins for single-wavelength DR or FR at 100–200G/lane when cost, maturity, and one-chip integration dominate. You get low chirp relative to a DML, a mature supply chain, and a bias-versus-temperature calibration that the industry already knows how to qualify. You pay for an InP process, EAM aging as a second wear mechanism, and a calibration table that must be verified at temperature corners.

When a silicon MZM wins. A silicon MZM wins when the modulator must sit on the PIC beside other functions and the link wants a broad, flat passband with no wavelength lock. You get CMOS-compatible integration and a wide optical bandwidth. You pay for die area, drive swing, RF co-design of the traveling-wave electrode, and a bias-control loop that holds quadrature. Validation must cover driver matching and bias-rail behavior across temperature.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.

FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.

DML Directly modulated laser: modulate bias current; high chirp, low cost.

InP Indium phosphide: III-V semiconductor for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

PIC Photonic integrated circuit: waveguides, modulators, detectors, and couplers on one chip (typically SOI at 220 nm Si).

CMOS Complementary metal-oxide-semiconductor: mainstream IC process; TIAs and SerDes at 16–3 nm nodes.

When a ring wins. A ring wins when many wavelengths must fit on one die, in dense WDM and co-packaged engines. It is small and WDM-native. You pay for a wavelength-lock loop, heater power, thermal-crosstalk validation, and a resonance-alignment failure mode the others do not have. The validation burden is the decision: if you cannot afford lock-range and crosstalk testing, you cannot afford the ring.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Thermal crosstalk Heater or self-heating on one ring shifts neighbors' resonances; worst case is full-bank traffic at max case T.

How to say it aloud. "Constraint first. EML for mature single-λ DR/FR. Silicon MZM for broadband PIC integration. Ring for dense WDM with the lock and heater cost accepted." Offer the validation-burden line as the closer.

A.10.12 How do you decide whether a field issue is performance, reliability, or manufacturability?

Wrong bucket, wrong owner, wasted weeks. Classify on the ticket before failure analysis starts (Section 7.12).

Bucket 1: performance. Ask: did the unit ever meet spec under the field condition? If the design or operating point never closed the budget at that corner, it is performance. The fix is retune, derate, a thermal redesign, or a honest spec change. Shipping more of the same design will not help.

Bucket 2: reliability. Ask: did it meet spec at ship and then degrade with time or stress? That is reliability. The fix is a life model, a screen, derating, burn-in, or field replacement. Telemetry trends (bias rising at constant power, actuator heading toward rail) usually show this before the hard failure.

Bucket 3: manufacturability. Ask: does the failure cluster by lot, date code, site, or process step rather than by time? That is manufacturability. The fix is containment, supplier corrective action (8D), and an ATP or SPC change. A lot-correlated escape is not a physics mystery; it is a process-control gap.

8D Eight disciplines (8D) corrective action: contain, root-cause, correct, and prevent with the supplier; close only with verification evidence.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

How telemetry answers all three before a pull. Install age, lot correlation, and trend shape (sudden, gradual, or corner- dependent) usually pick the bucket before a unit is removed. Classify first, then pull hardware with a plan that matches the bucket. Performance issues need a design owner. Reliability issues need a life-model owner. Manufacturability issues need a supplier and process owner.

How to say it aloud. "Ever meet spec? Then degrade with time? Or cluster by lot? Those three questions pick the owner." Offer a one-line example of each bucket from your experience if you have one.

Practice case: *Section B.5.*

A.10.13 What would you put in fleet telemetry, and why?

Telemetry exists to catch margin erosion early and to triage without pulling hardware. Log what discriminates hypotheses, not every register on the chip (Sections 7.8 and 7.12).

Per-lane observables. Transmit and receive optical power, laser bias current, and pre-FEC BER with FEC error histograms. Bias rising at constant power is aging. Power dropping at constant bias is the optical path. Clustered errors mean bursts (MPI, connector, ESD event) rather than Gaussian noise. Per-lane data is what makes the sibling-lane method possible in the field.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

Module observables. Temperature, supply rails, TEC or heater drive, and lock-loop error. Actuators near their rails are consumed margin even while performance still passes. A railed TEC today is a BER fail next summer. Alarm on actuator headroom, not only on hard optical thresholds. Read state over CMIS.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

Events and context. LOS and LOL history with timestamps, resets, and firmware versions, preserved through reboots so intermittents leave evidence. Lot and date code, install age, and rack position, so one query separates unit, lot, and environment. Without context, every incident looks unique.

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

Alarm philosophy. Alarm on trends and disagreements, such as monitor versus expected power or bias versus a peer lane, not only on hard thresholds. Hard thresholds catch dead units. Trends catch dying ones. The point of fleet telemetry is the dying ones.

How to say it aloud. "Per-lane power, bias, and pre-FEC BER; module temperature and actuator drive; events with context; alarms on trends." Justify each item with the hypothesis it separates, then stop.

LLM practice

Summary. The worked answers cover laser requirements, the validation ladder, hot BER with stable power, aging versus calibration, second-source qual, weak lane isolation, sensitivity shift, BER floor, HTOL credibility, replayable test data, modulator choice, field triage buckets, and fleet telemetry. Speak the frame, walk the steps, close on the decision and control.

Paste to an LLM and answer aloud. Run a 45-minute mock interview. Pick six of the rehearsal questions at random. After each answer, ask one follow-up that forces a measurement choice or a supplier/fleet decision. Score me as Staff-level only if I name the decision unlocked, not only the instrument used.

A.11 Must-know abbreviations (drill list)

Definitions live in Chapter G. Do not maintain a second glossary here. Drill the expansions cold, then say one measurement, stress, or decision each term unlocks.

Optical / debug core. ATP, APC, BER/BERT, CMIS, DCA, EML, ER, FEC/KP4, FIT, HTOL, LIV, LOS/LOL, MPI, OMA, ORL, RIN, RLM, SECQ, TDECQ, TEC, TIA, VOA.

Reliability core. Arrhenius/E_a, bathtub, burn-in versus HTOL, DPA, ESD, GR-468, HAST, HTSL, JESD47, MTBF, FIT/DPPM.

Manufacturing core. FAIR, golden unit, gauge R&R, NPI/PVT, first-pass yield, SPC, ATP, NFF/RMA, 8D/CAPA.

Form-factor and architecture hooks. DFB, DML, DR/FR, EAM, ELSFP, MZM, MRM, OSFP/QSFP-DD, PIC/SOI, VCSEL, WDM.

LLM practice

Summary. Know the drill list cold. Expand the term, then say why it matters in a debug or ship decision. Prefer Chapter G if the definition is fuzzy.

Paste to an LLM and answer aloud. Drill abbreviations. Give me ten random terms mixing optical debug, reliability, and manufacturing. For each, I must expand it in one sentence and give one measurement, stress, or failure mode it connects to. Fail me if I only expand the letters or confuse burn-in with HTOL.

A.12 Scoring rubric (0–2)

Score each dimension 0 (missing), 1 (named but thin), or 2 (used to drive the next action). Staff-level answers rarely leave zeros on scope, measurement, decision, and recurrence.

Scope Unit / lot / vendor / fleet / plant named before deep debug.

Hypotheses Competing mechanisms ranked; update spoken after evidence.

Measurement Named instrument or observable that cuts the tree.

Plane / access Reference plane stated; black-box versus engineering access honored.

Causal discipline Leading mechanism until controlled confirmation; no surviving-hypothesis-as-confirmed-mechanism.

Decision Ship / stop / contain / redesign / continue named explicitly.

Recurrence ATP, sample, SPC, supplier, design, firmware, or telemetry control named.

Communication Short frame first; numbers and planes without buzzword soup.

Use this case-and-debug rubric for App B cases and fleet drills. For chapter-end spoken Interview Q&A, use the five-dimension rubric below.

A.12.1 Chapter-end spoken answers

Score each spoken answer from 0 to 2 on every dimension. Maximum score is 10.

Scope and assumptions 0: no requirement, plane, or context.
1: mentions the requirement.
2: states requirement, conditions or access, and the release decision.

Technical or causal reasoning 0: assertion or test list.
1: gives relevant technical reasoning.
2: connects risks, mechanisms, or dependencies without premature causality.

Evidence selection 0: lists instruments.
1: chooses a useful measurement.
2: explains why the measurement separates hypotheses or supports the decision.

Decision and remaining risk 0: no decision.
1: technical conclusion only.
2: includes release, containment, restriction, redesign, or control, and names remaining risk when relevant.

Communication 0: rambling or jargon-heavy.
1: correct but difficult to follow.
2: direct, structured, and appropriately qualified.

Interpretation: 0–4, review the chapter concepts; 5–7, technically competent but needs stronger structure; 8–9, strong interview response; 10, Strong Staff-style answer structure (a rehearsed perfect score is not proof of Staff-level performance on the job).

A.13 One-week interview preparation plan

Assume seven days, with the interview near the end of Day 7. Protect sleep. Each day has a Learn list (pointers only) and one speakable Output. If a day slips, cut new reading first, not stories or the mock. Prefer the measurement with the highest information gain per cost (Sections A.2 and B.1).

Day 1: Mental models. *Learn:* five ledgers (Section A.8.4), validation lifecycle (Table 7.2), margin thinking, reference planes, Staff pattern (Section A.1).
Output: Explain aloud, "How would you debug a BER failure?"

Day 2: Optical fundamentals. *Learn:* power, OMA, ER, RIN, BER, sensitivity (Chapter 3 and section A.8.9).
Output: Explain aloud, "Why does power not equal quality?"

Day 3: Validation. *Learn:* characterization, margin, interoperability, and where qualification and manufacturing sit in the lifecycle (Section A.8.5, section C.8, and table 7.2).
Output: Design a validation plan for a new optical module in two minutes.

Day 4: Reliability qualification. *Learn:* mechanism-driven qualification, HTOL versus burn-in, sample confidence (Chapter 8 and sections C.15 and D.3).
Output: Walk a laser-degradation qualification argument aloud.

Day 5: Manufacturing. *Learn:* measurement systems, ATP, yield, SPC, and scale-up (Chapter 9, section C.13, section 9.4.3, and table 9.2).
Output: Discuss how you would take a link to volume production.

Day 6: Architecture and failure analysis. *Learn:* IM/DD, WDM, AI networking (Chapters 1, 3, 6 and 10); fleet trees and cases (Chapter D and sections B.5 and C.9). Rehearse two true stories (Section A.9).
Output: Defend an architecture tradeoff; debug one fleet case without naming a component first.

Day 7: Mock interviews. *Learn:* three cheat sheets below; Top 25 index (Chapter E); thirty-second callouts (Chapter C); whiteboard trees (Chapter D).
Output: Full mock. Light story review only. Stop two to three hours before the call. Sleep.

If you have less than seven days. Compress: Day 1, Day 3 validation, Day 4 reliability, Day 5 manufacturing, Day 7 mock. Keep the cheat sheets and Top 25. Cut new chapter reading first.

A.14 Staff interview cheat sheets

Interview-simple flows only. Full trees stay in Chapter D.

A.14.1 Cheat Sheet A: Debugging flow

Five questions (eyebrow). Requirement? Scope? Mechanisms? Separating measurement (plane and access level)? Decision and recurrence control?

Problem → **Scope** → **Power?**

↓

Path (power ledger vs quality ledger)

↓

Contain → **Confirm** → **Prevent**

Full trees: Sections D.1 and D.4. Pick the next measurement by information value, not habit (Section B.1 and table B.1).

A.14.2 Cheat Sheet B: Validation / decision loop

$$\begin{array}{c}\textbf{Requirement} \rightarrow \textbf{Risk} \rightarrow \textbf{Mechanism}\\ \downarrow \\ \textbf{Measurement} \rightarrow \textbf{Evidence} \rightarrow \textbf{Decision} \rightarrow \textbf{Control}\end{array}$$

Staff pattern and lifecycle: Section A.1 and table 7.2.

Executive frame (one breath). Issue / Impact / Population / Evidence / Confidence / Containment / Mechanism status / Next decision. Details: Section B.3.

A.14.3 Cheat Sheet C: Qualification flow

$$\textbf{Mechanism} \rightarrow \textbf{Stress} \rightarrow \textbf{Observable}$$
$$\downarrow$$
$$\textbf{Acceptance} \rightarrow \textbf{Confidence} \rightarrow \textbf{Production control}$$

Evidence path and planning matrix: Sections D.3 and 8.3.

Key idea. I first want to understand the scope of the problem, then determine which margin ledger is being spent, choose the measurement that eliminates the largest number of hypotheses, make the product decision, and finally add the control that prevents the next escape.

A.15 Bad answer / good answer

Three pairs only. Grade yourself against the strong column, then open the matching playbook.

1. *Debug a bad optical link.* *Weak:* "I would check the cable, swap the module, and look at the eye."
Strong: Scope first (one lane / one host / fleet). Run the power-versus-quality fork. Name the ledger, the separating measurement, the decision, and the control (Sections C.1 and D.4).

2. *Validate a new module.* *Weak:* "I would use a DCA, a power meter, and a temperature chamber."
Strong: Walk the lifecycle stage that matches the risk: requirements, characterization, margin, interop, qual, manufacturing control (Section C.8 and table 7.2). Instruments are means, not the plan.

3. *Qual escape in the field.* *Weak:* "Qual was insufficient."
Strong: Classify the miss: wrong mechanism, wrong stress, wrong observable, wrong acceptance, or missing production control. Then name the fix to the evidence path (Sections C.10 and D.3).

Chapter B

Engineering case studies

This appendix is practice under incomplete information. The frameworks already live in Chapters A, C and D and Table 7.2. Here you apply them. Do not jump to the mechanism. Score yourself with Section A.12 after each case.

B.1 Information value

The best next measurement is not the most detailed measurement. It is the measurement that eliminates the most uncertainty at acceptable cost (Sections A.1 and A.2).

Measurement	Cost	Information gained
Telemetry query	Low	Population scope, trends, lot/host tags
Swap / remap experiment	Low	Ownership isolation (Tx / Rx / host / plant)
Attenuation / BER waterfall	Medium	Margin shape: shift versus floor
External optical eye	Medium	Signal quality at a named plane
Optical / package teardown	High	Physical mechanism confirmation

Table B.1. Information value is uncertainty removed per cost. Decision unlocked: which next measurement to buy. Prefer Level 0–2 until those measurements stop reordering beliefs (Section A.2).

Bad next step: "Run full optical characterization." Better: "Compare failing and passing units from the same lot," because that separates population effects quickly.

Why experienced engineers rank measurements by information value?

Because the best next test is the one that reorders beliefs the most per minute and per dollar, not the one that fills the most plot panels.

Engineering heuristic. Never spend an hour measuring something that a five-minute swap test can eliminate.

B.2 How interviewers evaluate answers

Use this interview-facing summary. Full detail remains in Section A.12 (plane/access, causal discipline, and recurrence are scored there too).

Axis	Score	Meaning
Scope	0	Jumps to a solution.
	1	Asks some questions.
	2	Defines population, timeline, and ownership.
Hypothesis quality	0	One explanation.
	1	Several guesses.
	2	Mechanism-based hypothesis tree.
Measurement choice	0	Lists tests.
	1	Lists useful measurements.
	2	Chooses by information value and access level.
Decision quality	0	No action.
	1	Technical fix only.
	2	Containment + correction + prevention.
Communication	0	Rambling.
	1	Technically correct but unstructured.
	2	Clear executive frame plus engineering depth.

B.3 Executive communication

Staff answers must travel upward. Use this frame:

Problem What is observed?

Impact Who or what is affected, and how large?

Evidence What was measured, at what plane and condition?

Confidence Observation, correlation, hypothesis, or confirmation?

Containment What stops growth now?

Next decision What unlocks the next ship or hold call?

Illustrative executive line (Case B.5 style): "About three percent of modules from lot X show rising pre-FEC BER after roughly 90 days. Average received power is stable. Evidence points toward analog-supply degradation correlated with that lot. Shipment from the lot is paused while supplier FA proceeds."

B.4 Supplier conversation

Structure supplier calls the same way you structure debug (Sections C.10, D.9 and 9.2):

```
Supplier conversation framework
  |
Observed issue
  |
Affected population
  |
Evidence (plane, condition, access)
  |
Requested analysis
  |
Containment
  |
Corrective action
  |
Verification of next lots
```

Avoid "the vendor caused this." Prefer: "The evidence indicates a supplier-process correlation requiring confirmation."

B.5 Case 1: Fleet-wide BER degradation after deployment

Problem statement. Thousands of optical modules are deployed. Pre-FEC BER rises gradually on a subset of links. Average received power is unchanged. Firmware and host platform are common. Failures correlate with one manufacturing lot. What do you do?

> **What this usually means.** Lot-scoped gradual BER rise with flat average power
>
> *Usually:* process, calibration, component supplier, or attach change inside that lot
>
> *Not:* independent random wear of unrelated serials with no shared history

Available information. Illustrative: rising pre-FEC BER over weeks; Rx power telemetry flat; same CMIS firmware train; one date code / lot dominates the Pareto; sibling lots on the same hosts look healthy.

Missing information. Whether cool-down recovers; whether the waterfall shifts or floors; component genealogy inside the lot; whether monitors agree with external meters; whether a process change landed in that lot.

Initial hypotheses. Transmitter quality, channel / plant, receiver, DSP, control loop, environment. Gross optical loss is deprioritized while average power holds (Section D.4). Lot correlation raises process, calibration, or component-supplier hypotheses above exotic new physics.

Decision tree. Walk Sections C.9, D.1 and D.9: scope and time behavior; power versus quality; contain if the population can grow; compare good versus bad lots before deep FA.

Measurements selected. Level 0: cohort query by lot, host, temperature, firmware (Table B.1). Level 1: golden swap and dwell on failing versus passing serials. Level 2: BER waterfall and external power at the faceplate. Level 3–4 only after ownership is assigned.

Results (staged). Population is lot-scoped. Average power remains flat on external meter. BER waterfalls show rising floors or soft floors on failing serials, not a uniform power shift. Good-lot siblings on the same host pass. Process history shows a solder / attach change on the failing lot (illustrative).

Updated hypothesis. Leading mechanism: analog supply instability from solder attachment degradation, producing signal-quality loss with stable average optical power. Still a leading hypothesis until physical confirmation.

Confirmed mechanism. Physical inspection and electrical FA confirm solder attachment failure on the analog supply path, correlating with the BER signature. That is confirmation, not the first sentence you say in the interview.

Containment. Pause shipment and further deployment of the affected lot; raise monitoring on installed serials; replace highest-risk units per impact model (Section D.16).

Corrective action. Supplier process corrective action on the attach step; incoming inspection or process monitor for the signature; split RMA codes by lot and mechanism class.

Qualification / ATP / telemetry update. Add or tighten the production proxy that would have caught the signature; review whether qual stress covered this mechanism and sample diversity (Sections C.13 and D.3). Telemetry: alarm on BER trend with stable power for that cohort.

Fleet prevention. Burn down the installed cohort; keep the control owner until rates fall; feed the lesson into Section 7.1.12.

Score yourself. Use Sections A.12 and B.2. Fail the drill if you named a component before scope and containment.

Executive summary. Problem: gradual BER rise. Impact: one manufacturing lot in fleet. Evidence: stable average power; lot Pareto; FA confirms solder/supply path. Confidence: confirmed for returned units; fleet correlation strong. Containment: lot hold. Next decision: supplier CA verification on next lots.

Deep dive. Sections C.1, C.9, C.10, D.9 and 11.2.

B.6 Case 2: New optical module supplier qualification

Problem statement. A second-source supplier proposes an "identical" module. How do you qualify?

Available information. Incumbent module is shipping. Supplier offers datasheet alignment and a small set of engineering samples. Target hosts and cable plant are known.

Missing information. Multi-lot distributions; process corners; reliability mechanisms and E_a justification; ATP correlation; interop matrix coverage; pilot plan.

Initial hypotheses. Not "it will work if the datasheet matches." Equivalence must be defined across functional, performance, environmental, reliability, and manufacturing dimensions (Sections C.7 and C.15).

Decision tree. Freeze requirements first (Table 7.2). Then walk bring-up through margin, interop, mechanism-based qualification, manufacturing validation, controlled pilot, and fleet monitoring. A hero sample is not a gate.

Measurements selected. Black-box: BER, throughput, interop on claimed hosts, margin corners, power and sensitivity distributions across lots. Engineering access only when black-box evidence cannot decide. Reliability: stresses tied to named mechanisms (Section 8.3).

Results (staged). Illustrative: means look close to the incumbent; spreads are wider on RIN or hot BER; one host/firmware combo fails interop; HTOL needs a process-specific E_a argument the supplier has not yet closed.

Updated hypothesis. Supplier can meet nominal function but has not yet demonstrated distributional equivalence or mechanism-justified life evidence.

Confirmed mechanism. Not applicable until a specific fail mode appears. The decision now is hold or restrict, not a confirmed-mechanism narrative.

Containment. Do not open volume. Limit to lab or controlled pilot serials only.

Corrective action. Require multi-lot data, closed interop matrix, and FAIR before PVT/MP language is used.

Qualification / ATP / telemetry update. Correlate supplier ATP to your proxy; split field RMA codes by vendor from day one; define pilot exit before broader ship (Section 8.3).

Fleet prevention. Separate vendor codes in telemetry and RMA so a second-source escape cannot hide inside a merged bucket.

Score yourself. Fail if you qualified on one golden sample or copied the incumbent datasheet as the requirement.

Executive summary. Problem: second-source equivalence claim. Impact: single-vendor risk reduction versus escape risk. Evidence: incomplete lot distributions and open interop / life items. Confidence: insufficient for open volume. Containment: no volume ship. Next decision: multi-lot package and pilot exit criteria.

Deep dive. Sections C.7, C.15, 8.3 and 9.2 and table 7.2.

B.7 Case 3: Temperature-dependent BER failure

Problem statement. A module works at room temperature and fails at high temperature. What do you investigate?

What this usually means. Fails hot, may recover cool, average power looks stable

Usually: thermal margin, wavelength or lock, bias tables, receiver noise, mechanics

Not: a closed wear-out FIT claim from room-temperature ship data alone

Available information. Illustrative: pre-FEC BER rises above target at high case temperature; average optical power telemetry looks stable; cool-down may or may not recover (not yet stated).

Missing information. Recovery on cool-down; waterfall shape; wavelength walk; actuator headroom; host rail noise at temperature; whether siblings fail the same way.

Initial hypotheses. Do not immediately blame the laser. Build a tree:

```
Temperature-dependent BER hypothesis tree
  |
Temperature increase
  |
Laser efficiency / ER collapse
  |
Wavelength drift
  |
Receiver noise
  |
TIA bandwidth
  |
DSP adaptation
  |
Mechanical alignment
  |
Power supply / PSRR
```

Decision tree. Sections C.4 and D.4 and section 7.1.6: scope; power versus quality; measure remaining margin at the failing temperature first.

Measurements selected. Level 0–1: BER/FEC and telemetry at failing T; cool-down trial; sibling compare. Level 2: waterfall, external power, wavelength if telemetry weak. Level 3 (if access): bias sweep, external eye, OSA, TEC/heater codes (Section A.2).

Results (staged). Illustrative path A: cool-down recovers; bias sweep restores the external eye; calibration table segment boundary is wrong. Path B: actuator railed; thermal design lacks headroom. Path C: waterfall floors; ORL or RIN path needs work. Pick measurements that reorder these beliefs quickly.

Updated hypothesis. Leading mechanism follows the path that survived the staged cuts. Speak the update aloud after each measurement.

Confirmed mechanism. Only after controlled confirmation (bias restore, thermal redesign proof, or ORL dependence). Until then it remains leading.

Containment. Restrict deployment temperature or hosts if fleet risk exists; do not wait for perfect FA before ownership actions.

Corrective action. Retune tables, fix thermal design, or screen the ORL-sensitive population, as dictated by the confirmed path.

Qualification / ATP / telemetry update. Put the loaded hot corner in ATP; watch control-ledger headroom in telemetry.

Fleet prevention. Alarm on hot BER with stable power and railed actuators.

Score yourself. Fail if the first sentence was “the laser died” without a hypothesis tree.

Executive summary. Problem: BER fail at high temperature, power stable. Impact: envelope risk. Evidence: (state path A/B/C after your measurements). Confidence: leading until confirmed. Containment: temperature or SKU restrict if needed. Next decision: table, thermal, or ORL control with ATP corner.

Deep dive. Sections C.4 and 11.12 and sections A.10.3 and 7.1.6.

B.8 Case 4: Qualification escape

Problem statement. The product passed qualification. About six months later, field failures appear. Why did qualification miss it?

Available information. Illustrative: qual report shows HTOL and environmental passes on engineering lots; field fails show a mechanism or environment not stressed, or a lot/site not represented in the qual sample.

Missing information. Exact field mechanism; whether production process matches qual hardware; whether the observable used in qual could see the failure; sample diversity (lots, date codes, sites, corners).

Initial hypotheses. Do not stop at "qualification was insufficient." Prefer mechanism-level explanations:

Wrong failure mechanism Stress did not accelerate the real wear-out.

Insufficient sample diversity Wrong lots, sites, or corners (Section 8.3).

Measurement gap No observable tracked the degrading signature.

Production difference Qual hardware or process differed from volume.

Fleet condition Real environment exceeded qual assumptions.

Decision tree. Sections C.13, D.3 and 11.13: name the missed uncertainty; choose containment; update the evidence path and production proxy.

Measurements selected. Reproduce field signature at a named plane; compare qual versus production genealogy; check whether ATP would separate good versus bad today; FA/DPA only after black-box ownership.

Results (staged). Illustrative: field fails cluster on a production site absent from qual; or humidity-driven corrosion appears though damp heat used a different observable; or qual used hand-built engines while volume uses a new attach process.

Updated hypothesis. State which miss class is leading. Keep others alive until ruled out.

Confirmed mechanism. After FA and process compare, name the confirmed mechanism and the miss class (design, process, supplier, test escape, or still unknown).

Containment. Hold affected lots/sites; expand monitoring on the field cohort.

Corrective action. Fix the process or design as owned; do not only rewrite the qual report.

Qualification / ATP / telemetry update. Add the stress or observable that would have caught it; widen sample strategy; add the cheapest production control that separates the signature (Section C.13).

Fleet prevention. Keep cohort burn-down and a decision owner until rates fall; feed Section 7.1.12.

Score yourself. Fail if your only sentence was “qualification was insufficient” without a miss class and a control update.

Executive summary. Problem: field fails after passed qual. Impact: (rate / cohort). Evidence: (miss class + FA). Confidence: state confirmation level. Containment: lot/site hold. Next decision: updated qual proxy and ATP/SPC owner.

Deep dive. Sections C.13, D.3, 8.2, 8.3 and 11.13.)

Chapter C

Thirty-second interview frameworks

This appendix is the interview quick-reference. Deep technical explanations live in the body chapters and Chapter A. Each playbook uses one format: question, assumptions, 30-second answer, key concepts, common mistakes, and deep-dive reference. Memorize the callouts labeled "30-second answer." Expand only where the interview asks. Name access level (Section A.2) and use Section D.16 for claim, population, plane, and decision.

C.1 BER increased with stable power

Interview question.

> Pre-FEC BER rose while average received power held steady. How do you debug it?

What the interviewer is testing. Debugging discipline: scope before components, power-versus-quality fork, and ending on a decision with recurrence control.

Assumptions to state. Average received-power telemetry is trusted; the issue reproduces; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm scope, time behavior, and that average-power telemetry is trusted at a named plane.

30-second answer (memorize). Stable average power reduces the likelihood of gross optical-loss problems, so I move into signal quality. I first scope the population and timing behavior, then separate transmitter, channel, receiver, and DSP hypotheses using the highest-value measurement available at the lowest access level. If engineering access exists I add deeper measurements such as eyes or bias behavior. The final action depends on whether the issue requires containment, repair, supplier correction, or qualification updates.

```
Elevated BER, power stable
  |
Scope analysis (how large?)
  |
Sudden or gradual?
  |
Signal-quality path (black-box first)
  |
Tx quality clean?
  |-- NO --> stay on Tx
  |-- YES --> channel / Rx / DSP
  |
Correlation analysis (which cohort?)
  |
Decision + recurrence control
```

Key concepts. Stable average power deprioritizes gross insertion loss but does not eliminate fast fluctuation, intermittent contact, monitor averaging, or reflection-dependent effects. Scope sets severity first; correlation after isolation unlocks contain or lot action. A bias sweep requires engineering access and is not universally first (Sections D.1 and D.11).

Measurements. Power meter $\rightarrow$ power ledger intact? $\rightarrow$ optical path or SI.
Black-box BER/FEC $\rightarrow$ quality path? $\rightarrow$ next isolate.
External optical eye / bias (if access) $\rightarrow$ setpoint? $\rightarrow$ retune.
Golden swap $\rightarrow$ Tx or Rx? $\rightarrow$ owner.
BER waterfall $\rightarrow$ shift or floor? $\rightarrow$ sensitivity vs noise.

Typical follow-ups.

- Why measure power before the DCA?
- Why not start with RIN?
- What if cool-down recovers the BER?

Common mistakes.

- "The receiver is bad because optical power is fine." Power does not represent noise, timing, distortion, or control behavior.
- Naming the laser first.
- Calling "bad eye" a confirmed mechanism.
- Listing instruments without a decision or recurrence control.

Thirty-second close. Average power is stable, so I deprioritize gross loss, scope the population, chase signal quality on the black-box surface first, isolate Tx / channel / Rx / DSP, then close with the control that catches the next escape.

Deep dive. Full prose: Section A.10.7 and section 11.2. Ledgers: Section A.8.4. Pattern: Section A.1.

C.2 Received power decreased

Interview question.

> Received optical power dropped and the link is failing. How do you debug it?

What the interviewer is testing. Power-ledger triage: external reference planes, monitor honesty, and population-driven containment.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm the power drop with an external meter at a named plane before trusting module monitors.

30-second answer (memorize). Power moved, so stay on the power ledger. First scope the failure. Confirm with an external meter at a named plane, then bisect source enable, coupling, connectors, MUX loss, and monitor-PD / APC honesty. Therefore I would contain if lot-correlated, clean or replace the plant if local, and update ATP or hygiene rules so it does not recur.

```
Power down
  |
Scope
  |
External meter @ named plane
  |
```

```
Monitor vs external agree?
  |-- NO --> APC / monitor-PD / cal
  |-- YES --> source / coupling / connector / MUX
  |
Population?
  |-- lot/vendor --> contain + FA
  |-- single --> plant or unit repair
  |
Decision + control
```

Key concepts. A power drop is not automatically a dying laser. Monitor corruption and dirty plant fake laser failure. The external meter is the cut that protects you from trusting a lying APC loop.

Measurements. External power meter → true power? → optical path.
Inspect / ORL → plant dirty or reflective? → clean / replace.
LIV → device changed? → replace vs setpoint.
Lot query → population? → contain vs unit fix.

Typical follow-ups.

- Why not trust the module's reported Tx power?
- When do you stop ship versus clean the connector?

Common mistakes.

- Trusting monitor-PD without an external meter.
- Treating one dirty connector as a fleet laser problem.

Thirty-second close. I confirm power at a named plane with an external meter, bisect the optical path, contain if lot-correlated, and update the screen that missed it.

Deep dive. Power fork and instruments: Section A.8.3. APC and calibration: Section 5.11.

C.3 One weak lane

Interview question.

> A single lane in a four-lane transceiver has elevated BER while received power on that lane looks normal. How would you isolate it?

What the interviewer is testing. Lane isolation: sibling comparison, optical-versus-electrical bisect, and lot versus unit ownership.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm whether one lane is an outlier versus a multi-lane gradient or pattern.

30-second answer (memorize). First scope: one lane, one unit, or a pattern across the lot? Compare sibling lanes, then optical-versus-electrical swap to split path from driver or TIA. Therefore I would fix assembly or the array element that owns the fault, and add the screen that would have caught it.

```
One weak lane
  |
Siblings OK?
  |-- NO --> shared supply / thermal / host
  |-- YES --> lane-local
        |
   Opt vs elec swap
        |-- optical --> FAU / coupling / PIC lane
        |-- electrical --> driver / TIA / SerDes
        |
   Lot pattern?
        |
   Decision: rework / screen / supplier
```

Key concepts. Sibling comparison is the cheapest population cut inside one module. Optical-versus-electrical swap prevents weeks of laser FA on a TIA lane (Section A.10.6).

Measurements. Sibling BER/power $\longrightarrow$ shared or local? $\longrightarrow$ scope.
Lane swap $\longrightarrow$ optical or electrical? $\longrightarrow$ owner.
LIV on weak lane $\longrightarrow$ device or coupling? $\longrightarrow$ FAU vs die.

Typical follow-ups.

- Why compare siblings before opening the module?
- What lot pattern would make you call the supplier today?

Common mistakes.

- Rewriting the module spec for one assembly escape.
- Skipping the electrical path when power looks fine.

Thirty-second close. I compare siblings, swap optical versus electrical, then own the assembly or array element and the ATP row that would have caught it.

Deep dive. Worked answer: Section A.10.6. FAU and parallel optics appear throughout Chapter 11.

C.4 High-temperature failures

Interview question.

> BER worsens at high temperature but average power is stable. What do you do?

What the interviewer is testing. Corner debugging: measure where it fails, separate reversible control/calibration from permanent damage.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm scope and whether cool-down recovers before blaming the laser.

30-second answer (memorize). Power held, so leave the power ledger. First scope the failure and whether cool-down recovers. At the failing temperature, read externally visible remaining margin: BER/FEC, telemetry, retrains, and control headroom. With engineering access, add bias sweep, OSA, and external optical eye. Therefore I would fix the table or thermal design and put that loaded corner in the ATP.

```
Hot BER, power stable
  |
Scope + cool-down recovers?
  |-- YES --> operating point / table / control
  |-- NO  --> aging / permanent damage
  |
At failing T: black-box margin first
  |-- access --> external eye / bias / OSA
  |
Control ledger exhausted?
  |-- YES --> thermal design / tuning range
  |-- NO  --> cal table / wavelength / Rx
```

```
   |
Decision: retune + ATP corner
```

Key concepts. Temperature failures are often control or calibration, not a dead laser. The decision uses externally visible remaining margin; internal physics is optional (Section A.10.3 and section D.11).

Measurements. Cool-down test → reversible? → thermal vs aging.
Black-box BER/FEC at hot → remaining margin? → gate.
Bias sweep at hot (if access) → optimum moved? → table.
Actuator codes → control authority left? → thermal design.
OSA → spectral alignment? → lock / filter.

Typical follow-ups.

- Why insist on measuring at the failing temperature?
- How do you say "TEC maxed" in product language?

Common mistakes.

- Debugging only at ambient.
- Saying TEC current hit max instead of control ledger exhausted.

Thirty-second close. I measure externally visible remaining margin at the failing temperature, ask whether cool-down recovers, fix table or thermal design, and put that loaded corner in ATP.

Deep dive. Section A.10.3 and section 11.12. Control ledger: Section A.8.4.

C.5 Laser aging versus calibration drift

Interview question.

> How do you distinguish laser aging from calibration drift?

What the interviewer is testing. Mechanism separation with external references and reversibility, not telemetry narrative alone.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm whether recalibration recovers the symptom (black-box) before opening LIV.

30-second answer (memorize). Physical aging often changes a baseline: LIV, power, spectrum, RIN, sensitivity, or drive. Calibration drift changes the operating point while the device remains substantially healthy. Start black-box: BER/FEC, telemetry, and whether recalibration recovers. Recalibration recovery updates probability; it is not proof. With engineering access, compare external LIV and other physical baselines to ship data. Therefore I would route aging to life/derate/replace and drift to table control plus an ATP loaded-corner check.

```
Symptom (bias up / BER up)
  |
Black-box telemetry first
  |
Degraded?
  |
Recal recovers?
  |-- YES --> raise P(calibration drift); not proof
  |-- NO  --> raise P(physical aging / damage)
  |
Need mechanism split?
  |-- YES --> engineering access / external baselines
  |
Physical baselines vs ship data
  |-- moved --> aging --> life / derate / replace
  |-- OK ----> deeper FA or table owner
```

Key concepts. Telemetry can look identical for aging and drift. Recalibration recovery updates belief; external baselines confirm the owner (Section A.10.4 and section D.11).

Measurements. Recal trial → recovers? → updates P(drift).
External LIV / power / spectrum (if access) → baseline moved?
Monitor vs meter → APC honesty? → monitor-PD.
Stress-hours plot → monotonic or step? → confirms.

Typical follow-ups.

- What is the silent failure mode if the monitor lies?
- Who owns aging versus who owns the table?

Common mistakes.

- Treating recalibration recovery as proof of healthy silicon.
- Sending a table bug to the reliability team.

Thirty-second close. I start black-box, treat recal recovery as evidence not proof, then compare physical baselines when access exists; aging goes to life actions and drift to table control plus a loaded-corner screen.

Deep dive. Section A.10.4 and sections 5.10 and 5.11.

C.6 Second component source

Interview question.

> How would you qualify a second laser, photodiode, driver, TIA, or PIC supplier?

What the interviewer is testing. Component-boundary judgment: requirements at the component interface, wafer/lot distributions, integration compatibility, process-specific reliability, supplier ATP correlation, and module-level confirmation.

Assumptions to state. The component boundary and ownership are defined; module confirmation is still required before open volume; engineering access to source samples may be available.

First thing I would check. Freeze the component-boundary requirements slice, not the incumbent datasheet.

30-second answer (memorize). Freeze requirements at the component boundary, not the incumbent datasheet. Characterize multi-lot distributions for LIV, SMSR, RIN, wavelength, and process corners. Demonstrate integration compatibility in the host module, run mechanism-appropriate reliability, correlate supplier ATP to your proxy, then confirm on module samples. Therefore I would gate open volume on FAIR, ATP correlation, and split RMA codes, not on a hero die.

```
Second component source
    |
Component requirements freeze
    |
Multi-lot distributions
    |
```

```
Integration in host module
  |
Mechanism-appropriate reliability
  |
Supplier ATP correlation
  |
Module confirmation + split RMA
```

Key concepts. Component and complete-product second sources have different access, interop burden, and owners (Sections D.8, D.16 and 9.2).

Measurements. Wafer/lot distributions → process spread? → risk.
Module confirmation → integration OK? → continue.
Supplier ATP vs your proxy → escape risk? → ship gate.

Typical follow-ups.

- What is measured at the component boundary versus the module connector?
- What changes if the second source is only a laser die?

Common mistakes.

- Treating a hero sample as a distribution.
- Skipping module confirmation after die-level pass.

Thirty-second close. I qualify at the component boundary with multi-lot distributions, then confirm in-module with FAIR, ATP correlation, and split RMA.

Deep dive. Section A.10.5 and section 9.2.

C.7 Second module or cable source

Interview question.

> How would you qualify a second transceiver, AOC, or cable-assembly supplier?

What the interviewer is testing. Complete-product second source: black-box nominal behavior, margin, host/peer interop, environmental and reliability evidence, manufacturing readiness, controlled pilot, and fleet RMA separation.

Assumptions to state. The product is bookended unless engineering samples are contracted; customer-visible metrics dominate; open volume waits on pilot exit.

First thing I would check. Define what equivalence means for the customer-visible product before measuring a golden sample.

30-second answer (memorize). Freeze the customer-visible requirements slice. Walk architecture feasibility, bring-up, characterization, margin, interoperability, reliability qualification, manufacturing validation, controlled pilot, mass production readiness, then fleet monitoring with feedback. Request engineering access only when black-box evidence is insufficient. Therefore I would gate open volume on FAIR, ATP correlation, pilot exit, and split RMA codes.

```
Second module / cable source
   |
Requirements freeze
   |
Bring-up + nominal + margin
   |
Interop + env/reliability
   |
Manufacturing + ATP readiness
   |
Controlled pilot
   |
Fleet monitoring + split RMA
```

Key concepts. Module second-source burden is interop and black-box margin across the envelope, not die physics first (Section D.11 and section A.8.5).

Measurements. Multi-lot black-box margin $\longrightarrow$ process spread? $\longrightarrow$ risk.
Host/peer matrix $\longrightarrow$ interop cliffs? $\longrightarrow$ restrict.
Pilot cohort $\longrightarrow$ assumptions hold? $\longrightarrow$ open MP.

Typical follow-ups.

- Why not clone the incumbent datasheet?
- What changes between a pluggable and an AOC second source?

Common mistakes.

- Qualifying a hero module.
- Merging supplier RMA codes.

Thirty-second close. I walk the product lifecycle on the customer-visible surface, then gate volume on FAIR, ATP correlation, pilot exit, and split RMA.

Deep dive. Section A.10.5 and section 9.2.

C.8 Validation plan for a new transceiver

Interview question.

> How would you validate a new optical transmitter from bring-up through production?

What the interviewer is testing. Validation-program judgment: sequencing, ownership, stage gates, dependencies, exit criteria, schedule and resource judgment.

Assumptions to state. Requirements and owners exist; stage exits are binary enough to fail; engineering access is requested only when a gate cannot decide.

First thing I would check. Freeze the requirements slice and named risks before listing instruments.

30-second answer (memorize). I start by defining the system requirements and failure risks. Then I build the validation plan around reducing uncertainty: architecture demonstrates the budgets can close, bring-up demonstrates basic operation, characterization builds the behavioral model, margin testing determines headroom, interoperability demonstrates system compatibility, qualification demonstrates reliability, manufacturing validation demonstrates repeatability, and controlled deployment plus telemetry confirms field assumptions. Every stage has an exit criterion tied to a decision.

```
Requirements definition
  |
Architecture review
  |
Bring-up -> Characterization
  |
Margin validation -> Interoperability
  |
Reliability qualification
  |
Manufacturing validation
  |
Controlled pilot -> Mass production
  |
Fleet monitoring -> Feedback
```

Key concepts. This answer is about program sequencing and ownership. Qualification *evidence* construction lives in Sections C.15 and D.3. The ladder stages are the gates (Section A.8.5 and section D.2).

Measurements. Each ladder stage → named uncertainty removed → continue / redesign / tighten ATP / stop ship.
Margin sweeps → which ledger dies first? → telemetry alarms.

Typical follow-ups.

- Which stage removes combination risk?
- How does margin budgeting change what you instrument in the fleet?

Common mistakes.

- Treating bring-up as population proof.
- Running HTOL without a named mechanism.

Thirty-second close. I walk the ladder as staged uncertainty reduction and refuse any measurement that answers no new question about remaining margin.

Deep dive. Sections A.8.7 and A.10.2 and table 7.2.

C.9 Fleet issue

Interview question.

> Field telemetry shows rising pre-FEC BER on a subset of racks. How do you triage?

What the interviewer is testing. Fleet ownership: scope, trend, bucket, and contain-versus-monitor under incomplete evidence.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Scope population and trend (unit / lot / vendor / rack / fleet) before pulling hardware.

> **30-second answer (memorize).** First scope: unit, lot, vendor, rack, datacenter, or fleet? Ask trend and change history. Classify performance versus reliability versus manufacturability before pulling hardware. Therefore I would contain if growing and supplier-specific, or monitor-only if tiny, flat, and no customer impact, with an owner on the next control.

> **Engineering heuristic.** Contain a growing lot-scoped escape before you finish the physics story. The next day of ship can cost more than the next day of FA.

```
Fleet symptom
  |
Scope ladder
  |
Rate / trend / customer impact
  |-- tiny, flat, no impact --> monitor only
  |-- growing / supplier --> contain now
  |
Bucket: performance / reliability / manufacturability
  |
Decision + owner + telemetry control
```

Key concepts. Fleet economics and scope pick the owner before FA. Pulling units without a bucket wastes the only failing state you had (Sections A.6 and 7.12).

Measurements. Telemetry query → scope and trend? → contain vs monitor.
Lot/date correlation → manufacturability? → supplier.
Golden host in rack → environment vs module? → owner.

Typical follow-ups.

- When is monitor-only the correct Staff answer?
- What do you do before the mechanism is known?

Common mistakes.

- Treating one rack as the fleet.
- Waiting for DPA before containing a growing lot.

Thirty-second close. I scope first, pick the bucket, contain if growing and supplier-specific, and name the owner of the next control.

Deep dive. Section A.6 and table 7.6.

C.10 Supplier escape

Interview question.

> A new date code from Supplier B fails the hot corner at about 3%. What do you do?

What the interviewer is testing. Staff judgment under uncertainty: contain today, keep FA open, own recurrence control.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Provisionally contain the growing population, then refine scope.

30-second answer (memorize). Contain first, then own the loop. Stop shipment of Supplier B affected lots, scope the deployed population, compare failing versus healthy units, open joint FA with the supplier, drive corrective action, expand ATP or process control, verify the next lot, and keep fleet monitoring. The customer keeps ownership of evidence quality and the ship decision. Therefore I would contain today rather than wait for SEM before acting.

```
Escape detected
  |
Contain lot / pause deploy
  |
Scope deployed population
  |
Compare failing vs healthy
  |
Joint FA with supplier
  |
Corrective action
  |
ATP / process-control update
  |
Verify next lot
  |
Fleet monitoring (customer owns ship gate)
```

Key concepts. Containment, suspected mechanism, confirmed mechanism class, and recurrence control are separate steps. The system owner keeps responsibility for evidence quality and verifying the fix (Sections A.6 and D.9).

Measurements. Lot genealogy → exposure? → quarantine list.
ATP hot corner → escape path? → production gate.
DPA sample → mechanism class? → permanent fix.

Typical follow-ups.

- Why contain before FA completes?
- What residual risk do you name to leadership?

Common mistakes.

- Waiting for a confirmed mechanism before stop-ship.
- Blocking all vendors for one supplier's lot.

Thirty-second close. I contain Supplier B today, scope exposure, open joint FA, expand ATP, verify the next lot, keep fleet monitoring, and keep ownership of the ship gate.

Deep dive. Section A.6 and table A.1.

C.11 BER floor

Interview question.

> Why can a link show a BER floor that more launch power does not fix?

What the interviewer is testing. Separating waterfall shift from floor, then choosing measurements for the leading non-power-limited impairment.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Confirm floor versus waterfall shift with a VOA sweep at a named plane.

30-second answer (memorize). A floor means additional received power no longer removes the dominant impairment. Confirm floor versus shift on a waterfall, then test for signal-dependent noise, reflections or MPI, pattern dependence, crosstalk, timing or CDR limits, and DSP or equalization limits. Therefore I would fix the limiting mechanism rather than raise OMA.

```
BER stops improving with power
  |
Power no longer limiting
  |
Test for:
  |-- signal-dependent noise (RIN)
  |-- reflections / MPI
  |-- deterministic / pattern distortion
  |-- crosstalk / timing / CDR / DSP-EQ
  |
Choose measurement from leading hypothesis
  |
Fix mechanism, not launch
```

Key concepts. A floor is a diagnostic pattern, not a single mechanism. The waterfall shape is the fast cut between sensitivity shift and a non-power-limited impairment (Sections A.8.9 and A.10.8).

Measurements. Waterfall $\rightarrow$ shift or floor? $\rightarrow$ next hunt.
Controlled ORL $\rightarrow$ reflection-driven path? $\rightarrow$ plant.
Supply noise / FEC histogram $\rightarrow$ PSRR or burst? $\rightarrow$ owner.

Typical follow-ups.

- Why not increase OMA alone?
- How do FEC histograms look for MPI bursts?

Common mistakes.

- Raising launch into a floor.
- Equating every floor with RIN before bisecting.

Thirty-second close. I confirm floor versus shift on a waterfall, then remove the non-power-limited impairment rather than raise launch power.

Deep dive. Sections A.10.8 and 11.2.1 and section 4.3.

C.12 Intermittent failures

Interview question.

> The link fails intermittently and often passes when retested. How do you proceed?

What the interviewer is testing. Evidence preservation, reproduction strategy, and refusing premature NFF closure.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. Preserve the failing state and telemetry before reseat, clean, or reboot.

30-second answer (memorize). Preserve the failing state and telemetry before you reseat, clean, or reboot. Scope time and change: dwell, temperature, vibration, firmware, connector. Prefer burst/FEC histograms and long dwell over a single golden retest. Therefore I would contain if lot-correlated, tighten dwell/ATP if escape, and refuse to close an NFF without a reproduction plan.

```
Intermittent
   |
Preserve state / telemetry
   |
Triggers: T, time, mate, vibration, FW
   |
Reproduce with dwell / stress
   |
Scope population
   |
Decision: contain / ATP dwell / monitor
```

Key concepts. Intermittents die when the evidence is destroyed. NFF is often a triage failure, not a healthy part (Section 7.12).

Measurements. FEC histogram → burst vs Gaussian? → MPI / intermittent.
Time sync + event capture → state preserved? → FA.
Fixture / firmware identity → shared cause? → scope.
Dwell BER → reproduces? → control choice.
Mate/demate → connector? → hygiene / replace.

Typical follow-ups.

- What do you refuse to do on first touch?
- How does high NFF change your answer?

Common mistakes.

- Reseating before capture.
- Closing NFF without a reproduction plan.

Thirty-second close. I preserve state before reseating, reproduce with dwell, and refuse NFF without a reproduction and control plan.

Deep dive. Section 7.12 and table 11.1.

C.13 Production recurrence-control update

Interview question.

> A failure escaped to the field that production controls did not catch. How do you update recurrence control?

What the interviewer is testing. Choosing among ATP, sampling, SPC, supplier process, design, firmware, and telemetry controls; not assuming a new 100% screen.

Assumptions to state. The escape path can be named; a failing unit or equivalent can be replayed; measurement capability is known.

First thing I would check. Name the escape path and the measurement that would have caught it.

30-second answer (memorize). Name the escape path and the measurement that would have caught it at a named plane and corner. Choose the most economical control: new ATP item, tighter limit, sampled audit, SPC variable, supplier process correction, design or firmware guard, or telemetry alarm. Only select a 100% screen when separation, cycle time, measurement capability, and cost justify it. Therefore I would ship the chosen control with an owner and a metric.

```
Escape
   |
Which uncertainty ATP missed?
   |
New measurement / corner / limit
   |
Guard band from repeatability
   |
Station correlation
   |
Reaction plan + owner
   |
Ship ATP change
```

Key concepts. Recurrence control is not always an ATP change. A limit nobody can trace gets waived under ship pressure (Section A.10.2 and section D.9).

Measurements. Escape unit replay → missed corner? → new test.
Gage R&R → repeatability? → guard band.
Golden units across stations → correlation? → ship.

Typical follow-ups.

- How do you set the guard band?
- What is the reaction plan when the new limit fails?

Common mistakes.

- Adding a test with no reaction plan.
- Limits from one hero unit.

Thirty-second close. I name the missed uncertainty, choose ATP / sample / SPC / supplier / design / firmware / telemetry, demonstrate separation, and ship the control with an owner.

Deep dive. Production readiness stage in Section A.8.5.

C.14 Telemetry design

Interview question.

> What would you put in fleet telemetry, and why?

What the interviewer is testing. Instrumenting ledgers that unlock triage decisions, not logging every register.

Assumptions to state. Schema and firmware versions are known; retention and owners exist; missing-data behavior is defined; alarms have a decision owner.

First thing I would check. Name the decision each telemetry field must unlock before adding registers.

30-second answer (memorize). Log what discriminates hypotheses: per-lane power, bias, pre-FEC BER and FEC histograms; module temperature and actuator drive; LOS/LOL with context. Require timestamp accuracy, sampling cadence, aggregation window, units, calibration/scaling, missing-data behavior, firmware/schema version, serial/lot/platform, event trigger, retention, and a decision owner. Alarm on trends and disagreements, not only hard thresholds. Therefore I would instrument the ledgers margin testing said die first.

Add a register only if it changes contain, derate, RMA, or FA ownership. Full tradeoff: Section 7.12.

```
Telemetry purpose: early margin erosion
  |
Per-lane: power, bias, pre-FEC BER
  |
Module: T, TEC/heater, rails, lock
  |
Events + lot/age/rack context
  |
Alarms: trends / disagreements
```

Key concepts. Telemetry exists to triage without pulling hardware and to catch dying units before dead ones (Section A.10.13).

Measurements. Bias at constant power $\rightarrow$ aging? $\rightarrow$ reliability bucket.
Power at constant bias $\rightarrow$ optical path? $\rightarrow$ plant.
Actuator near rail $\rightarrow$ control margin? $\rightarrow$ thermal design.

Typical follow-ups.

- Why alarm on actuator headroom?
- What context fields make lot queries possible?

Common mistakes.

- Logging every register with no hypothesis.
- Hard thresholds only, no trends.

Thirty-second close. I instrument the ledgers that discriminate hypotheses and alarm on trends and disagreements that unlock triage.

Deep dive. Section A.10.13 and section 7.8.

C.15 Qualification planning

Interview question.

> How do you plan qualification for a new IM/DD module?

What the interviewer is testing. Qualification-evidence judgment: mechanism coverage, stress selection, sample strategy, acceptance criteria, confidence, production correlation, release evidence. This playbook is a scoped evidence path, not a second full lifecycle; use Sections C.8 and D.2 for the stage order.

Assumptions to state. Requirements name mechanisms; sample strategy and confidence are stated; production proxies are cheaper than full GR-468 replay; release is a ship / restrict / reject decision.

First thing I would check. Freeze the requirements slice and named mechanisms before listing stresses.

30-second answer (memorize). I qualify from requirements to mechanisms, not from a ritual list. What does the system require? What can fail? How do we accelerate those mechanisms? What observable changes? What defines pass/fail? How much evidence is enough? How does production control maintain quality? Therefore I would gate ship on remaining margin after named stresses with a stated sample strategy and confidence, plus a production proxy (Sections D.3 and 8.3).

```
Qualification interview framework
  |
Requirements: what does the system require?
  |
Failure mechanisms: what can fail?
  |
```

```
Acceleration: how do we stress those mechanisms?
   |
Measurement: what observable changes?
   |
Acceptance: what defines pass/fail?
   |
Confidence: how much evidence is enough?
   |
Production control: how do we maintain quality?
```

Key concepts. This is a scoped evidence path inside the lifecycle, not a second lifecycle (Sections C.8 and D.2). Stress consumes margin; qual measures what remains (Sections A.8.6 and A.8.7). Sample strategy depends on failure-rate target, confidence, cost, and population variation, not a fixed "we test 20 units."

Measurements. Margin sweeps $\rightarrow$ cliff distance? $\rightarrow$ derate or redesign.
HTOL $\rightarrow$ wear-out hypothesis? $\rightarrow$ FIT.
ATP correlation $\rightarrow$ factory control? $\rightarrow$ volume.

Typical follow-ups.

- What does "remaining margin" mean in one sentence?
- When do you ask for Tx-only or Rx-only samples?

Common mistakes.

- HTOL without mechanism or E_a.
- Qualifying internals the customer never observes.

Thirty-second close. I budget which stresses spend which ledgers, run HTOL only with a named mechanism, and gate ship on remaining margin.

Deep dive. Section A.8.5 and sections 5.13 and 8.3.

C.16 Unknown failure

Interview question.

> You must make a ship decision this week, but the physical mechanism is still unknown. What do you do?

What the interviewer is testing. Deciding with today's evidence: contain, owner, control, and residual risk before mechanism certainty.

Assumptions to state. The failure reproduces; telemetry is calibrated unless challenged; the product is bookended; engineering access is unavailable unless requested.

First thing I would check. State evidence, scope, and residual risk before waiting for a physical mechanism.

30-second answer (memorize). State evidence, confidence weights, and residual risk. Decide with today's evidence: contain the scoped population, keep a healthy path shipping, open FA, and add the ATP or telemetry control that would catch the next escape. Therefore I would not wait for certainty before ownership actions.

```
Unknown mechanism
  |
Evidence + scope + rate/trend
  |
Priors: common modes first
  |
Decision today (contain / ship / derate)
  |
FA path + control + owner
  |
Update when mechanism closes
```

Key concepts. The job is the best decision with today's evidence. Unknown mechanism is Framework 15 of Staff judgment, not a reason to freeze (Section A.6).

Measurements. Whatever removes the most uncertainty per hour toward the ship decision: scope query, golden swap, external power, hot-corner ATP sample.
DPA later $\rightarrow$ mechanism class? $\rightarrow$ permanent fix.

Typical follow-ups.

- What do you tell leadership about residual risk?
- Which measurement is worth delaying the decision for?

Common mistakes.

- Freezing all action until SEM.
- Making a ship call with no control and no owner.

Thirty-second close. I decide with today's evidence, contain the scoped population, keep a healthy path shipping, and name the control and residual risk.

Deep dive. Sections A.4 and A.6 and table A.1.

C.17 Tradeoff interview questions

Staff follow-ups often stop asking "what test?" and start asking "given constraints, what do you choose?" Answer each with benefit, downside, and decision criteria. Point at the matching tradeoff homes; do not invent a new framework.

Would you choose more margin or lower power? *Benefit of margin:* reach, temperature, aging, contamination tolerance.
Downside: laser power, heat, efficiency, sometimes lifetime.
Criteria: allocate margin where uncertainty is highest; do not maximize every ledger (Section 5.19 and chapter 7).

Would you add more telemetry? *Benefit:* faster fleet triage and earlier margin erosion detection.
Downside: firmware, storage, alarm fatigue.
Criteria: each field needs a decision owner and a reaction plan (Sections C.14 and 7.12).

Would you run more qualification? *Benefit:* confidence and fewer late escapes.
Downside: schedule, cost, delayed learning.
Criteria: prioritize by risk × uncertainty × impact; stop when remaining risk has a production control (Section 8.3).

Would you add a second supplier? *Benefit:* supply resilience and pricing leverage.
Downside: validation, interop, and manufacturing differences.
Criteria: qualify on concentration risk and evidence, not ideology (Sections C.7 and 9.2).

Would you increase ATP coverage? *Benefit:* escape detection earlier in the flow.
Downside: cycle time, cost, false rejects.
Criteria: cheapest control that reliably detects the named mechanism: 100%, sample, SPC, or supplier process (Section 9.4).

Key idea. Open the matching framework, deliver the thirty-second box, walk the tree, end on the decision and the control. When the interviewer asks a tradeoff question, name benefit, downside, and criteria. Philosophy is in Chapter A; this appendix is how you speak it under pressure.

Chapter D

Engineering decision trees

This appendix is the wall chart. It collects the universal reasoning frameworks used throughout the book, independent of any interview question. Open it when you need the pattern without the surrounding prose. Interview pressure maps are in Chapter C; the drill philosophy is in Chapter A.

Debugging and qualification are the same philosophy at different times. Debugging asks which margin was exhausted. Qualification asks how much margin remains after the expected stresses. Both end on an engineering decision and a recurrence control.

D.1 Universal debugging decision tree

```
Universal debugging decision tree
  |
Problem
  |
Scope and time behavior
  |
Stable average received power?
  |-- NO  --> Power ledger / optical path
  |-- YES --> Signal-quality path
              Tx / channel / Rx / DSP
              noise / timing / spectral / control
  |
Highest-value isolate
  |
Leading mechanism
  |
Controlled confirmation
  |
Decision + owner + timeline + reversibility
  |
Follow-up control (ATP / SPC / telemetry / supplier)
```

Stable average power deprioritizes gross optical loss but does not eliminate fast power fluctuations, clipping, monitor averaging, or reflection-dependent effects. A surviving hypothesis is only the leading mechanism until confirmed by reproduction, controlled swap, stress dependence, rollback, or physical evidence (Sections 4.8, D.16 and 7.11).

Why experienced engineers ask about scope before touching the lab?

Because scope eliminates enormous parts of the hypothesis space. Unit, lot, vendor, site, and fleet each point at different owners and different next measurements.

Engineering heuristic. If two explanations fit equally well, prefer the one that requires the fewest independent failures.

D.2 Validation flow (Steps 1–11)

```
Validation flow (same as tab:ladder)
  |
Step 1: Define requirements
  |
Step 2: Review architecture
  |
Step 3: Bring up hardware
  |
Step 4: Characterize behavior
  |
Step 5: Validate margin and interoperability
  |
Step 6: Qualify reliability
  |
Step 7: Validate manufacturing
  |
Step 8: Controlled pilot
  |
Step 9: Ramp mass production
  |
Step 10: Monitor the fleet
  |
Step 11: Feed learning / next revision
```

Same order as Table 7.2, not a second Staff loop. Do not treat a bring-up pass as margin evidence, or an HTOL pass as production readiness (Sections 7.1 and 8.3 and section A.8.5).

Step 6 (qualify reliability): mechanism $\longrightarrow$ stress $\longrightarrow$ observable $\longrightarrow$ acceptance $\longrightarrow$ confidence. See Chapter 8.

Step 7 (validate manufacturing): production reference $\longrightarrow$ representative builds $\longrightarrow$ trusted measurement $\longrightarrow$ yield/ATP/SPC $\longrightarrow$ ramp. See Chapter 9.

D.3 Qualification evidence tree

```
Qualification evidence tree
   |
Requirement
   |
Named failure mechanism
   |
Stress that accelerates that mechanism
   |
Representative sample and confidence
   |
Observable before and after stress
   |
Acceptance criterion
   |
Production proxy
   |
Ship / restrict / reject
```

Use inside the reliability gate of Section D.2. The lifecycle says *when*; this tree says *what evidence* is defensible (Sections D.16 and 8.3).

Why experienced engineers insist on mechanism → stress → observable?

Because a stress without a named mechanism is theater, and an observable without an acceptance rule cannot support a ship decision.

D.4 Power-versus-quality diagnostic fork

```
Power-versus-quality diagnostic fork
   |
BER or link degrade
   |
Stable average received power?
   |-- NO  --> Power ledger / optical path
   |           laser / coupling / connector / fiber / MUX /
     ↪ monitor
   |-- YES --> Signal-quality path
               Tx / channel / Rx / DSP
               noise / timing / spectral / control
   |
Highest-value measurement
   |
Leading mechanism -> controlled confirmation
   |
Decision + recurrence control
```

One check before retuning equalizers or bias tables (Sections D.1 and 4.8).

Engineering heuristic. Name the power-versus-quality branch before you name a laser, TIA, or connector. Parts without a branch are guesses.

D.5 Scope and population

```
Scope and population
   |
Initial symptom
   |
Scope analysis (how large?)
   |-- lane / module / host / rack / site
   |-- PN / firmware / lot / vendor / fleet
   |
Technical isolation
   |
Correlation analysis (which cohort?)
   |-- lot / date code / FW / cal / platform / location
   |
Containment and corrective action
```

Scope sets severity and priors. Correlation after isolation unlocks contain, pause, replace, or supplier escalate (Sections 7.12 and 9.3).

Engineering heuristic. Population behavior is usually more informative than one failing unit. A cohort plot often beats another hour on the same sample.

D.6 Sudden versus gradual

```
Failure onset
   |
Sudden?
   |-- config / handling / firmware / mechanical / power event
   |
Gradual?
   |-- aging / drift / margin erosion / contamination / cal
     ↪ movement
   |
Prioritize measurements (priors, not conclusions)
```

Sudden and gradual are priors that reorder the bench, not confirmed-mechanism claims (Chapter 11).

D.7 Transmitter, channel, or receiver

```
Power or quality path chosen
   |
Golden swap / loopback
   |-- Tx --> eye / LIV / bias / wavelength / RIN
   |-- Channel --> connector / fiber / MUX / ORL
   |-- Rx --> sensitivity / TIA / EQ / host
   |
Evidence
   |
Owner + decision
```

Bisect domains before opening packages (Sections 7.10 and 11.2).

D.8 Supplier qualification

```
Supplier qualification
   |
Requirements (customer-visible)
   |
Characterization (multi-lot distributions, not hero units)
   |
Margin under stress
   |
Qualification (named mechanisms)
   |
Production readiness (FAIR, ATP, SPC)
   |
Fleet monitoring (RMA, telemetry)
```

Customer view measures external behavior. Vendor view owns internals (Section A.8.7 and section 9.2).

D.9 Supplier escape containment flow

```
Supplier escape containment flow
   |
Escape detected
   |
Provisional containment
   |
Scope analysis
   |
Refine contained population
   |
Investigate / confirm mechanism
   |
Corrective action
   |
Recurrence control
   |
Decision closure: owner / timeline / reversibility / follow-up
```

Engineering heuristic. Contain first when the population can grow. Perfect mechanism stories do not unship yesterday's lot.

Use provisional containment when the population can grow, then refine the hold after scope analysis. The system owner keeps responsibility for evidence quality and verifying corrective action (Sections D.16, 7.12 and 9.5).

D.10 Margin-consumption flow

```
Margin–consumption flow
  |
Nominal system margin
  |
Temperature debit
  |
Voltage / power–quality debit
  |
Channel / connector debit
  |
Manufacturing variation
  |
Aging / wear
  |
Interoperability variation
  |
Remaining margin
  |
Above deployment requirement?
  |–– YES ––> proceed
  |–– NO  ––> redesign / restrict / recalibrate / reject
```

This is a conceptual margin flow. Measure net margin at a defined reference plane and avoid double-counting overlapping penalties (Sections 5.19 and 7.7).

D.11 Black-box versus engineering access

```
Bookended product
  |
End–to–end qualification
  |–– BER / FEC / telemetry / sensitivity
  |–– environment / interoperability
  |
Enough confidence to decide?
  |–– YES ––> deployment decision
  |–– NO  ––> request engineering access
              Tx–only / Rx–only / breakout / diagnostics
              |
              Isolate margin
```

An optical eye is measured externally with suitable access. Do not assume the module reports a conventional eye unless that capability exists (Section A.8.7).

D.12 Recurrence-control loop

```
Production escape
  |
Contain risk now
  |
Investigate mechanism
```

```
   |
Correct process
   |
Update screening or ATP
   |
Verify next lot
   |
Monitor fleet
```

Containment, confirmed mechanism, and prevention are three different actions (Section 9.5).

D.13 Production feedback loop

```
Design requirements
   |
Qualification
   |
ATP
   |
Production data
   |
Fleet data
   |
Failure analysis
   |
Updated limits or screens
   |
Next production cycle
```

Production validation is replayable and decision-oriented (Sections 9.2 and 9.4).

D.14 Measurement-selection loop

```
Current evidence
   |
Rank hypotheses
   |
Choose measurement that removes
the largest number of hypotheses
   |
Update beliefs
   |
Enough evidence to decide?
   |-- NO  --> next measurement
   |-- YES --> take action
```

Key idea. Great engineers perform the minimum measurement that eliminates the largest number of hypotheses.

D.15 Unknown failure

```
Observation (facts, not hopes)
   |
Hypotheses (short ranked list)
   |
Highest-value measurement
   |
Evidence (what died)
   |
Decision with today's confidence
   |
Continue measuring?
   |-- YES --> next measurement
   |-- NO  --> control + owner + residual risk
```

Unknown mechanism is not a freeze. Decide with weighted evidence (Sections A.6 and D.14).

D.16 Evidence block

Use this block for validation reports, qualification plans, supplier reviews, fleet incidents, and interview answers. Speak evidence language in order: observation ("I measured..."), correlation ("these units share..."), hypothesis ("this suggests..."), confirmation ("I reproduced..."), decision ("I will..."). Do not say "data proves" until confirmation exists.

```
Evidence block
   |
Claim: what are we trying to establish?
   |
Population: units, lots, hosts, environments
   |
Measurement: metric + reference plane + condition
   |
Access: Level 0--4 (\Cref{sec:interview-access-levels})
   |
Condition: T, V, pattern, ORL, traffic, dwell
   |
Evidence strength: n, repeatability, confidence
   |
Decision: what action is justified?
   |
Control: how recurrence or drift is detected
```

Decision closure. Every tree that ends in an action should close with:

```
Decision closure
   |
Decision: what action now?
   |
Owner: who is accountable?
   |
```

```
Timeline: when must it complete?
   |
Reversibility: how hard to undo?
   |
Follow-up control: ATP / SPC / telemetry / supplier CA
```

Example (supplier defect): stop shipment of the scoped lot; owner Operations plus Quality; follow-up is supplier corrective action and the production or fleet control that would catch the next escape.

Chapters C, 7 to 9 and 11 point here rather than restating the list.

D.17 Before you start: three checklists

Before debugging

- ☐ Scope (lane / module / host / rack / site / lot / vendor / fleet)
- ☐ Time behavior (sudden or gradual)
- ☐ Stable average received power? (power vs signal-quality path)
- ☐ Highest-value isolate chosen
- ☐ Leading mechanism stated (not yet confirmed RC)
- ☐ Decision unlocked with owner, timeline, and recurrence control

Before qualification

- ☐ Nominal function demonstrated
- ☐ Operating margin known
- ☐ Environmental and reliability stresses mapped to mechanisms
- ☐ Interoperability headroom retained on supported combos
- ☐ Manufacturing variation and ATP escape detection covered
- ☐ Controlled pilot exit criteria and fleet telemetry owners set

Before production

- ☐ Requirements and margin demonstrated
- ☐ Environment and interop verified
- ☐ Reliability risk addressed with named confidence
- ☐ Lot / site / date-code variation characterized
- ☐ ATP / sample / SPC controls classified
- ☐ Supplier controls reviewed; pilot exit and fleet monitor ready

Key idea. These trees are the book's reusable method. Chapters supply the physics and the numbers. This appendix supplies the order of thought.

Chapter E

Optical Systems Staff Engineer Interview Questions

Index only. Each item is one speakable question plus a pointer to the worked answer, playbook, or chapter home. Do not invent new topics here; expand in those homes.

E.1 Debugging

1. Pre-FEC BER rose while average received power held steady. How do you debug it? Section C.1 and section A.10.7
2. Received power decreased. What do you check first? Sections C.2 and D.4
3. A single lane is weak in a multi-lane module. How do you isolate it? Section C.3 and section A.10.6
4. BER worsens at high temperature but average power is stable. What next? Sections B.7 and C.4 and section A.10.3
5. The failure is intermittent. How do you scope and trap it? Sections C.12 and D.6

E.2 Validation

6. How would you validate a new optical transmitter from bring-up through margin? Section C.8, section A.10.2, and table 7.2
7. How would you set laser requirements for a new IM/DD link? Section A.10.1 and table 5.4

8. Received power is unchanged but required receiver power increased. Why? Sections A.8.9 and A.10.7

9. Why can a link show a BER floor that more launch power does not fix? Section C.11 and section A.10.8

10. Which measurement do you choose next when access is limited? Sections A.2 and B.1

E.3 Qualification

11. How would you qualify reliability? Chapter 8 and sections C.15 and D.3

12. How do you build a qualification plan that maps mechanisms to stress? Sections C.15, D.3 and 8.3

13. What makes an HTOL projection credible? Section A.10.9 and chapter 8

14. How do you distinguish laser aging from calibration drift? Section C.5 and section A.10.4

15. What evidence closes a qualification decision? Sections D.3 and D.16

E.4 Manufacturing

16. How would you validate manufacturing? Chapter 9 and tables 9.1 and 9.2

17. How would you respond to a yield drop? Chapters 9 and 11 and section 9.6

18. When would you update ATP after a field or lab escape? Sections C.13, D.13 and 9.5

19. Which data must an automated test save so a failure can be replayed? Section A.10.10 and chapter 9

20. How do you decide whether a field issue is performance, reliability, or manufacturability? Section A.10.12 and table 9.1

E.5 Supplier

21. How would you qualify a second laser or PIC supplier? Section C.6, section A.10.5, and chapters 8 and 9

22. How would you qualify a second transceiver or cable-assembly supplier? Sections C.7 and D.8 and chapter 9

23. A supplier lot looks correlated with a fleet cohort. What do you do? Sections B.5, C.9 and D.5

E.6 Architecture

24. When would you choose an EML, silicon MZM, or ring modulator? Section A.10.11 and chapter 3
25. How do optics choices constrain AI datacenter networking cost and power? Chapters 1 and 10

Key idea. Speak the question home first: scope, ledger, measurement, decision, control. Then open the matching playbook only if the interviewer asks for depth.

Chapter F

How Staff Engineers Think

This appendix is judgment, not physics. The machinery lives elsewhere: validation, qualification, and manufacturing (Chapters 7 to 9), networking constraints (Chapter 10), the failure handbook (Chapter 11), the Staff loop and access levels (Sections A.1 and A.2), cases and tradeoff drills (Chapter B and section C.17), and the wall-chart trees (Chapter D). Read this when you need the habits that turn that machinery into decisions under pressure.

F.1 Engineering under uncertainty

You rarely get a confirmed mechanism before a ship, hold, or contain call. Uncertainty is the normal state. The job is to decide with today's evidence, name residual risk, and keep the FA path open (Section A.6).

Do not confuse "I do not know the confirmed mechanism" with "I cannot act." Scope, population, trend, and a reversible control are often enough for today's call. Waiting for SEM photos while a bad lot keeps shipping is a process failure, not scientific humility.

Principle: Decide with the evidence you have; name what you still do not know.

F.2 Reversible decisions

Prefer actions you can undo cheaply: lot holds, derates, monitor-only watches, temporary ATP tightenings, firmware guards. Irreversible moves (public recall, permanent architecture change, destructive FA on the last failing unit) need stronger evidence and a named owner.

Ask: What is the cost of being wrong either way? If a hold costs a week of schedule and a miss costs a customer outage, hold. If a hold strands a healthy supply line and the rate is flat and tiny, monitor with a tripwire (Table A.1).

Principle: Buy reversibility when uncertainty is high; spend irreversibility only when evidence demands it.

F.3 Choosing measurements

Pick the next measurement by information value per cost and access level, not by instrument prestige (Sections A.2 and B.1). Climb Level 0–2 until they stop re-ordering beliefs. Name the decision the measurement unlocks before you request Level 3–4 access.

A good next measurement separates the largest useful hypothesis set. A bad next measurement confirms what you already believe or spends the sample before you know the question.

Principle: Measure to unlock an action, not to collect comfort.

F.4 Owning risk

Every product call has an owner, a residual risk statement, and a control. “We are still looking” is not ownership. Containment, correction, and recurrence control are three different jobs; name who holds each one (Sections A.1 and C.16).

Risk ownership includes saying when monitor-only is correct. Tiny, flat rates with no customer impact can stay on a watch list if the tripwire and review cadence are real. Hope without a tripwire is not a plan.

Principle: No decision without an owner, a residual risk line, and a control.

F.5 Evidence vs confidence

Separate observation, correlation, hypothesis, and confirmation. Strong confidence on weak evidence is a junior failure mode; weak confidence on strong evidence is how fleets ship escapes.

Hero samples do not answer ship. Population data, versioned ATP, and a life model do. One failing unit can open FA; it cannot by itself justify a fleet-wide architecture rewrite (Section A.2.1 and table 7.2).

Principle: Match the strength of the claim to the strength of the evidence.

F.6 Communicating bad news

Bad news travels upward in a fixed frame: problem, impact, evidence, confidence, containment, next decision (Section B.3). Lead with the decision leaders must make. Put mechanism depth behind the frame, not in front of it.

Do not soften numbers to protect feelings. Do not bury the hold recommendation in a paragraph of physics. If shipment should stop, say so in the first two sentences, then show the evidence.

Principle: Clarity beats comfort. Name the call, then the proof.

F.7 Supplier interactions

Treat supplier conversations like debug with a second party across the table (Sections B.4, C.10 and 9.2). Scope first: lot, site, date code, plane, condition. Ask for evidence that would change your ship call. Offer the measurements you will accept as closure.

Second sources are not ideology. Qualify them when concentration risk exceeds fleet tolerance, and only on evidence you would trust for the first source (Sections C.7 and C.17).

Principle: Same loop for suppliers as for silicon: scope, evidence, decision, control.

F.8 Fleet thinking

One unit is a story. A lot, rack, vendor, or time trend is a decision. Scope before you generalize (Sections C.9 and 7.12). Telemetry earns its keep only when each field has a decision owner and a reaction plan (Section C.14).

Ask what fraction of the population is affected, whether the rate is rising, and whether healthy paths can keep shipping while the bad population is held. Fleet judgment is population math plus reversible containment, not a louder lab story.

Principle: Decide on populations and trends, not on anecdotes.

F.9 Avoiding confirmation bias

State the leading hypothesis, then name the observation that would kill it. If you cannot name a falsifier, you are defending a narrative, not running a debug (Section A.7 and chapter D).

Prefer measurements that can reorder belief over measurements that decorate the current story. When two mechanisms predict the same symptom, design the cheapest test that splits them. Do not collect three confirming eyes when one power-plane split would end the debate.

Principle: Hunt for disproof as hard as you hunt for support.

F.10 Learning from escapes

An escape that passed qual is a process miss, not only a part miss (Sections C.10 and D.9). Classify it: wrong stress, wrong sample, wrong limit, wrong screen, or wrong assumption about use. Close with a versioned control in ATP, SPC, telemetry, or supplier process so the same path cannot ship tomorrow.

Learning is incomplete until the factory or fleet can catch the mechanism without heroics. FA closure without a production catch is a report, not a fix.

Principle: Every escape ends in a named control that would have caught it.

Key idea. Junior questions ask what test to run. Senior questions ask what uncertainty remains. Staff questions ask what decision is due and what minimum evidence unlocks it. Use the Staff loop (Section A.1), climb access levels on purpose (Section A.2), and end every call with owner, residual risk, and control.

Chapter G

Abbreviations and terminology

This glossary merges OIF CEI-448G Framework terms (OIF-FD-CEI-448G-01.0, September 2025) [14] with optical, reliability, and manufacturing terms used in the book (including the must-know list in Chapter A). Entries are alphabetical. Market-position claims belong in dated chapter sidebars, not here.

G.1 Glossary

448G A generic name for an expected technology enabling data rates (including overhead) of approximately 448 Gbps per lane.

8D / CAPA Eight-discipline problem solving and corrective and preventive action. Structured containment, confirmed mechanism, correction, and recurrence control for supplier or production failures.

ACC Active copper cable. ACCs are a type of active copper cable used in data centers. ACCs primarily use Redriver chips and Continuous Time Linear Equalization (CTLE) to amplify and equalize the signal and provide longer reach than DACs. ACCs generally are lower power and cost than AECs (another type of active copper cable), but provide shorter reach than AECs.

ADC An analog-to-digital converter is a system that converts an analog signal into a digital signal.

AEC Active electrical cable. AECs are a type of active copper cable used in data centers. AECs use retimer chips with clock and data recovery (CDR) to reshape and retransmit signals which generally offer longer reach and better signal integrity compared to ACCs, but at a higher cost.

AI Artificial intelligence.

ALS Automatic laser shutdown. A safety mechanism that cuts laser output when fiber continuity is lost. Modern systems prefer APR with automatic restart over full shutdown.

APC Automatic power control. A feedback loop from a monitor photodiode that holds average optical output power constant against temperature drift and aging.

APD Avalanche photodiode. A photodiode with internal multiplication gain. Any sensitivity benefit versus PIN depends on multiplication gain, bandwidth, excess-noise factor, receiver design, and BER target; do not treat 5–9 dB as universal.

Application spaces Portions of equipment or network architecture that could benefit from having a defined set of interconnection parameters.

APR Automatic power reduction. Holds laser output at or below Hazard Level 1M on fiber break and probes for re-mate with safe low-power pulses (ITU-T G.664).

Arrhenius / E_a Life-acceleration model. Temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$. E_a must be justified for the named mechanism on that process.

ASE Amplified spontaneous emission. Broadband optical noise generated by optical amplifiers (SOA, EDFA); adds to the receiver noise floor.

ASIC An application-specific integrated circuit is an integrated circuit (IC) customized for a particular use, rather than intended for general-purpose use.

ATP Acceptance test plan. Production test limits, methods, reference planes, and reaction rules for ship or hold decisions.

AUI Attachment unit interface. The electrical serial lane set between a host ASIC and the module cage connector (e.g. 400GAUI-4).

Backplane A group of electrical connections used as a backbone to connect several printed circuit boards together to make up a switch, computing or storage system.

Bathtub curve Reliability failure-rate shape: infant mortality (falling), useful life (roughly constant FIT), then wear-out (rising). Burn-in targets the left; Arrhenius life targets the right.

BER Bit error ratio is a measure of the number of bit errors that occur during data transmission, expressed as a ratio of erroneous bits to the total number of bits transmitted.

BERT Bit-error-ratio tester. An instrument or ASIC function that generates PRBS patterns and counts bit errors for pre-FEC and post-FEC BER measurement.

BiCMOS A semiconductor process combining bipolar transistors (high f_T) with CMOS on one die; used for high-speed TIAs and modulator drivers.

BOM Bill of materials. The component and assembly cost of a module or system; DSP presence is a large BOM driver.

BoW Bunch of wires is a die-to-die interface specification. It is part of the open compute project (OCP) and, like UCIe, is used to connect chiplets within a package.

Burn-in Production or sample screen that removes infant-mortality parts before ship. Distinct from HTOL life projection.

Cabled host An implementation where twinaxial cable is used instead of PCB traces expressly for the insertion loss benefits or 3D architecture.

CDR Clock and data recovery, a component that re-establishes the timing of a signal that may have degraded due to impairments on a transmission line, the retimed signal is able to continue further to its destination.

CEI Common Electrical IO, an OIF Implementation Agreement containing clauses defining electrical interface specifications, each optimized for various reaches at minimal power.

CMIS Common Management Interface Specification.

CMIS-LT CMIS-based link training provides a message set and exchange mechanism for out-of-band link training or tuning between a pluggable module and a host.

CMIS-VCS CMIS versatile control set extends the base CMIS standard to allow for more advanced and flexible signal integrity (SI) capabilities.

CMOS Complementary metal-oxide-semiconductor. Common IC fabrication process; TIAs and SerDes are built at nodes such as 16 nm to 3 nm.

COBO Consortium for On-Board Optics. Standardized mid-board optical engine placement; an on-board (OBO) architecture alternative to faceplate pluggables and CPO.

COD Catastrophic optical damage. A sudden, irreversible failure of a laser facet under thermal or optical overstress.

Coded modulation Coded modulation is a technique in digital communication that combines error-control coding with modulation to enhance reliability and efficiency. Its main goal is to balance bandwidth efficiency, power efficiency, and error probability.

COM Channel operating margin. A statistical SNR metric for the fully equalized electrical link; CEI go/no-go is typically COM $\geq$ 3 dB.

COUPE Compact Universal Photonic Engine. TSMC's SoIC-X hybrid-bonded EIC-on-PIC packaging platform for co-packaged optics; mass production in 2026.

CPC Co-packaged copper is an emerging interconnect technology where copper cables are directly attached to the top of an Application-Specific Integrated Circuit (ASIC) or other high-speed integrated circuit within a single package. This design minimizes the length of high-speed electrical signals on the printed circuit board (PCB) and package, addressing the limitations of traditional copper traces at increasingly high data rates (e.g., 224G and above).

CPO Co-packaged optics. An electrical to optical device intended to be mounted on the host package.

CTLE Continuous time linear equalizer.

CW Continuous wave. Unmodulated laser output; external modulators (MZM, ring, EAM) encode data onto a CW source.

CW-WDM Continuous-wave wavelength division multiplexing. A multi-source agreement for multi-wavelength CW laser sources feeding WDM photonic engines.

CXL Compute Express Link. A coherent memory interconnect protocol built on the PCIe PHY (CXL 4.0 on PCIe 7.0); potential optical-reach application.

DAC Direct attach copper cable. A high-speed cable assembly made of copper twinaxial cable with fixed passive transceiver modules on each end. DAC cables enable direct, electrical connections between networking devices over short distances.

DBR Distributed Bragg reflector. A laser architecture where the grating sits outside the gain region, enabling tunable single-mode output.

DCA Digital communication analyzer. A sampling oscilloscope used for PAM4 eye diagrams, TDECQ, OMA, and RLM measurements.

DDM Digital diagnostic monitoring. Per-lane telemetry in CMIS: Tx/Rx optical power, laser bias, module temperature, LOS/LOL flags.

DER Detector error ratio.

DFB Distributed feedback laser. A laser with a grating along the active region that selects one longitudinal mode; a common CW or directly modulated source.

DFE Decision feedback equalizer. An equalizer by adding a filtered version of previous symbol estimates to the original filter output.

DML Directly modulated laser. Modulates bias current to encode data; simple and efficient but chirp-limited over dispersive fiber.

DPA Destructive physical analysis. Cross-section, SEM, or EDX on failed units to confirm facet, solder, FAU, or die failure modes.

DPPM Defective parts per million. Always state the denominator and window (incoming, outgoing, or escaped field). Distinct from FIT.

DR Datacenter reach. An IEEE 802.3 single-mode fiber class, typically 500 m at 1310 nm; a common reach for AI scale-out IM/DD optics.

DSP Digital signal processing.

DVT Design Validation Test. Proves the design meets requirements across corners, margin, and a frozen life plan before volume tooling (Table 9.2).

DWDM Dense wavelength division multiplexing. WDM with channel spacing $\leq$100 GHz; used in metro/long-haul and dense CPO architectures.

EAM Electro-absorption modulator. A voltage-controlled absorption region on InP; low chirp compared with direct modulation; paired with a DFB in an EML.

EECQ Electrical eye closure quaternary. The electrical analog of TDECQ, quoted at host-referenced test points TP1a and TP4a.

ELS External laser source. A CW laser module feeding a co-packaged optical engine; superset of ELSFP and CW-WDM module forms.

ELSFP External Laser Small Form-Factor Pluggable. An OIF-defined faceplate-pluggable CW laser module for CPO with a 24-pin card edge, CMIS management, and blind-mate MT optics.

EML Externally modulated laser. A DFB laser integrated with an EAM on one InP chip; a common transmitter architecture for 100–200G/lane pluggables.

ENOB Effective number of bits. A measure of the dynamic performance of an analog-to-digital converter (ADC).

EO Electro-optic. Conversion from voltage to optical phase or intensity; EO bandwidth is the primary speed metric for a modulator.

EOL End of life. The defined wear-out criterion (threshold rise, slope drop, or hard fail) used in HTOL projection and derating.

ER Extinction ratio. Linear $\mathrm{ER}_{\mathrm{lin}} = P_1/P_0$ (PAM4 outer uses P_3/P_0). In decibels, $\mathrm{ER}_{\mathrm{dB}} = 10\log_{10}(\mathrm{ER}_{\mathrm{lin}})$. Higher ER widens OMA for a given average power and trades against chirp and driver swing.

ESD Electrostatic discharge. Handling or assembly damage to drivers or TIAs; sudden hard fail, not Arrhenius wear-out. Qual uses HBM/CDM models.

EVT Engineering Validation Test. Early bring-up on engineering samples: first light, CMIS, basic LIV/SMSR/RIN, one closing BER (Table 9.2).

Faceplate A plate, cover, or bezel on the front of a device which may contain I/O ports.

FAIR First-article inspection report. Re-qualify after tooling, site, epi, or firmware change before open volume.

FAU Fiber array unit. A precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

FEC Forward error correction gives a receiver the ability to correct errors without needing a reverse channel to request retransmission of data.

FFE Feed forward equalizer.

FIT Failures in time. Failures per 10^9 device-hours. State component versus module versus link FIT, predicted versus observed, and the observation window. Distinct from DPPM.

FR Far reach. An IEEE 802.3 single-mode fiber class, typically 2 km; tighter chirp and dispersion budget than DR.

FSR Free spectral range. The wavelength or frequency span between adjacent resonances of a ring resonator or etalon.

GBd The baud rate is the number of electrical transitions per second, also called symbol rate. Giga Baud is 1×10^9 symbols per second.

Gbps Gigabits per second. The throughput or data rate of a port or piece of equipment. Gbps is 1×10^9 bits per second.

Gearbox A component used for managing and manipulating data streams, primarily by converting multiple serial data streams at one rate to multiple streams at another rate,

Golden unit / host Known-good reference used to bisect station, fixture, host, module, and fiber. Essential in debug and ATP correlation.

GR-468 Telcordia optoelectronic device qualification framework covering HTOL, environmental, and mechanical stresses.

Hamming codes A type of linear error-correcting code used in digital communication and data storage systems. It enhances data integrity by detecting and correcting single-bit errors that may occur during transmission or storage.

HAST Highly accelerated stress test. Humidity plus temperature and bias; exercises corrosion and delamination.

HBM High-bandwidth memory. Stacked DRAM on an interposer beside the accelerator die; memory bandwidth sets the decode roofline.

HCF Hollow-core fiber. Guides light primarily in air ($n \approx 1$), giving roughly 33% lower latency than solid silica SMF.

HPC High performance compute.

HPDC High-performance data center. A general term for data centers designed for high-compute workloads like AI/HPC.

HTOL High-temperature operating life. An accelerated reliability stress test (GR-468); Arrhenius modeling projects field wear-out from HTOL results.

HTSL High-temperature storage life. Unbiased bake; separates storage mechanisms from biased HTOL wear-out.

IA Implementation Agreements, what the OIF names their defined interface specifications.

IBTA InfiniBand Trade Association. Maintains the InfiniBand Architecture spec (NDR 400G, XDR 800G/port); reuses QSFP/OSFP optics with Ethernet.

IC Integrated circuit.

IEEE Institute of Electrical and Electronics Engineers. IEEE 802.3 owns Ethernet PHY/MAC rates, KP4 FEC, and IM/DD optical PMD clauses.

IL Insertion loss. Optical power lost traversing a component (connector, fiber, MUX); quoted in dB.

IMDD Intensity modulation direct detection is a method where the intensity of a light source is modulated by an electrical signal. This modulated light then travels through an optical medium (like a fiber optic cable) and is directly detected by a photo detector at the receiving end. This is a common and relatively straightforward technique for transmitting information over optical links.

InP Indium phosphide. A III-V semiconductor material for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.

I/O Input Output, a common name for describing a port or ports on equipment.

ISI Intersymbol interference.

JESD47 JEDEC IC qualification stress suite. Silicon-side counterpart to GR-468 for drivers and TIAs.

KP4 FEC Reed–Solomon RS(544,514) FEC defined in IEEE 802.3 Clause 91; used on many PAM4 Ethernet PHYs. See also RS-FEC.

LDPC A low-density parity-check code is a linear error correcting code, a method of transmitting a message over a noisy transmission channel. An LDPC is constructed using a sparse Tanner graph. LDPC codes are capacity-approaching codes.

LIV Light–current–voltage. The fundamental laser characterization curve: plots optical power and voltage versus bias current to read threshold, slope, kinks, and rollover.

LOL Loss of lock. A CDR or SerDes flag indicating the clock recovery circuit has lost symbol timing.

LOS Loss of signal. A receiver or host flag indicating optical input power has fallen below the detect threshold.

LPO Linear pluggable optic is a technology used in optical transceivers that simplifies the design of pluggable optical modules by removing the traditional Digital Signal Processor (DSP) and Clock Data Recovery (CDR) chips. Instead, LPO uses a direct-drive linear approach where the signal path is considered linear, relying on the capabilities of the Application Specific Integrated Circuit (ASIC) in the host system (like a switch or Network Interface Card) to perform signal conditioning and equalization.

LR Long reach. CEI LR specifies backplane/midplane and copper cable electrical interfaces.

LRO Linear receive optic (also called RTLR). A module with a retimer/DSP only on the transmit path; the receive path is linear into the host SerDes.

MAC Media access control. The Ethernet data-link sublayer that builds frames, handles addressing and CRC. MAC rate is payload throughput.

MACsec IEEE 802.1AE media access control security. Provides line-rate encryption, frame integrity, and data origin authentication at Layer 2; transparent to the optical PMD.

MCM Multi chip module, a specialized electronic package where multiple integrated circuits (ICs), semiconductor dies or other discrete components are packaged onto a unifying substrate, facilitating their use as a single component (as though a larger IC).

MDI Medium dependent interface. The optical connector where a module meets the fiber; test points TP2 (Tx launch) and TP3 (Rx input) sit at the MDI.

MEMS Micro-electro-mechanical systems. Tilting mirror arrays used in optical circuit switches; millisecond switching at $\sim$2 dB loss.

Mid-board optics an optical transceiver that is mounted on a PCBA away from the PCBA edge, close to a switch ASIC to reduce the amount of PCBA trace loss between an ASIC and the optical transceiver. This is in contrast to the common practice today of locating optical transceivers at the PCBA edge.

Midplane Some backplanes are constructed with slots for connecting to devices on both sides and are referred to as midplanes.

MLC Multi-level coding. Coded modulation paired with PAM6/PAM8; OIF 448G workshop models often add a strong outer RS code at $\sim$12% overhead.

MLSD Maximum likelihood sequence detection is a mathematical algorithm to extract useful data out of a noisy data stream.

MMF Multimode fiber. Fiber supporting many spatial modes (50/125 μm); used with VCSELs at 850–940 nm for SR links (OM3/OM4/OM5 grades).

MoE Mixture of experts. A transformer architecture that routes tokens to specialist subnetworks; drives all-to-all collective traffic on the fabric.

MP Mass production. Sustained volume after PVT: DPPM, RMA ownership, ECO control (Table 9.2).

MPI Multipath interference. Coherent beating from reflective paths; often floors BER and clusters FEC errors.

MPO Multi-fiber push-on. A standard multi-fiber connector for parallel optics (8, 12, 16, 24, or 32 fibers per ferrule).

MR Medium reach. CEI MR specifies chip-to-chip electrical interfaces.

MRM Microring modulator. A resonant silicon modulator; compact and WDM-native but usually needs active wavelength control. Common in CPO engines at 200G/channel class.

MSA Multi-source agreement. An industry specification for interoperable products (e.g. LPO MSA, CW-WDM MSA, 100G Lambda MSA).

MTBF Mean time between failures. For constant failure rate, MTBF $= 10^9$/FIT hours. Fleet math usually uses FIT times population.

MZM Mach–Zehnder modulator. A push-pull interferometer; broadband, low chirp. Built in silicon, TFLN, or III-V platforms.

NFF No fault found. An RMA unit that passes all tests on return; high NFF rate points at triage or intermittent connector faults.

NG Next generation.

NIC Network interface card. A host adapter connecting an accelerator or CPU to the scale-out fabric (Ethernet or InfiniBand).

NPC Near-package copper. NPC uses a copper cable to bring the front panel signal to a location close to the host silicon to minimize the host PCB losses. It reduces PCB losses by bringing the signals to a connector on the PCB close to the ASIC whereas CPC (Co-packaged copper) brings the signal to a connector on the ASIC package.

NPI New product introduction. Gate sequence EVT $\rightarrow$ DVT $\rightarrow$ Qual $\rightarrow$ PVT $\rightarrow$ Pilot $\rightarrow$ MP that maps onto the validation ladder (Tables 7.2 and 9.2).

NPO Near-package optics. Similar to CPO (Co-packaged optics) and NPC (Near-package copper), NPO is an electrical to optical device intended to be mounted on the host PCB at a location adjacent to the host silicon to minimize host PCB traces to minimize electrical signaling requirements.

NRZ (PAM2) Non return to zero, a binary code in which 1s are represented by one significant condition (usually a positive voltage) and 0s are represented by some other significant condition (usually a negative voltage), with no other neutral or rest condition.

OBO On-board optics. Optical engines mounted mid-board on the host PCB (COBO and related approaches); an architecture alternative to faceplate pluggables and CPO.

OCS Optical circuit switch. A Layer-1 switch that steers light without O-E-O conversion. Transparent to digital format and rate within the supported wavelength band, optical power, polarization, loss, crosstalk, and switching constraints.

OE Optical engine.

OMA Optical modulation amplitude. For binary NRZ, OMA $= P_1 - P_0$. For PAM4, outer OMA is $P_3 - P_0$; do not use $P_1 - P_0$ as the general PAM4 definition. Distinct from average optical power.

ORL Optical return loss. $\mathrm{ORL_{dB}} = -10\log_{10}(P_{\mathrm{reflected}}/P_{\mathrm{incident}})$. Higher ORL means less reflected power and is generally better. Low ORL can raise laser RIN and contribute to burst errors.

OSA Optical spectrum analyzer. An instrument that measures wavelength, SMSR, side-mode structure, and spectral width of a laser source.

OSFP Octal Small Form-factor Pluggable. A high-power faceplate module cage for 800G/1.6T/3.2T; larger thermal envelope than QSFP-DD.

O-to-E and E-to-O Optical to electrical interface and Electrical to optical interface, a component that converts an optical signal to an electrical signal or vice versa.

PAM Pulse amplitude modulation, a form of signal modulation where the message information is encoded in the amplitude of a series of signal pulses. For optical links it refers to intensity modulation.

PAM4 Pulse amplitude modulation-4 is a two-bit modulation that takes two bits at a time and maps the signal amplitude to one of four possible levels.

PAM6 Pulse-amplitude modulation with six levels. Information-theoretically, $\log_2 6 \approx 2.585$ bits per uncoded symbol. Practical systems require a defined coding or mapping method; do not treat "2.5 bits" as a complete system rate.

PAM8 A digital signal modulation scheme that encodes information by varying the amplitude of a pulse across eight distinct voltage levels. Each of these eight levels can represent 3 bits of data.

PCB/PCBA Printed circuit board (PCB) assembly, an assembly of electrical components built on a rigid glass-reinforced epoxy-based board.

PCS Physical coding sublayer. The IEEE 802.3 sublayer that performs 64B/66B line coding, 256B/257B transcoding, scrambling, and RS-FEC encoding.

PD Photodiode. A semiconductor device that converts photons to photocurrent (PIN, APD, or UTC variants).

PHY Physical layer. The combined PCS, PMA, and PMD that maps MAC frames to signals on the wire or fiber.

PIC Photonic integrated circuit. A chip integrating waveguides, modulators, detectors, and couplers (typically on SOI at 220 nm silicon).

PIN A p-intrinsic-n photodiode with no internal gain and low excess noise; Ge-on-Si waveguide PIN is a common short-reach detector.

PMA Physical medium attachment. The IEEE 802.3 sublayer that serializes, lane-multiplexes, and recovers clock. The SerDes lives here.

PMD Physical medium dependent. The IEEE 802.3 sublayer that modulates and detects light: the optical transceiver (laser, driver, PD, TIA).

PRBS Pseudo-random binary sequence. A deterministic test pattern that exercises all bit transitions; used for BER and eye measurements.

PSRR Power supply rejection ratio. A circuit's ability to suppress supply-rail noise; critical for laser bias drivers to avoid injecting RIN.

PUE Power usage effectiveness. Total facility power divided by IT equipment power; typical hyperscale values 1.1–1.3.

PVT Production Validation Test. Proves the factory can build the qualified design repeatedly at yield with ATP and SPC (Table 9.2).

QSFP-DD Quad Small Form-factor Pluggable Double Density. A faceplate module cage carrying eight electrical lanes for 400G/800G/1.6T.

RDMA Remote direct memory access. Zero-copy network transfer between accelerator memories; carried over RoCE (Ethernet) or native InfiniBand.

Repeater A low-latency electronic device that receives a signal and retransmits it. Repeaters are used to extend transmissions so that the signal can cover a longer distance. Besides signal equalization, clock and data recovery (CDR) functions could be also added to remove jitter from received signals effectively.

RIN Relative intensity noise. The laser's amplitude-noise spectral density (dB/Hz). Intrinsic RIN is a source property; RIN_xOMA is a standard-specific stressed-transmitter metric. RIN can create a BER floor when signal-proportional intensity noise dominates; not every BER floor is RIN-limited.

RLM Relative level mismatch. A measure of PAM4 level spacing uniformity (1.0 = perfectly even); CEI typically requires $\geq$0.95.

RMA Return merchandise authorization. A field-failed unit returned to the supplier for failure analysis. Distinct RMA codes keep FIT accounting honest.

RoCE RDMA over Converged Ethernet (CE) is a network protocol which allows remote direct memory access (RDMA) over an Ethernet network.

RS-FEC Reed–Solomon forward error correction. A block code on fixed-size symbols (for example KP4 RS(544,514) on 10-bit symbols). Distinct from the Ethernet RS sublayer.

RS sublayer Reconciliation Sublayer. Adapts the MAC to the PHY across the xMII. Distinct from RS-FEC.

RTLR (Retimed Transmit, Linear Receive) also generically referred to as Linear receive optic (LRO), is a type of optical transceiver technology used primarily in high-speed data center and networking applications, especially within AI clusters. The RTLR naming convention is within OIF. LRO is characterized by the presence of a Digital Signal Processor (DSP) solely on the transmit path, while the receive path operates with a linear, non-retimed architecture. This differs from fully retimed optical modules that use DSPs on both transmit and receive paths, and also from Linear Pluggable Optics (LPO) that eliminate DSPs entirely.

Scale-out Fabric connecting racks, systems, or scale-up domains (for example Ethernet or InfiniBand cluster networking).

Scale-up Tightly coupled accelerator fabric inside a scale-up domain (memory-coherent or high-bandwidth GPU/XPU interconnect). Not "adding RAM to one server."

SDO standard development organizations.

SECQ Stressed eye closure quaternary. A receiver-side metric that applies calibrated optical stress and reports remaining margin before BER threshold.

SerDes A Serializer/Deserializer is a pair of functional blocks commonly used in high-speed communications to transfer data over a relatively low number of lanes.

SFP Small form-factor pluggable connector is a modular, hot-pluggable interface used in networking devices to connect to various types of fiber optic or copper cables. It uses a PCB card edge interface.

SI Signal integrity is a set of measures of the quality of an electrical signal.

SiGe Silicon-germanium BiCMOS. A high-f_T semiconductor process for TIAs and modulator drivers operating at 100–200+ GHz bandwidth.

SiPh Silicon photonics. Waveguides and modulators on silicon-on-insulator; CMOS fab-compatible and widely used for DR/FR and CPO engines.

SMF Single-mode fiber. Fiber with a $\sim$9 μm core carrying one spatial mode; G.652.D (standard) and G.657 (bend-insensitive) for datacenter plant.

SMSR Side-mode suppression ratio. The power difference (dB) between the dominant lasing mode and the strongest side mode on an OSA.

SNDR Signal-to-noise-and-distortion ratio is a measurement of the purity of a signal.

SNR Signal-to-noise ratio.

SOA Semiconductor optical amplifier. A III-V inline optical gain block; adds ASE noise. Used as a receiver preamplifier for sensitivity.

SOI Silicon on insulator. A wafer substrate for silicon photonics: a thin ($\sim$220 nm) silicon layer on buried oxide confines the optical mode.

SPC Statistical process control. Tracks process distributions and trends to catch drift before escapes ship.

SR Short reach. An IEEE 802.3 multimode fiber class, typically $\leq$100 m over OM4; VCSEL at 850 nm.

TBD To be determined.

Tbps Terabits per second. The throughput or data rate of a port or piece of equipment. Tbps is 1×10^{12} bits per second.

TCM Trellis coded modulation is a technique that combines convolutional coding and modulation to improve data transmission efficiency over bandwidth-limited channels, like telephone lines. It achieves this by intelligently integrating the encoding and modulation processes, increasing the distance between signal points in the constellation to enhance error correction without expanding bandwidth.

TCO Total cost of ownership. Acquisition (BOM, yield) plus lifetime energy plus service cost (FIT $\times$ replacement + labor).

TDECQ Transmitter and dispersion eye closure quaternary. The headline PAM4 transmitter-quality metric; measured after a reference receiver and bounded FFE, including a test fiber.

TEC Thermoelectric cooler. A Peltier device that holds a laser junction or ring at a controlled temperature against case thermal variations.

TECQ Transmitter eye closure quaternary. Same measurement as TDECQ without the test fiber; used alongside TDECQ in LPO MSA OMA tables.

TFLN Thin-film lithium niobate. A sub-micrometer LN layer on a silicon or silica handle; EO bandwidth can exceed 110 GHz and is a candidate path for native 224 GBd PAM4.

TIA Transimpedance amplifier. Converts photodiode current to voltage with low input-referred noise; co-packaging with the PD minimizes capacitance and noise.

TME Test and measurement equipment.

TP Test point. A defined reference plane where a link measurement is taken (TP0 through TP5, plus TP1a/TP4a).

TRO Transmit retimed optic. A module with DSP only on the transmit (electrical-to-optical) path; the receive path is linear.

TWE Traveling-wave electrode. An RF transmission line on a modulator that velocity-matches the electrical and optical group indices for high EO bandwidth.

TWI Two-wire interface. An I2C-like serial bus (SCL + SDA) used for CMIS module management communication.

Twinax copper cable A type of copper cable similar to coaxial cable, but with two inner conductors instead of one.

UALink Ultra Accelerator Link. An open scale-up fabric specification for direct accelerator memory access (200G/lane class; $\sim$400G/lane next).

UCIe Universal chiplet interconnect express is an open industry standard that defines the interconnect between chiplets, or small component dies, within a single package.

UEC Ultra Ethernet Consortium. Builds an open Ethernet-based stack (UET transport, PHY) optimized for AI/HPC scale-out with cluster-scale congestion control.

UI Unit interval. One symbol period ($= 1/R_{\mathrm{sym}}$); jitter budgets and eye diagrams are referenced in UI.

UTC / MUTC Uni-traveling-carrier (modified UTC) photodiode. Uses electron-only transport for >200 GHz bandwidth and high saturation current; linear/LPO niche.

VCSEL Vertical-cavity surface-emitting laser. 850 nm class source for multimode short-reach links.

VLC Vertical line card. A new line card design in which vertical I/O connectors and ASIC are mounted side by side, reducing the signal trace distance.

VNA Vector network analyzer. Measures electrical or electro-optic *S*-parameters versus frequency for bandwidth and reflection.

VOA Variable optical attenuator. A calibrated loss element used for sensitivity sweeps and stressed-receiver testing.

VSR Very short reach. CEI VSR specifies chip-to-module electrical interfaces.

WDM Wavelength division multiplexing. Sending multiple wavelengths on one fiber; CWDM (20 nm spacing) or DWDM ($\leq$100 GHz spacing).

XPO eXtra-dense Pluggable Optics. A liquid-cooled, 12.8 Tb/s faceplate pluggable module (Arista MSA, OFC 2026); front-panel serviceable at CPO-class density.

XSR Extra short reach. CEI XSR specifies die-to-optical engine (D2OE) and die-to-die (D2D) electrical interfaces.

G.2 Symbols and units

dB Decibel. A relative power or ratio unit: $10 \log_{10}(P_a/P_b)$ for power ratios.

dBm Absolute optical or electrical power relative to 1 mW: $10 \log_{10}(P/1\ \mathrm{mW})$.

GBd Gigabaud. 10^9 symbols per second (symbol rate).

Gb/s Gigabits per second. Bit rate; for PAM4, roughly $2\times$ baud before coding details.

UI Unit interval. One symbol period ($= 1/R_{\mathrm{sym}}$).

BER Bit error ratio. Erroneous bits divided by bits observed; state pre-FEC or post-FEC.

SER Symbol error ratio. Erroneous symbols divided by symbols observed; related to but not identical to BER.

OMA Optical modulation amplitude (power units, often dBm when converted). See glossary entry.

Average optical power Mean optical power over the pattern; not OMA.

Wavelength spacing Channel separation in nm; not constant when frequency spacing is constant.

Frequency spacing Channel separation in GHz (for example LAN-WDM or DWDM grids).

FIT Failures per 10^9 device-hours. See glossary entry.

DPPM Defective parts per million. See glossary entry.

pJ/bit Picojoules per bit. Energy per bit; always name the measurement boundary.

G.3 Do not confuse

BER / SER / FLR BER counts bits; SER counts symbols; FLR (frame-loss ratio) is a post-FEC packet/frame metric. State which one and the confidence window.

OMA / average optical power OMA is the outer swing; average power is the mean. Same dBm number is not interchangeable.

TDECQ / SECQ / EECQ TDECQ is Tx optical with test fiber; TECQ omits that fiber; SECQ is stressed Rx optical; EECQ is electrical eye closure at host test points.

DVT / PVT / EVT EVT is early engineering validation; DVT proves the design; PVT proves the factory can build it (Table 9.2).

FIT / DPPM FIT is a rate per device-hours; DPPM is a fraction of units defective in a population and window.

CPO / NPO / OBO CPO mounts optics on the host package; NPO is adjacent on the PCB; OBO/COBO is mid-board on the PCB.

LPO / LRO / TRO / RTLR LPO is linear both directions; LRO/RTLR retimes Tx only (linear Rx); TRO retimes Tx (linear Rx) under that naming. Draw the host electrical Tx/Rx path rather than expanding acronyms alone (Chapter 10).

RS sublayer / RS-FEC RS sublayer is Reconciliation; RS-FEC is Reed–Solomon forward error correction in the PCS.

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[74] Cignal AI, *The Linear Drive Market Opportunity* (2023) and OFC 2025 show report. Market view: LPO stays a small share, especially at 800G, because the installed fabric is DSP-based; 100G/lane 800GbE LPO late to market; at 1.6T, LRO favored over LPO on power/thermal. Provisional (analyst reports).

[75] Eoptolink, *Industry-first 200G-per-lane LPO; 100G/lane 800G LPO entering mass production*, March 2024. 200G/λ four-channel LPO, no DSP/CDR; Gen2 100G/lane 800G and 400G SMF (OSFP, QSFP-DD, QSFP112), full TP2 compliance claimed. Vendor, provisional.

[76] Macom, *Macom to showcase 200G-per-lane products at OFC*, March 2024. PURE DRIVE 200 Gb/s LPO, extensible to 212 Gb/s/lane for 1.6T; linear TIA and driver as the key blocks. Vendor, provisional.

[77] Marvell, *Marvell introduces 1.6 Tbps LPO chipset*, December 2024. 200G/lane TIA and laser-driver chipset for 800G and 1.6T LPO, targeting scale-up XPU compute fabrics. Vendor, provisional.

[78] Alphawave Semi and InnoLight, *Low-latency LPO with a PCIe 6.0 subsystem for AI infrastructure*, March 2024. 64 Gb/s/lane PCIe 6.0 controller+PHY over InnoLight LPO OSFP optics (OFC 2024); later a 128 Gb/s/lane PCIe 7.0-ready platform. Vendor, provisional.

[79] LPO MSA, *100G-DR-LPO Specification*, rev. 1.0, 19 March 2025. 100 Gb/s/lane PAM4 SMF to 500 m; builds on IEEE 802.3 and OIF CEI-112G-LINEAR. TDECQ/TECQ ≤3.4 dB; OMA coupled to max(TECQ,TDECQ); host RS(544,514) FEC. Template for linear-module optical limits; 224G linear modules follow CEI-224G-Linear electrically while optical PMDs track 802.3dj.

[80] Telcordia. *GR-468-CORE: Generic Reliability Assurance Requirements for Optoelectronic Devices*.

[81] Telcordia. *GR-1221-CORE: Generic Reliability Assurance Requirements for Passive Optical Components*, Issue 2 (January 1999). Passive-component companion to GR-468: connectors, couplers, WDM filters, splitters, and isolators. Same stress families (damp heat, temperature cycle, mechanical, aging) scored on insertion and return loss rather than LIV.

[82] Broadcom Inc. *Tomahawk 6 – Davisson: 102.4 Tb/s co-packaged-optics Ethernet switch* (press release and briefs). October 2025.

[83] NVIDIA. *Spectrum-X and Quantum-X silicon photonics co-packaged-optics switches* (GTC 2025; Hot Chips 2025, Shainer).

[84] TSMC. *Compact Universal Photonic Engine (COUPE): SoIC-X electronic/ photonic integration; mass-production milestone*. 2026.

[85] Optical Internetworking Forum. OIF Implementation Agreement for 400ZR. 400 Gb/s coherent DWDM pluggable; 15 W module power budget; DCI reach to ∼120 km; interoperable QSFP-DD/OSFP coherent optics.

[86] Optical Internetworking Forum. CMIS 5.3 Implementation Agreement (OIF-CMIS-05.3). Host-to-module management over two-wire bus; module and data-path state machines; DDM, VDM, CDB; link training; ELSFP and CPO extensions.

[87] DMTF, OpenConfig, and the SONiC community. DMTF Redfish (DSP0266), OpenConfig models and gNMI, and SONiC. Box- and fleet-level management and telemetry above OIF CMIS. Redfish: RESTful server/switch management API (v1.0, August 2015). OpenConfig: vendor-neutral YANG models and streaming telemetry. SONiC: open network OS with dataplane telemetry. Where per-module optical monitors surface at cluster scale.

[88] IEC. IEC 60825-1:2014/AMD1:2021 and IEC 60825-2:2021. Laser product classes (1, 1M, 3R, 3B, 4) and OFCS hazard levels at accessible fiber ports.

[89] ITU-T. Recommendation ITU-T G.664 (02/2012). Optical safety procedures; APR with automatic restart; reduce to Hazard Level 1M within 3 s of continuity loss.

[90] ITU-T. Recommendation ITU-T G.694.1 (10/2020), Spectral grids for WDM applications: DWDM frequency grid and ITU-T G.694.2 (12/2003), CWDM wavelength grid. G.694.1 anchors the DWDM grid at 193.1 THz with 12.5/25/50/100 GHz spacings and integer multiples; G.694.2 defines the 20 nm CWDM grid, 18 channels from 1271 to 1611 nm. Datacom DWDM locks to G.694.1; CWDM4 uses four O-band lines of G.694.2.

[91] ITU-T. Recommendation ITU-T G.652 (11/2016), Characteristics of a single-mode optical fibre and cable and ITU-T G.657 (08/2024), Characteristics of a bending-loss insensitive single-mode optical fibre and cable. G.652.D sets the standard single-mode fiber used in short-reach links: attenuation $\approx$0.2 dB/km at 1550 nm, $\approx$0.32 dB/km at 1310 nm, and zero-dispersion wavelength near 1310 nm. G.657 defines bend-insensitive fiber for tight-radius routing in patch panels and shelves; category A stays G.652-compatible, category B trades some standard-fiber interoperability for tighter bend radii.

[92] ISO/IEC and TIA. ISO/IEC 11801:2017, Information technology: Generic cabling for customer premises and the TIA-492 series, Detail Specifications for Optical Fibers. ISO/IEC 11801 defines the cabled fiber performance classes used in datacenter patch plant: OM1–OM5 for multimode and OS1/OS2 for single-mode. TIA-492 gives the matching detailed fiber specifications (for example TIA-492AAAD corresponds to OM4); TIA-568 adopts the same OM/OS class names for structured cabling.

[93] JEDEC Solid State Technology Association. JESD47M, Stress-Test-Driven Qualification of Integrated Circuits. Published August 2025. Baseline acceptance-test flow (temperature cycle, HTOL/HTSL, autoclave/HAST, mechanical shock and vibration) for qualifying an IC as a new product, a product family, or after a process change. The silicon-side counterpart to Telcordia GR-468-CORE for optoelectronics.

[94] JEDEC Solid State Technology Association and ESD Association. ANSI/ESDA/JEDEC JS-001-2024, Human Body Model (HBM) ESD Sensitivity Testing and JS-002-2025, Charged Device Model (CDM) ESD Sensitivity Testing. Component-level ESD classification for driver, TIA, and DSP silicon. Supersede the older JEDEC-only JESD22-A114 (HBM) and JESD22-C101 (CDM) test methods.

[95] JEDEC Solid State Technology Association. JESD78F.02, IC Latch-Up Test. Published November 2023. Overvoltage and ± 100 mA current-injection test method for parasitic-thyristor latch-up susceptibility in CMOS, bipolar, and BiCMOS ICs; the failure mode COD is to a laser facet.

[96] Automotive Electronics Council. AEC-Q100 Rev-J, Failure Mechanism Based Stress Test Qualification for Integrated Circuits. Automotive-grade IC qualification built on the

JEDEC JESD47/JESD22 methods with tightened ESD targets and temperature grades (Grade 3: -40 to 85°C through Grade 0: -40 to 150°C). Not required for datacenter optics, but a useful borrow when a driver or TIA die also ships in an automotive part number and the datasheet cites a Q100 grade.

[97] IEC. IEC 61754-7-1:2014, Fibre optic connector interfaces – Type MPO connector family: one fibre row and IEC 61754-7-2:2017, Type MPO connector family: two fibre rows. Mechanical interface standard for the MPO/MT connector family: 6.4 mm $\times$ 2.5 mm rectangular ferrule, guide-pin alignment, 8/12/16/24-fiber counts. Superseded the single-part IEC 61754-7:2008 to add active-device receptacles and up-angled plugs.

[98] IEC. IEC 61300-2-2:2009, Basic test and measurement procedures – Mating durability and IEC 61300-3-35:2022, Visual inspection of fibre optic connectors and fibre-stub transceivers. IEC 61300-2-2 defines the mate/unmate cycling test method that connector datasheets rate against; TIA-568.3 sets 500 cycles as the structured-cabling durability floor, and MPO/MTP-class connectors are commonly rated well above 1000. IEC 61300-3-35 grades endface scratches, pits, and debris against zones on the fiber core and cladding for pass/fail inspection.

[99] Concatenated RS-BCH coded modulation latency study (incl. KP4). Performance-complexity-latency trade-offs of concatenated codes, arXiv:2402.09364, 2024; and vendor RS(544,514) implementation notes. KP4 RS(544,514) decode latency in practice $\approx$20–100 ns depending on implementation and clock; encode is comparable. Academic/vendor figures, provisional.

[100] Vendor switch datasheets (Ethernet and InfiniBand). 800GbE 51.2 Tb/s SONiC switch, $\sim$560 ns port-to-port; NVIDIA Quantum-X800 InfiniBand advertises sub-100 ns port-to-port with in-network compute. Per-hop O-E-O switch latency: a few hundred ns for cut-through Ethernet, under $\sim$100 ns for InfiniBand. Vendor orientation, provisional.

[101] Y. Chen, M. N. Petrovich, E. Numkam Fokoua, *et al*. (Southampton/Microsoft). Record-low-loss hollow-core fibre (double nested antiresonant nodeless fibre, DNANF). *Nature Photonics*, September 2025: 0.091 dB/km at 1550 nm (below the $\sim$0.14 dB/km silica floor), $\approx$0.2 dB/km over a 66 THz window; air guiding gives group index near 1.0, $\sim$45% faster propagation (about a third lower latency) than silica. Microsoft reports Azure network deployment. Record result, provisional.

[102] Quantum-dot mode-locked laser (QD-MLL) frequency combs for datacom. High-efficiency QD lasers as comb sources for DWDM, *Appl. Sci.* 12(4):1836 (2022): among DFB arrays, nonlinear microcombs, and semiconductor MLLs, QD-MLLs currently offer the best efficiency, simplicity, and size for a single-device comb. O-band IM/DD demos: a 1.3 µm InAs/InGaAs QD-MLL comb carrying 14$\times$100 Gb/s PAM4 over 10 km SMF at $\sim$284 fJ/bit; and an isolator-free QD-MLL comb reported for interconnects beyond 3.2 Tb/s (arXiv:2506.02402, 2025). Research demos, provisional.

[103] Integrated Kerr/soliton microcomb multi-wavelength sources. Integrated multi-port multi-wavelength coherent optical source, *Nat. Commun.* 16 (2025): a Kerr microcomb with a monolithically integrated demultiplexer that autonomously locks to and tracks the comb lines, aimed at beyond-Tb/s links. A microcomb turns one pump into hundreds of evenly spaced lines on a chip, but per-line power is low (pump-conversion efficiency is modest), so a booster or per-line SOA is usually required (Section 6.6.2). Research demos, provisional.

[104] Semiconductor optical amplifiers for datacom booster/distribution use. Commercial O-band SOA modules quote $\sim$15 dB gain, noise figure $\leq$7 dB, and $\leq$1.5 dB polarization-dependent gain (Anritsu SOA datasheet); the quantum NF floor is 3 dB. O-band quantum-dot SOAs grown on a CMOS-compatible silicon substrate (*ACS Photonics*,

2019) give wide O-band gain with low noise for co-packaged integration. An SOA restores launch power after comb generation and fan-out, at the cost of added ASE. Vendor/academic figures.

[105] Co-packaged and near-package copper, copper's answer to the same reach wall that drives CPO. Molex co-packaged copper (CPC): on-substrate routing validated at 224 Gb/s PAM4 with a stated roadmap to 448G, aimed at short scale-up runs where power, latency, and cost favor copper over optics. The Impress compression substrate connector and mating cable support 224 Gb/s PAM4 and beyond. Marvell 224G long-reach SerDes reports $\sim$4 pJ/bit and targets CPC cables, near-packaged optics, and CPO. Near-package copper (NPC) mates the connector near the ASIC rather than onto the package substrate; shared NPC/NPO sockets have been proposed. Vendor orientation, provisional.

[106] FiberMall. OSFP thermal management: data center cooling guide. 2026. A 32-port 800G switch dissipates >1 kW, with optical modules $\sim$50% of the thermal load; observed power runs 15–25% above rated. Vendor orientation, provisional.

[107] Cao, *et al*. Simulation and experimental investigation of liquid-cooling thermal management for high-bandwidth co-packaged optics. Front. Optoelectron., 2025. Power density and on-package thermal crosstalk limit CPO reliability; liquid cooling investigated to hold optical-engine and laser temperature.

[108] Semtech. Low-power 1.6T datacom transceivers and the path to 3.2T. Webinar recap, 2025. Linear modules cited at $\sim$50–60% less power than fully retimed; a 500,000-GPU cluster estimated to save >100 MW and $\sim$\$100 million/year in electricity. Vendor orientation, provisional.

[109] Epoch AI. Total cost of ownership of a one-gigawatt AI data center. Data insight, 2025. A 1 GW AI site estimated near \$38 billion up-front capex and $\sim$\$0.9 billion/year opex. Analyst breakdown, provisional.